CHAPTER

23

COMMUNICATIONS

CHAPTER 23
COMMUNICATIONS

Subject/Page	Date	COC
23-EFFECTIVE PAGES		
1 thru 6	Sep 15/2026	
23-CONTENTS		
1	Jan 15/2025	
2	May 15/2025	
3	May 15/2026	
4	May 15/2026	
5	May 15/2026	
6	May 15/2026	
7	May 15/2026	
8	May 15/2026	
9	May 15/2026	
10	May 15/2026	
11	May 15/2026	
12	May 15/2026	
13	May 15/2026	
14	May 15/2026	
15	May 15/2026	
16	May 15/2026	
17	May 15/2026	
18	BLANK	
23-00-00		
901	Jan 15/2025	
902	May 15/2024	
903	May 15/2024	
904	May 15/2024	
905	May 15/2024	
906	May 15/2024	
907	May 15/2024	
908	May 15/2024	
909	May 15/2024	
910	Jan 15/2025	
911	Jan 15/2025	
912	BLANK	
23-11-00		
201	May 15/2025	
202	May 15/2025	
203	May 15/2025	
23-11-00 (cont)		
204	May 15/2025	
23-11-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	Jan 15/2025	
505	Jan 15/2025	
506	Jan 15/2025	
23-11-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2025	
405	May 15/2025	
406	BLANK	
23-11-51		
401	May 15/2024	
402	May 15/2024	
23-11-51		
601	May 15/2026	
602	May 15/2024	
603	May 15/2024	
604	BLANK	
23-11-61		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	Sep 15/2024	
406	May 15/2024	
407	Jan 15/2025	
408	May 15/2024	
409	May 15/2024	
410	BLANK	
23-12-00		
501	May 15/2024	
502	Jan 15/2026	
23-12-00 (cont)		
503	Jan 15/2026	
504	Jan 15/2026	
23-12-11		
401	May 15/2025	
402	May 15/2025	
403	May 15/2026	
R 404	Sep 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	BLANK	
23-12-11		
601	May 15/2026	
602	May 15/2026	
603	May 15/2026	
604	BLANK	
23-12-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
23-12-41		
201	May 15/2024	
202	May 15/2024	
23-12-41		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

23-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

Subject/Page	Date	COC
23-15-00		
201	Jan 15/2025	
202	Jan 15/2025	
203	Jan 15/2025	
204	Jan 15/2025	
205	Jan 15/2025	
206	Jan 15/2025	
23-15-00		
501	Jan 15/2026	
502	Jan 15/2026	
503	Jan 15/2026	
504	Jan 15/2026	
505	Jan 15/2026	
506	Jan 15/2026	
507	Jan 15/2026	
508	BLANK	
23-15-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2026	
405	Jan 15/2025	
406	BLANK	
23-15-12		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Jan 15/2025	
405	Jan 15/2025	
406	BLANK	
23-15-25		
401	Sep 15/2025	
402	Sep 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
R 406	Sep 15/2026	
23-24-00		
201	Jan 15/2026	
202	Jan 15/2026	
203	Jan 15/2026	
204	Jan 15/2026	
205	Jan 15/2026	
206	Jan 15/2026	
207	Jan 15/2026	
208	Jan 15/2026	
209	Jan 15/2026	
210	Jan 15/2026	
211	Jan 15/2026	
212	Jan 15/2026	
23-24-00		
401	Sep 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Sep 15/2025	
405	Sep 15/2024	
406	Sep 15/2025	
407	Sep 15/2025	
408	BLANK	
23-24-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2025	
504	May 15/2024	
505	May 15/2025	
506	May 15/2024	
23-24-02		
401	May 15/2025	
402	May 15/2025	
403	Jan 15/2025	
R 404	Sep 15/2026	
405	May 15/2025	
R 406	Sep 15/2026	
23-24-03		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
23-27-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
23-27-00		
501	May 15/2024	
502	Jan 15/2025	
503	Jan 15/2025	
504	Jan 15/2025	
505	Jan 15/2025	
506	Jan 15/2025	
507	Jan 15/2025	
508	Jan 15/2025	
509	Jan 15/2025	
510	BLANK	
23-27-33		
201	Jan 15/2025	
202	Jan 15/2025	
203	Jan 15/2025	
204	BLANK	
23-27-33		
401	Jan 15/2025	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
23-27-35		
201	May 15/2024	
202	May 15/2024	
R 203	Sep 15/2026	
O 204	Sep 15/2026	

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23-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

Subject/Page	Date	COC
23-27-35		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
23-28-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
23-28-00		
R 501	Sep 15/2026	
R 502	Sep 15/2026	
R 503	Sep 15/2026	
504	BLANK	
23-28-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Sep 15/2024	
405	Sep 15/2024	
406	BLANK	
23-31-00		
501	Jan 15/2025	
502	Jan 15/2025	
503	Jan 15/2025	
504	Jan 15/2025	
505	May 15/2025	
506	Sep 15/2025	
507	May 15/2025	
508	May 15/2025	
509	May 15/2025	
23-31-00 (cont)		
510	May 15/2025	
23-31-01		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	BLANK	
23-31-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2025	
404	BLANK	
23-31-02		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	BLANK	
23-31-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
23-31-03		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	BLANK	
23-31-03		
401	May 15/2026	
402	May 15/2024	
403	May 15/2024	
404	May 15/2026	
405	May 15/2026	
406	BLANK	
23-31-05		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	BLANK	
23-31-05		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
23-31-07		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
209	May 15/2024	
210	BLANK	
23-31-07		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
23-41-00		
R 501	Sep 15/2026	
O 502	Sep 15/2026	
O 503	Sep 15/2026	
504	BLANK	

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23-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

Subject/Page	Date	COC
23-42-00		
501	May 15/2024	
502	May 15/2024	
23-42-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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23-42-01		
501	May 15/2024	
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23-42-03		
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202	Jan 15/2025	
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205	Jan 15/2025	
206	Jan 15/2025	
207	Jan 15/2025	
208	Jan 15/2025	
23-42-03		
401	May 15/2025	
402	May 15/2025	
403	May 15/2025	
404	May 15/2024	
405	May 15/2025	
406	May 15/2025	
407	May 15/2025	
408	May 15/2025	
409	May 15/2025	
410	BLANK	
23-42-03		
501	May 15/2025	
502	May 15/2024	
23-42-03		
701	May 15/2024	
702	BLANK	
23-42-04		
401	May 15/2024	
402	May 15/2024	
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23-43-00		
501	May 15/2024	
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23-43-02		
401	Sep 15/2025	
402	Sep 15/2025	
403	Sep 15/2025	
404	Sep 15/2025	
405	Sep 15/2025	
406	Sep 15/2025	
23-43-03		
401	Jan 15/2026	
402	Jan 15/2026	
403	May 15/2024	
404	Jan 15/2026	
405	Jan 15/2026	
406	BLANK	
23-51-00		
501	Sep 15/2024	
502	Sep 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
R 507	Sep 15/2026	
R 508	Sep 15/2026	
23-51-01		
401	Jan 15/2026	
402	Jan 15/2026	
403	May 15/2024	
404	Jan 15/2026	
405	Jan 15/2026	
406	Jan 15/2026	
23-51-02		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
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23-51-02		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
23-51-03		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
23-51-04		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	May 15/2026	
23-51-04		
501	May 15/2026	
502	May 15/2024	
503	May 15/2024	
504	BLANK	

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23-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

Subject/Page	Date	COC
23-51-05		
401	Sep 15/2024	
402	Sep 15/2024	
403	May 15/2024	
404	Sep 15/2024	
405	Sep 15/2024	
406	Sep 15/2024	
23-61-00		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2025	
407	Jan 15/2025	
408	May 15/2025	
409	May 15/2025	
410	May 15/2025	
411	May 15/2026	
412	May 15/2025	
413	May 15/2025	
414	May 15/2025	
415	Jan 15/2025	
R 416	Sep 15/2026	
417	Jan 15/2025	
R 418	Sep 15/2026	
419	May 15/2025	
420	May 15/2025	
421	May 15/2025	
422	May 15/2025	
R 423	Sep 15/2026	
O 424	Sep 15/2026	
R 425	Sep 15/2026	
426	May 15/2026	
427	May 15/2026	
428	May 15/2026	
429	May 15/2026	
23-61-00 (cont)		
430	May 15/2026	
23-61-00		
601	May 15/2024	
602	Sep 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2026	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	
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611	May 15/2024	
612	May 15/2024	
613	May 15/2024	
614	May 15/2024	
23-71-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2025	
205	Sep 15/2025	
206	May 15/2025	
207	May 15/2025	
208	May 15/2024	
209	May 15/2024	
210	BLANK	
23-71-00		
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503	Jan 15/2025	
504	Jan 15/2025	
505	Sep 15/2025	
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508	Sep 15/2025	
23-71-11		
401	May 15/2024	
402	Jan 15/2025	
403	May 15/2024	
404	Jan 15/2025	
405	May 15/2025	
406	BLANK	
23-71-12		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
23-71-13		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
23-71-21		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
209	May 15/2024	
210	May 15/2024	
23-71-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
23-71-21		
501	Sep 15/2024	
502	Sep 15/2024	

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23-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
23-71-21 (cont)		
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
R 506	Sep 15/2026	
O 507	Sep 15/2026	
508	May 15/2024	
23-71-21		
701	May 15/2024	
702	BLANK	
23-75-00		
201	May 15/2024	
202	May 15/2024	
23-75-00		
501	May 15/2024	
502	BLANK	
23-75-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
23-75-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	BLANK	

Subject/Page	Date	COC

Subject/Page	Date	COC

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23-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
COMMUNICATION - DDG MAINTENANCE PROCEDURES	23-00-00		901	PTF ALL
DDG 23-24-01 Fixed Emergency Locator Transmitter (ELT) Inoperative - Preparation TASK 23-00-00-040-807			901	PTF ALL
DDG 23-24-01 Fixed Emergency Locator Transmitter (ELT) Inoperative - Restoration TASK 23-00-00-440-807			902	PTF ALL
DDG 23-24-01-01 Remote Emergency Locator Transmitter (ELT) Switch Inoperative - Preparation TASK 23-00-00-040-813			903	PTF ALL
DDG 23-24-01-01 Remote Emergency Locator Transmitter (ELT) Switch Inoperative - Restoration TASK 23-00-00-440-813			904	PTF ALL
DDG 23-51-04-01 Control Wheel Push-To-Talk (PTT) Switches Inoperative - Preparation TASK 23-00-00-040-808			904	PTF ALL
DDG 23-51-04-01 Control Wheel Push-To-Talk (PTT) Switches Inoperative - Restoration TASK 23-00-00-440-808			905	PTF ALL
DDG 23-51-04-02 Audio Control Panel (ACP) Push-To-Talk (PTT) Switches Inoperative - Preparation TASK 23-00-00-040-809			906	PTF ALL
DDG 23-51-04-03 Glareshield Panel Push-To-Talk (PTT) Switches Inoperative - Preparation TASK 23-00-00-040-810			907	PTF ALL
DDG 23-51-04-03 Glareshield Panel Push-To-Talk (PTT) Switches Inoperative - Restoration TASK 23-00-00-440-810			908	PTF ALL
DDG 23-71-01-02 Aircraft with Recorder Independent Power Supply (RIPS) Inoperative (EASA) - Preparation TASK 23-00-00-040-811			910	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 23-71-01-02 Aircraft with Recorder Independent Power Supply (RIPS) Inoperative (EASA) - Restoration TASK 23-00-00-440-811			910	PTF ALL
HIGH FREQUENCY (HF) COMMUNICATION SYSTEM - MAINTENANCE PRACTICES	23-11-00		201	PTF ALL
High Frequency Communication System - Deactivation TASK 23-11-00-040-801			201	PTF ALL
High Frequency Communication System - Activation TASK 23-11-00-440-801			204	PTF ALL
HIGH FREQUENCY (HF) COMMUNICATION SYSTEM - ADJUSTMENT/TEST	23-11-00		501	PTF ALL
HF Communication System - Operational Test TASK 23-11-00-710-801			501	PTF ALL
HF Communication System - System Test TASK 23-11-00-730-801			503	PTF ALL
HF TRANSCEIVER - REMOVAL/INSTALLATION	23-11-21		401	PTF ALL
HF Transceiver Removal TASK 23-11-21-000-801			401	PTF ALL
HF Transceiver Installation TASK 23-11-21-400-801			404	PTF ALL
HF ANTENNA - REMOVAL/INSTALLATION	23-11-51		401	PTF ALL
HF Antenna Removal TASK 23-11-51-000-801			401	PTF ALL
HF Antenna Installation TASK 23-11-51-400-801			402	PTF ALL
HF ANTENNA - INSPECTION/CHECK	23-11-51		601	PTF ALL
HF Antenna Electrical Bond Check TASK 23-11-51-760-801			601	PTF ALL
HF ANTENNA COUPLER - REMOVAL/INSTALLATION	23-11-61		401	PTF ALL
HF Antenna Coupler Removal TASK 23-11-61-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
HF Antenna Coupler Installation TASK 23-11-61-400-801			405	PTF ALL
VERY HIGH FREQUENCY (VHF) COMMUNICATION SYSTEM - ADJUSTMENT/TEST	23-12-00		501	PTF ALL
VHF Communication System - Operational Test TASK 23-12-00-710-801			501	PTF ALL
VHF Communication System - System Test TASK 23-12-00-730-801			502	PTF ALL
VHF COMMUNICATION ANTENNA - REMOVAL/INSTALLATION	23-12-11		401	PTF ALL
VHF Communication Antenna Removal TASK 23-12-11-000-801			401	PTF ALL
VHF Communication Antenna Installation TASK 23-12-11-400-801			404	PTF ALL
VHF COMMUNICATION ANTENNA - INSPECTION	23-12-11		601	PTF ALL
VHF Communication Antenna - Inspection TASK 23-12-11-210-801			601	PTF ALL
VHF COMMUNICATION TRANSCEIVER - REMOVAL/INSTALLATION	23-12-21		401	PTF ALL
VHF Communication Transceiver Removal TASK 23-12-21-000-801			401	PTF ALL
VHF Communication Transceiver Installation TASK 23-12-21-400-801			405	PTF ALL
RADIO TUNING PANEL - MAINTENANCE PRACTICES	23-12-41		201	PTF ALL
Radio Tuning Panel - INOP Display Toggle TASK 23-12-41-860-801			201	PTF ALL
RADIO TUNING PANEL - REMOVAL/INSTALLATION	23-12-41		401	PTF ALL
Radio Tuning Panel (RTP) Removal TASK 23-12-41-000-801			401	PTF ALL
Radio Tuning Panel (RTP) Installation TASK 23-12-41-400-801			404	PTF ALL

23-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SATCOM SYSTEM - MAINTENANCE PRACTICES	23-15-00		201	PTF ALL
Satcom System - Deactivation TASK 23-15-00-040-801			201	PTF ALL
Satcom System - Activation TASK 23-15-00-440-801			201	PTF ALL
Satcom System - SDU Software Installation with ONS TASK 23-15-00-470-801			202	PTF ALL
Satcom System - SDU Configuration Check TASK 23-15-00-700-801			203	PTF ALL
Satcom System - SCM Configuration Check TASK 23-15-00-700-802			204	PTF ALL
SATELLITE COMMUNICATIONS (SATCOM) SYSTEM - ADJUSTMENT/TEST	23-15-00		501	PTF ALL
Satellite Communications (SATCOM) System - BITE Test at the Flight Management Computer Control Display Unit TASK 23-15-00-740-801			501	PTF ALL
Satellite Communications (SATCOM) System - BITE Test at the Satellite Data Unit TASK 23-15-00-740-802			502	PTF ALL
Satellite Communications (SATCOM) System - Operational Test TASK 23-15-00-710-801			502	PTF ALL
Satellite Communications (SATCOM) System - System Test TASK 23-15-00-710-802			506	PTF ALL
SATELLITE DATA UNIT - REMOVAL/INSTALLATION	23-15-01		401	PTF ALL
Satellite Data Unit Removal TASK 23-15-01-000-801			401	PTF ALL
Satellite Data Unit Installation TASK 23-15-01-400-801			404	PTF ALL
SDU CONFIGURATION MODULE - REMOVAL/INSTALLATION	23-15-12		401	PTF ALL
SDU Configuration Module Removal TASK 23-15-12-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SDU Configuration Module Installation TASK 23-15-12-400-801			404	PTF ALL
SATCOM ANTENNA - REMOVAL/INSTALLATION	23-15-25		401	PTF ALL
SATCOM Antenna Removal TASK 23-15-25-000-802			401	PTF ALL
SATCOM Antenna Installation TASK 23-15-25-400-802			404	PTF ALL
EMERGENCY LOCATOR TRANSMITTER SYSTEM - MAINTENANCE PRACTICES	23-24-00		201	PTF ALL
Emergency Locator Transmitter Deactivation TASK 23-24-00-040-802			201	PTF ALL
Emergency Locator Transmitter Activation TASK 23-24-00-440-802			202	PTF ALL
ELT Programming TASK 23-24-00-860-801			202	PTF ALL
Emergency Locator Transmitter Battery Replacement TASK 23-24-00-960-801			210	PTF ALL
EMERGENCY LOCATOR TRANSMITTER - REMOVAL/INSTALLATION	23-24-00		401	PTF ALL
Emergency Locator Transmitter Removal TASK 23-24-00-000-801			401	PTF ALL
Emergency Locator Transmitter Installation TASK 23-24-00-400-801			404	PTF ALL
ELT/Navigation Interface Unit Removal TASK 23-24-00-000-802			405	PTF ALL
ELT/Navigation Interface Unit Installation TASK 23-24-00-400-802			406	PTF ALL
EMERGENCY LOCATOR TRANSMITTER - ADJUSTMENT/TEST	23-24-00		501	PTF ALL
Emergency Locator Transmitter Operational Test TASK 23-24-00-710-801			501	PTF ALL
Emergency Locator Transmitter System Test TASK 23-24-00-730-801			503	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
EMERGENCY LOCATOR TRANSMITTER ANTENNA - REMOVAL/INSTALLATION	23-24-02		401	PTF ALL
Emergency Locator Transmitter Antenna - Removal TASK 23-24-02-000-801			401	PTF ALL
Emergency Locator Transmitter Antenna - Installation TASK 23-24-02-400-801			404	PTF ALL
EMERGENCY LOCATOR TRANSMITTER CONTROL PANEL - REMOVAL/INSTALLATION	23-24-03		401	PTF ALL
Emergency Locator Transmitter Control Panel - Removal TASK 23-24-03-000-801			401	PTF ALL
Emergency Locator Transmitter Control Panel - Installation TASK 23-24-03-400-801			403	PTF ALL
AIRCRAFT COMMUNICATION AND REPORTING SYSTEM (ACARS) - MAINTENANCE PRACTICES	23-27-00		201	PTF ALL
Aircraft Communication and Reporting System - Deactivation TASK 23-27-00-040-801			201	PTF ALL
Aircraft Communication and Reporting System - Activation TASK 23-27-00-400-801			204	PTF ALL
ACARS SYSTEM - ADJUSTMENT/TEST	23-27-00		501	PTF ALL
ACARS - Operational Test TASK 23-27-00-740-802			501	PTF ALL
ACARS - System Test TASK 23-27-00-700-802			501	PTF ALL
ACARS COMMUNICATIONS MANAGEMENT UNIT (CMU) - MAINTENANCE PRACTICES	23-27-33		201	PTF ALL
ACARS Communications Management Unit (CMU) Software Configuration Check TASK 23-27-33-700-801			201	PTF ALL
ACARS Software Installation with Onboard Network System (ONS) TASK 23-27-33-470-802			202	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ACARS COMMUNICATIONS MANAGEMENT UNIT - REMOVAL/INSTALLATION	23-27-33		401	PTF ALL
ACARS Communications Management Unit (CMU) Removal TASK 23-27-33-020-801			401	PTF ALL
ACARS Communications Management Unit (CMU) Installation TASK 23-27-33-420-801			404	PTF ALL
AIRPLANE PERSONALITY MODULE (APM) - MAINTENANCE PRACTICES	23-27-35		201	PTF ALL
ACARS Airplane Personality Module (APM) Data Verification Check TASK 23-27-35-700-801			201	PTF ALL
ACARS Airplane Personality Module (APM) Software Installation TASK 23-27-35-700-802			202	PTF ALL
ACARS AIRPLANE PERSONALITY MODULE - REMOVAL/INSTALLATION	23-27-35		401	PTF ALL
ACARS Airplane Personality Module (APM) Removal TASK 23-27-35-020-801			401	PTF ALL
ACARS Airplane Personality Module (APM) Installation TASK 23-27-35-420-801			405	PTF ALL
SELCAL SYSTEM - MAINTENANCE/PRACTICES	23-28-00		201	PTF ALL
SELCAL System - Deactivation TASK 23-28-00-040-801			201	PTF ALL
SELCAL System - Activation TASK 23-28-00-440-801			204	PTF ALL
SELCAL Program Switch Module Switch Settings TASK 23-28-00-860-801			204	PTF ALL
SELCAL - ADJUSTMENT/TEST	23-28-00		501	PTF ALL
SELCAL System - Operational Test TASK 23-28-00-700-801			501	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SELCAL DECODER - REMOVAL/INSTALLATION	23-28-11		401	PTF ALL
SELCAL Decoder Unit Removal TASK 23-28-11-020-801			401	PTF ALL
SELCAL Decoder Unit Installation TASK 23-28-11-420-801			404	PTF ALL
PASSENGER ADDRESS SYSTEM - ADJUSTMENT/TEST	23-31-00		501	PTF ALL
Passenger Address System - Operational Test TASK 23-31-00-740-801			501	PTF ALL
Passenger Address System - System Test TASK 23-31-00-730-801			505	PTF ALL
PASSENGER ADDRESS AMPLIFIER - MAINTENANCE PRACTICES	23-31-01		201	PTF ALL
Passenger Address Amplifier - Deactivation TASK 23-31-01-040-801			201	PTF ALL
Passenger Address Amplifier - Activation TASK 23-31-01-440-801			203	PTF ALL
PASSENGER ADDRESS AMPLIFIER - REMOVAL/INSTALLATION	23-31-01		401	PTF ALL
Passenger Address (PA) Amplifier Removal TASK 23-31-01-000-801			401	PTF ALL
Passenger Address (PA) Amplifier Installation TASK 23-31-01-400-801			403	PTF ALL
PASSENGER SERVICE UNIT SPEAKER - MAINTENANCE PRACTICES	23-31-02		201	PTF ALL
Passenger Service Unit Speaker - Deactivation TASK 23-31-02-040-801			201	PTF ALL
Passenger Service Unit Speaker - Activation TASK 23-31-02-440-801			203	PTF ALL
PASSENGER SERVICE UNIT SPEAKER - REMOVAL/INSTALLATION	23-31-02		401	PTF ALL
Passenger Service Unit (PSU) Speaker Removal TASK 23-31-02-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Passenger Service Unit (PSU) Speaker Installation TASK 23-31-02-400-801			405	PTF ALL
ATTENDANT SPEAKER - MAINTENANCE PRACTICES	23-31-03		201	PTF ALL
Attendant Speaker - Deactivation TASK 23-31-03-040-801			201	PTF ALL
Attendant Speaker - Activation TASK 23-31-03-440-801			203	PTF ALL
ATTENDANT SPEAKER - REMOVAL/INSTALLATION	23-31-03		401	PTF ALL
Attendant Speaker Removal TASK 23-31-03-000-801			401	PTF ALL
Attendant Speaker Installation TASK 23-31-03-400-801			404	PTF ALL
LAVATORY SPEAKER - MAINTENANCE PRACTICES	23-31-05		201	PTF ALL
Lavatory Speaker - Deactivation TASK 23-31-05-040-801			201	PTF ALL
Lavatory Speaker - Activation TASK 23-31-05-440-801			203	PTF ALL
LAVATORY SPEAKER - REMOVAL/INSTALLATION	23-31-05		401	PTF ALL
Lavatory Speaker Removal TASK 23-31-05-000-801			401	PTF ALL
Lavatory Speaker Installation TASK 23-31-05-400-801			407	PTF ALL
PRE-RECORDED ANNOUNCEMENT MACHINE (PRAM) - MAINTENANCE PRACTICES	23-31-07		201	PTF ALL
Pre-Recorded Announcement Machine (PRAM) Maintenance Modes TASK 23-31-07-820-801			201	PTF ALL
Pre-Recorded Announcement Machine (PRAM) Content Installation TASK 23-31-07-470-801			202	PTF ALL
Pre-Recorded Announcement Machine (PRAM) - Deactivation TASK 23-31-07-040-801			207	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Pre-Recorded Announcement Machine (PRAM) - Activation TASK 23-31-07-440-801			209	PTF ALL
PRE-RECORDED ANNOUNCEMENT MACHINE (PRAM) - REMOVAL/INSTALLATION	23-31-07		401	PTF ALL
Pre-Recorded Announcement Machine (PRAM) Removal TASK 23-31-07-000-801			401	PTF ALL
Pre-Recorded Announcement Machine (PRAM) Installation TASK 23-31-07-400-801			405	PTF ALL
SERVICE INTERPHONE SYSTEM - ADJUSTMENT/TEST	23-41-00		501	PTF ALL
Service Interphone Operational Test TASK 23-41-00-710-801			501	PTF ALL
CABIN INTERPHONE SYSTEM - ADJUSTMENT/TEST	23-42-00		501	PTF ALL
Cabin Interphone System - Operational Test TASK 23-42-00-710-801			501	PTF ALL
Cabin Interphone system - Attendant Panel Installation Test TASK 23-42-00-700-801			502	PTF ALL
ATTENDANT HANDSET - REMOVAL/INSTALLATION	23-42-01		401	PTF ALL
Attendant Handset Removal TASK 23-42-01-000-801			401	PTF ALL
Attendant Handset Installation TASK 23-42-01-400-803			403	PTF ALL
Attendant Handset Cradle Removal TASK 23-42-01-000-803			403	PTF ALL
Attendant Handset Cradle Installation TASK 23-42-01-400-801			406	PTF ALL
Attendant Handset Cord Removal TASK 23-42-01-000-802			407	PTF ALL
Attendant Handset Cord Installation TASK 23-42-01-400-802			407	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ATTENDANT HANDSET - ADJUSTMENT/TEST	23-42-01		501	PTF ALL
Attendant Handset Installation Test TASK 23-42-01-700-802			501	PTF ALL
ATTENDANT CONTROL PANEL - MAINTENANCE PRACTICES	23-42-03		201	PTF ALL
Attendant Control Panel (ACP) - Software Loading TASK 23-42-03-470-801			201	PTF ALL
Attendant Control Panel (ACP) - Fault Data Report TASK 23-42-03-970-803			203	PTF ALL
Attendant Control Panel (ACP) - Configuration Check TASK 23-42-03-700-801			205	PTF ALL
Attendant Control Panel (ACP) - Configuration Data Report TASK 23-42-03-970-804			206	PTF ALL
Attendant Control Panel (ACP) - Erasing Mass Storage Area (MSA) TASK 23-42-03-070-801			207	PTF ALL
Attendant Control Panel (ACP) - LCD Touch Screen Display Check TASK 23-42-03-210-801			207	PTF ALL
ATTENDANT LCD CONTROL PANEL - REMOVAL/INSTALLATION	23-42-03		401	PTF ALL
Cabin Attendant LCD Control Panel Removal TASK 23-42-03-000-801			401	PTF ALL
Cabin Attendant LCD Control Panel Installation TASK 23-42-03-400-801			406	PTF ALL
ATTENDANT CONTROL PANEL - ADJUSTMENT/TEST	23-42-03		501	PTF ALL
Attendant Control Panel - Operational Test TASK 23-42-03-710-801			501	PTF ALL
Attendant Control Panel (ACP) - BITE Procedure TASK 23-42-03-740-801			502	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ATTENDANT CONTROL PANEL - CLEANING/PAINTING	23-42-03		701	PTF ALL
Attendant Control Panel (ACP) Touchscreen Cleaning TASK 23-42-03-100-802			701	PTF ALL
INLINE FILTER - REMOVAL/INSTALLATION	23-42-04		401	PTF ALL
Inline Filter Removal TASK 23-42-04-000-801			401	PTF ALL
Inline Filter Installation TASK 23-42-04-400-801			404	PTF ALL
FLIGHT AND GROUND CREW CALL SYSTEM - ADJUSTMENT/TEST	23-43-00		501	PTF ALL
Flight and Ground Crew Call System - Operational Test TASK 23-43-00-710-801			501	PTF ALL
GROUND CREW CALL HORN - REMOVAL/INSTALLATION	23-43-02		401	PTF ALL
Ground Crew Call Horn Removal TASK 23-43-02-000-801			401	PTF ALL
Ground Crew Call Horn Installation TASK 23-43-02-420-801			405	PTF ALL
NOSE WHEEL WELL INTERPHONE BOX - REMOVAL/INSTALLATION	23-43-03		401	PTF ALL POST SB 737-24-1236
Nose Wheel Well Interphone Box Removal TASK 23-43-03-000-801			401	PTF ALL POST SB 737-24-1236
Nose Wheel Well Interphone Box Installation TASK 23-43-03-400-801			404	PTF ALL POST SB 737-24-1236
FLIGHT INTERPHONE SYSTEM - ADJUSTMENT/TEST	23-51-00		501	PTF ALL
Flight Interphone System Operational Test TASK 23-51-00-710-801			501	PTF ALL
Oxygen Mask Microphone Operational Test TASK 23-51-00-710-802			507	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
REMOTE ELECTRONICS UNIT - REMOVAL/INSTALLATION	23-51-01		401	PTF ALL
Remote Electronics Unit Removal TASK 23-51-01-000-801			401	PTF ALL
Remote Electronics Unit Installation TASK 23-51-01-400-801			404	PTF ALL
AUDIO CONTROL PANEL - MAINTENANCE PRACTICES	23-51-02		201	PTF ALL
Audio Control Panel - Mic Select Switch Lamp Replacement TASK 23-51-02-300-801			201	PTF ALL
AUDIO CONTROL PANEL - REMOVAL/INSTALLATION	23-51-02		401	PTF ALL
Audio Control Panel Removal TASK 23-51-02-000-801			401	PTF ALL
Audio Control Panel Installation TASK 23-51-02-400-801			405	PTF ALL
FLIGHT INTERPHONE SPEAKER - REMOVAL/INSTALLATION	23-51-03		401	PTF ALL
Flight Interphone Speaker Removal TASK 23-51-03-000-801			401	PTF ALL
Flight Interphone Speaker Installation TASK 23-51-03-400-801			403	PTF ALL
PUSH TO TALK CONTROL WHEEL SWITCH - REMOVAL/INSTALLATION	23-51-04		401	PTF ALL
Push-To-Talk (PTT) Control Wheel Switch Removal TASK 23-51-04-000-801			401	PTF ALL
Push To Talk (PTT) Control Wheel Switch Installation TASK 23-51-04-400-801			404	PTF ALL
PUSH TO TALK CONTROL WHEEL SWITCH - ADJUSTMENT/TEST	23-51-04		501	PTF ALL
Catchball Adjustment TASK 23-51-04-720-801			501	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
GLARESHIELD PUSH TO TALK SWITCH - REMOVAL/INSTALLATION	23-51-05		401	PTF ALL
Glareshield Push To Talk (PTT) Switch Removal TASK 23-51-05-000-801			401	PTF ALL
Glareshield Push To Talk (PTT) Switch Installation TASK 23-51-05-400-801			405	PTF ALL
STATIC DISCHARGERS - REMOVAL/INSTALLATION	23-61-00		401	PTF ALL
Static Discharger Removal TASK 23-61-00-000-801			401	PTF ALL
Static Discharger Installation TASK 23-61-00-400-801			406	PTF ALL
Static Discharger Base with Screws Removal TASK 23-61-00-000-802			408	PTF ALL
Static Discharger Base with Screws Installation TASK 23-61-00-400-802			410	PTF ALL
Static Discharger Base with Rivets Removal TASK 23-61-00-000-803			415	PTF ALL
Static Discharger Base with Rivets Installation TASK 23-61-00-400-803			418	PTF ALL
Static Discharger Base with Adapter Plate Assembly Removal TASK 23-61-00-000-804			422	PTF ALL
Static Discharger Base with Adapter Plate Assembly Installation TASK 23-61-00-400-804			425	PTF ALL
STATIC DISCHARGER - INSPECTION/CHECK	23-61-00		601	PTF ALL
Static Discharger Inspection TASK 23-61-00-210-801			601	PTF ALL
Static Discharger Resistance Measurement TASK 23-61-00-760-801			606	PTF ALL
VOICE RECORDER SYSTEM - MAINTENANCE PRACTICES	23-71-00		201	PTF ALL
Voice Recorder System Deactivation TASK 23-71-00-040-801			201	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Voice Recorder System Activation TASK 23-71-00-440-801			202	PTF ALL
Make a Copy of the CVR Data with the HHMPI TASK 23-71-00-970-801			202	PTF ALL
Make a Copy of the CVR Data with the EHHDLU TASK 23-71-00-970-802			204	PTF ALL
Make a copy of the Cockpit Voice Recorder (CVR) data with the Data Acquisition Ruggedized Tool (DART) TASK 23-71-00-970-804			207	PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X OR 2100-1925-2X
VOICE RECORDER SYSTEM - ADJUSTMENT/TEST	23-71-00		501	PTF ALL
Voice Recorder System - Operational Test TASK 23-71-00-710-801			501	PTF ALL
Voice Recorder System - Audio Fidelity Test TASK 23-71-00-730-801			502	PTF ALL
Voice Recorder System Test TASK 23-71-00-730-802			505	PTF ALL
VOICE RECORDER - REMOVAL/INSTALLATION	23-71-11		401	PTF ALL
Voice Recorder Removal TASK 23-71-11-000-801			401	PTF ALL
Voice Recorder Installation TASK 23-71-11-400-801			404	PTF ALL
VOICE RECORDER CONTROL PANEL - REMOVAL/INSTALLATION	23-71-12		401	PTF ALL
Voice Recorder Control Panel Removal TASK 23-71-12-000-801			401	PTF ALL
Voice Recorder Control Panel Installation TASK 23-71-12-400-801			403	PTF ALL
VOICE RECORDER SWITCH PANEL - REMOVAL/INSTALLATION	23-71-13		401	PTF ALL
Voice Recorder Switch Panel Removal TASK 23-71-13-000-801			401	PTF ALL
Voice Recorder Switch Panel Installation TASK 23-71-13-400-801			403	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
UNDERWATER LOCATOR BEACON - MAINTENANCE PRACTICES	23-71-21		201	PTF ALL
Dukane Underwater Locator Beacon Battery - Replacement TASK 23-71-21-960-801			201	PTF ALL
L-3 Underwater Locator Beacon Battery - Replacement TASK 23-71-21-960-803			207	PTF ALL
UNDERWATER LOCATOR BEACON - REMOVAL/INSTALLATION	23-71-21		401	PTF ALL
Underwater Locator Beacon Removal TASK 23-71-21-000-801			401	PTF ALL
Underwater Locator Beacon Installation TASK 23-71-21-400-801			404	PTF ALL
UNDERWATER LOCATOR BEACON - ADJUSTMENT/TEST	23-71-21		501	PTF ALL
Underwater Locator Beacon Test with a 42A12 Test Set TASK 23-71-21-730-801			501	PTF ALL
Underwater Locator Beacon Test with a PL1 Test Set TASK 23-71-21-730-802			502	PTF ALL
Underwater Locator Beacon Test with a PL3 Test Set TASK 23-71-21-730-803			503	PTF ALL
Underwater Locator Beacon Test with a TS200 Test Set TASK 23-71-21-730-806			504	PTF ALL
Underwater Locator Beacon Test with a L-3AR Test Set TASK 23-71-21-730-807			505	PTF ALL
Underwater Locator Beacon Test with a TS500 Test Set TASK 23-71-21-730-808			506	PTF ALL
Underwater Locator Beacon Operational Test with a TS500 Test Set TASK 23-71-21-710-801			507	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 23
COMMUNICATIONS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Underwater Locator Beacon Voltage Test with a TS500 Test Set TASK 23-71-21-720-801			508	PTF ALL
UNDERWATER LOCATOR BEACON - CLEANING/PAINTING	23-71-21		701	PTF ALL
Underwater Locator Beacon Cleaning TASK 23-71-21-100-801			701	PTF ALL
FLIGHT DECK ENTRY VIDEO SURVEILLANCE SYSTEM - MAINTENANCE PRACTICES	23-75-00		201	PTF ALL
Video Surveillance System - Deactivation TASK 23-75-00-040-801			201	PTF ALL
Video Surveillance System - Activation TASK 23-75-00-440-801			201	PTF ALL
FLIGHT DECK ENTRY VIDEO SURVEILLANCE SYSTEM - ADJUSTMENT/TEST	23-75-00		501	PTF ALL
Flight Deck Entry Video Surveillance System - System Test TASK 23-75-00-730-801			501	PTF ALL
VIDEO CAMERAS - REMOVAL/INSTALLATION	23-75-01		401	PTF ALL
Video Camera Removal TASK 23-75-01-000-801			401	PTF ALL
Video Camera Installation TASK 23-75-01-400-801			404	PTF ALL
CAMERA CONTROL PANEL ASSEMBLY-REMOVAL/INSTALLATION	23-75-02		401	PTF ALL
Camera Control Panel Removal TASK 23-75-02-000-801			401	PTF ALL
Camera Control Panel Installation TASK 23-75-02-400-801			403	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMMUNICATION - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks are used to prepare the airplane for flight with certain systems/components inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. These are the tasks for the components in the communication system:

(1) DDG 23-24-01 Fixed Emergency Locator Transmitter (ELT) Inoperative - Preparation

(2) DDG 23-24-01 Fixed Emergency Locator Transmitter (ELT) Inoperative - Restoration

(3) DDG 23-24-01-01 Remote Emergency Locator Transmitter (ELT) Switch Inoperative - Preparation

(4) DDG 23-24-01-01 Remote Emergency Locator Transmitter (ELT) Switch Inoperative - Restoration

(5) DDG 23-51-04-01 Control Wheel Push-To-Talk (PTT) Switches Inoperative - Preparation

(6) DDG 23-51-04-01 Control Wheel Push-To-Talk (PTT) Switches Inoperative - Restoration

(7) DDG 23-51-04-02 Audio Control Panel (ACP) Push-To-Talk (PTT) Switches Inoperative - Preparation

(8) DDG 23-51-04-03 Glareshield Panel Push-To-Talk (PTT) Switches Inoperative - Preparation

(9) DDG 23-51-04-03 Glareshield Panel Push-To-Talk (PTT) Switches Inoperative - Restoration

(10) DDG 23-71-01-02 Aircraft with Recorder Independent Power Supply (RIPS) Inoperative (EASA) - Preparation

(11) DDG 23-71-01-02 Aircraft with Recorder Independent Power Supply (RIPS) Inoperative (EASA) - Restoration

TASK 23-00-00-040-807

2. DDG 23-24-01 Fixed Emergency Locator Transmitter (ELT) Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Emergency Locator Transmitter (ELT) inoperative.

B. References

Reference	Title
23-24-00-000-801	Emergency Locator Transmitter Removal (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. ELT System Deactivation

SUBTASK 23-00-00-040-008

(1) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

SUBTASK 23-00-00-010-003

(2) Open the ceiling panel for access to the ELT unit (station 794).

SUBTASK 23-00-00-040-005

(3) Remove the ELT unit (TASK 23-24-00-000-801).

SUBTASK 23-00-00-410-003

(4) Close the ceiling panel for access to the ELT unit.

SUBTASK 23-00-00-040-006

(5) Install an INOP placard adjacent to the switch of the ELT control panel in the flight compartment.

——— END OF TASK ———

TASK 23-00-00-440-807

3. DDG 23-24-01 Fixed Emergency Locator Transmitter (ELT) Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Emergency Locator Transmitter (ELT) inoperative.

B. References

Reference	Title
23-24-00-400-801	Emergency Locator Transmitter Installation (P/B 401)
23-24-00-710-801	Emergency Locator Transmitter Operational Test (P/B 501)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. ELT System Restoration

SUBTASK 23-00-00-010-004

(1) Open the ceiling panel for access to the ELT unit (station 794).

SUBTASK 23-00-00-420-006

(2) Install the ELT unit (TASK 23-24-00-400-801).

SUBTASK 23-00-00-440-005

(3) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-00-00-710-002

(4) Do this task: Emergency Locator Transmitter Operational Test, TASK 23-24-00-710-801

SUBTASK 23-00-00-410-004

(5) Close the ceiling panel for access to the ELT unit (station 794).

SUBTASK 23-00-00-020-009

(6) Remove the INOP placard from the ELT control panel in the flight compartment.

——— END OF TASK ———

TASK 23-00-00-040-813

4. DDG 23-24-01-01 Remote Emergency Locator Transmitter (ELT) Switch Inoperative - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the Emergency Locator Transmitter (ELT) switch inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

C. ELT Control Panel Deactivation

SUBTASK 23-00-00-040-010

(1) If the remote ELT switch is failed and the ELT system is activated, do this task: DDG 23-24-01 Fixed Emergency Locator Transmitter (ELT) Inoperative - Preparation, TASK 23-00-00-040-807.

SUBTASK 23-00-00-010-006

(2) Open the ceiling panel for access to the ELT (station 794).

SUBTASK 23-00-00-440-014

(3) On the front of the ELT, do these steps:

(a) Set the OFF/ON switch to the OFF position.

SUBTASK 23-00-00-410-007

(4) Close the ceiling panel for access to the ELT.

SUBTASK 23-00-00-440-009

(5) On the ELT control panel, do these steps:

(a) Set the ARM/ON switch to the ARM position.

(b) Install the INOP placard adjacent to the ARM/ON switch.

SUBTASK 23-00-00-440-010

(6) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-00-00-440-813

5. DDG 23-24-01-01 Remote Emergency Locator Transmitter (ELT) Switch Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Emergency Locator Transmitter (ELT) switch inoperative.

B. References

Reference	Title
23-24-00-710-801	Emergency Locator Transmitter Operational Test (P/B 501)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. ELT Control Panel Activation

SUBTASK 23-00-00-010-007

(1) Open the ceiling panel for access to the ELT (station 794).

SUBTASK 23-00-00-440-015

(2) On the front of the ELT, do these steps:

(a) Set the OFF/ON switch to the ON position.

SUBTASK 23-00-00-410-008

(3) Close the ceiling panel for access to the ELT.

SUBTASK 23-00-00-440-012

(4) On the ELT control panel, do these steps:

(a) Remove the INOP placard from to the ARM/ON switch.

(b) Set the ARM/ON switch to the ARM position.

SUBTASK 23-00-00-700-002

(5) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

SUBTASK 23-00-00-710-003

(6) Do this task: Emergency Locator Transmitter Operational Test, TASK 23-24-00-710-801.

——— END OF TASK ———

TASK 23-00-00-040-808

6. DDG 23-51-04-01 Control Wheel Push-To-Talk (PTT) Switches Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Control Wheel Push-To-Talk (PTT) switches inoperative.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
23-51-04-000-801	Push-To-Talk (PTT) Control Wheel Switch Removal (P/B 401)
23-51-04-400-801	Push To Talk (PTT) Control Wheel Switch Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. PTT Switches Deactivation

SUBTASK 23-00-00-010-005

(1) Remove the PTT switch on the control wheel to get access to the GROUND wire (TASK 23-51-04-000-801).

SUBTASK 23-00-00-020-003

(2) Disconnect, cap, and stow the GROUND (BLK/BRN) wire from the PTT switch.

SUBTASK 23-00-00-420-003

(3) Install the deactivated PTT switch (TASK 23-51-04-400-801).

SUBTASK 23-00-00-930-001

(4) Install an INOP placard adjacent to the Control Wheel PTT switch.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 23-00-00-860-006

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT

END OF TASK

TASK 23-00-00-440-808

7. DDG 23-51-04-01 Control Wheel Push-To-Talk (PTT) Switches Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Control Wheel Push-To-Talk (PTT) switches inoperative.

B. References

Reference	Title
23-51-04-000-801	Push-To-Talk (PTT) Control Wheel Switch Removal (P/B 401)
23-51-04-400-801	Push To Talk (PTT) Control Wheel Switch Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Prepare for Restoration

SUBTASK 23-00-00-860-003

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT

D. PTT Switches Restoration

SUBTASK 23-00-00-020-004

(1) Remove the INOP placard from the PTT switch.

SUBTASK 23-00-00-020-005

(2) Remove the PTT switch on the control wheel (TASK 23-51-04-000-801).

SUBTASK 23-00-00-020-006

(3) Remove the cap from the GROUND (BLK/BRN) wire for the PTT switch.

(a) Reconnect the GROUND (BLK/BRN) wire.

SUBTASK 23-00-00-420-004

(4) Install the PTT switch back onto the control wheel (TASK 23-51-04-400-801).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 23-00-00-860-004

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT

——— END OF TASK ———

TASK 23-00-00-040-809

8. DDG 23-51-04-02 Audio Control Panel (ACP) Push-To-Talk (PTT) Switches Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Audio Control Panel (ACP) Push-To-Talk (PTT) switches inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right

C. Verify the Inoperative Flight Crew Audio Control Panel (ACP) PTT Switch is Failed Open

SUBTASK 23-00-00-700-001

(1) Do the steps that follow to confirm that the ACP PTT switch is inoperative open and the related Control Wheel PTT switch operates normally:

(a) Push the PA microphone selector switch on the related ACP.

1) Make sure that the PA MIC light comes on.

(b) Speak into the related microphone without pushing any PTT switch.

1) Make sure that you hear no voice on the PA system.

(c) Push the inoperative PTT switch and speak into the related microphone.

1) Make sure that you hear no voice at the other pilot's headset.

(d) Push the related pilot's Control Wheel PTT switch and speak into the related microphone.

1) Make sure that you hear your voice clearly at the other pilot's headset.

——— END OF TASK ———

TASK 23-00-00-040-810

9. DDG 23-51-04-03 Glareshield Panel Push-To-Talk (PTT) Switches Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Glareshield Panel Push-To-Talk (PTT) switches inoperative.

B. References

Reference	Title
23-51-05-000-801	Glareshield Push To Talk (PTT) Switch Removal (P/B 401)
23-51-05-400-801	Glareshield Push To Talk (PTT) Switch Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the PTT Deactivation

SUBTASK 23-00-00-860-007

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	24	C00085	AUDIO OBS

E. PTT Switches Deactivation

SUBTASK 23-00-00-020-007

(1) Remove the PTT switch on the glareshield panel to get access to the wire (TASK 23-51-05-000-801).

SUBTASK 23-00-00-040-007

(2) Disconnect, cap, and stow the GROUND wire from the PTT switch.

SUBTASK 23-00-00-410-005

(3) Install the deactivated PTT switch (TASK 23-51-05-400-801).

SUBTASK 23-00-00-860-008

(4) Install an INOP placard adjacent to the Glareshield Panel PTT switch.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-00-00-860-009

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

——— END OF TASK ———

TASK 23-00-00-440-810

10. DDG 23-51-04-03 Glareshield Panel Push-To-Talk (PTT) Switches Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Glareshield Panel Push-To-Talk (PTT) switches inoperative.

B. References

Reference	Title
23-51-05-000-801	Glareshield Push To Talk (PTT) Switch Removal (P/B 401)
23-51-05-400-801	Glareshield Push To Talk (PTT) Switch Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the PTT Restoration

SUBTASK 23-00-00-860-010

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

E. PTT Switches Restoration

SUBTASK 23-00-00-860-011

(1) Remove the INOP placard from the PTT switch.

SUBTASK 23-00-00-020-008

(2) Remove the PTT switch on the glareshield panel to get access to the wire (TASK 23-51-05-000-801).

SUBTASK 23-00-00-420-005

(3) Remove the cap from the GROUND wire for the PTT switch.

(a) Reconnect the GROUND wire.

SUBTASK 23-00-00-410-006

(4) Install the restored PTT switch back onto the glareshield panel (TASK 23-51-05-400-801).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-00-00-860-012

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-00-00-040-811

11. DDG 23-71-01-02 Aircraft with Recorder Independent Power Supply (RIPS) Inoperative (EASA) - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Recorder Independent Power Supply (RIPS) inoperative.

(2) This procedure is only for aircraft using the EASA MMEL. DDG Maintenance procedures with no regulatory authority identification in the procedure title apply to both FAA and EASA MMELs.

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Recorder Independent Power Supply (RIPS) Deactivation

SUBTASK 23-00-00-040-001

(1) Do this step minimum of 15 minutes before the departure:

(a) This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 23-00-00-910-001

(2) Install the INOP placards adjacent to these components:

- Cockpit Voice Recorder's control panel, on the P5 pilot's overhead panel
- VOICE RCDR/RIPS (C01948) circuit breaker, on the P18-2 panel.

——— END OF TASK ———

TASK 23-00-00-440-811

12. DDG 23-71-01-02 Aircraft with Recorder Independent Power Supply (RIPS) Inoperative (EASA) - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Recorder Independent Power Supply (RIPS) inoperative.

(2) This procedure is only for aircraft using the EASA MMEL. DDG Maintenance procedures with no regulatory authority identification in the procedure title apply to both FAA and EASA MMELs.

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Recorder Independent Power Supply (RIPS) Activation

SUBTASK 23-00-00-440-001

(1) This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 23-00-00-910-002

(2) Remove the INOP placards from these components:

- Cockpit Voice Recorder's control panel, on the P5 pilot's overhead panel
- VOICE RCDR/RIPS (C01948) circuit breaker, on the P18-2 panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIGH FREQUENCY (HF) COMMUNICATION SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A deactivation of the High Frequency (HF) communication system

(2) An activation of the HF communication system.

TASK 23-11-00-040-801

2. High Frequency Communication System - Deactivation

(Figure 201)

A. General

(1) This task removes electrical power to the High Frequency (HF) communications system.

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

C. Access Panels

Number	Name/Location
822	Aft Cargo Door

D. High Frequency Communication System Deactivation

NOTE: The airplane will have single or dual HF transceivers installed depending on customer requirements. Reference the airplane wire diagram manual.

SUBTASK 23-11-00-860-016

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

E. High Frequency Communication - Tryout

NOTE: This tryout is to make sure that the HF communications system is in a zero energy state.

SUBTASK 23-11-00-860-017

(1) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-11-00-860-018

(2) Do the following steps at the HF-1 or HF-2 (if applicable) control panel on the P8 aft electronic panel:

(a) If installed, set the mode selector switch to the USB position.

(b) Push the applicable HF switch light (HF 1 or HF 2), on the No. 1 radio tuning panel (RTP-1).

SUBTASK 23-11-00-010-003

(3) Get access to the HF transceivers, do this step:

(a) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 23-11-00-710-002

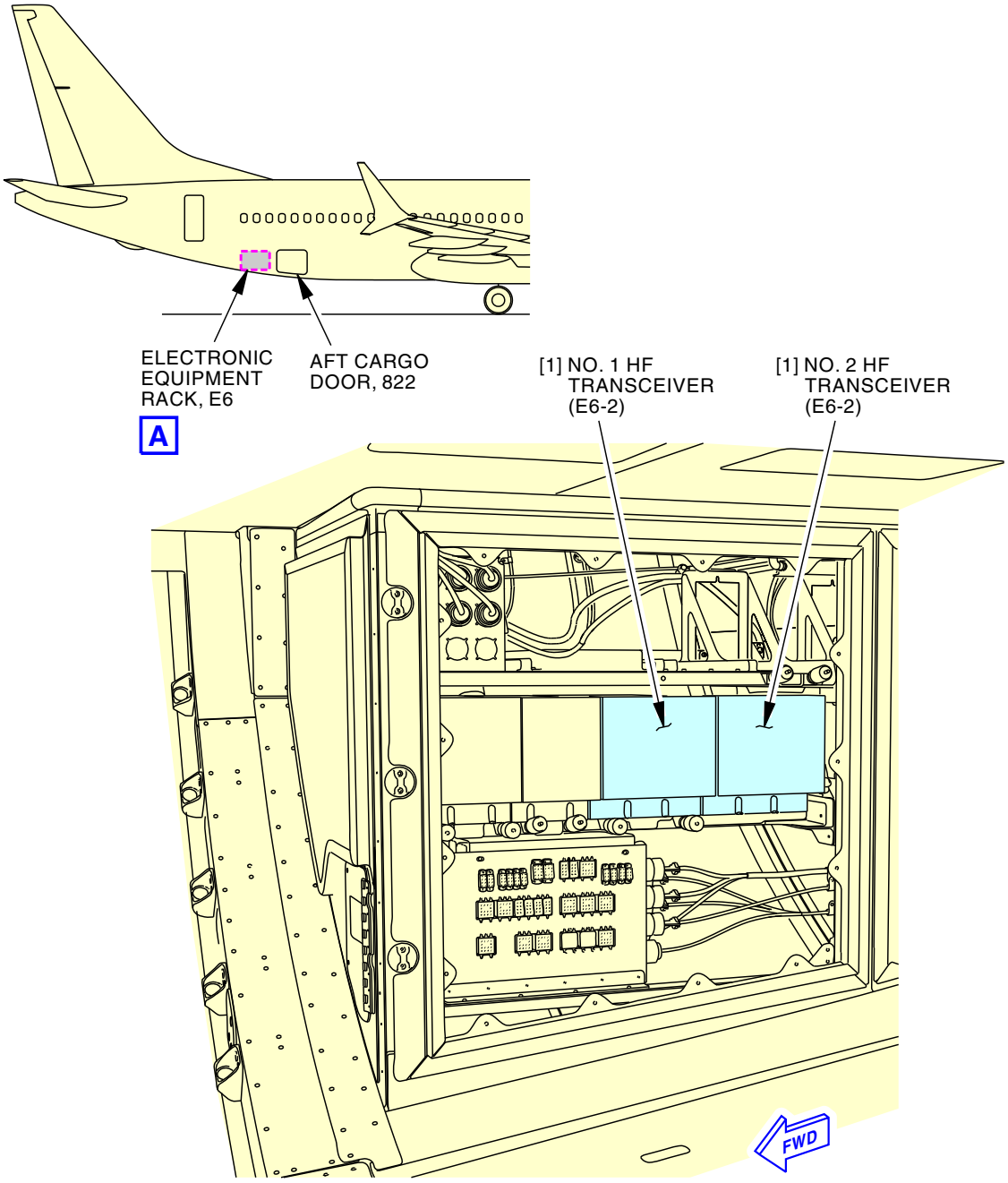
(4) Do the following steps at the HF-1 and HF-2 (if applicable) transceiver:

(a) Push and hold the TEST switch on the transceiver front panel.

1) Make sure that none of the lights on the front panel come on.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRONIC EQUIPMENT RACK, E6

A

2572093 S0000617099_V1

HF Transceiver
Figure 201/23-11-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-11-00-440-801

3. High Frequency Communication System - Activation

(Figure 201)

A. General

(1) This task adds electrical power to the High Frequency (HF) communication system.

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

C. Access Panels

Number	Name/Location
822	Aft Cargo Door

D. High Frequency Communication System Activation

WARNING DO NOT OPERATE THE HF SYSTEM WHILE THE AIRPLANE IS REFUELED OR DEFUELED. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

NOTE: The airplane will have single or dual HF transceivers installed depending on customer requirements. Reference the airplane wire diagram manual.

SUBTASK 23-11-00-860-019

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-11-00-410-003

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIGH FREQUENCY (HF) COMMUNICATION SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) HF Communication System - Operational Test

(2) HF Communication System - System Test.

B. There are two HF communication systems installed on the airplane.

C. The No. 1 and No. 2 HF transceivers are located in the aft cargo compartment on the electronic equipment rack E6.

D. The No. 1, No. 2, and No. 3 radio tuning panels (RTPs) are located on the aft electronic panel, P8, in the flight compartment.

TASK 23-11-00-710-801

2. HF Communication System - Operational Test

A. General

(1) The operational test is a BITE test of the HF transceiver.

(a) The BITE test does a check of the internal functions of the HF transceiver. The BITE test also does a check of the ARINC 429/719 frequency tuning input data to the HF transceiver.

(2) The operational test for the HFS-900D transceivers is a BITE test which does a check of the internal functions of the HF transceiver. The BITE test also does a check of the ARINC 429/719 frequency tuning input data to the HF transceiver. The BITE test does not include a check of the HF antenna couplers, coaxial cables, or the HF antenna. The HF communication system does not transmit during the BITE test.

(3) The same sequence of steps is used for the No. 1 or No. 2 HF communication system.

(4) HF Antenna Coupler Tuning

(a) A tune tone is generated while the HF antenna coupler tunes to a new frequency. The tune tone can be continuous or pulsed (determined by the coupler/transceiver configuration). Some couplers save previously tuned frequencies in memory. If a coupler is set to a frequency in memory, then the tune time can be very short and you may not hear the tune tone. If you hear the tune tone for more than 7 seconds (15 seconds maximum), then it is an indication of a tuning failure.

(b) Rockwell Collins Digital HF Antenna Coupler (Model CPL-920D, Part Number 822-0987-XXX):

1) The tune time for a frequency not in memory is 2 to 4 seconds typical, 7 seconds maximum.

2) If HFS-700 or HFS-900 transceiver is installed:

a) The tune time for a frequency saved in memory is 1 second typical.

b) Up to 100 frequencies can be saved in memory.

c) The tune tone is pulsed. Basically, the tuning tone is audible, but if using a previously tuned channel, the tune tone may not be audible.

3) If HFS-900D transceiver is installed:

a) The tune time for a frequency saved in memory is 200 milliseconds typical.

b) The tune tone is continuous.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) At the first tuning after a cold start, the tune tone is always audible regardless of whether this frequency is stored (average 2 to 4 seconds, 7 seconds maximum).

5) When either HF antenna or HF transceiver is replaced, the frequency memory is reset.

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
822	Aft Cargo Door

D. HF Communication System Operational Test

SUBTASK 23-11-00-800-002

WARNING DO NOT OPERATE THE HF SYSTEM WHILE YOU REFUEL OR DEFUEL THE AIRPLANE. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(1) Do not operate the HF system while a fuel operation is done on the airplane.

SUBTASK 23-11-00-860-013

(2) Push the applicable HF switch light (HF 1 or HF 2), on the No. 1 radio tuning panel (RTP-1), on the aft electronic panel, P8.

(a) Make sure that the switch light comes on.

(b) If the AM lamp is on, push the AM switch once to select the USB mode.

SUBTASK 23-11-00-010-001

(3) To gain access to the HF transceivers, do this step:

Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 23-11-00-010-002

(4) Open the access panel to the equipment rack.

SUBTASK 23-11-00-860-014

(5) Make sure that the internal blower fan on the HF transceiver operates when you transmit.

E. HF Transceiver BITE Test

SUBTASK 23-11-00-740-001

(1) Push the TEST switch on the front panel.

(a) Release the TEST switch on the front panel.

(b) Make sure that these conditions occur:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) The LRU STATUS and CONTROL INPUT LEDs turn red for 1 to 3 seconds.

NOTE: The KEY INTERLOCK LED flashes red and then off.

2) The LRU STATUS LED turns green and the other LEDs turn red for 1 to 3 seconds.

3) All LEDs go off for 1 to 7 seconds.

4) The LRU STATUS LED turns green and the other LEDs stay off. After 30 seconds, all LEDs go off.

NOTE: The three LEDs that come on red after the self-test is completed shows a defective condition.

F. Put the Airplane Back to Its Initial Condition

SUBTASK 23-11-00-410-001

(1) Close the access panel to the equipment rack.

SUBTASK 23-11-00-410-002

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

TASK 23-11-00-730-801

3. HF Communication System - System Test

A. General

(1) This task includes these tests:

(a) An Operational Test

(b) A HF Communication Test.

(2) Make sure that the flight interphone system operates. The communication test uses these flight interphone system components: audio control panels (ACPs), headsets, microphones, and push-to-talk (PTT) switches.

(3) Make sure that the airplane is not in or near any large metal structures. HF communication while the airplane is on the ground can be degraded by airplane position or blocked by RF signal attenuation due to ground equipment and structures. If the quality of the transmitted and received voice signals is not satisfactory, move the airplane to a better position.

(4) Make sure that you use approved test frequencies when you speak with the radio tower operator.

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 23-11-00-800-001

WARNING DO NOT OPERATE THE HF SYSTEM WHILE YOU REFUEL OR DEFUEL THE AIRPLANE. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(1) Do not operate the HF system while a fuel operation is done on the airplane.

D. Operational Test

SUBTASK 23-11-00-710-001

(1) Do this task: HF Communication System - Operational Test, TASK 23-11-00-710-801.

E. HF Communication Test

SUBTASK 23-11-00-730-001

(1) Do a test of the HF-1 communication system:

(a) Connect a headset/boom microphone to the captain's jack panel.

(b) Do these steps to prepare the captain's audio control panel (ACP) for HF-1 system operation:

1) Push and release the HF-1 switch.

a) Make sure that the switch light comes on.

2) Push and release the HF-1 volume control.

a) Make sure that the volume control indicator light comes on.

3) Set the volume control to the middle position.

4) Set the MASK/BOOM switch (if installed) to the BOOM position.

(c) Do these steps at the No. 1 radio tuning panel (RTP-1):

1) Push the HF 1 switch light.

a) Make sure that the switch light comes on.

2) Push the AM switch light for AM or USB mode of operation.

a) Make sure that the switch light is on for AM, or off for USB.

3) Turn the SENS control clockwise for maximum gain.

4) Set the STANDBY frequency window to an approved test frequency.

5) Push the display transfer switch.

a) Make sure that the STANDBY frequency moves to the ACTIVE frequency window.

(d) Push and release the captain's push-to-talk (PTT) switch.

1) Make sure that you hear a 1 kHz tune-in-progress tone in the headset.

NOTE: A continuous or pulsed tone indicates that the coupler is tuning to a new frequency. The coupler tune tone will sound no longer than 15 seconds. The average coupler tune time is approximately 1 to 7 seconds.

NOTE: Some coupler types are able to tune quickly when previously used frequencies are selected, in which case the tune tone may be only a momentary beep.

(e) Do these steps to do a voice communication test with a radio tower operator or with another aircraft utilizing HF channel:

1) Push and hold the PTT switch while you speak.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that you hear the sidetone in the headset while you speak.

b) Make sure that the LRU STATUS, KEY INTERLOCK, and CONTROL FAIL lights on the front panel of the HF transceiver are not on.

2) Release the PTT switch while you listen.

a) Make sure that the quality of the transmitted and received voice is satisfactory.

b) Make sure that the sound of the received voice decreases and increases when you turn the SENS control counterclockwise and clockwise on the RTP.

c) Make sure that the sound of the received voice changes when you turn the HF-1 volume control on the captain's ACP with no change in voice quality.

(f) If more than one mode of operation (AM and USB) is available at your location, do this step:

1) Push the AM switch light to the other mode of operation and make a voice transmission.

SUBTASK 23-11-00-730-002

(2) Do a test of the HF-2 communication system:

(a) Connect a headset/boom microphone to the first officer's jack panel.

(b) Do these steps to prepare the first officer's ACP for HF-2 system operation:

1) Push and release the HF-2 switch.

a) Make sure that the switch light comes on.

2) Push and release the HF-2 volume control.

a) Make sure that the volume control indicator light comes on.

3) Set the volume control to the middle position.

4) Set the MASK/BOOM switch (if installed) to the BOOM position.

(c) Do these steps at RTP-2:

1) Push the HF 2 switch light.

a) Make sure that the switch light comes on.

2) Push the AM switch light for AM or USB mode of operation.

a) Make sure that the switch light is on for AM, or off for USB.

3) Turn the SENS control clockwise for maximum gain.

4) Set the STANDBY frequency window to an approved test frequency.

5) Push the display transfer switch.

a) Make sure that the STANDBY frequency moves to the ACTIVE frequency window.

(d) Push and release the first officer's PTT switch.

1) Make sure that you hear a 1 kHz tune-in-progress tone in the headset.

NOTE: A continuous or pulsed tone indicates that the coupler is tuning to a new frequency. The coupler tune tone will sound no longer than 15 seconds. The average coupler tune time is approximately 1 to 7 seconds.

NOTE: Some coupler types are able to tune quickly when previously used frequencies are selected, in which case the tune tone may be only a momentary beep.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Do these steps to do a voice communication test with a radio tower operator or with another aircraft utilizing HF channel:

1) Push and hold the PTT switch while you speak.

a) Make sure that you hear the sidetone in the headset while you speak.

b) Make sure that the LRU STATUS, KEY INTERLOCK, and CONTROL FAIL lights on the front panel of the HF transceiver are not on.

2) Release the PTT switch while you listen.

a) Make sure that the quality of the transmitted and received voice is satisfactory.

b) Make sure that the sound of the received voice decreases and increases when you turn the SENS control counterclockwise and clockwise on the RTP.

c) Make sure that the sound of the received voice changes when you turn the HF-2 volume control on the first officer's ACP with no change in voice quality.

(f) If more than one mode of operation (AM and USB) is available at your location, do this step:

1) Push the AM switch light to the other mode of operation and make a voice transmission.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-11-00-860-008

(1) Do these steps at the P8 panel:

(a) Push and release the HF-1 switch on the captain's ACP.

1) Make sure that the switch light goes off.

(b) Push and release the HF-2 switch on the first officer's ACP.

1) Make sure that the switch light goes off.

(c) Push and release the HF-1 volume control on the captain's ACP.

1) Make sure that the volume control indicator light goes off.

(d) Push and release the HF-2 volume control on the first officer's ACP.

1) Make sure that the volume control indicator light goes off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HF TRANSCEIVER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) HF Transceiver Removal

(2) HF Transceiver Installation.

B. The No. 1 and No. 2 HF transceivers are located in the aft cargo compartment on the electronic equipment rack E6.

TASK 23-11-21-000-801

2. HF Transceiver Removal

(Figure 401)

A. General

(1) This task has instructions to remove the HF transceiver from the equipment rack.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 23-11-21-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-11-21-010-002

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 23-11-21-010-003

(3) Open the applicable access panel to the equipment rack.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-11-21-910-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(4) Before you touch the HF transceiver [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

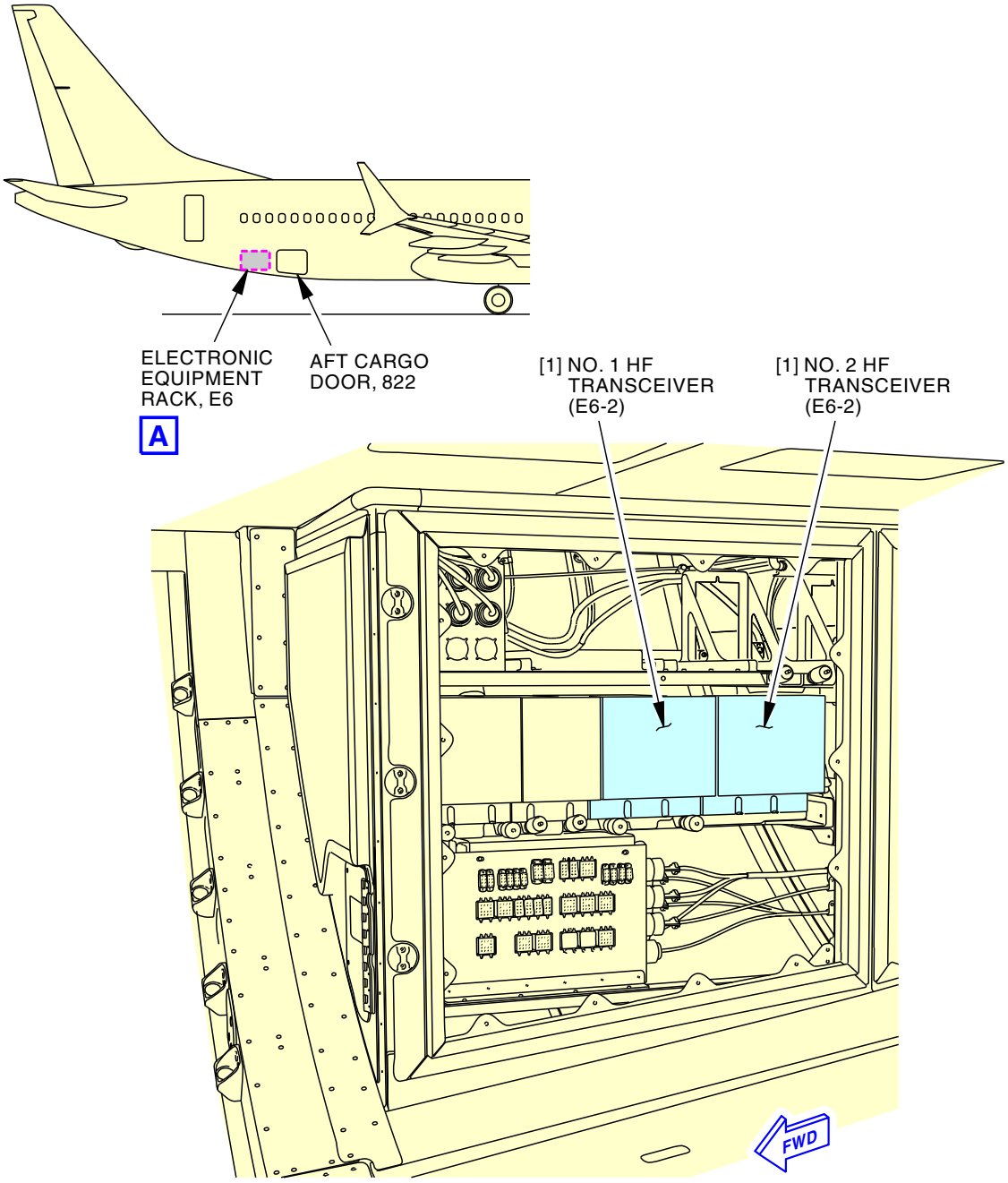
F. HF Transceiver Removal

SUBTASK 23-11-21-020-002

(1) Remove the HF transceiver [1] (TASK 20-10-07-000-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRONIC EQUIPMENT RACK, E6

A

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HF Transceiver Installation
Figure 401/23-11-21-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-11-21-400-801

3. HF Transceiver Installation

(Figure 401)

A. General

(1) This task has instructions to install the High Frequency (HF) transceiver to the equipment rack.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	HF transceiver	23-11-21-01-310	PTF ALL

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
822	Aft Cargo Door

F. HF Transceiver Installation

SUBTASK 23-11-21-910-002

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before you touch the HF transceiver [1], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-11-21-420-002

(2) Install the HF transceiver [1] (TASK 20-10-07-400-801).

G. HF Transceiver Installation Test

SUBTASK 23-11-21-910-003

WARNING

DO NOT OPERATE THE HF SYSTEM WHILE YOU REFUEL OR DEFUEL THE AIRPLANE. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

WARNING

MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(1) Do not operate the HF system while a fuel operation is done on the airplane.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-11-21-860-004

(2) Close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-11-21-850-002

(3) Push the applicable HF switch light (HF 1 or HF 2), on the No. 1 radio tuning panel (RTP-1), on the aft electronic panel, P8.

(a) Make sure that the switch light comes on.

SUBTASK 23-11-21-730-001

(4) Make sure that the internal blower fan on the HF transceiver operates.

PTF ALL; AIRPLANES WITH P/N 822-0990-004

SUBTASK 23-11-21-740-001

(5) At the front panel of the transceiver for the applicable HF communication system (HF-1 or HF-2), push and release the TEST switch to do the Built-In-Test Equipment (BITE) test of that system.

PTF ALL

(a) Make sure that these conditions occur:

1) All three Light Emitting Diode (LED)s on the HF transceiver front panel come on red for one to three seconds.
2) The Line Replaceable Unit (LRU) STATUS LED changes to green.
3) The KEY INTERLOCK and CONTROL FAIL LEDs stay red for one to three seconds.
4) All three LEDs go off for one to seven seconds.
5) The LRU STATUS LED comes on green for about 30 seconds.

NOTE: Any of the three LEDs that come on red after the self-test is completed indicates a defective condition.

6) The KEY INTERLOCK and CONTROL FAIL LEDs stay off.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 23-11-21-410-002

(1) Close the applicable access panel to the equipment rack.

SUBTASK 23-11-21-410-003

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HF ANTENNA - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) HF Antenna Removal

(2) HF Antenna Installation.

B. The HF antenna is part of the leading edge of the vertical stabilizer.

TASK 23-11-51-000-801

2. HF Antenna Removal

A. General

(1) This task has instructions to remove the HF antenna.

B. References

Reference	Title
55-33-11-000-801	Vertical Stabilizer (Fin) Leading Edge Removal (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
322	Vertical Fin - Removable Fin Leading Edge

D. Prepare for the Removal

SUBTASK 23-11-51-860-001

WARNING REMOVE THE ELECTRICAL POWER FOR THE HF COMMUNICATION SYSTEM BEFORE YOU REMOVE THE HF COMMUNICATION ANTENNA. HF SIGNALS CAN CAUSE ELECTRICAL SHOCK AND INJURIES TO PERSONS.

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

E. HF Antenna Removal

SUBTASK 23-11-51-020-001

(1) Remove the part of the leading edge of the vertical stabilizer that contains the HF antenna (TASK 55-33-11-000-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-11-51-400-801

3. HF Antenna Installation

A. General

(1) This task has instructions to install the HF Antenna.

B. References

Reference	Title
23-11-00-730-801	HF Communication System - System Test (P/B 501)
55-33-11-400-801	Vertical Stabilizer (Fin) Leading Edge Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
322	Vertical Fin - Removable Fin Leading Edge

D. HF Antenna Installation

SUBTASK 23-11-51-420-001

(1) Install the part of the leading edge of the vertical stabilizer that contains the HF antenna (TASK 55-33-11-400-801).

E. HF Antenna Installation Test

SUBTASK 23-11-51-860-003

WARNING REMOVE THE ELECTRICAL POWER FOR THE HF COMMUNICATION SYSTEM BEFORE YOU REMOVE THE HF COMMUNICATION ANTENNA. HF SIGNALS CAN CAUSE ELECTRICAL SHOCK AND INJURIES TO PERSONS.

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-11-51-730-001

(2) Do this task: HF Communication System - System Test, TASK 23-11-00-730-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HF ANTENNA - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) HF Antenna Electrical Bond Check.

B. The HF antenna is part of the leading edge of the vertical stabilizer.

C. You must remove the HF antenna coupler(s) before you do the electrical bond check.

(1) The HF antenna coupler(s) are installed adjacent to the HF antenna in the vertical stabilizer.

TASK 23-11-51-760-801

2. HF Antenna Electrical Bond Check

(Figure 601)

A. General

(1) This task has instructions to do a HF antenna electrical bond check.

B. References

Reference	Title
23-11-61-000-801	HF Antenna Coupler Removal (P/B 401)
23-11-61-400-801	HF Antenna Coupler Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
322	Vertical Fin - Removable Fin Leading Edge

E. HF Antenna Electrical Bond Check

SUBTASK 23-11-51-010-001

CAUTION YOU MUST PREPARE EACH HF ANTENNA COUPLER FOR THE REMOVAL. IF YOU DO NOT PREPARE FOR THE REMOVAL, YOU CAN CAUSE DAMAGE TO INTERNAL PARTS OF THE HF ANTENNA COUPLER.

(1) Remove the HF antenna coupler (TASK 23-11-61-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-11-51-760-001

(2) Use an intrinsically safe approved bonding meter, COM-1550 to measure the resistance at the HF antenna coupler mount and the airplane structure (SWPM 20-20-00, Paragraph 21).

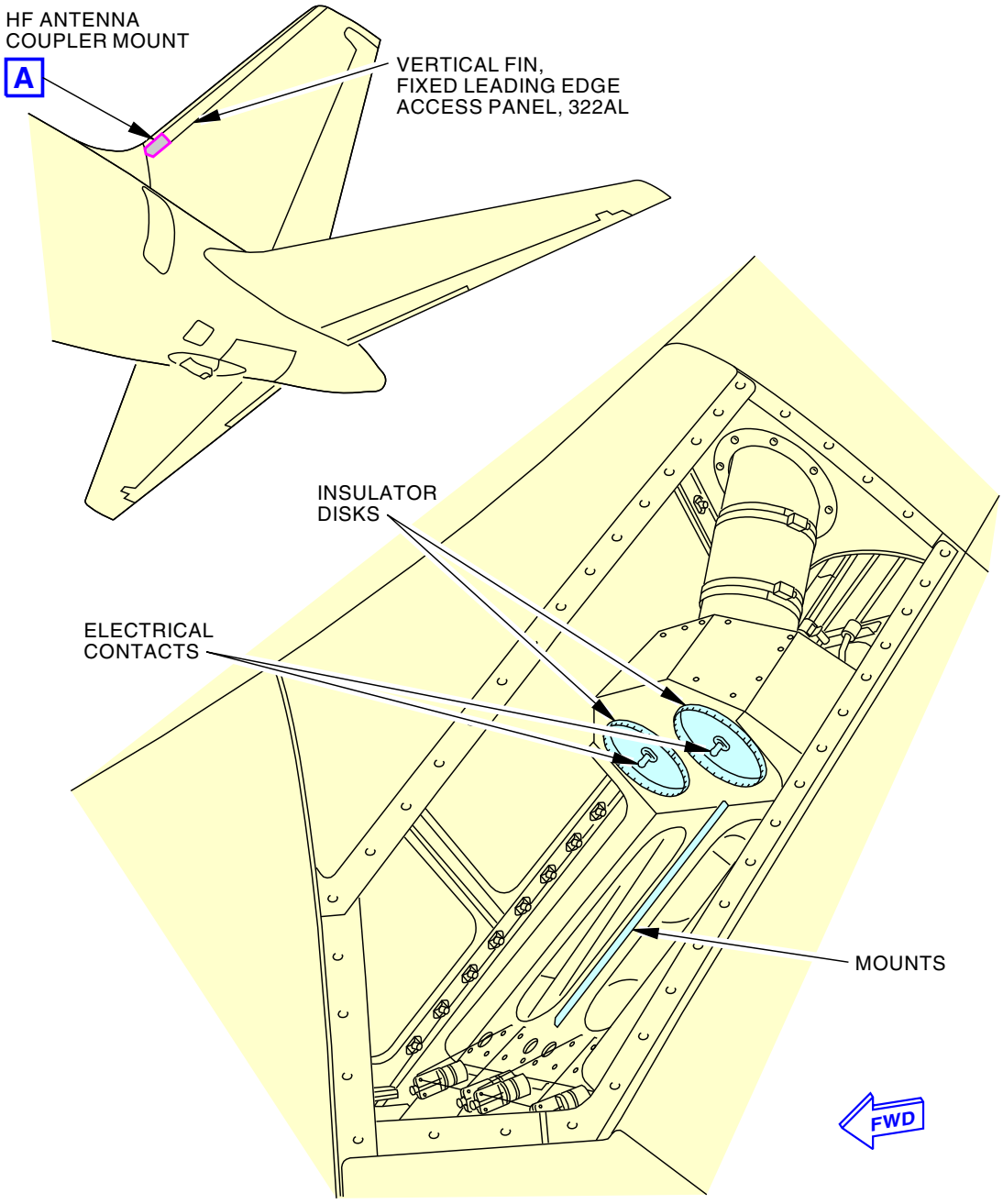
(a) Make sure that the resistance is 10 milliohms or less.

SUBTASK 23-11-51-410-001

(3) Install the HF antenna coupler (TASK 23-11-61-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HF Antenna Electrical Bond Check
Figure 601/23-11-51-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HF ANTENNA COUPLER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) HF Antenna Coupler Removal

(2) HF Antenna Coupler Installation.

B. The No. 1 and No. 2 HF antenna couplers are located in the antenna coupler mount in the lower leading edge of the vertical stabilizer.

TASK 23-11-61-000-801

2. HF Antenna Coupler Removal

(Figure 401)

A. General

(1) This task has instructions to remove the HF antenna coupler.

B. References

Reference	Title
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
322	Vertical Fin - Removable Fin Leading Edge

E. Access Panels

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge
322AR	Vertical Fin, Fixed Leading Edge

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 23-11-61-860-001

WARNING BEFORE YOU OPEN THE COUPLER ACCESS PANEL, REMOVE ELECTRICAL POWER FROM EACH HF COMMUNICATION SYSTEM. HF SIGNALS CAN CAUSE A SHOCK, CAN BURN YOU, AND CAN CAUSE INJURIES TO PERSONNEL.

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-11-61-010-002

(2) Remove the applicable access panels:

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge
322AR	Vertical Fin, Fixed Leading Edge

NOTE: The HF-1 antenna coupler is installed on the left side of the vertical stabilizer. The HF-2 antenna coupler is installed on the right side of the vertical stabilizer.

SUBTASK 23-11-61-020-001

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Before you touch the HF antenna coupler [2], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

G. HF Antenna Coupler Removal

SUBTASK 23-11-61-020-002

(1) Remove the lockwire [3] from the electrical connector [5] (TASK 20-10-44-000-801).

SUBTASK 23-11-61-020-003

(2) Disconnect the coaxial connector [4] and the electrical connector [5] from the HF antenna coupler [2].

SUBTASK 23-11-61-020-004

(3) Loosen and disengage the hold-down nuts.

SUBTASK 23-11-61-020-005

CAUTION OBEY THESE STEPS WHEN YOU REMOVE THE HF ANTENNA COUPLER. IF YOU IGNORE THESE STEPS, DAMAGE TO THE HF ANTENNA COUPLER CAN OCCUR.

(4) Remove the HF antenna coupler [2]:

(a) Always keep the bottom side of the coupler in the down position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do not set the coupler on its end (with the handle up).

(c) If you rotate or invert the coupler, this can cause damage to the internal parts.

SUBTASK 23-11-61-420-001

(5) Install protective dust caps on the HF antenna coupler [2], coaxial connector [4], and electrical connector [5] (TASK 20-40-12-400-803).

SUBTASK 23-11-61-420-002

(6) If the airplane will return-to-service with one or more HF trays empty, install the MS20995C32 lockwire, G01048, on each unused hold-down fitting as follows:

NOTE: The HF tray is located in a high vibration area.

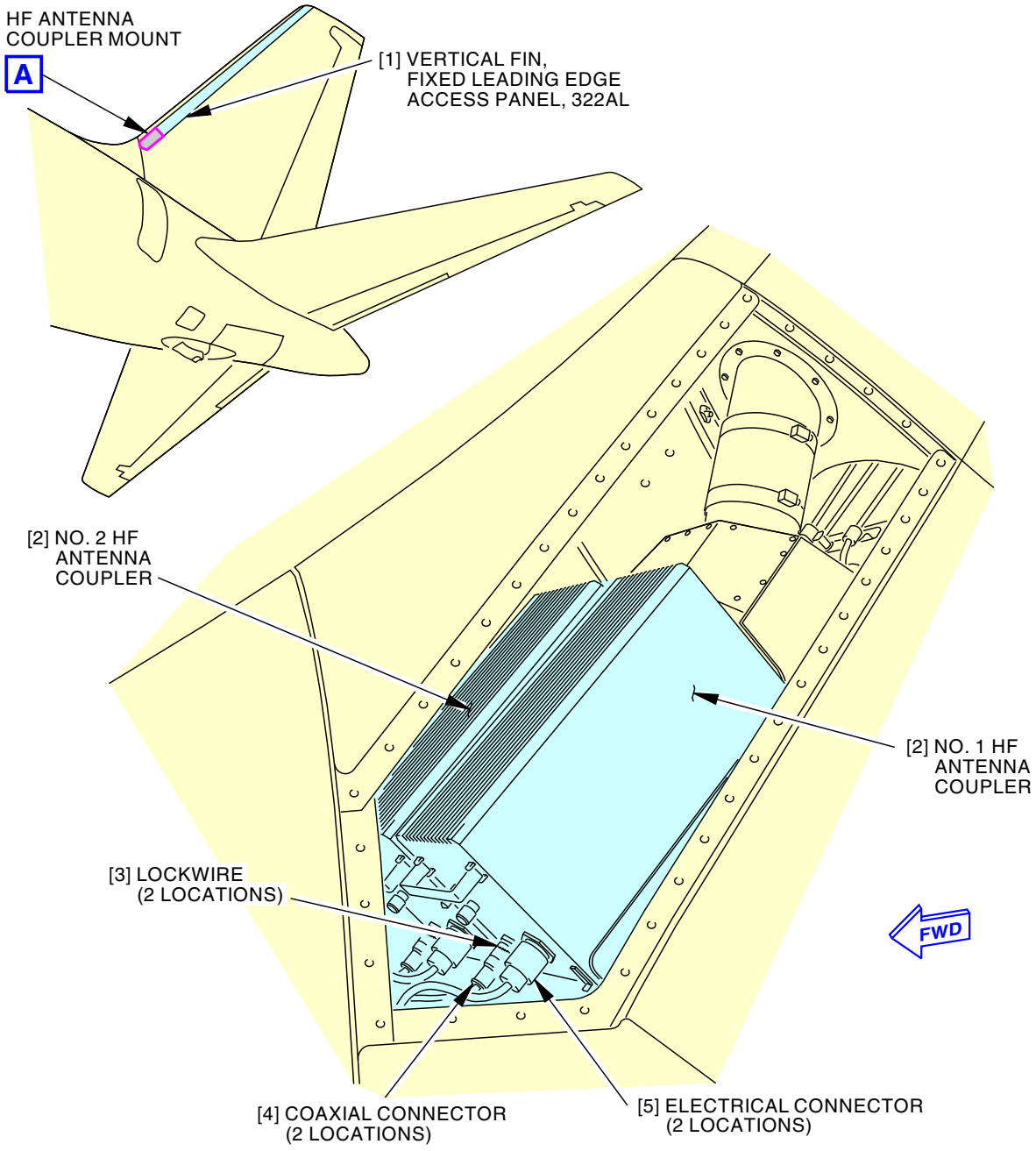
(a) Rotate the hold-down fitting upward, and hold in place.

(b) Tighten down the wing nut until it contacts with the HF tray.

(c) Install the MS20995C32 lockwire, G01048, using the pre-drilled holes in the hold-down fitting, and the holes in the HF tray (TASK 20-10-44-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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HF Antenna Coupler Installation
Figure 401/23-11-61-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-11-61-400-801

3. HF Antenna Coupler Installation

(Figure 401)

A. General

(1) This task has instructions to install the High Frequency (HF) antenna coupler.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
B50135	Solvent - 1,1,1 Trichloroethane	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51448	Abrasive - Aluminum Oxide Paper, 220 grit or finer	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	HF antenna coupler	23-11-61-05-015	PTF ALL
		23-11-61-05-110	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
322	Vertical Fin - Removable Fin Leading Edge

F. Access Panels

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge
322AR	Vertical Fin, Fixed Leading Edge

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. HF Antenna Coupler Installation

SUBTASK 23-11-61-420-003

WARNING BEFORE YOU OPEN THE COUPLER ACCESS PANEL, REMOVE ELECTRICAL POWER FROM EACH HF COMMUNICATION SYSTEM. HF SIGNALS CAN CAUSE A SHOCK, CAN BURN YOU, AND CAN CAUSE INJURIES TO PERSONNEL.

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-11-61-420-008

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(2) Before you touch the HF antenna coupler [2], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-11-61-710-001

(3) Examine the coupler mount.

(a) Make sure that there is no corrosion or damage from electrical arcing at the rear connection.

SUBTASK 23-11-61-420-004

(4) Install the HF antenna coupler [2].

(a) Engage and hand tighten the hold down nuts.

SUBTASK 23-11-61-710-002

(5) Examine the connectors for bent or broken pins, and for dirt or foreign objects.

(a) If you find a problem, do the repair of the damage you found.

SUBTASK 23-11-61-420-005

(6) Connect the coaxial connector [4] and electrical connector [5] to the HF antenna coupler [2].

SUBTASK 23-11-61-420-006

(7) Install the lockwire [3] on the coaxial connector [4] (TASK 20-10-44-400-801).

SUBTASK 23-11-61-420-007

(8) If the airplane will return-to-service with one or more HF trays empty, install MS20995C32 lockwire, G01048, on each unused hold-down fitting as follows.

NOTE: The HF tray is located in a high vibration area.

(a) Rotate the hold-down fitting upward, and hold in place.

(b) Tighten down the wing nut until it contacts with the HF tray.

(c) Install MS20995C32 lockwire, G01048, using the pre-drilled holes in the hold-down fittings, and the holes in the HF tray (TASK 20-10-44-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-11-61-100-001

(9) Remove remaining sealant from access panels and mating surfaces before installing.

(a) Use plastic or hardwood tools that do not scratch surfaces and that do not prevent paint or sealant bond.

(b) It is possible to sharpen the scrapers with 220 grit or finer abrasive paper, G51448.

(c) Before you use a sharpened scraper, rub it with a cotton wiper, G00034, to remove loose sanding particles.

SUBTASK 23-11-61-100-002

(10) Clean and prepare surfaces to be sealed with solvent, B50135, or solvent, B00184, or solvent, B00666.

SUBTASK 23-11-61-410-004

(11) Install the applicable access panels:

Number	Name/Location
322AL	Vertical Fin, Fixed Leading Edge
322AR	Vertical Fin, Fixed Leading Edge

SUBTASK 23-11-61-390-001

(12) Apply aerodynamic sealant, A02315, at the panel joints.

(a) Make sure that the cured sealant is aligned within the tolerance of 0 in. (0.000 mm) at the surface to 0.010 in. (0.254 mm) below the surface.

(b) Do not cover fasteners with aerodynamic sealant.

SUBTASK 23-11-61-440-002

(13) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

H. HF Antenna Coupler Installation Test

SUBTASK 23-11-61-800-001

WARNING DO NOT OPERATE THE HF SYSTEM WHILE YOU REFUEL OR DEFUEL THE AIRPLANE. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(1) Do not operate the HF system while a fuel operation is done on the airplane.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-11-61-860-009

(2) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-11-61-720-001

(3) Do these steps at the P8 panel:

(a) Connect a headset/boom microphone to the captain's jack panel.

(b) Do these steps to prepare the captain's Audio Control Panel (ACP) for HF system operation:

1) Push and release the applicable HF-1 or HF-2 switch.

a) Make sure that the switch light comes on.

2) Push and release the applicable HF-1 or HF-2 volume control.

a) Make sure that the volume control indicator light comes on.

3) Set the volume control to the middle position.

4) Set the MASK/BOOM switch (if installed) to the BOOM position.

(c) Do these steps at the No. 1 radio tuning panel (RTP-1):

1) Push the applicable HF switch light (HF 1 or HF 2).

a) Make sure that the switch light comes on.

2) Push the AM switch light for AM or USB mode of operation.

a) Make sure that the switch light is on for AM, or off for USB.

3) Turn the SENS control fully clockwise.

4) Set the STANDBY frequency window to an approved test frequency.

5) Push the display transfer switch.

a) Make sure that the STANDBY frequency moves to the ACTIVE frequency window.

(d) Push and release the captain's Push-To-Talk (PTT) switch.

1) Make sure that you hear a 1 kHz tune-in-progress tone in the headset.

NOTE: A continuous or pulsed tone indicates that the coupler is tuning to a new frequency. The coupler tune tone will sound no longer than 15 seconds. The average coupler tune time is approximately 1 to 7 seconds.

NOTE: Some coupler types are able to tune quickly when previously used frequencies are selected, in that case the tune tone may be only a momentary beep.

(e) Do these steps to do a voice communication test with a radio tower operator:

1) Push and hold the PTT switch while you speak.

a) Make sure that you hear the sidetone in the headset while you speak.

2) Release the PTT switch while you listen.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that the quality of the transmitted and received voice is satisfactory.

b) Make sure that the sound of the received voice changes when you turn the applicable HF-1 or HF-2 volume control, on the captain's ACP, with no change in voice quality.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 23-11-61-860-008

(1) Do these steps at the P8 panel:

(a) Push and release the applicable HF-1 or HF-2 switch, on the captain's ACP.

1) Make sure that the switch light goes off.

(b) Push and release the applicable HF-1 or HF-2 volume control, on the captain's ACP.

1) Make sure that the volume control indicator light goes off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VERY HIGH FREQUENCY (VHF) COMMUNICATION SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An Operational Test of the VHF communication transceiver.

(2) A System Test of the VHF communication system.

B. There are three VHF communication systems installed on the airplane. These are: VHF-1, VHF-2, and VHF-3.

C. The No. 1, No. 2 and No. 3 radio tuning panels are located on the aft electronic panel, P8, in the flight compartment.

TASK 23-12-00-710-801

2. VHF Communication System - Operational Test

A. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

C. Procedure

SUBTASK 23-12-00-010-001

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 23-12-00-720-002

(2) Do the steps that follow for the BITE test:

(a) Push and release the TEST switch on the applicable VHF communication transceiver front panel to do the BITE test for each VHF communication system (VHF-1, VHF-2, or VHF-3).

1) Make sure that these conditions occur:

a) All three LEDs on the VHF transceiver front panel will turn on red for about two seconds.

b) The LRU STATUS LED turns green and the CONTROL FAIL and ANTENNA FAIL LEDs will stay red for about two seconds.

c) All three LEDs will go off for about two seconds.

d) The LRU STATUS LED comes on green and the CONTROL FAIL and ANTENNA FAIL LEDs remain off.

NOTE: Any of the three LEDs that come on red after the self-test is completed indicates a defective condition.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Put the Airplane Back to Its Initial Condition

SUBTASK 23-12-00-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 23-12-00-730-801

3. VHF Communication System - System Test

A. General

(1) Make sure that the flight interphone system operates. The communication test uses these flight interphone system components: Audio Control Panels (ACP)s, headsets, microphones, and Push-To-Talk (PTT) switches.

(2) Make sure that the airplane is not in or near any large metal structures. Very High Frequency (VHF) communication while the airplane is on the ground can be degraded by airplane position or blocked by Radio Frequency (RF) signal attenuation due to ground equipment and structures. If the quality of the transmitted and received voice signals is not satisfactory, move the airplane to a better position.

(3) Make sure you use approved test frequencies when you speak with the radio tower operator.

B. References

Reference	Title
23-27-00-700-802	ACARS - System Test (P/B 501)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. BITE Test

SUBTASK 23-12-00-710-001

(1) Do the Built-In-Test Equipment (BITE) test of the VHF communication system, do this task: VHF Communication System - Operational Test, TASK 23-12-00-710-801.

E. RTP to RTP Interface Test

SUBTASK 23-12-00-730-006

(1) Do these steps at the Radio Tuning Panel (RTP)s:

(a) Push the VHF-1 switch on each RTP.

1) Make sure that three RTPs show the same ACTIVE and STANDBY frequencies.

(b) Set RTP-1 to a new frequency.

1) Make sure that RTP-2 and RTP-3 show the new frequency.

(c) Set RTP-2 to a new frequency.

1) Make sure that RTP-1 and RTP-3 show the new frequency.

(d) Set RTP-3 to a new frequency.

1) Make sure that RTP-1 and RTP-2 show the new frequency.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Make sure that the crosstuning lights are on, in three RTPs.

NOTE: The crosstuning light is above the display transfer switch, between the frequency displays.

(f) Push the VHF-2 switch on RTP-2.

(g) Push the VHF-3 switch on RTP-3.

(h) Make sure that the crosstuning lights are off, in three RTPs.

F. VHF Communication Test

SUBTASK 23-12-00-730-002

(1) Do a test of the VHF-1 communication system:

(a) Connect a headset/boom microphone to the captain's jack panel.

(b) Do these steps to prepare the captain's Audio Control Panel (ACP) for VHF-1 system operation:

1) Push and release the VHF-1 switch.

a) Make sure that the switch light comes on.

2) Push and release the VHF-1 volume control.

a) Make sure that the volume control indicator light comes on.

3) Set the volume control to the middle position.

4) Set the MASK/BOOM switch (if installed) to the BOOM position.

(c) Set the RTP-1 to an approved test frequency for VHF-1.

(d) Do these steps to do a voice communication test with a radio tower operator:

1) Push and hold a PTT switch while you speak.

a) Make sure that the sidetone in the headset is present while you speak.

2) Release the PTT switch while you listen.

a) Make sure that the quality of the transmitted and received voice is satisfactory.

b) Make sure that the sound of the received voice changes when you turn the VHF-1 volume control on the captain's ACP with no change in voice quality.

SUBTASK 23-12-00-730-003

(2) Do a test of the VHF-2 communication system:

(a) Connect a headset/boom microphone to the first officer's jack panel.

(b) Do these steps to prepare the first officer's ACP for VHF-2 system operation:

1) Push and release the VHF-2 switch.

a) Make sure that the switch light comes on.

2) Push and release the VHF-2 volume control.

a) Make sure that the volume control indicator light comes on.

3) Set the volume control to the middle position.

4) Set the MASK/BOOM switch (if installed) to the BOOM position.

(c) Set the RTP-2 to an approved test frequency for VHF-2.

(d) Do these steps to do a voice communication test with a radio tower operator:

1) Push and hold a PTT switch while you speak.

a) Make sure you hear the sidetone in the headset while you speak.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Release the PTT switch while you listen.
   a) Make sure that the quality of the transmitted and received voice is satisfactory.
   b) Make sure that the sound of the received voice changes when you turn the VHF-2 volume control on the first officer's ACP with no change in voice quality.

SUBTASK 23-12-00-730-004

(3) Do a test of the VHF-3 communication system:
   (a) Connect a headset/boom microphone to the captain's jack panel.
   (b) Do these steps to prepare the captain's ACP for VHF-3 system operation:
      1) Push and release the VHF-3 switch.
         a) Make sure that the switch light comes on.
      2) Push and release the VHF-3 volume control.
         a) Make sure that the volume control indicator light comes on.
      3) Set the volume control to the middle position.
      4) Set the MASK/BOOM switch (if installed) to the BOOM position.
   (c) Set the RTP-2 to an approved test frequency for VHF-3.
   (d) Do these steps to do a voice communication test with a radio tower operator:
      1) Push and hold a PTT switch while you speak.
         a) Make sure you hear the sidetone in the headset while you speak.
      2) Release the PTT switch while you listen.
         a) Make sure that the quality of the transmitted and received voice is satisfactory.
         b) Make sure that the sound of the received voice changes when you turn the VHF-3 volume control on the captain's ACP with no change in voice quality.

SUBTASK 23-12-00-730-005

(4) Do the data test of the VHF-3 communication system, do this task: ACARS - System Test, TASK 23-27-00-700-802

NOTE: If the VHF Link Test passes, the VHF-3 communication system is operational.

G. Put the Airplane Back to Its Initial Condition

SUBTASK 23-12-00-860-008

(1) At each ACP, push and release the applicable VHF switches.
   (a) Make sure all the switch lights go off.

SUBTASK 23-12-00-860-009

(2) At each ACP, push and release the applicable VHF volume controls.
   (a) Make sure all the volume control indicator lights go off.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VHF COMMUNICATION ANTENNA - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) VHF Communication Antenna Removal

(2) VHF Communication Antenna Installation

B. The VHF communications antennas are installed at centerline on top and bottom of the airplane.

TASK 23-12-11-000-801

2. VHF Communication Antenna Removal

(Figure 401)

A. General

(1) This procedure has steps to remove a VHF Communication Antenna [1].

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

C. Location Zones

Zone	Area
123	Forward Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left

D. Prepare for the VHF Communication Antenna Removal

SUBTASK 23-12-11-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

E. VHF Communication Antenna Removal

SUBTASK 23-12-11-020-001

(1) Remove the VHF Communication Antenna [1]:

(a) Remove the 10 bolts [2] that attach the VHF Communication Antenna [1] to the airplane.

CAUTION

REMOVE THE AERODYNAMIC FILLET SEAL CAREFULLY WITH THE SEALANT REMOVAL TOOL. DAMAGE TO THE AIRPLANE SKIN OR THE COAXIAL CABLE CAN OCCUR.

(b) Hold the VHF Communication Antenna [1], use a sealant removal tool, COM-2481 to break the fillet seal around the base of the antenna.

CAUTION

DO NOT PULL ON THE COAXIAL CABLE. CAREFULLY MOVE THE ANTENNA. DISCONNECT THE COAXIAL CONNECTOR. THIS WILL PREVENT DAMAGE TO THE COAXIAL CABLE.

(c) Move the VHF Communication Antenna [1] away from the airplane to get access to the coaxial connector [4] on the coaxial cable.

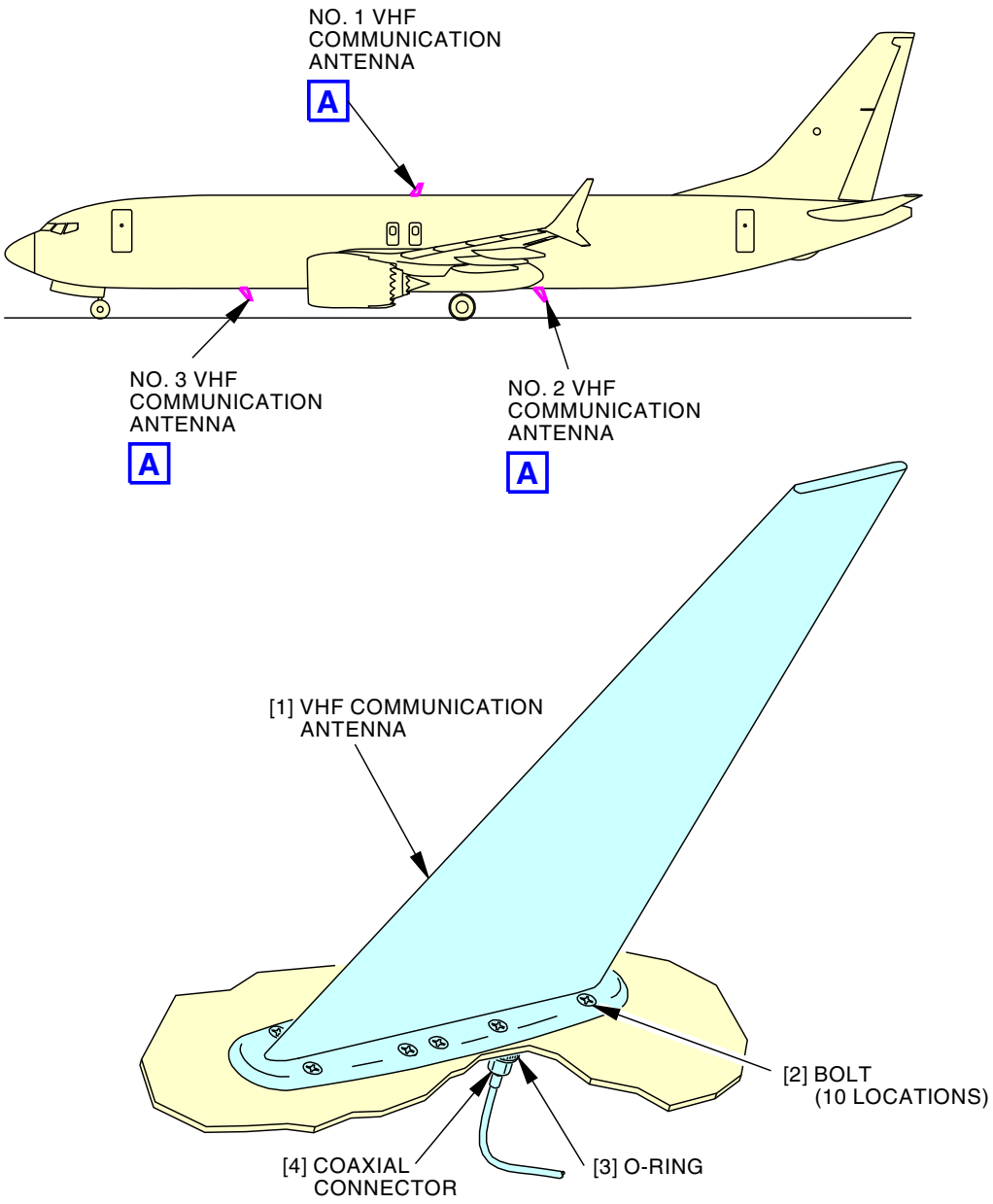
(d) Disconnect the coaxial connector [4] and o-ring [3] from the VHF Communication Antenna [1].

1) For the top VHF Communication Antenna [1], make sure that the coaxial connector [4] does not go back through the hole in the fuselage. Make sure that you attach the coaxial connector [4] to something to prevent this.

(e) Remove the VHF Communication Antenna [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VHF COMMUNICATION ANTENNA (EXAMPLE)

A

2411668 S00061528570_V1

VHF Communication Antenna Installation
Figure 401/23-12-11-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-12-11-400-801

3. VHF Communication Antenna Installation

(Figure 401)

A. General

(1) This task is the installation of the VHF Communication Antenna [1].

B. References

Reference	Title
23-12-00-730-801	VHF Communication System - System Test (P/B 501)
51-21-31-350-801	Removal and Control of Corrosion for Aluminum and Aluminum Alloys (P/B 701)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
51-22-00-390-806	Fillet Seal Application (P/B 201)
SL 20-043	Deferred Application of Aero-Sealant in Antenna Installations

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
B01026	Solvent - FCC-55	
B01054	Solvent - Methyl Ethyl Ketone and sec-Butyl Alcohol Blend - (MEK:secButyl Alcohol - 42:58 Percent)	BAC5750, ASTM D740/ASTM D1007
C50005	Coating - Chemical Conversion - Bonderite M-CR 1200S Aero (Formerly Alodine 1200S)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
D00652	Lubricant - O-Ring - Parker Super-O-Lube	
G00009	Compound - Organic Corrosion Inhibiting	BMS3-23
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	VHF Communication Antenna	23-12-11-01-010	PTF ALL
		23-12-11-01-035	PTF ALL
		23-12-11-01-165	PTF ALL
3	O-ring	23-12-11-01-015	PTF ALL
		23-12-11-01-040	PTF ALL
		23-12-11-01-170	PTF ALL

F. Location Zones

Zone	Area
123	Forward Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left

G. Prepare for the VHF Antenna Communication Installation

SUBTASK 23-12-11-100-001

(1) Clean the airplane mating surface.

(a) Remove sealant with a sealant removal tool, COM-2481.

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Clean the airplane mating surface with a cotton wiper, G00034, that is moist with solvent, B00148, solvent, B00666, solvent, B00184, FCC-55 solvent, B01026, or solvent, B01054.

(c) Use a clean cotton wiper, G00034, to clean the airplane mating surface.

1) Make sure that the airplane mating surface is dry.

SUBTASK 23-12-11-100-002

(2) If the mating surface has corrosion or other damage, or if the airplane mating surface has no alodine, then do these steps to prepare the airplane mating surface:

NOTE: An alternative installation procedure is to apply corrosion inhibiting compound, G00009, on an airplane mating surface that has no alodine.

(a) If there is corrosion, do this task: Removal and Control of Corrosion for Aluminum and Aluminum Alloys, TASK 51-21-31-350-801.

(b) Apply a layer of Bonderite M-CR 1200S Aero coating, C50005, to the airplane mating surface (TASK 51-21-41-370-802).

SUBTASK 23-12-11-420-001

(3) Install a new o-ring [3].

(a) If it is necessary, lubricate the o-ring [3] with Parker Super-O-Lube lubricant, D00652.

(b) Align the o-ring [3] in the groove.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. VHF Antenna Communication Installation

SUBTASK 23-12-11-420-002

(1) Install the VHF Communication Antenna [1].

(a) Connect the coaxial connector [4] to the VHF Communication Antenna [1].

(b) Apply a layer of sealant, A00247, to the shank and threads of 9 of the 10 bolts [2].

(c) Put the VHF Communication Antenna [1] in its position. Then install 9 of the 10 bolts [2].

1) Tighten the bolts [2] to 80 in-lb (9.0 N·m).

NOTE: Keep one of the ten bolts [2] out to do an electrical bond check.

SUBTASK 23-12-11-700-001

(2) Measure the resistance between the VHF Communication Antenna [1] base and the airplane skin with a intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00, Paragraph 21).

NOTE: Use the empty bolt hole to get access to the VHF Communication Antenna [1] base.

(a) Make sure that the resistance is 0.001 ohm or less.

SUBTASK 23-12-11-420-003

(3) Apply a layer of sealant, A00247, to the shank and threads of the last bolt [2].

(a) Install the last bolt [2].

1) Tighten the bolt [2] to 80 in-lb (9.0 N·m).

SUBTASK 23-12-11-390-001

(4) Apply sealant, A00247, around the base of the VHF Communication Antenna [1] to make a corrosion fillet seal (TASK 51-22-00-390-806).

NOTE: Operators can defer the application of the aero-sealant in the antenna installation until next “A” check or overnight maintenance stop to prevent a flight delay. The deferral of the application of sealant must not exceed of eight weeks (SL 20-043).

(a) Let the sealant dry for the correct cure time.

I. VHF Communication Antenna Installation Test

SUBTASK 23-12-11-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

SUBTASK 23-12-11-730-001

(2) Do this task: VHF Communication System - System Test, TASK 23-12-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VHF COMMUNICATION ANTENNA - INSPECTION

1. General

A. This procedure has this task:

(1) An inspection of the Very High Frequency (VHF) Communication Antenna.

TASK 23-12-11-210-801

2. VHF Communication Antenna - Inspection

(Figure 601)

A. General

(1) This task provides the steps to inspect the Very High Frequency (VHF) communication antenna.

B. References

Reference	Title
23-12-11-000-801	VHF Communication Antenna Removal (P/B 401)
23-12-11-400-801	VHF Communication Antenna Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
C00033	Coating - Protective Enamel, Flexibility Use	BMS10-60 Type II
G50316	Cloth - Clean, Dry, Lint-free, White, Cotton	

D. Location Zones

Zone	Area
123	Forward Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left

E. VHF Communication Antenna Inspection

SUBTASK 23-12-11-210-001

(1) Visually inspect VHF antenna for damage or unusual wear.

(a) Make sure that all mating surfaces are clean and carefully fitted.

1) If you find any indications of cracking or delamination, replace the VHF antenna:

- VHF Communication Antenna Removal, TASK 23-12-11-000-801
- VHF Communication Antenna Installation, TASK 23-12-11-400-801

(b) Make sure that the paint erosion on the leading edge is less than 1 in. (25.40 mm) back.

1) If the paint erosion is down to the primer, repaint the antenna as follows:

NOTE: Figure 601 shows an example of caution markings for the bottom VHF antenna. Caution markings for the upper VHF antenna are the same. Follow airline requirements for VHF antenna marking. Color and pattern can be different.

a) Clean the surface of the VHF antenna with solvent, B00148, and clean, dry cotton cloth, G50316.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

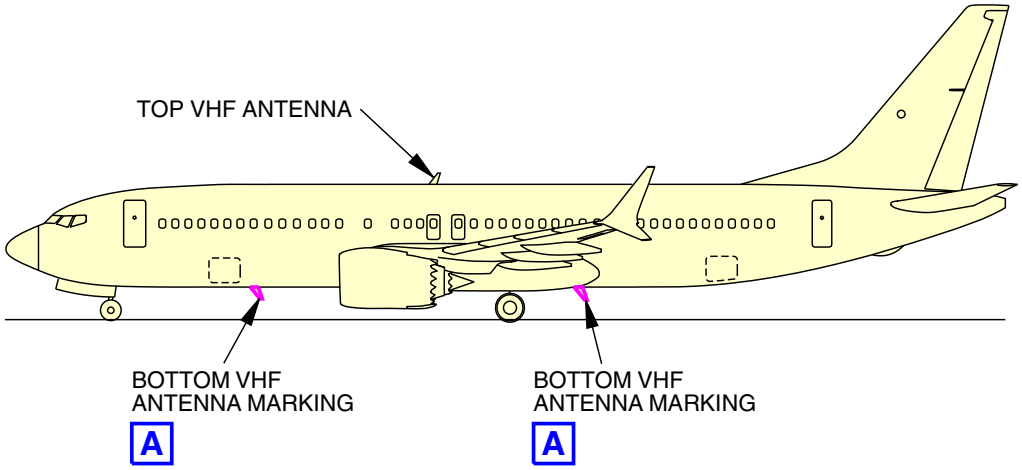
DO NOT USE METAL-BASED PAINT, OR METALLIC TAPE ON THE VHF ANTENNA. A HIGH VSWR WILL OCCUR. THIS CAN CAUSE DAMAGE TO THE VHF COMM TRANSCEIVER.

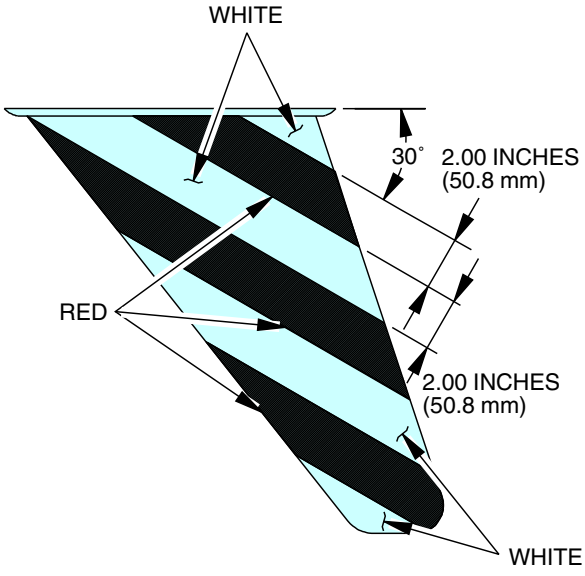
b) Apply coating, C00033, Enamel or equivalent.

c) If applicable, apply paint stripes on the bottom or top VHF antenna (Figure 601).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





**BOTTOM VHF ANTENNA MARKING
(PAINT STRIPES, EXAMPLE)**

A

2411669 S00061528571_V3

**VHF Communication Antenna Inspection
Figure 601/23-12-11-990-803**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VHF COMMUNICATION TRANSCEIVER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) VHF Communication Transceiver Removal

(2) VHF Communication Transceiver Installation

B. The VHF Communication Transceivers are located in the main equipment center.

TASK 23-12-21-000-801

2. VHF Communication Transceiver Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the VHF Communication Transceivers [1].

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the VHF Communication Transceiver Removal

SUBTASK 23-12-21-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

SUBTASK 23-12-21-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-12-21-020-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

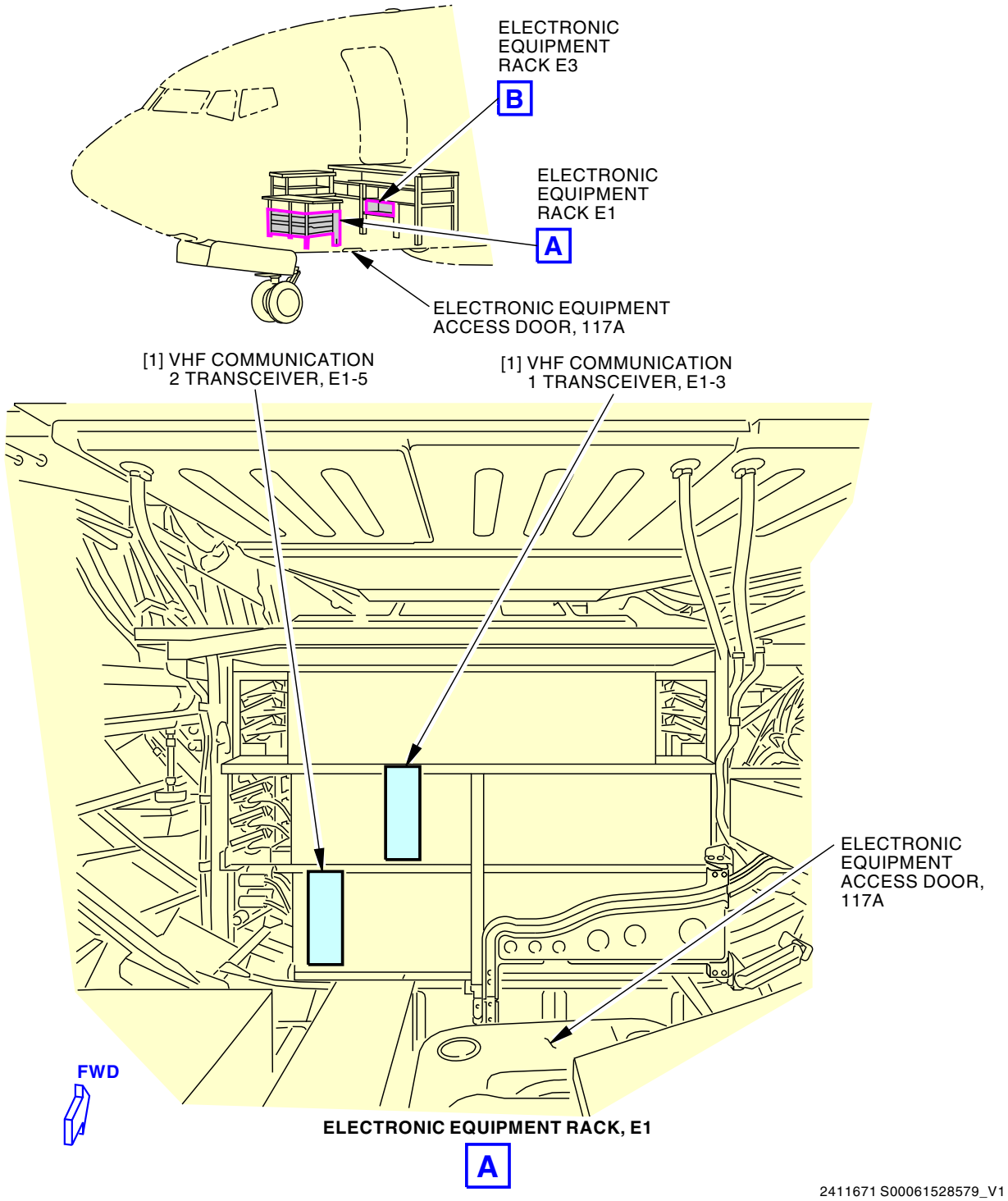
F. VHF Communication Transceiver Removal

SUBTASK 23-12-21-020-002

(1) Remove the VHF Communication Transceiver [1], do this task: E/E Box Removal, TASK 20-10-07-000-801.

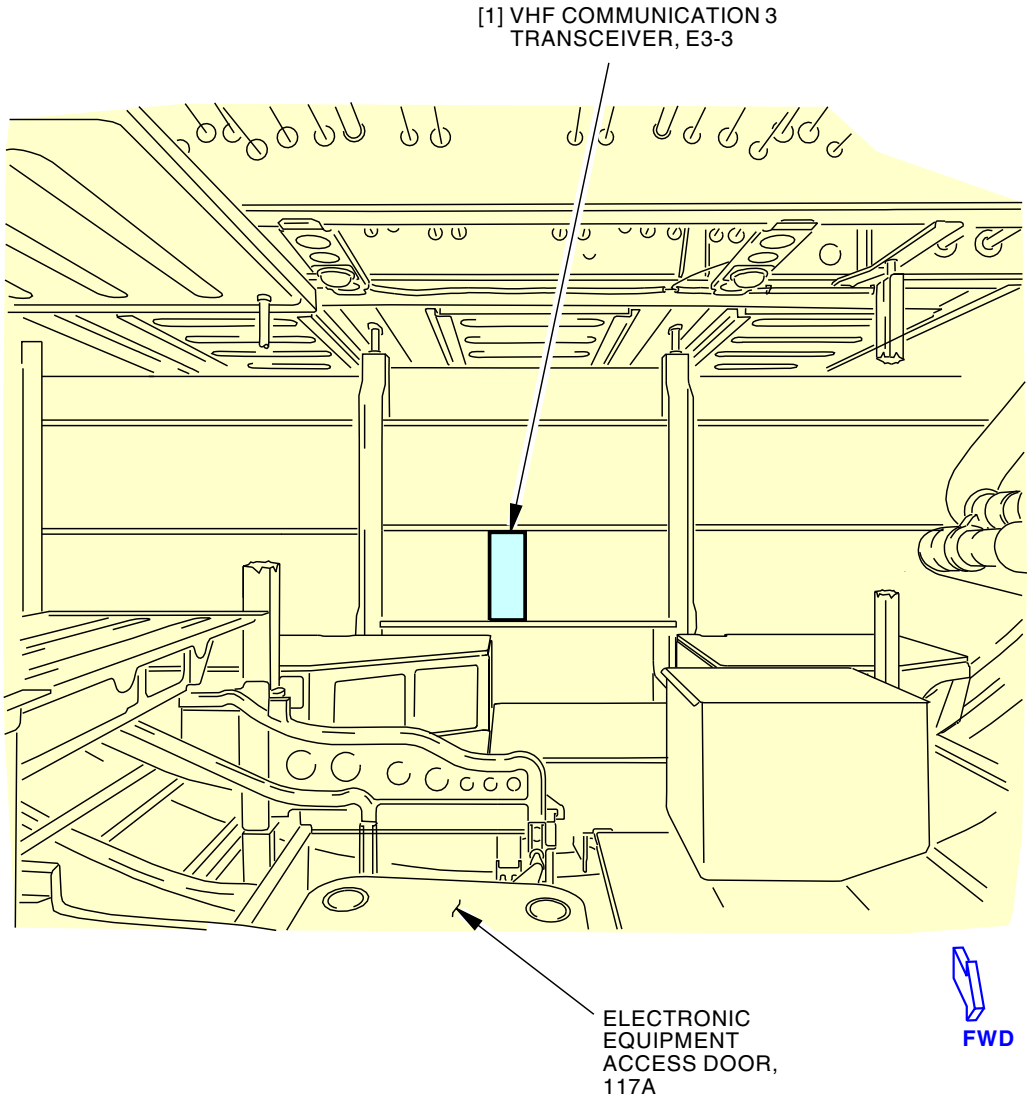
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VHF Communication Transceiver Installation
Figure 401/23-12-21-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRONIC EQUIPMENT RACK, E3

B

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VHF Communication Transceiver Installation
Figure 401/23-12-21-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-12-21-400-801

3. VHF Communication Transceiver Installation

(Figure 401)

A. General

(1) This procedure has steps to install the VHF Communication Transceivers [1].

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-12-00-710-801	VHF Communication System - Operational Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	VHF Communication Transceiver	23-12-21-04-005	PTF ALL
		23-12-21-05-005	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. VHF Communication Transceiver Installation

SUBTASK 23-12-21-420-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-12-21-420-002

(2) Install the VHF Communication Transceiver [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

SUBTASK 23-12-21-860-002

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. VHF Communication Transceiver Installation Test

SUBTASK 23-12-21-860-003

(1) Do this task: VHF Communication System - Operational Test, TASK 23-12-00-710-801.

H. Put the Airplane Back to Its Initial Condition

SUBTASK 23-12-21-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RADIO TUNING PANEL - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Radio Tuning Panel (RTP) INOP Display Toggle

TASK 23-12-41-860-801

2. Radio Tuning Panel - INOP Display Toggle

A. General

(1) The Radio Tuning Panel (RTP) [1] will show INOP in active and standby displays.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Radio Tuning Panel - INOP Display Toggle

SUBTASK 23-12-41-860-001

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

SUBTASK 23-12-41-850-001

(2) Push the Radio Tuning Panel (RTP) [1] VHF1 switch.

(a) Make sure that a frequency is shown in ACTIVE and STANDBY displays.

SUBTASK 23-12-41-850-002

(3) Push the Radio Tuning Panel (RTP) [1] VHF 2 switch.

(a) Make sure that a frequency is shown in ACTIVE and STANDBY displays.

SUBTASK 23-12-41-850-003

(4) Push the Radio Tuning Panel (RTP) [1] VHF 3 switch.

(a) Make sure that INOP is shown in ACTIVE and STANDBY displays.

NOTE: INOP is shown in 10 seconds after power is ON.

SUBTASK 23-12-41-860-002

(5) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	12	C00471	COMMUNICATIONS VHF 3

SUBTASK 23-12-41-850-004

(6) Push the Radio Tuning Panel (RTP) [1] VHF 3 switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the switch is held.

SUBTASK 23-12-41-850-005

(7) Push the Radio Tuning Panel (RTP) [1] Transfer switch.

(a) Make sure that the switch is held.

SUBTASK 23-12-41-860-003

(8) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	12	C00471	COMMUNICATIONS VHF 3

SUBTASK 23-12-41-860-004

(9) Make sure that the ACTIVE and STANDBY displays show a frequency.

(a) Release the Radio Tuning Panel (RTP) [1] Transfer switch.

(b) Release the Radio Tuning Panel (RTP) [1] VHF 3 switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RADIO TUNING PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Radio Tuning Panel (RTP) Removal

(2) Radio Tuning Panel (RTP) Installation

B. The No. 1 and No. 2 Radio Tuning Panels (RTPs) are located in the aft electronic panel, P8, in the flight compartment.

C. The No. 3 Radio Tuning Panel (RTP) is located in the aft electronic panel, P8, in the flight compartment.

TASK 23-12-41-000-801

2. Radio Tuning Panel (RTP) Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the Radio Tuning Panel (RTP) [1].

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Radio Tuning Panel (RTP) Removal

SUBTASK 23-12-41-860-005

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
D	12	C00471	COMMUNICATIONS VHF 3
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-12-41-860-006

(2) Turn the PANEL BRIGHT control on the aft electronic panel, P8, to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-12-41-020-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

E. Radio Tuning Panel (RTP) Removal

SUBTASK 23-12-41-020-002

(1) Remove the Radio Tuning Panel (RTP) [1]:

(a) Loosen the four quarter-turn fasteners [2].

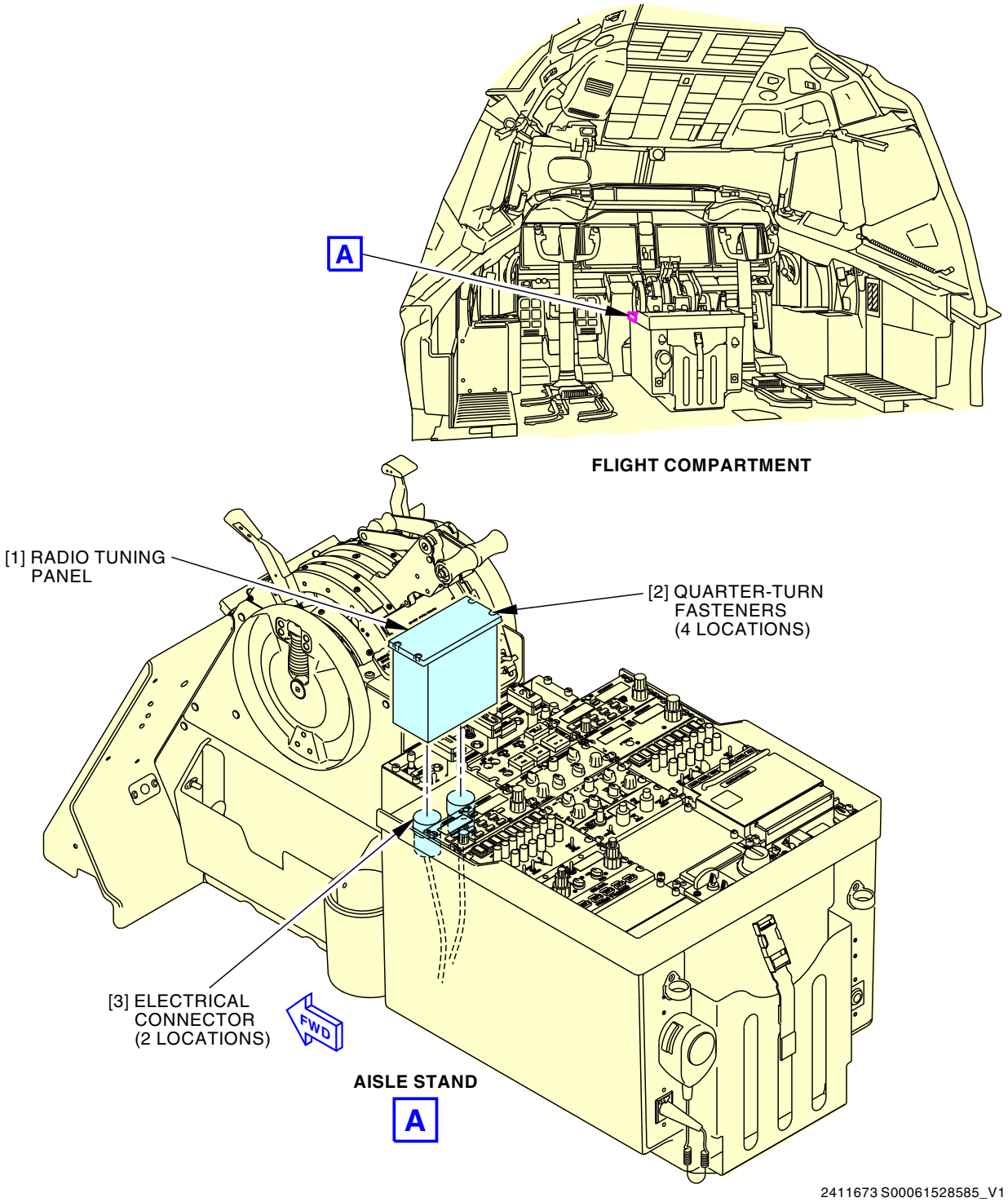
(b) Lift the Radio Tuning Panel (RTP) [1] from the P8 panel to get access to the two electrical connectors [3].

(c) Disconnect the two electrical connectors [3].

(d) Install protective covers on the two electrical connectors [3], do this task: Conductive Dust Cap and Connector Cover Installation, TASK 20-40-12-400-803.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Radio Tuning Panel (RTP) Installation
Figure 401/23-12-41-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-12-41-400-801

3. Radio Tuning Panel (RTP) Installation

(Figure 401)

A. General

(1) This procedure has steps to install the Radio Tuning Panel (RTP) [1].

B. References

Reference	Title
20-40-12-000-803	Conductive Dust Cap and Connector Cover Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Radio Tuning Panel (RTP)	23-12-41-04-010	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Radio Tuning Panel (RTP) Installation

SUBTASK 23-12-41-420-001

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-12-41-420-002

(2) Install the Radio Tuning Panel (RTP) [1]:

(a) Remove the protective covers from the two electrical connectors [3], do this task: Conductive Dust Cap and Connector Cover Removal, TASK 20-40-12-000-803.

(b) Examine the two electrical connectors [3] for bent or broken pins, dirt, and damage.

(c) Connect the two electrical connectors [3].

(d) Put the Radio Tuning Panel (RTP) [1] in its position on the P8 panel.

(e) Tighten the four quarter-turn fasteners [2].

F. Radio Tuning Panel (RTP) Installation Test

SUBTASK 23-12-41-860-008

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
E	11	C00839	COMMUNICATIONS HF 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-12-41-860-009

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	12	C00471	COMMUNICATIONS VHF 3

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-12-41-860-010

(3) Turn the PANEL BRIGHT control on the P8 panel to the midrange position.

(a) Make sure that the light changes on the Radio Tuning Panel (RTP) [1].

SUBTASK 23-12-41-860-011

WARNING DO NOT OPERATE THE COMMUNICATION SYSTEM DURING A REFUEL OPERATION. AN EXPLOSION CAN OCCUR AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE PERSONNEL STAY A MINIMUM OF 6 FEET (1.8288 METERS) AWAY FROM THE COMMUNICATION ANTENNA WHEN THE SYSTEM TRANSMITS. RADIO FREQUENCY ENERGY FROM THE COMMUNICATION ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(4) Set the Radio Tuning Panel (RTP) [1] to a test frequency.

(a) Make sure the VHF COMM 1 TRANSCEIVER operates.

SUBTASK 23-12-41-860-012

(5) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1
E	11	C00839	COMMUNICATIONS HF 1

SUBTASK 23-12-41-860-013

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-12-41-850-006

(7) Turn the PANEL BRIGHT control on the P8 panel to the midrange position.

(a) Make sure that the light changes on the Radio Tuning Panel (RTP) [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-12-41-850-007

WARNING DO NOT OPERATE THE COMMUNICATION SYSTEM DURING A REFUEL OPERATION. AN EXPLOSION CAN OCCUR AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE PERSONNEL STAY A MINIMUM OF 6 FEET (1.8288 METERS) AWAY FROM THE COMMUNICATION ANTENNA WHEN THE SYSTEM TRANSMITS. RADIO FREQUENCY ENERGY FROM THE COMMUNICATION ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(8) Set the Radio Tuning Panel (RTP) [1] to a test frequency.

(a) Make sure the VHF COMM 2 TRANSCEIVER operates.

SUBTASK 23-12-41-860-014

(9) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

SUBTASK 23-12-41-860-015

(10) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	12	C00471	COMMUNICATIONS VHF 3

SUBTASK 23-12-41-850-008

(11) Turn the PANEL BRIGHT control on the P8 panel to the midrange position.

(a) Make sure that the light changes on the Radio Tuning Panel (RTP) [1].

SUBTASK 23-12-41-850-009

WARNING DO NOT OPERATE THE COMMUNICATION SYSTEM DURING A REFUEL OPERATION. AN EXPLOSION CAN OCCUR AND CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

WARNING MAKE SURE PERSONNEL STAY A MINIMUM OF 6 FEET (1.8288 METERS) AWAY FROM THE COMMUNICATION ANTENNA WHEN THE SYSTEM TRANSMITS. RADIO FREQUENCY ENERGY FROM THE COMMUNICATION ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(12) Set the Radio Tuning Panel (RTP) [1] to a test frequency.

(a) Make sure the VHF COMM 3 TRANSCEIVER operates.

SUBTASK 23-12-41-860-016

(13) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	11	C00839	COMMUNICATIONS HF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2
D	2	C00857	COMMUNICATIONS HF 2

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SATCOM SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Satcom System Deactivation

(2) Satcom System Activation

(3) Satcom System SDU Software Installation with Onboard Network System (ONS)

(4) Satcom System SDU Configuration Check

(5) Satcom System Satellite Data Unit Configuration Module (SCM) Configuration Check.

B. For general information for software installation times and data loaders, do this task: On-Airplane Software Installation, TASK 20-15-11-470-801.

TASK 23-15-00-040-801

2. Satcom System - Deactivation

A. General

(1) This procedure removes electrical power to the SATCOM system.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-15-00-860-013

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

D. Satcom System - Tryout

NOTE: This tryout is to make sure that the Satcom System is in a zero energy state.

SUBTASK 23-15-00-710-018

(1) Do these steps on the Control Display Unit (CDU):

(a) Push the MENU key.

(b) Make sure that <SAT does not show on the MENU.

——— END OF TASK ———

TASK 23-15-00-440-801

3. Satcom System - Activation

A. General

(1) This procedure adds electrical power to the Satcom System.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Procedure

WARNING: MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FEET (3M) FROM THE SATCOM ANTENNA DURING THE TEST. RF ENERGY CAN CAUSE INJURIES TO PERSONNEL.

SUBTASK 23-15-00-860-015

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

——— END OF TASK ———

TASK 23-15-00-470-801

4. Satcom System - SDU Software Installation with ONS

A. General

(1) This procedure has steps to install software to the Satellite Data Unit (SDU) with the Onboard Network System (ONS).

(a) The SDU must contain these pieces of software:

1) SDU Operational Program Software (OPS)

2) SECURE Owner Requirements Table (ORT) software

3) USER ORT software

(2) A Control Display Unit (CDU) is necessary for this procedure.

(3) The Satellite Data Unit Configuration Module (SCM) must be installed.

(4) The airplane must be on the ground with the engines shutdown before you can install software.

(5) For the SDU to be an approved installation, the correct software part numbers must be installed.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Procedure

SUBTASK 23-15-00-860-016

(1) Do these steps at the P61 data loader control panel:

(a) Set the DATA LOAD SELECTOR switch to L CAPT UPR position.

(b) Set the SYSTEM SELECT switch to the SATCOM position.

SUBTASK 23-15-00-470-001

(2) Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

(a) Install the software one at a time in this order:

1) SDU OPS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) SECURE ORT

3) USER ORT

SUBTASK 23-15-00-860-017

(3) Do these steps at the P61 data loader control panel:

(a) Set the DATA LOAD SELECTOR switch to C SINGLE SYS position.

(b) Set the SYSTEM SELECT switch to the NORMAL position.

SUBTASK 23-15-00-860-023

(4) Open and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

SUBTASK 23-15-00-700-001

(5) Do this task: Satcom System - SDU Configuration Check, TASK 23-15-00-700-801.

——— END OF TASK ———

TASK 23-15-00-700-801

5. Satcom System - SDU Configuration Check

A. General

(1) The configuration check makes sure that the Satellite Data Unit (SDU) has the correct software installed.

(a) The SDU must contain these pieces of software:

1) SDU Operational Program Software (OPS)

2) SECURE Owner Requirements Table (ORT) software

3) USER ORT software

(2) A Control Display Unit (CDU) is necessary for this procedure.

(3) The Satellite Data Unit Configuration Module (SCM) must be installed.

(4) The airplane must be on the ground with the engines shutdown before you do the check.

(5) For the SDU to be an approved installation, the correct software part numbers must be installed.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Software Configuration Check

SUBTASK 23-15-00-700-003

(1) Do these steps at a CDU to do a configuration check of the OPS, the SECURE ORT, and the USER ORT software for the ICS-300 SDU:

(a) Push the MENU key.

(b) Select SAT.

(c) Push the NEXT PAGE key.

(d) Select PART NUM.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Push the NEXT PAGE key twice to access page 3 of 7.

1) Make sure that the SECURE ORT and USER ORT software part numbers shown match the load list or installation plan.

(f) Push the PREV PAGE key three times to access page 7 of 7.

1) Make sure that the OPS software part number shown matches the load list or installation plan.

SUBTASK 23-15-00-470-003

(2) If the software part numbers are not correct, do this task: Satcom System - SDU Software Installation with ONS, TASK 23-15-00-470-801.

——— END OF TASK ———

TASK 23-15-00-700-802

6. Satcom System - SCM Configuration Check

A. General

(1) The configuration check makes sure that the Satellite Data Unit Configuration Module (SCM) has the correct customer configuration.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-15-00-860-025

(1) Read the procedure carefully before you make changes to the SAT USER ORT CONFIG page.

NOTE: You can cause a malfunction of the SATCOM system if you install the incorrect configuration.

SUBTASK 23-15-00-860-026

(2) Use the Control Display Unit (CDU) to get access to the SAT USER ORT CONFIG page:

(a) Push the MENU key to show the MENU page.

(b) Select the Line Select Key (LSK) adjacent to <SAT to show the SAT-PHONE 1/2 page.

(c) Push the NEXT PAGE key to show the SAT-PHONE 2/2 page.

(d) Select the LSK adjacent to CONFIG> to show the SAT CONFIG 1/2 page.

(e) Enter the USER ORT password “ICG1234”.

NOTE: This is a password protected page. It is the responsibility of the operator to record and to keep the password if it is changed. The SATCOM supplier may charge a fee for recovery of a lost password.

(f) Select the LSK adjacent to USER ORT> to show the SAT USER ORT CONFIG page.

1) Make sure that the CDU scratch-pad does not show one of these messages:

- NOT ALLOWED
- INVALID PASSWORD

a) If NOT ALLOWED shows, make sure that the LG ALT DIS + CONT circuit breaker is closed, and the airplane is in ground mode.

b) If INVALID PASSWORD shows, make sure that the password is correct.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> If the problem continues, then contact the SATCOM supplier.

SUBTASK 23-15-00-210-001

(3) Make sure that the SAT USER ORT configuration agrees with the airline operational requirements.

(a) If the SAT USER ORT configuration is different from the airline operational requirements, follow the applicable steps in this procedure to change the SAT USER ORT configuration.

NOTE: The LSK adjacent to <DEFAULT can be selected to return the SAT USER ORT configuration to supplier defaults.

SUBTASK 23-15-00-730-001

(4) If there are fault indicators, select the LSK adjacent to CLEAR FAULTS> to remove the indicators.

SUBTASK 23-15-00-730-002

(5) If it is necessary to change the default USER ORT password, then do the steps that follow:

NOTE: These steps let the user change the default USER ORT password to a customized password. A customized password must be pre-approved by the airlines engineering with a valid airplane engineering order.

(a) On the SAT USER ORT CONFIG page, enter the new password (up to 8 digits with 0-9 and A-Z) on the CDU scratch-pad.

NOTE: It is the responsibility of the operator to record and to keep the password if it is changed. The SATCOM supplier may charge a fee for recovery of a lost password.

(b) Select the LSK adjacent to <SET USER PW.

(c) Select the LSK adjacent to SAVE>.

NOTE: The CDU shows SAVE IN PROG on the left side of the scratch-pad. SAVE COMPLETE shows when the save is done.

(d) Select the LSK adjacent to <RESTART then <CONFIRM to reboot the CDU with the changed configuration.

SUBTASK 23-15-00-730-003

(6) If it is necessary to change the Aircraft Communications Addressing And Reporting System (ACARS) service provider, then do the steps that follow:

(a) On the SAT USER ORT CONFIG page, select the LSK adjacent to <ACARS SRV to toggle through the five available options:

- GLOBALINK
- ARINC DIRECT
- HONEYWELL GDC
- SITA
- OTHER

(b) Select the LSK adjacent to SAVE> to save changes.

NOTE: The CDU shows SAVE IN PROG on the left side of the scratch-pad. SAVE COMPLETE shows when the save is done.

(c) Select the LSK adjacent to <RESTART then <CONFIRM to reboot the CDU with the changed configuration.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-15-00-730-004

(7) If it is necessary to change the CALL DISPLAY, then do the steps that follow:

NOTE: The CALL DISPLAY type shows either the name or number of the caller. When the CALL DISPLAY is set BY NAME, the CDU will display the call recipient name if located in the directory for ATG calls. The caller user ID is displayed for GTA calls. When the CALL DISPLAY is set BY NUMBER, the CDU displays the number, which may be a phone number or short code. Speed dial numbers are not displayed on the CDU.

(a) On the SAT USER ORT CONFIG page, select the LSK adjacent to <CALL DISPLAY to toggle between BY NUMBER and BY NAME.

(b) Select the LSK adjacent to SAVE> to save changes.

NOTE: The CDU shows SAVE IN PROG on the left side of the scratch-pad. SAVE COMPLETE shows when the save is done.

(c) Select the LSK adjacent to <RESTART then <CONFIRM to reboot the CDU with the changed configuration.

SUBTASK 23-15-00-730-005

(8) If it is necessary to change the Entry Control, then do the steps that follow:

NOTE: The scratch-pad number entry control type has two configurations, ALL NUMBERS and SHORT CODES. When ENTRY CONTROL is set to ALL NUMBERS, the scratch-pad accepts long numbers, short codes, speed dial numbers and station extensions. When ENTRY CONTROL is set to SHORT CODES, the scratch-pad accepts short codes, speed dial numbers and station extensions only.

(a) On the SAT USER ORT CONFIG page, select the LSK adjacent to <ENTRY CONTROL to toggle between ALL NUMBERS and SHORT CODE.

(b) Select the LSK adjacent to SAVE> to save changes.

NOTE: The CDU shows SAVE IN PROG on the left side of the scratch-pad. SAVE COMPLETE shows when the save is done.

(c) Select the LSK adjacent to <RESTART then <CONFIRM to reboot the CDU with the changed configuration.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SATELLITE COMMUNICATIONS (SATCOM) SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks for the satellite communications (SATCOM) system:

(1) A BITE test at the flight management computer control display unit

(2) A BITE test at the satellite data unit

(3) An operational test

(4) A system test.

TASK 23-15-00-740-801

2. Satellite Communications (SATCOM) System - BITE Test at the Flight Management Computer Control Display Unit

A. General

(1) The BITE test at the flight management computer Control Display Unit (CDU) does the same test as the BITE test at the Satellite Data Unit (SDU).

(2) The BITE test does a check of these components and functions:

(a) SDU

(b) Satellite Data Unit Configuration Module (SCM)

(3) The SATCOM system does a power-up sequence during the BITE test. If the SATCOM system is logged-on, it will log-off. When the BITE test is complete, the SATCOM system will log-on again (automatic mode).

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-15-00-740-006

(1) Do these steps to do the BITE test at a CDU:

(a) Open this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

(b) Wait for a minimum of 20 seconds.

(c) Make sure that SAT does not show on the CDU.

(d) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

(e) Wait for the SDU to power-up.

NOTE: It can take approximately up to 90 seconds for the SDU to power up.

(f) Make sure that SAT shows on each CDU.

(g) Select SAT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Push the NEXT PAGE key.

1) Make sure that the BITE status shows PASS.

——— END OF TASK ———

TASK 23-15-00-740-802

3. Satellite Communications (SATCOM) System - BITE Test at the Satellite Data Unit

A. General

(1) The BITE test at the Satellite Data Unit (SDU) does the same test as the BITE test at the flight management computer Control Display Unit (CDU).

(2) The BITE test does a check of these components and functions:

(a) SDU

(b) Satellite Data Unit Configuration Module (SCM).

(3) The SATCOM system does a power-up sequence during the BITE test. If the SATCOM system is logged-on, it will log-off. When the BITE test is complete, the SATCOM system will log-on again (automatic mode).

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Procedure

SUBTASK 23-15-00-740-008

(1) Do a BITE test at the front panel of the SDU:

(a) Open and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

(b) Monitor the Light Emitting Diode (LED) while the SDU starts.

NOTE: It can take approximately up to 90 seconds for the SDU to start.

1) Make sure that the POWER light comes on in Green.

2) Make sure that the FAULT light comes on in Green.

——— END OF TASK ———

TASK 23-15-00-710-801

4. Satellite Communications (SATCOM) System - Operational Test

A. General

(1) The operational test includes these tests:

(a) A BITE test.

(b) A RF link test.

(c) A Voice Communication Test.

(d) A SATCOM link test with ACARS

(2) When you do this test, make sure that the airplane is not in or near a large metal structure. The SATCOM antenna must have a clear view to the applicable satellite. Buildings and other large objects can prevent a clear view.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) This voice communication test uses the captain's audio control panel, headset, and push-to-talk switch, but you can do the test with equipment at one of the other flight crew positions.

B. References

Reference	Title
23-27-00-700-802	ACARS - System Test (P/B 501)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare Procedure

SUBTASK 23-15-00-860-006

(1) Connect a headset with built in microphone to the Captain's mic/audio jack panel.

SUBTASK 23-15-00-860-007

(2) Push all the volume control switches to the OFF position on all the ACPs.

(a) Make sure that the related LED for the volume controls are OFF.

SUBTASK 23-15-00-860-008

(3) Make sure that the SAT-1 and SAT-2 volume controls of the Captain’s ACP are at the approximate midrange position.

SUBTASK 23-15-00-860-028

(4) Make sure that the ATC transporders are turned to the ON position.

E. BITE Test

SUBTASK 23-15-00-740-003

(1) Do this task: Satellite Communications (SATCOM) System - BITE Test at the Flight Management Computer Control Display Unit, TASK 23-15-00-740-801.

F. RF Link Test

SUBTASK 23-15-00-710-001

WARNING MAKE SURE PERSONS STAY A MINIMUM OF 10 FEET AWAY FROM THE ANTENNA DURING THE VOICE COMMUNICATION TEST. RF ENERGY CAN CAUSE INJURIES TO PERSONS.

(1) Do these steps to connect to the SATCOM system:

(a) Push the MENU key on the Left and Right Control Display Unit (CDU).

1) Make sure that <SAT shows on the two CDU.

(b) Push the Line Select Key (LSK) adjacent to <SAT on the Left and Right CDU.

1) Make sure that the SAT-PHONE page 1/2 page shows on the two CDU.

(c) Do these steps at one CDU:

1) Push the NEXT PAGE key to show SAT-PHONE page 2/2.

2) Push the LSK adjacent to <LOG.

3) Wait for the log-on status to show LOGGED ON.

NOTE: It can take up to 10 minutes for the SATCOM system to log-on automatically to the GES.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) If not logged on, push the LSK adjacent to LOGON>.

4) Continue the procedure after the SATCOM system LOGGED ON.

5) Push the NEXT PAGE key on the CDU to show the SAT LOGON 2/2 page.

a) Make sure that the MODEM status shows as follows:

<1> LBT-1 AVAILABLE

<2> LBT-2 AVAILABLE

<3> LBT-3 AVAILABLE

b) Make sure that the SIGNAL level of LBT-1, LBT-2, and LBT-3 stays above 3.

G. Voice Communication Test

SUBTASK 23-15-00-710-002

(1) Do these steps to enter the approved test telephone numbers to the SAT-1 and SAT-2 channels:

(a) Push the MENU key on the CDU.

(b) Push the LSK adjacent to <SAT to go into the SAT-PHONE page 1/2.

(c) Push the LSK adjacent to DIRECTORY to go into the SAT DIRECTORY INDEX page.

(d) Push the LSK adjacent to AOC> to go into the AOC DIR page.

(e) Push the LSK adjacent to an applicable category.

(f) Find the name for an approved test telephone number.

(g) Push the LSK adjacent to that telephone number.

NOTE: The left LSK will set the number on SAT-1. The right LSK will set the number on SAT-2.

1) Make sure that a name or a telephone number shows below the <MAKE CALL of SAT-1 and SAT-2 on SAT-PHONE page 1/2.

SUBTASK 23-15-00-710-003

(2) If you can not find an approved test telephone number from the SAT DIRECTORY pages, do these steps:

(a) Make sure that the CDU shows SAT-PHONE page 1/2.

(b) Enter a short code or use the prefix 001 followed by an approved test telephone number to the scratchpad.

NOTE: If you do not know a short code or an approved test telephone number for this test, speak with your engineering department.

(c) Push the LSK for the SAT-1 channel and SAT-2 channel to prepare the call.

(d) Make sure that the short code or telephone number shows below the <MAKE CALL of SAT-1 and SAT-2 channel.

SUBTASK 23-15-00-710-004

(3) Set the SATCOM call priority for SAT-1 and SAT-2 channels.

NOTE: The values for priority are PUB (public), LOW (low), HGH (high), SBB (Swift BB) and EME (emergency). Do not use priority EME (emergency) for test, because the priority EME is reserved for emergency and distress calls only, it will set the alarms to the ground earth stations (GES) and notifies the airline operators that an airplane is in an emergency or distress condition.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-15-00-710-005

(4) Make a call on the SAT-1 channel:

(a) Make sure that a telephone number shows for SAT-1 (SAT-PHONE page 1/2).

(b) Push the LSK adjacent to <MAKE CALL on the SAT-1 channel.

NOTE: The status of SAT-1 changes from READY - DIALING - RINGING - ANSWERED.

(c) When SAT-1 call is connected:

1) Make sure the SAT-1 status changes on the CDU (SAT PHONE page 1/2).

2) Make sure that the CALL lights for SAT1 come on at all the Audio Control Panel (ACP).

(d) Do these steps at the captain's ACP for SAT-1 call:

1) Push the SAT1 MIC switch to the ON position.

2) Push the SAT1 receiver volume control switch to the ON position.

3) Adjust the receiver volume control switch for the SAT1.

(e) Use the captain's headset to speak with the person on the other end of the call for the SAT-1 channel.

1) Push the Push-To-Talk (PTT) to use the microphone.

2) Tell the person at the other end to hold this line while you make a call on the SAT-2 channel.

SUBTASK 23-15-00-710-010

(5) Make a SATCOM call on the SAT-2 channel while the SAT-1 call is connected:

(a) Make sure that a telephone number shows for SAT-2 (SAT-PHONE page 1/2).

(b) Push the LSK adjacent to <MAKE CALL on the SAT-2 channel.

NOTE: The status of SAT-2 changes from READY - DIALING - RINGING - ANSWERED.

(c) When SAT-2 call is connected:

1) Make sure the SAT-1 status changes on the CDU (SAT PHONE page 1/2).

2) Make sure that the CALL lights for SAT2 come on at all the ACP.

3) Make sure that the two SATCOM channels (SAT-1 and SAT-2) stay connected at the same time.

(d) Do these steps at the captain's ACP for the SAT-2 call:

1) Push the SAT2 MIC switch.

2) Push the SAT2 receiver volume control switch to the ON position.

3) Adjust the receiver volume control switch for the SAT2.

SUBTASK 23-15-00-710-014

(6) Do a voice quality check on both SAT-1 and SAT-2 channels:

(a) Make sure that transmitted voice and received voice qualities are loud and clear.

(b) While you listen to the received voice. Slowly turn the applicable receiver volume control on the captain's ACP.

(c) Make sure that the sound level changes smoothly without a voice quality change.

SUBTASK 23-15-00-710-019

(7) Tell the person at the other end to call the SATCOM for this airplane after you complete the SAT-1 and SAT-2 calls.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-15-00-710-016

(8) To complete the SATCOM calls:

(a) Push the LSK adjacent to <END CALL for the SAT-1 channel.

(b) Push the LSK adjacent to <END CALL for the SAT-2 channel.

(c) Make sure that the CALL lights for the SAT1 and SAT2 go off when the call is completed.

SUBTASK 23-15-00-710-020

(9) Do a check of the following when the Ground to Air call is received:

(a) Make sure that a single tone is heard.

(b) Make sure that the applicable SAT call lights on the ACPs come on.

(c) Push the LSK adjacent to <ANSWER to answer the call on the CDU.

(d) Make sure that the applicable SAT status on the CDU shows ANSWERED.

(e) Make sure that transmitted voice and received voice qualities are loud and clear.

(f) Push the LSK adjacent to <END CALL to complete the call.

SUBTASK 23-15-00-710-021

(10) Make sure that ACARS passes the SATCOM link test (TASK 23-27-00-700-802).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 23-15-00-860-009

(1) Push the MENU key on the CDUs.

SUBTASK 23-15-00-860-010

(2) If the air data inertial reference system is not necessary, set the mode select switches on the mode select unit (P5) to the OFF position.

——— END OF TASK ———

TASK 23-15-00-710-802

5. Satellite Communications (SATCOM) System - System Test

A. General

(1) The system test includes these tests:

(a) A software configuration check

(b) An operational test.

(2) The software configuration check makes sure that the Satellite Data Unit (SDU) has correct software.

(3) When you do the system test, make sure that the airplane is not in or near a large metal structure. The SATCOM antenna must have a clear view to the applicable satellite. Buildings and other large objects can prevent a clear view.

(4) This system test uses the captain's audio control panel, headset, and push-to-talk switch, but you can do the test with equipment at one of the other flight crew positions.

B. References

Reference	Title
23-15-00-700-801	Satcom System - SDU Configuration Check (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right

D. Software Configuration Check

SUBTASK 23-15-00-740-005

(1) Do this task: Satcom System - SDU Configuration Check, TASK 23-15-00-700-801.

NOTE: You must know the correct software part numbers for the SDU. For the SDU to be an approved installation, the correct software part numbers must be installed.

E. Operational Test

SUBTASK 23-15-00-710-017

(1) Do this task: Satellite Communications (SATCOM) System - Operational Test, TASK 23-15-00-710-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-15-00-860-011

(1) Push the MENU key on the CDUs.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SATELLITE DATA UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the Satellite Data Unit (SDU).

(2) An installation of the SDU.

B. The SDU is installed in the E4-2 equipment rack. The E4-2 rack is in the Electrical and Electronics Compartment (EEC) on the right side.

TASK 23-15-01-000-801

2. Satellite Data Unit Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the Satellite Data Unit (SDU).

(2) It is the responsibility of the airlines to contact their SATCOM Service Provider to deactivate the Short Burst Data (SBD) and Voice services for the SDU when it is removed from the aircraft.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 23-15-01-860-005

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

SUBTASK 23-15-01-010-003

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Satellite Data Unit Removal

SUBTASK 23-15-01-020-002

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before you touch the satellite data unit [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 23-15-01-020-003

(2) To remove the satellite data unit [1], do this task: E/E Box Removal, TASK 20-10-07-000-801.

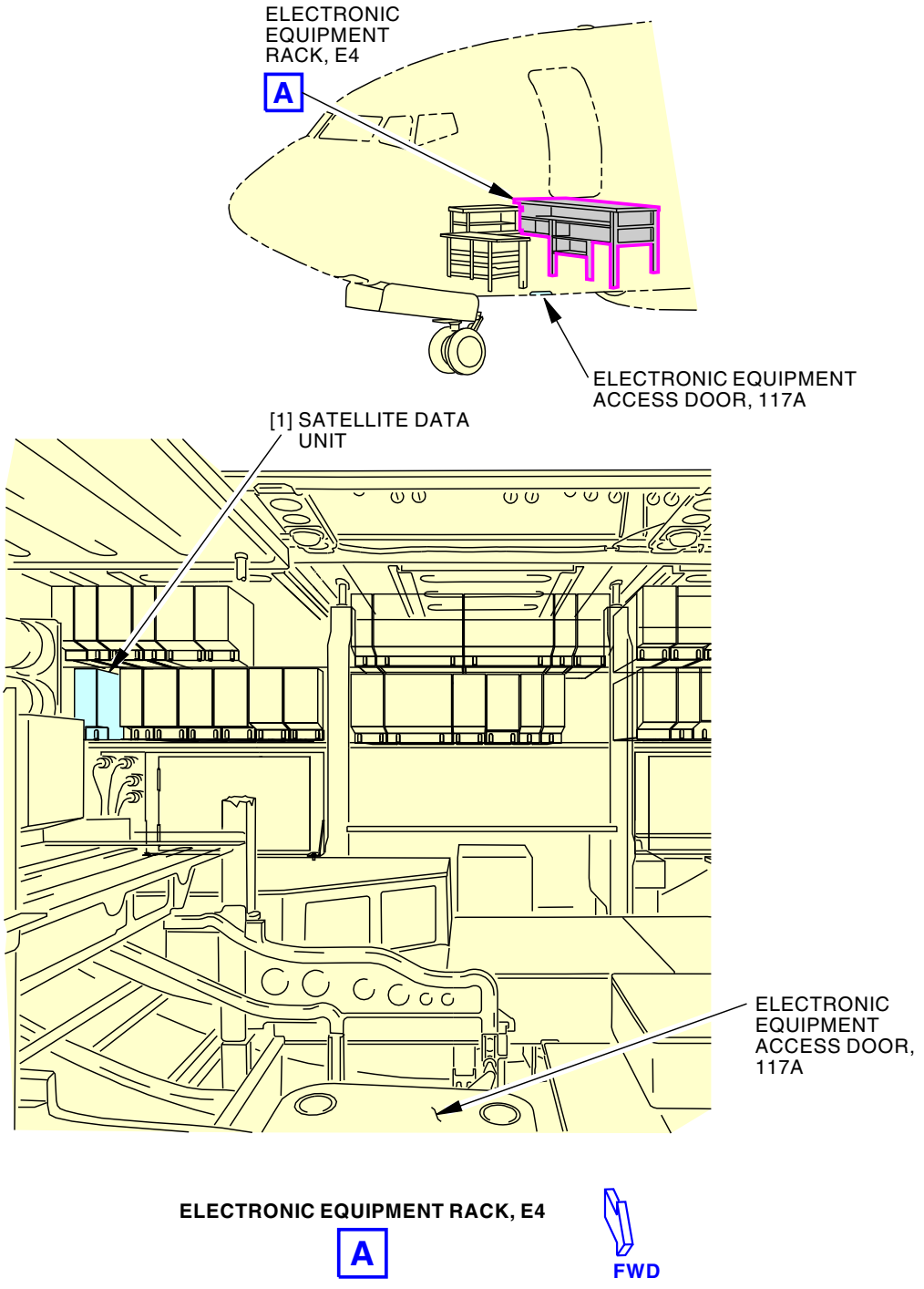
(a) Put the Electrostatic Discharge Sensitive (ESDS) protective cover on the electrical connector of the removed SDU.

SUBTASK 23-15-01-040-002

(3) If applicable, contact your designated SATCOM Service Provider to deactivate the SBD and Voice services for the removed Satellite Data Unit Configuration Module (SCM).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Satellite Data Unit Installation
Figure 401/23-15-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-15-01-400-801

3. Satellite Data Unit Installation

(Figure 401)

A. General

(1) This task includes the steps to install the Satellite Data Unit (SDU).

(2) After the SDU installation, verify the software to make sure that the correct software parts are installed to the SDU.

(3) It is the responsibility of the airlines to contact their SATCOM Service Provider to activate the Short Burst Data (SBD) and Voice services for the SDU prior to installation on the aircraft.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-15-00-470-801	Satcom System - SDU Software Installation with ONS (P/B 201)
23-15-00-710-802	Satellite Communications (SATCOM) System - System Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Satellite data unit	23-15-51-50-045	PTF ALL

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Satellite Data Unit Installation

SUBTASK 23-15-01-440-002

(1) If applicable, contact your designated SATCOM Service Provider to activate the SBD and Voice services for the Satellite Data Unit Configuration Module (SCM).

SUBTASK 23-15-01-020-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(2) Before you touch the satellite data unit [1], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-15-01-420-001

(3) To install the satellite data unit [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-15-01-860-002

(4) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

G. Installation Test

SUBTASK 23-15-01-470-002

(1) If a new SDU is installed, do this task: Satcom System - SDU Software Installation with ONS, TASK 23-15-00-470-801.

SUBTASK 23-15-01-730-001

(2) Do this task: Satellite Communications (SATCOM) System - System Test, TASK 23-15-00-710-802.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 23-15-01-410-005

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SDU CONFIGURATION MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the SCM

(2) An installation of the SCM.

B. The SCM is installed on the E5 equipment rack right aft stanchion. The E5 rack is in the Electrical and Electronics Compartment (ECC) on the right side.

TASK 23-15-12-000-801

2. SDU Configuration Module Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the Satellite Data Unit Configuration Module (SCM).

(2) It is the responsibility of the airlines to contact their SATCOM Service Provider to deactivate the Short Burst Data (SBD) and Voice services for the SCM when it is removed from the aircraft.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 23-15-12-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

SUBTASK 23-15-12-020-005

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. SDU Configuration Module Removal

SUBTASK 23-15-12-020-006

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Before you touch the SDU Configuration Module [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 23-15-12-020-003

(2) Do these steps to remove the SDU Configuration Module [1]:

(a) Loosen the screws [2] that attach the electrical connector [3] to the SDU Configuration Module [1].

(b) Disconnect the electrical connector [3] from the SDU Configuration Module [1].

(c) Remove the four mounting bolts [5] and washers [4].

(d) Remove the SDU Configuration Module [1] from its mounting bracket.

SUBTASK 23-15-12-910-001

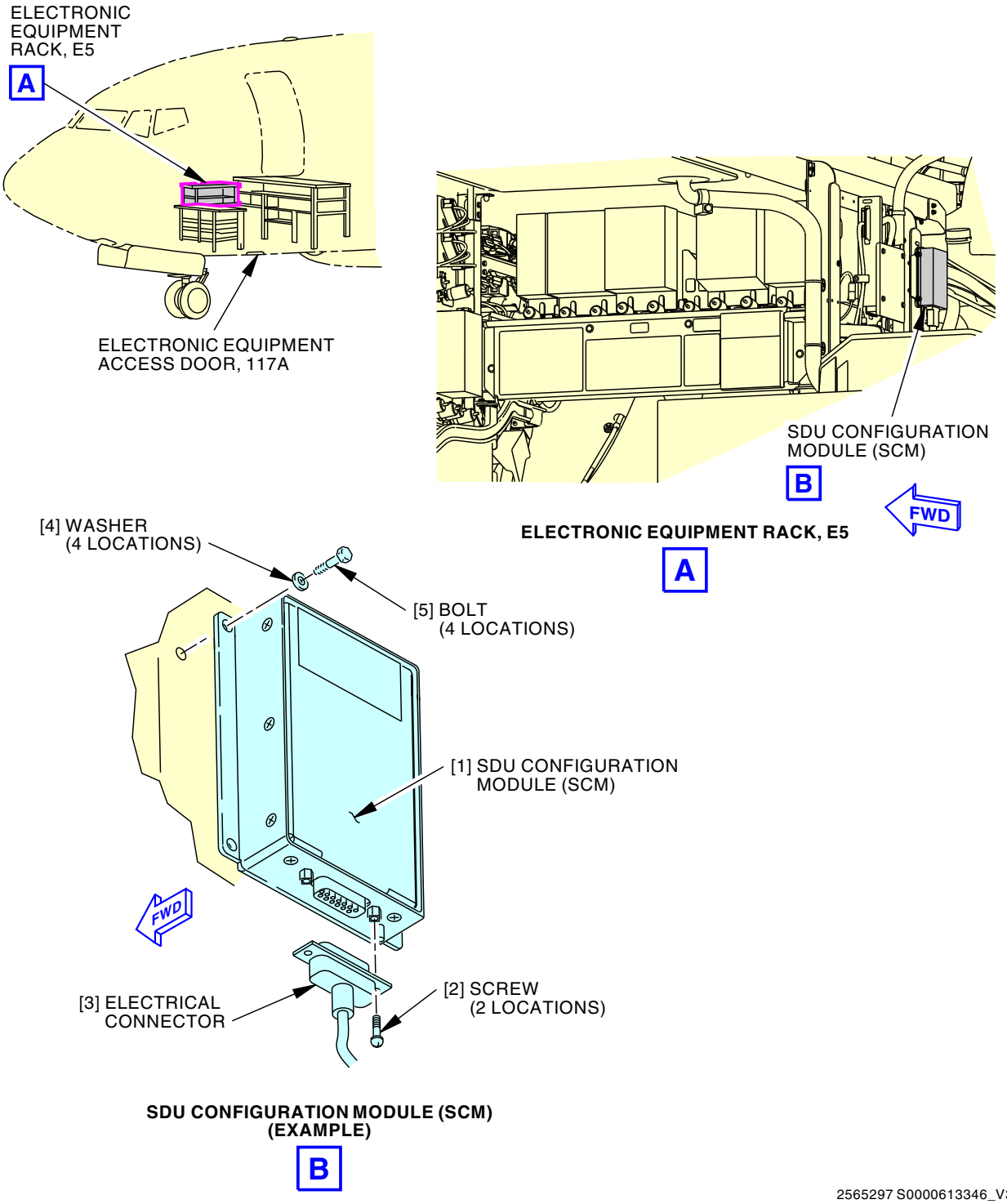
(3) Put the Electrostatic Discharge Sensitive (ESDS) protective covers on the connectors.

SUBTASK 23-15-12-040-002

(4) If applicable, contact your designated SATCOM Service Provider to deactivate the SBD and Voice services for the removed SCM.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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SDU Configuration Module Installation
Figure 401/23-15-12-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-15-12-400-801

3. SDU Configuration Module Installation

(Figure 401)

A. General

(1) This task includes the steps to install the Satellite Data Unit Configuration Module (SCM).

(2) It is the responsibility of the airlines to contact their SATCOM Service Provider to activate the Short Burst Data (SBD) and Voice services for the SCM prior to installing it on the aircraft.

B. References

Reference	Title
23-15-00-470-801	Satcom System - SDU Software Installation with ONS (P/B 201)
23-15-00-710-802	Satellite Communications (SATCOM) System - System Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	SDU Configuration Module	23-15-51-60-020	PTF ALL

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. SDU Configuration Module Installation

SUBTASK 23-15-12-440-002

(1) Contact your designated SATCOM Service Provider to activate the SBD and Voice services for the new SCM.

SUBTASK 23-15-12-420-001

(2) Install the SDU Configuration Module [1] as follows:

(a) Remove the Electrostatic Discharge Sensitive (ESDS) protective covers from the electrical connectors.

(b) Examine the electrical connectors for broken or bent pins, dirt, and damage.

(c) Put the SDU Configuration Module [1] in position on the mounting bracket.

(d) Install and tighten the mounting bolts [5] and washers [4] to attach the SDU Configuration Module [1].

(e) Connect the electrical connector [3] to the SDU Configuration Module [1].

(f) Tighten the screws [2] to attach the electrical connector.

SUBTASK 23-15-12-860-002

(3) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Installation Test

SUBTASK 23-15-12-470-002

(1) If a new SCM is installed, do this task: Satcom System - SDU Software Installation with ONS, TASK 23-15-00-470-801.

NOTE: After the Satellite Data Unit (SDU) software installation, the SDU automatically installs the software to the SCM.

SUBTASK 23-15-12-470-001

(2) Do this task: Satellite Communications (SATCOM) System - System Test, TASK 23-15-00-710-802.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 23-15-12-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SATCOM ANTENNA - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the SATCOM antenna

(2) An installation of the SATCOM antenna

TASK 23-15-25-000-802

2. SATCOM Antenna Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the SATCOM antenna.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-768	Sealant Removal Tool, Non-Metallic Part #: ST982L-9 Supplier: 81205 Part #: ST982LF-9 Supplier: 81205 Part #: ST982LH-A-1 Supplier: 81205

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the Removal

SUBTASK 23-15-25-860-005

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

E. SATCOM Antenna Removal

SUBTASK 23-15-25-100-001

(1) Remove the sealant from the heads of the bolts [13] with the sealant removal tool, SPL-768.

SUBTASK 23-15-25-020-007

(2) Remove the bolts [13] that attach the antenna [12] to the airplane.

SUBTASK 23-15-25-100-002

CAUTION

REMOVE THE AERODYNAMIC FILLET SEAL CAREFULLY WITH THE SEALANT REMOVAL TOOL. DAMAGE TO THE AIRPLANE SKIN OR THE COAXIAL CABLE CAN OCCUR.

(3) Remove the aerodynamic fillet seal around the base of the antenna [12] with the sealant removal tool, SPL-768.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-15-25-020-008

(4) Lift the antenna [12] to get access to the coaxial cables [14].

SUBTASK 23-15-25-020-009

(5) Disconnect the coaxial cables [14] from the antenna [12].

(a) Make sure that the coaxial cables [14] do not fall back through the hole.

(b) Temporarily attach the coaxial cables [14] to the airplane structure with tape.

SUBTASK 23-15-25-480-001

(6) Put protective covers on the coaxial cables [14] and antenna [12] electrical connector.

SUBTASK 23-15-25-020-012

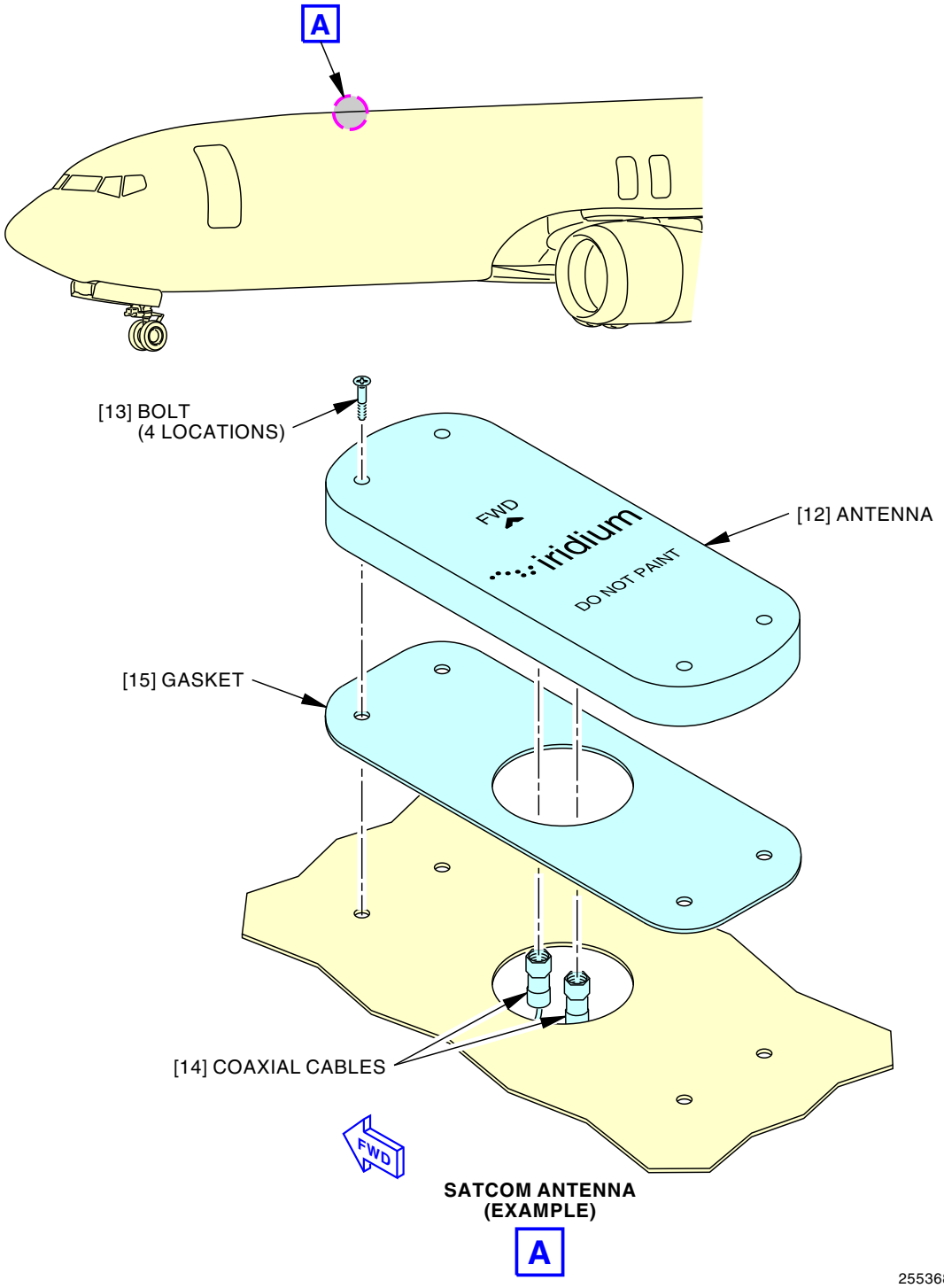
(7) Remove the antenna [12] from the airplane.

SUBTASK 23-15-25-020-013

(8) Remove and discard the gasket [15].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SATCOM Antenna Installation
Figure 401/23-15-25-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-15-25-400-802

3. SATCOM Antenna Installation

(Figure 401)

A. General

(1) This task includes the steps to install the SATCOM antenna to the aircraft fuselage.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
23-15-00-710-802	Satellite Communications (SATCOM) System - System Test (P/B 501)
51-22-00-390-809	Aerodynamic Smoother Application (P/B 201)
SL 20-043	Deferred Application of Aero-Sealant in Antenna Installations

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-768	Sealant Removal Tool, Non-Metallic Part #: ST982L-9 Supplier: 81205 Part #: ST982LF-9 Supplier: 81205 Part #: ST982LH-A-1 Supplier: 81205
STD-765	Scraper - Plastic

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00130	Alcohol - Isopropyl	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	Antenna	23-15-51-01-020	PTF ALL
15	Gasket	23-15-51-01-015	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
200	Upper Half of Fuselage

G. Prepare for the Installation

SUBTASK 23-15-25-860-006

(1) Make sure that this circuit breaker is open and has safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

H. SATCOM Antenna Installation

SUBTASK 23-15-25-100-003

(1) Remove the sealant from the mating surfaces of the airplane skin and antenna [12] with the sealant removal tool, SPL-768.

SUBTASK 23-15-25-110-001

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS) AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(2) Solvent clean the mating surfaces between the base of the antenna [12] and the airplane skin (TASK 20-10-34-110-802).

SUBTASK 23-15-25-020-014

(3) If a new antenna [12] will be installed, discard the O-ring supplied with the antenna [12].

SUBTASK 23-15-25-420-009

(4) Do these steps to install the gasket [15]:

(a) Remove protective film from the antenna [12] side of the gasket [15].

(b) Align the gasket [15] to the fastener and the connector holes of the antenna [12].

(c) Press the gasket [15] firmly onto the antenna [12] base.

(d) Remove protective film from the airplane side of the gasket [15].

SUBTASK 23-15-25-420-010

(5) Do these steps to connect the coaxial cables [14]:

(a) Remove the protective caps from the coaxial cables [14] and antenna [12].

(b) If it is necessary, remove the tape from the coaxial cables [14].

(c) Connect the coaxial cables [14] to the antenna [12].

SUBTASK 23-15-25-420-012

(6) Do these steps to install the antenna [12]:

(a) Put the antenna [12] in position on the airplane skin.

1) Make sure that the bolt holes on the antenna [12] align with the bolt holes on the airplane skin.

(b) Do these steps to install three of the four bolts [13]:

NOTE: The last bolt hole is used for the electrical bonding check.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Apply sealant, A00247, to the threads and shank of the three bolts [13].
2) Install the antenna [12] with the three bolts [13].
3) Tighten the three bolts [13] to 25.0 in-lb (2.8 N·m).
4) Wait 15 minutes.

NOTE: The gasket gel material can compress during this time.

5) Tighten the three bolts [13] to 30.0 in-lb (3.4 N·m).

(c) Use a non-intrinsically safe bonding meter, COM-614 to measure the resistance between the base of the antenna [12] and the airplane skin.

NOTE: The empty bolt hole may be used to get access to the base of the antenna [12].

1) Make sure that the electrical bonding resistance is 0.001 ohms (1.0 milliohms) or less.

(d) Apply sealant, A00247, to the threads and shank of the last bolt [13].

(e) Install the last bolt [13].

(f) Tighten the bolts [13] to 30.0 in-lb (3.4 N·m).

(g) Remove the unwanted sealant, A00247 from the bolts [13].

(h) Remove the unwanted gel squeeze out from the gasket [15] with a plastic scraper, STD-765.

1) If it is necessary, use a clean cotton wiper, G00034, moist (not soaked) with alcohol, B00130, to help remove the unwanted gel.

(i) Apply an aerodynamic fillet seal around the base of the antenna [12] with sealant, A00247 (TASK 51-22-00-390-809).

NOTE: Operators can defer the application of the aero-sealant in the antenna installation until next “A” check or overnight maintenance stop to prevent a flight delay. The deferral of the application of sealant must not exceed a maximum of eight weeks (SL 20-043).

1) Make sure that there are no gaps between the antenna [12] base and the aircraft skin.

SUBTASK 23-15-25-860-007

(7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	13	C04060	SATCOM SDU

I. Installation Test

SUBTASK 23-15-25-730-001

(1) Do this task: Satellite Communications (SATCOM) System - System Test, TASK 23-15-00-710-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY LOCATOR TRANSMITTER SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Emergency locator transmitter deactivation

(2) Emergency locator transmitter activation

(3) Emergency locator transmitter programming

(4) Emergency locator transmitter battery replacement.

TASK 23-24-00-040-802

2. Emergency Locator Transmitter Deactivation

A. General

(1) This task removes electrical power to the Emergency Locator Transmitter (ELT).

B. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Emergency Locator Transmitter Deactivation

NOTE: Circuit breaker C01278 can stay closed, if Master Caution Warning functionality is necessary for other systems such as Auxiliary Power Unit (APU) or thrust reversers.

SUBTASK 23-24-00-860-004-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT
C	14	C01278	MASTER CAUTION ANNUNCIATOR CONT 4

E. Emergency Locator Transmitter Tryout

NOTE: This tryout is to make sure that the ELT is in a zero energy state.

SUBTASK 23-24-00-860-025

(1) Do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

SUBTASK 23-24-00-700-003

(2) Do these steps to deactivate the ELT:

(a) Make sure that the switch on the ELT computer front panel is set to OFF.

(b) Loosen thumb screws that attach the front cover to the ELT.

(c) Disconnect the electrical connector from the ELT.

(d) Make sure that none of the LED lights come on.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-24-00-440-802

3. Emergency Locator Transmitter Activation

A. General

(1) This task adds electrical power to the Emergency Locator Transmitter (ELT).

B. References

Reference	Title
23-24-00-730-801	Emergency Locator Transmitter System Test (P/B 501)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Emergency Locator Transmitter Activation

SUBTASK 23-24-00-440-004

(1) Do these steps to activate ELT.

(a) Connect electrical connector to ELT.

(b) Position front cover to front of the ELT and tighten thumb screws.

(c) Make sure that the ELT control panel switch, on the P5 aft overhead panel in the flight compartment, is set to ARM.

SUBTASK 23-24-00-860-006-001

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT
C	14	C01278	MASTER CAUTION ANNUNCIATOR CONT 4

SUBTASK 23-24-00-730-007

(3) Do this task: Emergency Locator Transmitter System Test, TASK 23-24-00-730-801

SUBTASK 23-24-00-860-026

(4) Do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

——— END OF TASK ———

TASK 23-24-00-860-801

4. ELT Programming

(Figure 201)

A. General

(1) This procedure is for re-programming of the Emergency Locator Transmitter (ELT) with Aircraft 24-bit ID version of Standard Location Protocol. Use this procedure if the aircraft ID (or 24-bit International Civil Aviation Organization (ICAO) aircraft address code) has changed as due to a change of aircraft registry or ownership.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) This procedure includes the steps to set the Dual In-line Package (DIP) switches on the ELT system's Programmable Switch Module (PSM), M2115. The PSM is located at approximate STA 794, on the side of the ELT mounting tray.

(3) The 24-bit ICAO aircraft address transmitted by the ELT is set by the DIP switches on PSM, M2115. The first bit of the address corresponds to Switch 1 and so on sequentially through bit 24.

(4) If the ELT is going to be moved to an aircraft which is registered in a country other than the country for which the ELT was originally programmed, the ELT must be re-registered and the product label remarked to reflect the new country of registry. Contact your ELT supplier for details of marking and reprogramming.

B. References

Reference	Title
23-24-00-730-801	Emergency Locator Transmitter System Test (P/B 501)
34-21-00 P/B 201	AIR DATA INERTIAL REFERENCE SYSTEM - MAINTENANCE PRACTICES
FIM 23-24	Fault Isolation Manual
WDM 23-24-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-14302	Cable - ELT test (TNC to TPS) Part #: 611-6052 Supplier: 0Z3C6
COM-14421	Test Set - ELT Ramp Part #: 90000136 Supplier: 92606 Part #: 90000147 Supplier: 92606 Part #: ARTS-7000 Supplier: 0RDZ5 Part #: AVX-10K-CNS Supplier: 51190 Opt Part #: IFR 4000 Supplier: 51190

D. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

E. ELT Programming

SUBTASK 23-24-00-860-001

(1) Open this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

SUBTASK 23-24-00-010-001

(2) Do these steps to get access to the DIP switches [6] on the programmable switch module [1]:

(a) Open the applicable access panel to get access to the programmable switch module [1] for the ELT system.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Loosen the two captive screws [4] and washers [5] to remove the security cover [3] from the programmable switch module [1].

(c) Remove the switch seals [2] to get access to the DIP switches [6].

SUBTASK 23-24-00-860-002

(3) Examine the 24-bit DIP switch [6] setting on the programmable switch module [1].

(a) Make sure that the DIP switches [6] are set correctly with the 24-bit ICAO aircraft ID.

NOTE: The PSM for the ELT system which is configured to match the PSM connected to the aircraft's Mode S transponder system.

SUBTASK 23-24-00-860-003

(4) If the DIP switches [6] are not set correctly, then do these steps to set the DIP switches [6] (1 to 24) on the programmable switch module [1] (M2115) for the ELT system as follows:

NOTE: See WDM 23-24-11 for specific airplane DIP switches settings.

(a) Look at the ELT coding table in WDM 23-24-11 and find the airplane you are performing this task on.

(b) Read across the columns (24 binary codes) to find all switch settings.

(c) If the aircraft registry or ownership has changed, make sure that you have the correct ICAO aircraft address (ICAO Hexadecimal or Octal address) for reprogramming the ELT.

1) If necessary, use the code conversion tables below to convert the ICAO aircraft address from the Hexadecimal or Octal code to the binary code.

NOTE: See examples in Figure 201.

Hexadecimal to Binary Conversion table:

Hexadecimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Octal to Binary Conversion Table:

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

(d) Move the DIP switches [6] to the ON/OFF position according to the binary codes shown in the ELT coding table of WDM 23-24-11 or the 24-Bit ICAO aircraft address.

- For each switch where you find a “1”, set the switch to ON (red dot visible).
- For each switch where you find a “0”, set the switch to OFF (red dot not visible).

1) Make sure that all the DIP switches [6] are set correctly.

(e) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

NOTE: Closing the ELT Navigation Interface Unit (NIU) circuit breaker will initiate re-coding of the ELT by the NIU.

NOTE: Upon power-up, the ELT/NIU will overwrite the existing programming with the 24-bit address set on the Programmable Switch Module.

1) Monitor the Light Emitting Diode (LED) activity on the front panel of the ELT unit for a minimum of two minutes after you closed the circuit breaker.

a) Make sure that the LED comes on one time and goes off.

NOTE: This indicates the ELT re-coding is successful.

NOTE: If there is a problem with the programming, the LED will come on flashes. Rapid flashes of the ELT's LED indicates that the ELT is programmed with an incompatible protocol. In that case, the ELT/NIU will attempt to program the ELT with the aircraft's 24-bit address as set on the DIP switch module. The ELT will not accept this programming and after two minutes will indicate a programming failure by continuous rapid flashing of the ELT's LED.

<1> If the LED on the ELT flashes rapidly, then do the above steps again to make sure that the 24-bit aircraft address set correctly.

<a> Refer to the fault isolation procedure for the ELT system (FIM 23-24) if it is necessary.

F. ELT Coding Verification

SUBTASK 23-24-00-860-024

(1) Make sure that the Inertial Reference System (IRS) is aligned (AIR DATA INERTIAL REFERENCE SYSTEM - MAINTENANCE PRACTICES, PAGEBLOCK 34-21-00/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-24-00-480-001

(2) Prepare the ELT ramp test set:

(a) Install the ELT Ramp Test Set, COM-14421:

NOTE: This procedure is for using IFR 4000 ELT Ramp Test Set, COM-14421. If you use an equivalent test set other than IFR 4000, please refer to the applicable operation manual of that test set for detailed user instructions.

1) Disconnect the antenna coaxial cable connector D10583 from the ELT.

2) Disconnect the antenna coaxial cable connector D02905 from the ELT.

3) Use a ELT test cable, COM-14302, (TNC to TPS) to connect the ELT test set to the ELT unit:

a) Connect the TNC connector of the TNC to TPS cable to the “RF I/O” connector on the ELT Ramp Test Set, COM-14421.

b) Connect the TPS connector of the TNC to TPS cable to the Radio Frequency (RF) output common to D10583 on the ELT unit.

(b) Verify the 406 MHz (Megahertz) message or Beacon ID:

1) Press the POWER button on the test set to turn it on.

NOTE: The test set will start in VHF Omnidirectional Range (VOR) Mode.

2) Press the SETUP key on the test set.

3) Press the UP arrow key on the test set to select PORT: RF I/O.

4) Press the MODE key on the test set.

NOTE: VOR mode will be displayed.

5) Press the UP arrow key on the test set until the 406 BCN mode page is displayed.

SUBTASK 23-24-00-720-001

(3) Use an ELT Ramp Test Set, COM-14421, to verify the ELT coding (or BEACON ID):

(a) On the front panel of the ELT unit, move the toggle switch from "OFF" to "ON" for approximately 2 seconds and then back to "OFF".

(b) Make sure that the 406 BCN screen displays message.

(c) Make sure that the "BEACON ID" displayed on test set matches the "Hex ID" shown on WDM 23-24-11.

NOTE: The Beacon ID is always 15 hexadecimal characters. Valid characters are 0-9, A-F.

NOTE: This step requires that the ELT be coded with 24-bit address version of Standard Location Protocol. ELTs programmed with 24-bit address protocol will have the following fields on the ELT's product identification label blank:

- 15 DIGIT HEX ID
- COUNTRY
- THREE DIGIT CODE

Non-blank fields indicate that the ELT is programmed with an incompatible protocol. In that case, the ELT/NIU will attempt to program the ELT with the aircraft's 24-bit address as set on the DIP switch module. The ELT will not accept this programming and after two minutes will indicate a programming failure by continuous rapid flashing of the ELT's LED.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If the FMC supplies location data to the ELT, make sure that the "LAT" and "LONG" fields are not dashes "-".

NOTE: Dashes (-) will be displayed in the “LAT” and “LONG” fields if no position data is being received from the Flight Management Computer (FMC).

SUBTASK 23-24-00-720-004

(4) Record the following information from the test set:

- COUNTRY: _____ (Example: 316 CANADA)
- BEACON ID (15 digit Hex ID):__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ (Example: 34670 008BE FFBFF)

NOTE: The Beacon ID is always 15 hexadecimal characters. Valid characters are 0-9, A-F.

NOTE: This code is used to register the ELT. The last five characters encode position data, FFBFF is default “null” location used for registration purpose.

NOTE: If necessary, the “Beacon Message Decode Program” at the COSPAS-SARSAT website (http://www.cospas-sarsat.org/index.php?lang=en) can be use to decode the Beacon ID.

SUBTASK 23-24-00-720-002

(5) Record the BEACON ID on the HEX ID label installed adjacent to the ELT.

NOTE: Preferred method: Use a label-maker to print an appropriately sized label to fit the HEX ID field.

Alternate method: Write neatly using a Sharpie or other fine-point permanent marker. If needed, a replacement (BAC27WEX2107) label or equivalent can be installed over existing label.

NOTE: If the ELT was re-coded, the ELT may be required to re-register through your local authority.

(a) If it is necessary, record the THREE DIGIT CODE and COUNTRY on the label installed adjacent to the ELT.

SUBTASK 23-24-00-080-001

(6) Remove the ELT Ramp Test Set, COM-14421:

(a) Press the POWER button on the test set to turn it off.

(b) Disconnect the test cables from the ELT unit and the test set.

(c) Connect the antenna coaxial cable connector D10583 to the ELT unit.

(d) Connect the antenna coaxial cable connector D02905 to the ELT unit.

G. ELT System Test

SUBTASK 23-24-00-710-001

(1) Do this task: Emergency Locator Transmitter System Test, TASK 23-24-00-730-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 23-24-00-910-001

(1) Put the programmable switch module [1] back to its usual condition:

(a) Install the two switch seals [2] to cover the DIP switches [6].

(b) Put the security cover [3] on the programmable switch module [1].

1) Tighten the captive screws [4] and washers [5] to attach the security cover [3] on the programmable switch module [1].

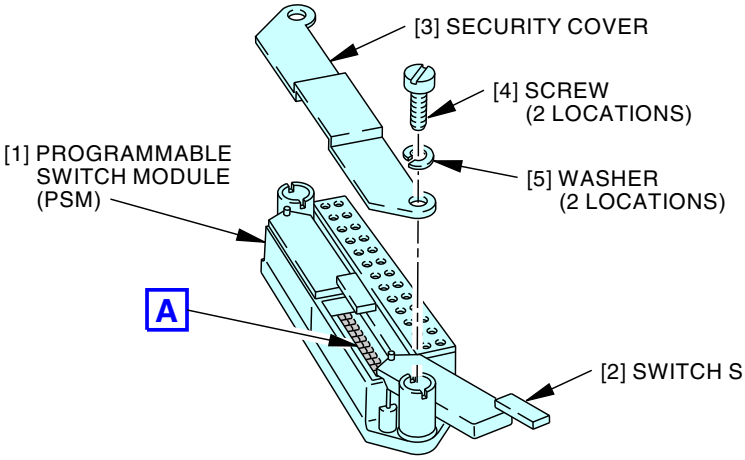
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-24-00-410-001

(2) Close the access panel that was opened for access to the programmable switch module [1].

——— END OF TASK ———

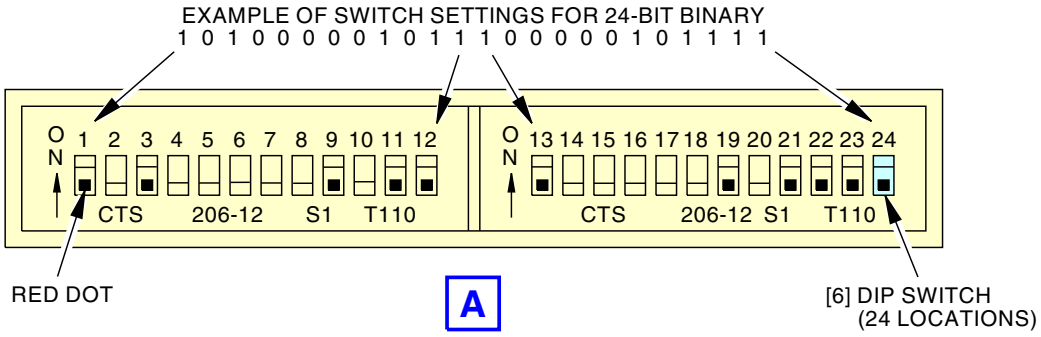
ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

"1" = SWITCH ON

"0" = SWITCH OFF



ICAO OCTAL ADDRESS TO 24 BIT BINARY																							
5			0			1			3			4			0			5			7		
1	0	1	0	0	0	0	0	1	0	1	1	1	0	0	0	0	0	1	0	1	1	1	1
MSB																							LSB

EXAMPLE NO. 1 CONVERTING ICAO ADDRESS 50134057 (OCTAL) TO BINARY 24-BIT ADDRESS

ICAO HEX ADDRESS TO 24 BIT BINARY																							
A				0				B				8				2				F			
1	0	1	0	0	0	0	0	1	0	1	1	1	0	0	0	0	0	1	0	1	1	1	1
MSB																							LSB

EXAMPLE NO. 2 CONVERTING ICAO ADDRESS A0B82F (HEXADECIMAL) TO BINARY 24-BIT ADDRESS

2411674 S00061528592_V2

ELT System - 24-Bit Aircraft Address Settings
Figure 201/23-24-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-24-00-960-801

5. Emergency Locator Transmitter Battery Replacement

(Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has steps to replace the battery for a Emergency Locator Transmitter (ELT).

B. References

Reference	Title
23-24-00-000-801	Emergency Locator Transmitter Removal (P/B 401)
23-24-00-400-801	Emergency Locator Transmitter Installation (P/B 401)

C. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Emergency Locator Transmitter Battery Replacement

SUBTASK 23-24-00-010-004-001

(1) Do this task: Emergency Locator Transmitter Removal, TASK 23-24-00-000-801.

SUBTASK 23-24-00-020-003-001

(2) Remove the battery pack assembly:

(a) Remove the screws [13] that attach the battery pack assembly to the ELT.

(b) Disconnect the electrical connectors [8].

SUBTASK 23-24-00-420-003-001

WARNING MAKE SURE THAT THE BATTERY WIRES ARE INSTALLED CORRECTLY. MAKE SURE THAT THE BATTERY PACK IS NOT DAMAGED. IF YOU IGNORE THESE CONDITIONS, SMOKE OR A FIRE CAN OCCUR. DAMAGE TO EQUIPMENT AND INJURIES TO PERSONNEL CAN OCCUR.

(3) Install the new battery pack assembly:

(a) Connect the electrical connectors [8].

1) Connect the ELT lead 8-pin connector to the header on the battery pack circuit board.

2) Connect the power harness 2-pin connector to the recessed connector inside the ELT body.

CAUTION DO NOT OPERATE THE ELT FOR MORE THAN 50 SECONDS. IF YOU OPERATE THE ELT CONTINUOUSLY FOR MORE THAN 50 SECONDS, IT WILL TRANSMIT A 406-MHZ EMERGENCY SIGNAL. THE SIGNAL TELLS EMERGENCY PERSONNEL TO DO A SEARCH FOR A MISSING AIRCRAFT. IF THE EMERGENCY SIGNAL OCCURS DURING THIS TEST, IMMEDIATELY TELL THE APPLICABLE EMERGENCY PERSONNEL.

(b) Set the ELT.

1) Move the OFF switch to the ON position.

2) Wait approximately one second.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Move the ON/OFF switch to the OFF position.

NOTE: These steps are necessary because the ELT could start to operate when you connect power to it.

(c) Put the battery pack in its installed position on the ELT.

1) Make sure that all gaskets are aligned correctly.

2) Make sure that wires are not between these areas where they can be pinched:

a) The areas between the standoffs and battery pack.

b) The edges between the ELT and battery pack.

3) Hold the ELT and battery pack together tightly.

(d) Loosely install the screws [13] that attach the battery pack to the ELT.

CAUTION

DO NOT APPLY MORE THAN THE MAXIMUM SPECIFIED TORQUE WHEN YOU TIGHTEN THE PARTS. DAMAGE TO THE PARTS CAN OCCUR IF YOU APPLY TOO MUCH TORQUE.

(e) Tighten the screws [13] in a diagonal pattern to 8 ±1 in-lb (0.90 ±0.12 N·m).

SUBTASK 23-24-00-900-001-001

(4) Replace the expiration date label for the ELT battery with the expiration date label supplied with the new ELT battery pack:

NOTE: The expiration date label for the ELT battery is located on the external side of the ELT (i.e. on the external side of the end cap or other visible location).

(a) Make sure that you install the new expiration date label on the external side of the ELT where persons can see it.

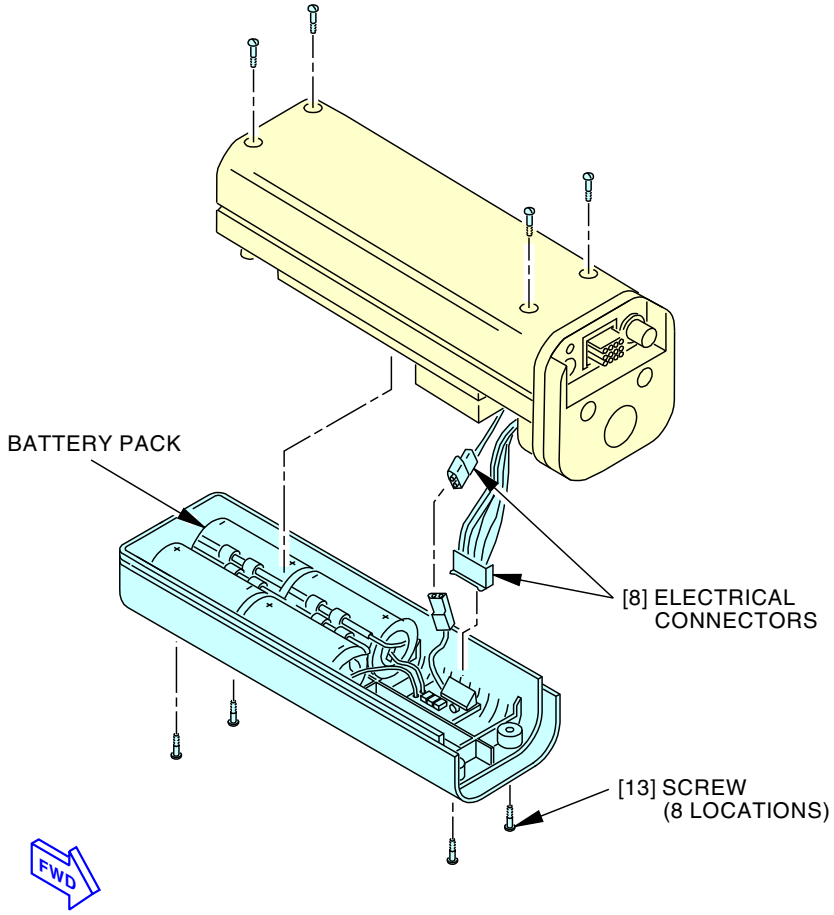
(b) Make sure that the expiration date on the new label shows that the new ELT battery pack is serviceable.

SUBTASK 23-24-00-410-004-001

(5) Do this task: Emergency Locator Transmitter Installation, TASK 23-24-00-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELT BATTERY PACK ASSEMBLY, M1523

C

2535844 S0000593637_V1

Emergency Locator Transmitter Battery Replacement
Figure 202/23-24-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY LOCATOR TRANSMITTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Emergency Locator Transmitter Removal

(2) Emergency Locator Transmitter Installation.

(3) ELT/Navigation Interface Unit Removal

(4) ELT/Navigation Interface Unit Installation.

TASK 23-24-00-000-801

2. Emergency Locator Transmitter Removal

(Figure 401)

A. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)

B. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

C. Emergency Locator Transmitter Removal

SUBTASK 23-24-00-865-001-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

SUBTASK 23-24-00-010-002-001

(2) Do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

SUBTASK 23-24-00-020-001-001

(3) Do these steps to remove the ELT [1]:

(a) Make sure the ELT switch is in the OFF position.

(b) Loosen the two thumb screws [7] that attach the front cover [6] to the ELT [1].

(c) Move the front cover [6] away from the ELT [1].

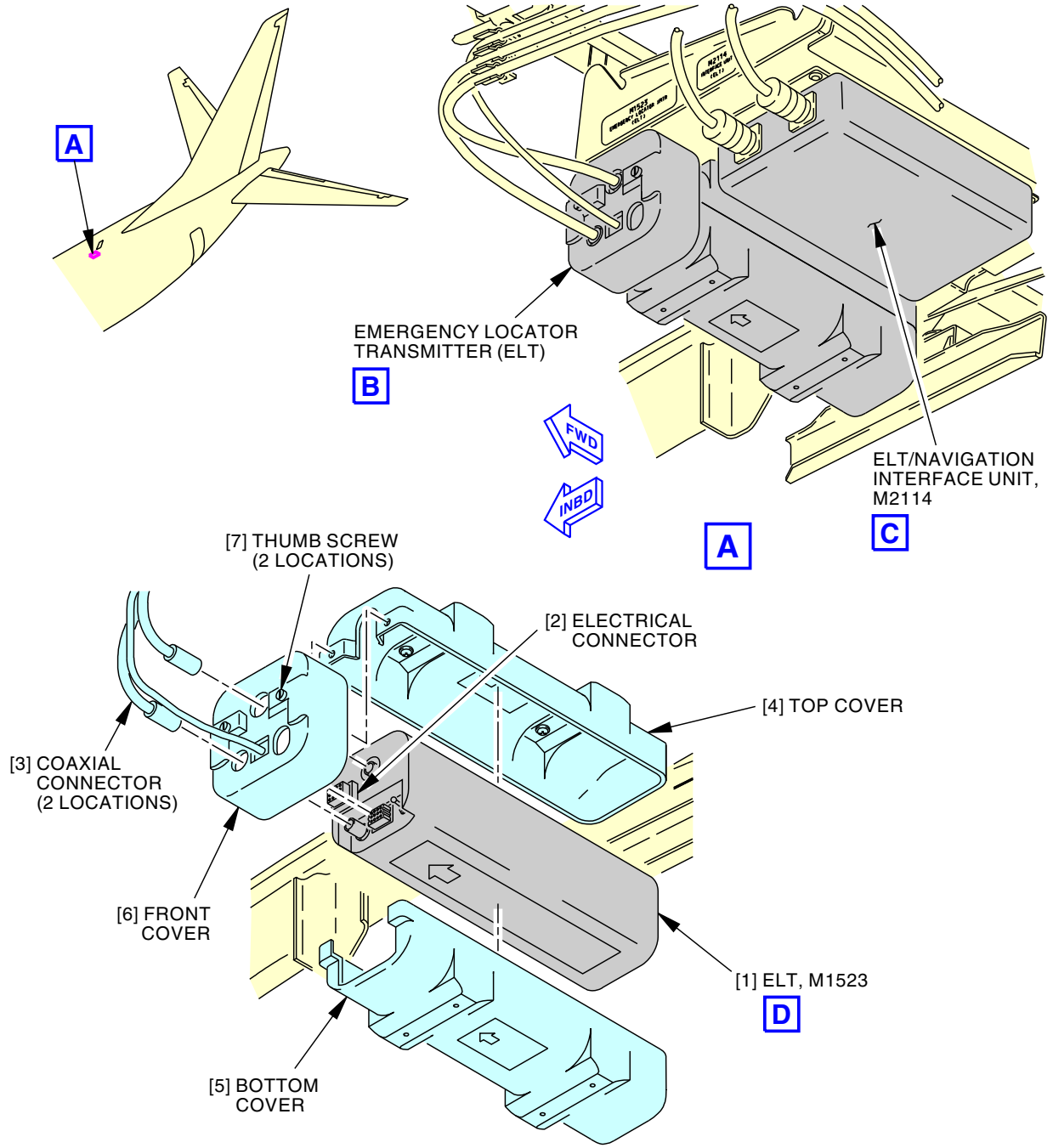
(d) Disconnect the electrical connector [2] and the coaxial connectors [3] from the ELT [1].

(e) Pull down and forward on the front of the ELT bottom cover [5] to release the interlocking slots and tabs at the rear of the ELT [1].

(f) Remove the ELT [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



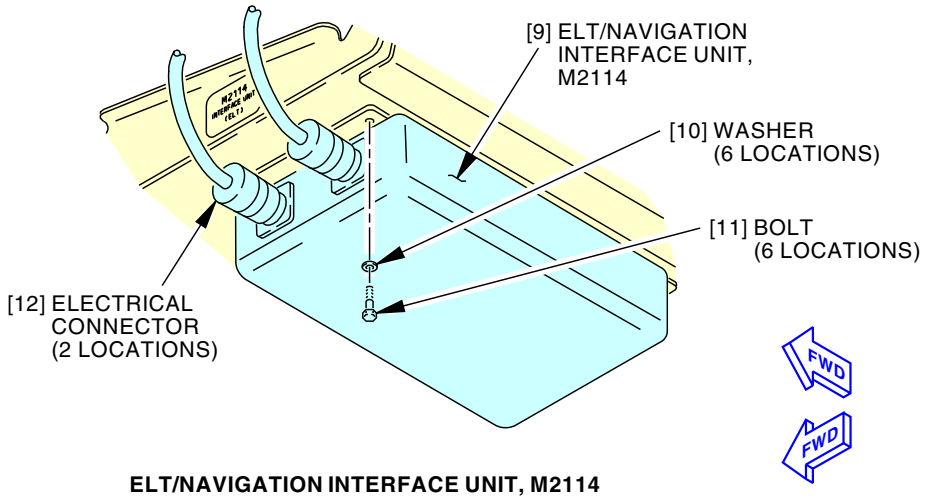
EMERGENCY LOCATOR TRANSMITTER (ELT) ASSEMBLY

B

2411676 S00061528598_V1

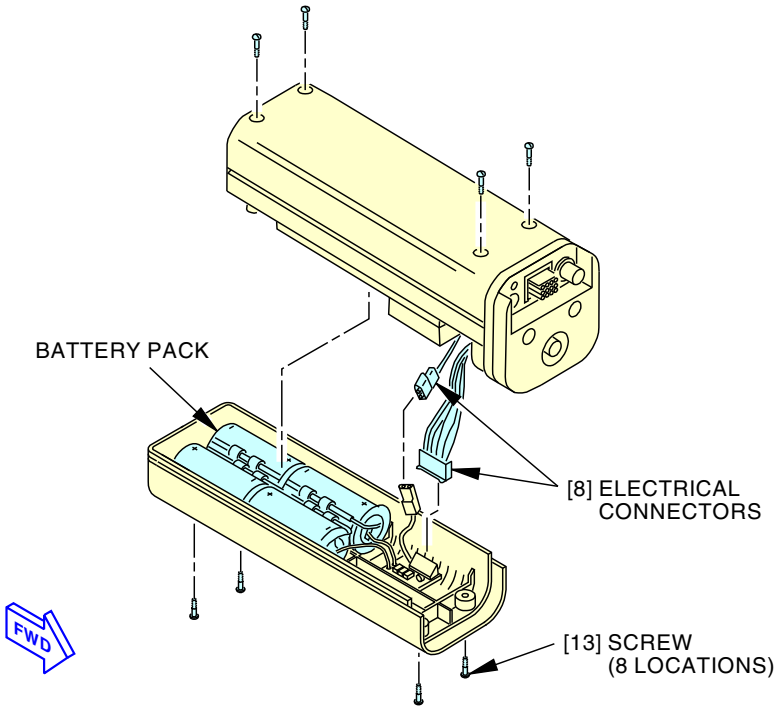
Emergency Locator Transmitter (ELT) Installation
Figure 401/23-24-00-990-803-001 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELT/NAVIGATION INTERFACE UNIT, M2114

C



ELT BATTERY PACK ASSEMBLY, M1523

D

2411677 S00061528599_V2

Emergency Locator Transmitter (ELT) Installation
Figure 401/23-24-00-990-803-001 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-24-00-400-801

3. Emergency Locator Transmitter Installation

(Figure 401)

A. References

Reference	Title
23-24-00-730-801	Emergency Locator Transmitter System Test (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	ELT	23-24-00-01-025	PTF ALL

C. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Emergency Locator Transmitter Installation

SUBTASK 23-24-00-865-002-001

(1) Make sure that this circuit breaker is open:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

SUBTASK 23-24-00-420-001-001

(2) Do these steps to install the ELT [1]:

NOTE: For all airplanes with an operational ELT/Nav Interface Unit (ELT/NIU) and Programmable Switch Module, the ELT 406 MHz message protocol must be Standard Location Protocol (long message) encoded with Aircraft 24-bit address to be compatible with the 24-bit ICAO code. The unique airplane's 24-bit ICAO address (Mode S Code) will automatically be encoded by the ELT/Nav Interface Unit into the ELT during its installation.

(a) Make sure the ELT switch is in the OFF position.

(b) Put the ELT [1] in its position.

(c) Put the ELT bottom cover [5] in its position.

(d) Connect the electrical connector [2] and the coaxial connectors [3] to the ELT [1].

(e) Attach the ELT front cover [6] to the tray with the two thumb screws [7].

E. ELT Installation Test

SUBTASK 23-24-00-861-001-001

(1) Make sure that the airplane has electrical power.

(a) If it is necessary, do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 23-24-00-740-001-001

(2) Do these steps to make sure that the ELT/NIU does an automatic programming to the ELT correctly:

NOTE: You must monitor the LED activity on the front panel of the ELT for a minimum of two minutes after the circuit breaker closed in the step follow.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

(b) Monitor the LED activity on the front panel of the ELT for a minimum two minutes.

(c) Make sure that the LED on the ELT does not flash rapidly. This is an indication that the ELT is programmed correctly.

1) If the LED on the ELT flashes rapidly two minutes after power applied to the ELT/NIU, refer to the IFIM and do the applicable procedure(s).

NOTE: For airplane's configuration with an operational ELT/NIU, no other message protocols are compatible, including Standard Location Protocol (long message) encoded with ELT serial identification. If an improperly programmed ELT is installed in an airplane configured with an operational ELT/NIU, two minutes after power is applied, an error condition will be annunciated with the LED on the ELT flashing rapidly. This indicates by one of the following error conditions: 1) The ELT is programed for a protocol other than Aircraft 24-bit address and the ELT/NIU has been strapped for a 24-bit address. Or 2) The RS-232 TX line from the ELT (pin 12) to the ELT/NIU (pin 10) is not connected.

SUBTASK 23-24-00-710-002-001

(3) Do this task: Emergency Locator Transmitter System Test, TASK 23-24-00-730-801.

NOTE: It is the responsibility of the airlines to report any change in the registered ELT information to their designated registration authority.

——— END OF TASK ———

TASK 23-24-00-000-802

4. ELT/Navigation Interface Unit Removal

Figure 401

A. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)

B. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

C. Removal Procedure

SUBTASK 23-24-00-010-003-001

(1) Do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

SUBTASK 23-24-00-040-001-001

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-24-00-020-002-001

(3) Do these steps to remove the ELT/Nav Interface Unit [9]:

(a) Disconnect the electrical connectors [12] from the ELT/Nav Interface Unit [9].

(b) Remove the six bolts [11] and six washers [10] that attach the ELT/Nav Interface Unit [9] to the airplane structure.

(c) Remove the ELT/Nav Interface Unit [9].

——— END OF TASK ———

TASK 23-24-00-400-802

5. ELT/Navigation Interface Unit Installation

Figure 401

A. References

Reference	Title
23-24-00-730-801	Emergency Locator Transmitter System Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	ELT/Nav Interface Unit	23-24-51-10-010	PTF ALL

C. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Installation Procedure

SUBTASK 23-24-00-860-007-001

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

SUBTASK 23-24-00-420-002-001

(2) Do these steps to install the ELT/Nav Interface Unit [9]:

(a) Put the ELT/Nav Interface Unit [9] in its position.

(b) Install the six bolts [11] and six washers [10] to attach the ELT/Nav Interface Unit [9] to the airplane structure.

(c) Connect the electrical connectors [12] to the ELT/Nav Interface Unit [9].

SUBTASK 23-24-00-440-001-001

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	1	C01495	ELT INTERFACE UNIT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. ELT Installation Test

SUBTASK 23-24-00-710-003-001

(1) Do this task: Emergency Locator Transmitter System Test, TASK 23-24-00-730-801.

NOTE: It is the responsibility of the airlines to report any change in the registered ELT information to their designated registration authority.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY LOCATOR TRANSMITTER - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An operational test of the emergency locator transmitter

(2) A system test of the emergency locator transmitter.

TASK 23-24-00-710-801

2. Emergency Locator Transmitter Operational Test

A. General

(1) This task gives instruction to do an operational test of the Emergency Locator Transmitter (ELT) system from the ELT control panel in the flight deck.

B. References

Reference	Title
23-12-00-730-801	VHF Communication System - System Test (P/B 501)
23-24-00 P/B 401	EMERGENCY LOCATOR TRANSMITTER - REMOVAL/INSTALLATION

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Prepare for the ELT System Operational Test

SUBTASK 23-24-00-860-009

(1) Make sure that the switch, on the front panel of the ELT unit, is in the OFF position.

NOTE: For the location of the ELT, refer to the ELT - Maintenance Practices procedure (PAGEBLOCK 23-24-00/401).

SUBTASK 23-24-00-860-010

(2) Set the No. 2 Very High Frequency (VHF) communication system (VHF COMM 2) to a frequency of 121.5 MHz (VHF Communication System - System Test, TASK 23-12-00-730-801).

NOTE: Any VHF COMM System with a bottom mounted antenna can be used for this test. If the VHF antenna is on the top, it is possible to hear the ELT signal even though the ELT antenna has failed.

SUBTASK 23-24-00-860-011

(3) Prepare one of the Audio Control Panels (ACP) in the flight compartment to listen for No. 2 VHF communication system:

(a) Push all audio monitor switches to the OFF position.

(b) Push the volume control for the VHF-2 selector switch.

1) Make sure that the VHF-2 selector switch light comes ON.

(c) Turn the volume control for the VHF-2 selector switch clockwise to the middle position.

(d) Push the SPKR volume control switch to the ON position.

1) Make sure that the SPKR volume control switch light comes ON.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Turn the SPKR volume control switch clockwise to the middle position.

SUBTASK 23-24-00-910-002

(4) Make sure that the ELT test is done in the first five minutes of the hour (Universal Time Coordinated (UTC)).

NOTE: Free Space Transmission Tests from the aircraft on 121.5/243 MHz are generally allowed any time within five minutes after each hour. Check your local regulations. THERE IS NO ALLOWANCE FOR TRANSMITTING A VALID 406 MHZ MESSAGE AT ANY TIME.

NOTE: The ELT will begin transmitting on 121.5/243 MHz immediately following activation and 406 MHz distress signals approximately 50 seconds after activation. If the ELT is inadvertently kept in the ON position for longer than approximately 50 seconds, you will hear a brief interruption of the 121.5 MHz signal while the 406 MHz message is transmitted. If this occurs, reset the ELT immediately and tell the local emergency authorities. This will prevent emergency search and rescue operations that are not necessary.

(a) If you do the ELT test outside of the first five minutes of the hour (UTC), do this step:

1) Speak with the local emergency authority (Air Traffic Control Tower or Flight Service Station) to let them know there will be an ELT Test Transmission on the Emergency Frequencies 121.5/243 MHz.

SUBTASK 23-24-00-910-003

WARNING MAKE SURE THAT YOU FOLLOW THE LOCAL ELT OPERATION REGULATIONS. IF YOU MAKE AN ACCIDENTAL TRANSMISSION, TELL THE AUTHORITIES. THIS WILL PREVENT EMERGENCY SEARCH OPERATIONS THAT ARE NOT NECESSARY.

(5) Follow the local ELT operation requirements.

E. ELT Operational Test at the ELT Control Panel

SUBTASK 23-24-00-710-004

(1) Do these steps for the ELT test at the ELT remote control panel in the flight compartment:

(a) If the ELT is activated accidentally, reset it:

1) To reset the ELT, move the switch, on the ELT's remote control panel, to the ON position for approximately 1 second.

a) If the switch is already in the ON position, move it to the ARM position.

2) Move the switch back to the ARM position.

CAUTION READ ALL THE STEPS THAT FOLLOW BEFORE YOU START THE ELT TEST PROCEDURE. YOU MUST BECOME FAMILIAR WITH THE INSTRUCTIONS. MAKE SURE THAT YOU USE A TIMING DEVICE TO MONITOR THE TIME DURING THE ELT TEST. YOU MUST COMPLETE THE ELT TEST PROCEDURE IN LESS THAN 50 SECONDS. THIS WILL HELP PREVENT ACCIDENTAL ACTIVATION OF THE ELT.

(b) Read all of the instructions below before you do the subsequent steps.

(c) Push and hold the ELT annunciator.

1) Make sure that the ELT annunciator comes ON.

(d) Release the ELT annunciator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the ELT annunciator goes OFF.

(e) Push and release the MASTER CAUTION reset button, on the P7 glareshield panel.

1) Make sure that the MASTER CAUTION and OVERHEAD annunciators are OFF.

CAUTION

MAKE SURE THAT YOU DO THE ELT LIVE TRANSMISSION TEST IN THE FIRST FIVE MINUTES OF THE HOURS (UTC). LET THE APPLICABLE LOCAL EMERGENCY AUTHORITY (AIR TRAFFIC CONTROL TOWER OR FLIGHT SERVICE STATION) FOR THE APPROVAL OF THIS TEST. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

DO NOT OPERATE THE ELT FOR MORE THAN 50 SECONDS. IF YOU OPERATE THE ELT CONTINUOUSLY FOR MORE THAN 50 SECONDS, IT WILL TRANSMIT A 406-MHZ EMERGENCY SIGNAL. THE SIGNAL TELLS EMERGENCY PERSONNEL TO DO A SEARCH FOR A MISSING AIRCRAFT. IF THE EMERGENCY SIGNAL OCCURS DURING THIS TEST, IMMEDIATELY TELL THE APPLICABLE EMERGENCY PERSONNEL.

(f) On the P5 ELT remote control panel at the pilot overhead panel, open the guard on the ELT ARM/ON switch.

(g) Set the ELT ARM/ON switch to the ON position for less than 5 seconds.

1) Make sure that you hear three sweeps of the ELT signal on the VHF COMM system.

2) Make sure that the ELT annunciator comes ON.

3) Make sure that the MASTER CAUTION and OVERHEAD annunciators, on the P7 glareshield panel, come ON.

(h) Set the ELT ARM/ON switch to the ARM position.

1) Make sure that you do not hear the ELT signal on the VHF COMM system.

2) Make sure that the ELT annunciator goes OFF.

(i) Close the guard on the ELT ARM/ON switch, on the ELT remote control panel.

(j) Push and release the MASTER CAUTION annunciator, on the P7 glareshield panel.

1) Make sure that the MASTER CAUTION and OVERHEAD annunciators, on the P7 glareshield panel, are OFF.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-24-00-910-004

(1) If it is necessary, tell the local emergency authority that the ELT test is completed.

——— END OF TASK ———

TASK 23-24-00-730-801

3. Emergency Locator Transmitter System Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instruction to do a system test of the Emergency Locator Transmitter (ELT) system. The system test includes the ELT operational test at the ELT control panel in the flight deck and ELT self-test at the ELT front panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
23-24-00 P/B 401	EMERGENCY LOCATOR TRANSMITTER - REMOVAL/INSTALLATION
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)
WDM 23-24-11	Wiring Diagram Manual

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Prepare for the ELT System Test

SUBTASK 23-24-00-860-013

(1) Make sure that the Air Data Inertial Reference Unit (ADIRU) is aligned, do one of these tasks: Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803.

NOTE: This step only applies to airplanes with Nav Flight Management System (FMS) ADIRU that send position data to the ELT via FMS.

SUBTASK 23-24-00-910-005

(2) Make sure that the ELT test is done in the first five minutes of the hour (Universal Time Coordinated (UTC)).

NOTE: Free Space Transmission Tests from the aircraft on 121.5/243 MHz are generally allowed any time within five minutes after each hour. Check your local regulations. THERE IS NO ALLOWANCE FOR TRANSMITTING A VALID 406 MHz MESSAGE AT ANY TIME.

NOTE: The ELT will begin transmitting on 121.5/243 MHz immediately following activation and 406 MHz distress signals approximately 50 seconds after activation. If the ELT is inadvertently kept in the ON position for longer than approximately 50 seconds, you will hear a brief interruption of the 121.5 MHz signal while the 406 MHz message is transmitted. If this occurs, reset the ELT immediately and tell the local emergency authorities. This will prevent emergency search and rescue operations that are not necessary.

(a) If you do the ELT test outside of the first five minutes of the hour (UTC), do this step:

1) Speak with the local emergency authority (Air Traffic Control Tower or Flight Service Station) to let them know there will be an ELT Test Transmission on the Emergency Frequencies 121.5/243 MHz.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-24-00-910-006

WARNING MAKE SURE THAT YOU FOLLOW THE LOCAL ELT OPERATION REGULATIONS. IF YOU MAKE AN ACCIDENTAL TRANSMISSION, TELL THE AUTHORITIES. THIS WILL PREVENT EMERGENCY SEARCH OPERATIONS THAT ARE NOT NECESSARY.

(3) Follow the local ELT operation requirements.

E. ELT Operational Test at the ELT Control Panel

SUBTASK 23-24-00-710-005

(1) Do this task: Emergency Locator Transmitter Operational Test, TASK 23-24-00-710-801.

F. ELT Self-Test at the Front Panel of the ELT Unit

SUBTASK 23-24-00-010-005

(1) Remove the ceiling panel to get access to the ELT, do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

NOTE: For the location of the ELT, refer to the ELT - Maintenance Practices procedure (PAGEBLOCK 23-24-00/401).

SUBTASK 23-24-00-730-001

(2) To do the self-test at the ELT front panel, do these steps:

(a) If the ELT is activated accidentally, reset it:

1) To reset the ELT, move the switch, on the ELT's remote control panel, to the ON position for approximately 1 second.

a) If the switch is already in the ON position, move it to the OFF position.

2) Move the switch back to the OFF position.

CAUTION READ ALL THE STEPS THAT FOLLOW BEFORE YOU START THE ELT TEST PROCEDURE. YOU MUST BECOME FAMILIAR WITH THE INSTRUCTIONS. MAKE SURE THAT YOU USE A TIMING DEVICE TO MONITOR THE TIME DURING THE ELT TEST. YOU MUST COMPLETE THE ELT TEST PROCEDURE IN LESS THAN 50 SECONDS. THIS WILL HELP PREVENT ACCIDENTAL ACTIVATION OF THE ELT.

(b) Read all of the instructions below before you do the subsequent steps.

CAUTION MAKE SURE THAT YOU DO THE ELT LIVE TRANSMISSION TEST IN THE FIRST FIVE MINUTES OF THE HOURS (UTC). LET THE APPLICABLE LOCAL EMERGENCY AUTHORITY (AIR TRAFFIC CONTROL TOWER OR FLIGHT SERVICE STATION) FOR THE APPROVAL OF THIS TEST. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION	DO NOT OPERATE THE ELT FOR MORE THAN 50 SECONDS. IF YOU OPERATE THE ELT CONTINUOUSLY FOR MORE THAN 50 SECONDS, IT WILL TRANSMIT A 406-MHZ EMERGENCY SIGNAL. THE SIGNAL TELLS EMERGENCY PERSONNEL TO DO A SEARCH FOR A MISSING AIRCRAFT. IF THE EMERGENCY SIGNAL OCCURS DURING THIS TEST, IMMEDIATELY TELL THE APPLICABLE EMERGENCY PERSONNEL.

(c) Move the switch, on the front panel of the ELT unit, to the ON position for approximately 1 second and then back to the OFF position.

NOTE: The ELT self-test includes a 406 MHz test transmission. This test transmission does not cause a search operation.

1) Make sure that the Light Emitting Diode (LED) on the ELT comes on for approximately 1 second pulse and goes off after the switch is moved to the OFF position.

NOTE: If the self-test detects a fault, the LED gives a coded signal AFTER the initial one second pulse. The LED will flash in order of importance with approximately one-half second to one second pause between error codes. The coded signal and related problems are as follows:

- If the LED flashes one time, this indicates a G-switch loop open failure.
- If the LED flashes three times or four times (depends on ELT part number), this indicates a transmitter problem (bad or unconnected coax, antenna problem, low power output or programming error).
- If the LED flashes five times, this indicates a problem with the navigation system connection. Please note that if the ELT/NAV Interface unit (or PPIU) is installed but is not connected to the navigation system to provide current latitude/longitude position updates, this is not a fault. Please refer to WDM 23-24-11 to determine the correct airplane configuration.
- If the LED flashes seven times, this indicates a battery problem.

2) If there was a self-test failure, refer to the IFIM and do the applicable procedure(s).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-24-00-910-007

(1) If it is necessary, tell the local emergency authority that the ELT test is completed.

SUBTASK 23-24-00-410-005

(2) Install the ceiling panel that gets access to the ELT, do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

SUBTASK 23-24-00-860-014

(3) Set the applicable mode select switch on the Inertial Reference System (IRS) Mode Select Unit (MSU) to the OFF position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY LOCATOR TRANSMITTER ANTENNA - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Emergency Locator Transmitter Antenna - Removal

(2) Emergency Locator Transmitter Antenna - Installation

TASK 23-24-02-000-801

2. Emergency Locator Transmitter Antenna - Removal

(Figure 401)

A. General

(1) This task has the steps to remove the Emergency Locator Transmitter (ELT) antenna.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

C. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. ELT Antenna Removal

SUBTASK 23-24-02-020-001

(1) Do these steps to remove the ELT antenna [1]:

(a) Remove the bolts [2] that attach the ELT antenna [1] to the airplane.

CAUTION BE CAREFUL WHEN YOU REMOVE THE FILLET SEAL WITH THE SEALANT REMOVAL TOOL. DAMAGE TO THE AIRPLANE SKIN CAN OCCUR.

(b) While you hold the ELT antenna [1], use a sealant removal tool, COM-2481 to break the fillet seal around the base of the antenna.

CAUTION DO NOT PULL ON THE COAXIAL CABLE. CAREFULLY MOVE THE ANTENNA. DISCONNECT THE COAXIAL CONNECTOR. THIS WILL PREVENT DAMAGE TO THE COAXIAL CABLE.

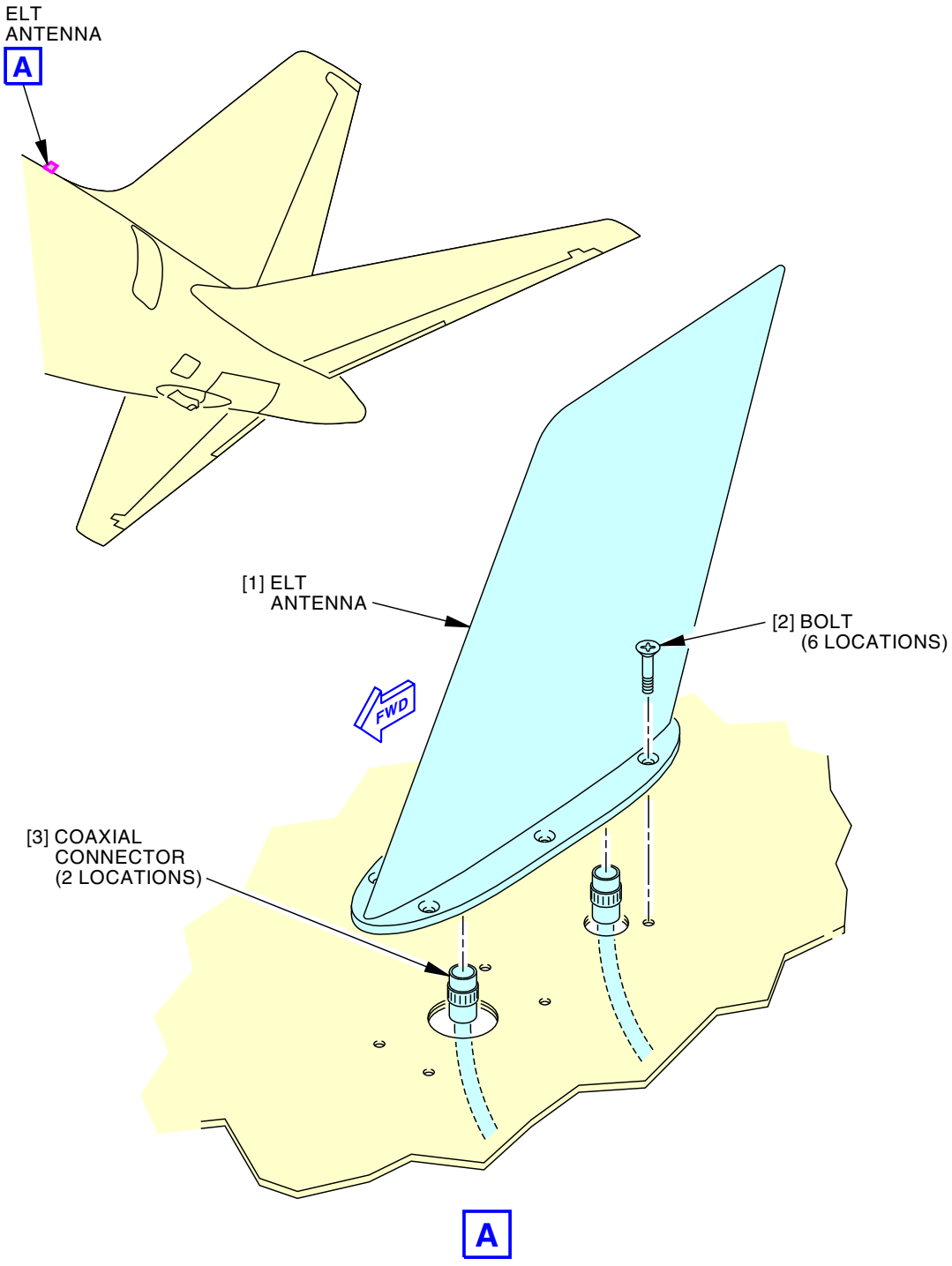
(c) Carefully lift the ELT antenna [1] to get access to the coaxial connectors [3].

(d) Disconnect the coaxial connectors [3] from the ELT antenna [1].

(e) Remove the ELT antenna [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Emergency Locator Transmitter Antenna Installation
Figure 401/23-24-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-24-02-400-801

3. Emergency Locator Transmitter Antenna - Installation

(Figure 401)

A. General

(1) This task has the steps to install the Emergency Locator Transmitter (ELT) antenna.

B. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
23-24-00-710-801	Emergency Locator Transmitter Operational Test (P/B 501)
51-21-00-160-801	Surface Preparation Of Aluminum Bonding Surfaces For Antenna Installations (P/B 701)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
51-22-00-390-806	Fillet Seal Application (P/B 201)
SL 20-043	Deferred Application of Aero-Sealant in Antenna Installations

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C50475	Coating - Chemical Conversion - Bonderite M-CR 600 Aero	BAC5719 Type II Class D

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	ELT antenna	23-24-02-04-012	PTF ALL

F. Location Zones

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

G. ELT Antenna Installation

SUBTASK 23-24-02-100-003

(1) Do these steps to clean the mating surfaces:

(a) Remove sealant from the airplane mating surface and ELT antenna [1] mating surface with a sealant removal tool, COM-2481.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Abrasive clean the airplane mating surface (TASK 51-21-00-160-801).

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS) AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(c) Solvent clean the airplane mating surface and ELT antenna [1] mating surface (TASK 20-10-34-110-802).

SUBTASK 23-24-02-620-001

(2) Apply a layer of coating, C00064, or Bonderite M-CR 600 Aero coating, C50475, to the airplane mating surface (TASK 51-21-41-370-802).

SUBTASK 23-24-02-420-001

(3) Do these steps to install the ELT antenna [1]:

(a) Connect the coaxial connectors [3] to the ELT antenna [1].

(b) Put the ELT antenna [1] in its position.

(c) Apply a layer of sealant, A00247, to the shank and threads of the bolts [2].

(d) Install the bolts [2].

(e) Tighten the bolts [2] to 35 in-lb (4 N·m).

(f) Use a non-intrinsically safe bonding meter, COM-614, to measure the resistance between the ELT antenna [1] base and the airplane skin (SWPM 20-20-00).

1) Make sure that the resistance is 0.005 ohm (5 milliohm) or less.

SUBTASK 23-24-02-390-001

(4) Do these steps to apply a fillet seal around the ELT antenna [1]:

NOTE: Operators can defer the application of the aero-sealant in the antenna installation until next “A” check or overnight maintenance stop to prevent a flight delay. The deferral of the application of sealant must not exceed a maximum of eight weeks (SL 20-043).

(a) Apply sealant, A00247, around the base of the ELT antenna [1] to make a corrosion fillet seal (TASK 51-22-00-390-806).

(b) Let the sealant dry for the correct cure time.

H. ELT Antenna Installation Test

SUBTASK 23-24-02-710-001

(1) Do this task: Emergency Locator Transmitter Operational Test, TASK 23-24-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY LOCATOR TRANSMITTER CONTROL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Emergency Locator Transmitter Control Panel - Removal

(2) Emergency Locator Transmitter Control Panel - Installation

B. The ELT control panel is installed on the forward overhead panel, P5, in the flight compartment.

TASK 23-24-03-000-801

2. Emergency Locator Transmitter Control Panel - Removal

(Figure 401)

A. General

(1) This task has the steps to remove the Emergency Locator Transmitter (ELT) control panel.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
23-24-00-040-802	Emergency Locator Transmitter Deactivation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. ELT Control Panel Removal

SUBTASK 23-24-03-020-001

(1) Do this task: Emergency Locator Transmitter Deactivation, TASK 23-24-00-040-802.

NOTE: This will prevent accidental activation of the ELT when the ELT control panel is removed.

NOTE: If the ELT is accidentally started. You can reset the toggle switch on the front of the ELT transmitter from OFF to ON position momentarily then back to the OFF position.

SUBTASK 23-24-03-860-001

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	14	C01278	MASTER CAUTION ANNUNCIATOR CONT 4
E	10	C00328	OVERHEAD 5V SECONDARY
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 23-24-03-020-003

(3) Do these steps to remove the ELT control panel [1] (TASK 20-40-12-000-802):

(a) Loosen the quarter-turn fasteners [2].

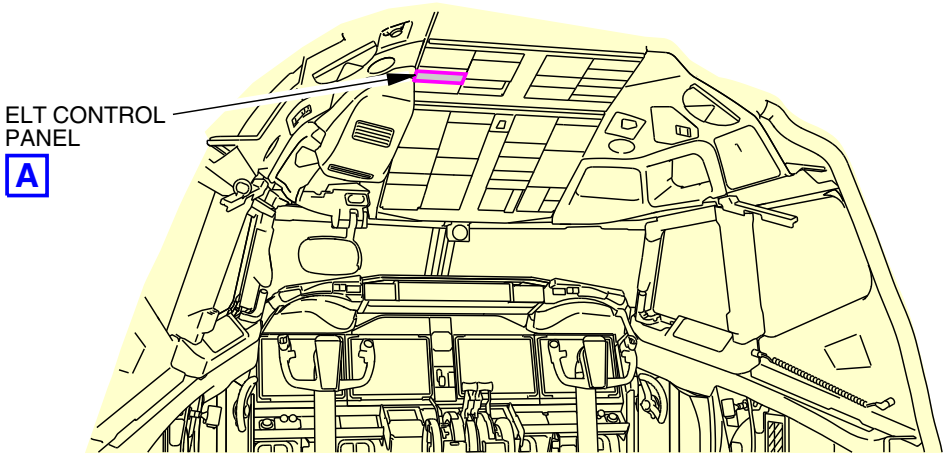
(b) Carefully lower the ELT control panel [1] to get access to the electrical connector [3].

(c) Disconnect the electrical connector [3] from the ELT control panel [1].

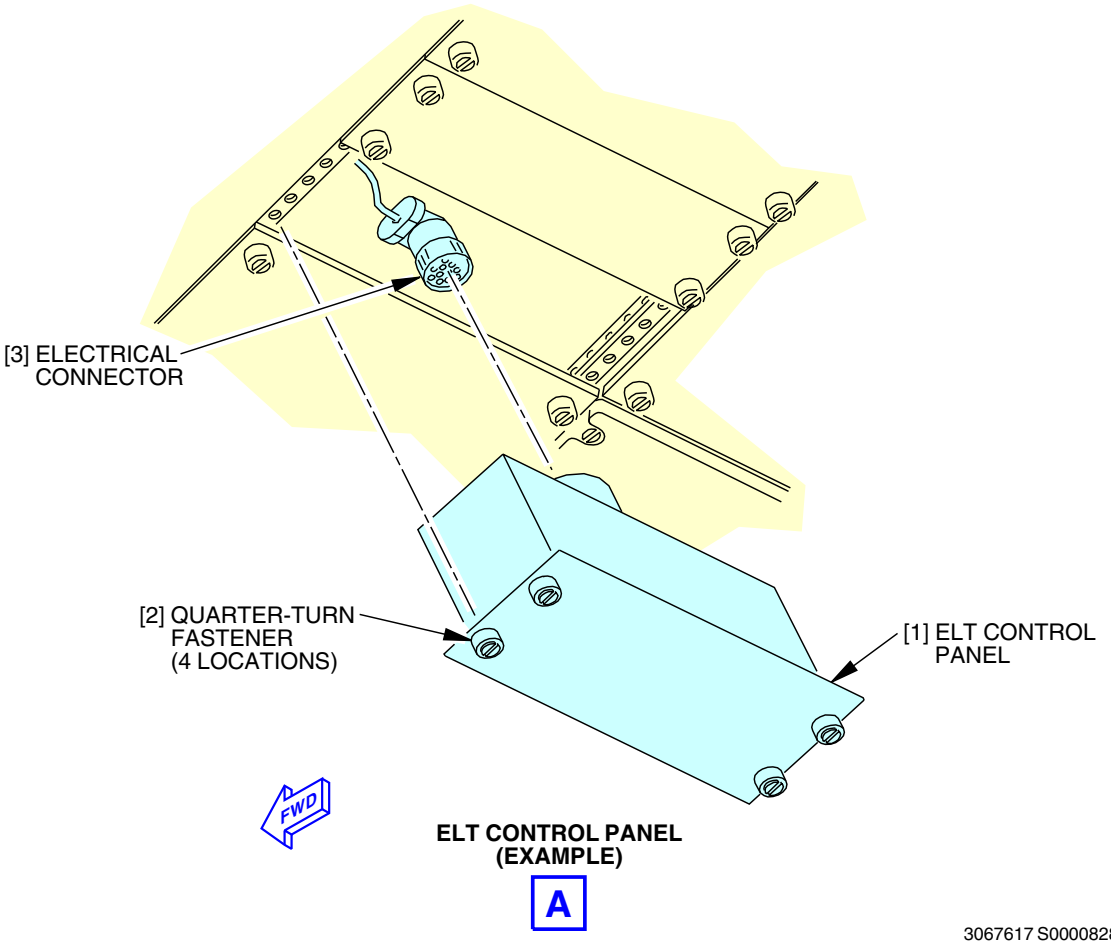
(d) Remove the ELT control panel [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



ELT CONTROL PANEL
(EXAMPLE)

A

3067617 S0000828868_V1

Emergency Locator Transmitter Control Panel Installation
Figure 401/23-24-03-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-24-03-400-801

3. Emergency Locator Transmitter Control Panel - Installation

(Figure 401)

A. General

(1) This task has the steps to install the Emergency Locator Transmitter (ELT) control panel.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-24-00-440-802	Emergency Locator Transmitter Activation (P/B 201)
23-24-00-710-801	Emergency Locator Transmitter Operational Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	ELT control panel	23-24-03-01-050	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left

E. ELT Control Panel Installation

SUBTASK 23-24-03-860-002

(1) Make sure that these circuit breakers are open:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	14	C01278	MASTER CAUTION ANNUNCIATOR CONT 4
E	10	C00328	OVERHEAD 5V SECONDARY
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 23-24-03-420-002

(2) Do these steps to install the ELT control panel [1] (TASK 20-40-12-400-802):

(a) Make sure that the ELT control panel [1] switch is set to the ARM position.

(b) Connect the electrical connector [3].

(c) Put the ELT control panel [1] in its position.

(d) Tighten the quarter-turn fasteners [2].

F. ELT Control Panel Installation Test

SUBTASK 23-24-03-860-003

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	14	C01278	MASTER CAUTION ANNUNCIATOR CONT 4
E	10	C00328	OVERHEAD 5V SECONDARY
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 23-24-03-420-003

(2) Do this task: Emergency Locator Transmitter Activation, TASK 23-24-00-440-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-24-03-860-004

(3) Do this task: Emergency Locator Transmitter Operational Test, TASK 23-24-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIRCRAFT COMMUNICATION AND REPORTING SYSTEM (ACARS) - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Aircraft Communication and Reporting System (ACARS) Deactivation

(2) Aircraft Communication and Reporting System (ACARS) Activation

TASK 23-27-00-040-801

2. Aircraft Communication and Reporting System - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power to the ACARS system.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

NOTE: The aircraft will have either a CMU or AMU installed. Reference the aircraft wire diagram to determine the system installed.

SUBTASK 23-27-00-860-019

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

E. ACARS - Tryout

NOTE: This tryout is to make sure that the ACARS system is in a zero energy state.

SUBTASK 23-27-00-860-020

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

SUBTASK 23-27-00-010-002

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 23-27-00-710-006

(3) Do the following steps at the ACARS CMU:

(a) Push and hold the TEST switch on the front panel.

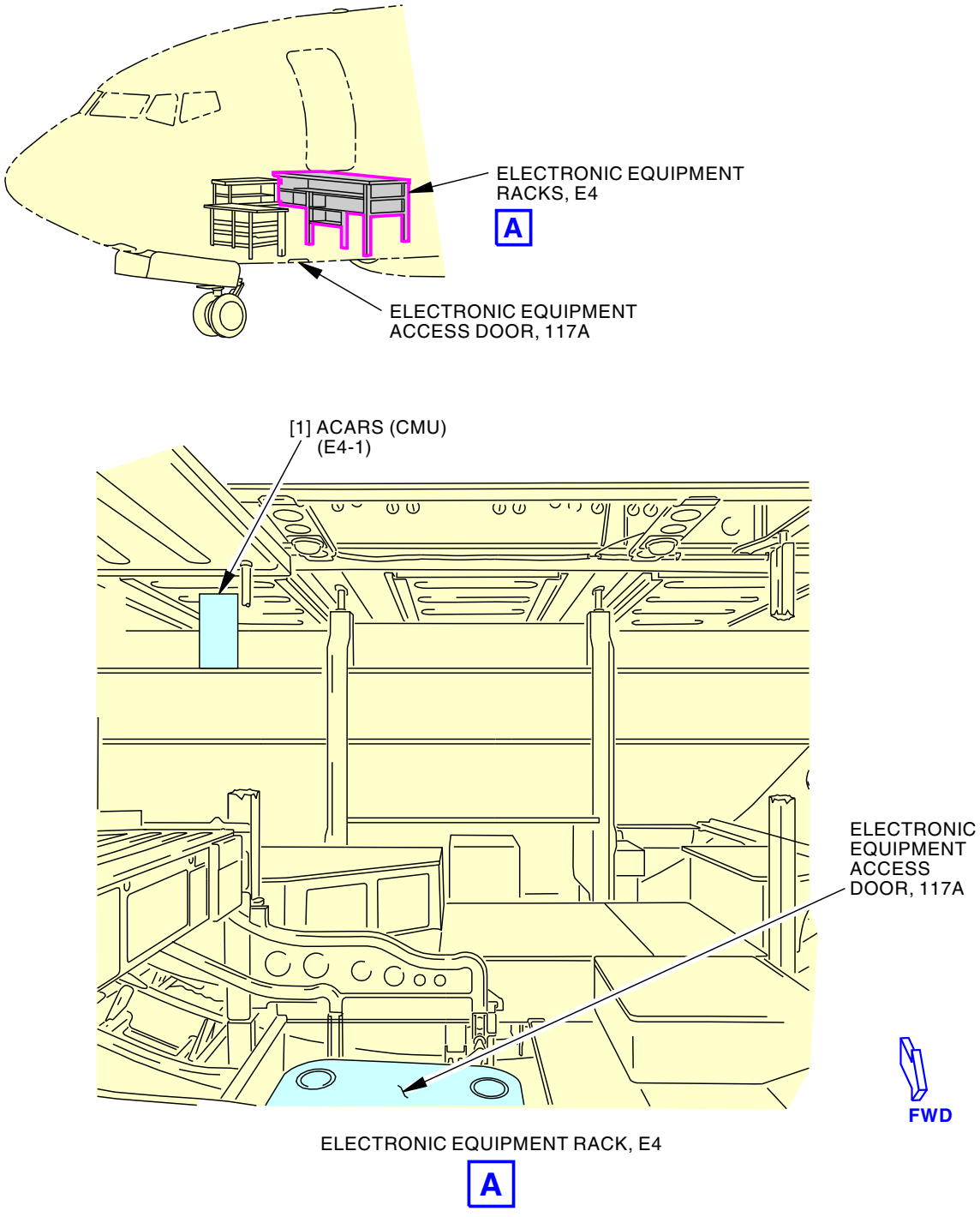
1) Make sure that all the lights on the front of the CMU do not come on.

(b) Release the switch and wait at least one minute.

1) Make sure that none of the lights momentarily come on.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aircraft Communications Addressing and Reporting System (ACARS)
Figure 201/23-27-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-27-00-400-801

3. Aircraft Communication and Reporting System - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the ACARS system.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

SUBTASK 23-27-00-860-021

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

SUBTASK 23-27-00-410-001

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACARS SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An operational test of the Aircraft Communications Addressing and Reporting System (ACARS)

(2) A system test of the ACARS

TASK 23-27-00-740-802

2. ACARS - Operational Test

A. References

Reference	Title
23-27-33-700-801	ACARS Communications Management Unit (CMU) Software Configuration Check (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

C. Procedure

SUBTASK 23-27-00-700-003

(1) Do this task: ACARS Communications Management Unit (CMU) Software Configuration Check, TASK 23-27-33-700-801.

SUBTASK 23-27-00-710-005

(2) Do these steps to do the ACARS Operational test

(a) Push and release the TEST switch on the ACARS Communications Management Unit (CMU) on shelf E4-1 in the E/E compartment.

(b) Make sure the SYSTEM FAIL indicator stays off.

(c) Make sure the LRU PASS/FAIL indicator is red for 2 seconds, then is green for 2 seconds, then is off for at least 2 seconds and no more than 5 seconds.

(d) If the test passes, the LRU PASS/FAIL indicator is green for 30 seconds, then is off.

(e) If the test fails, the LRU PASS/FAIL indicator is red for 30 seconds, then is off.

——— END OF TASK ———

TASK 23-27-00-700-802

3. ACARS - System Test

A. General

(1) The system test contains checks of all of the primary ACARS functions. The checks in the operational test are also in the system test.

(2) For the ACARS system to operate correctly, it must be linked to a remote ground station.

(3) For ACARS to link to the ground station through the VHF communication system, the airplane must be within range of the ground station. Also, the VHF antenna must have a clear view to the station tower.

(4) The ACARS ground station and network must also be operational.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(5) Make sure that the airplane is not in or near large metal structures. HF communication while the airplane is on the ground can be degraded by airplane position or blocked by RF signal attenuation because of ground equipment and structures.

(6) The system test is based on the default Boeing AOC software. If airline customized AOC software is loaded, there can be differences in CDU screen formats and screen prompts.

B. References

Reference	Title
23-27-33-020-801	ACARS Communications Management Unit (CMU) Removal (P/B 401)
23-27-33-470-802	ACARS Software Installation with Onboard Network System (ONS) (P/B 201)
23-27-35-700-801	ACARS Airplane Personality Module (APM) Data Verification Check (P/B 201)
31-65-00-970-801	System Maintenance Data Pages Function (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Procedure

SUBTASK 23-27-00-860-018

(1) Set the Captain's clock to the current Greenwich Mean Time (GMT).

SUBTASK 23-27-00-710-004

(2) Do this task: ACARS - Operational Test, TASK 23-27-00-740-802.

SUBTASK 23-27-00-700-002

(3) Do this task: ACARS Airplane Personality Module (APM) Data Verification Check, TASK 23-27-35-700-801.

SUBTASK 23-27-00-730-006

(4) Do the Initialization and Print test:

(a) Push the MENU function key on both CDUs.

(b) Push the LSK adjacent to the <ACARS prompt on both CDUs.

NOTE: DLNK can be displayed instead of ACARS. This depends on CMU core software.

1) Make sure that the ACARS-APPLICATION MENU shows on both displays.

NOTE: Use CDU-1 when it is necessary to use the ACARS menu functions during the system test. You can use CDU-2 when it is necessary to use the FMC functions.

(c) Push the LSK adjacent to the TECHNICAL> prompt.

1) Make sure that the ACARS-TECHNICAL MENU shows on the display.

(d) Push the LSK adjacent to the <SYSTEM CONFIG prompt.

1) Make sure that the ACARS-SYSTEM CONFIG page 1 shows on the display.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure that the correct ACARS hardware and software part numbers show on the display.

3) If the correct software part numbers do not show on the display, do this task: ACARS Software Installation with Onboard Network System (ONS), TASK 23-27-33-470-802 or replace the CMU with a unit that has the correct software loaded (TASK 23-27-33-020-801).

(e) Push the LSK adjacent to the *PRINT prompt.

1) Make sure that the printer prints the data that shows on the display.

NOTE: The print format is not the same as the display format.

(f) Push the LSK adjacent to the <RETURN prompt to go back to the ACARS-TECHNICAL MENU page.

SUBTASK 23-27-00-730-007

(5) Do the VHF-Link Test:

(a) Push the VHF-3 key on RTP-1, RTP-2 and RTP-3.

1) Make sure that ACARS or DATA shows in the RTP-1 ACTIVE display.

NOTE: If ACARS or DATA does not show in the ACTIVE display, then turn the RTP-1 frequency select knob until "ACARS" or "DATA" shows in the STANDBY display. Push the transfer switch on RTP-1 to move "ACARS" or "DATA" to the ACTIVE window of RTP-1.

(b) Push the LSK adjacent to the <LINK STATUS prompt.

1) Make sure that the ACARS-LINK STATUS page shows on the display.

2) Make sure that the FREQ shown adjacent to the VHF prompt below the MEDIA line is correct for the geographic area where the airplane is currently located.

3) Make sure that AOA COMM shows adjacent to the *VHF prompt on the display.

NOTE: If the airplane is located in an area where AOA VDL Mode 2 is not available, POA COMM shows instead of AOA COMM.

NOTE: If SCAN or NO-COMM shows, wait approximately 2 minutes before you do the next step.

(c) Push the LSK adjacent to the *VHF prompt on the line under the TEST prompt.

1) Make sure that the "*" adjacent to the VHF prompt goes away, then shows again after the link test is successful.

2) Make sure that “VHF IN PROG” shows on the line above the <RETURN prompt.

3) Make sure that “LINK TEST SUCCESSFUL” shows on the line above the <RETURN prompt.

(d) Push the transfer switch on RTP-1 to move ACARS or DATA to the STANDBY display and a voice frequency to the ACTIVE display.

1) Make sure that VOICE shows adjacent to the *VHF prompt on the display.

(e) Push the transfer switch on RTP-2 to move ACARS or DATA to the ACTIVE display and a voice frequency to the STANDBY display.

NOTE: Wait for the displays to change in approximately 2 minutes.

1) Make sure that AOA COMM shows adjacent to the *VHF prompt on the display.

NOTE: If the airplane is located in an area where AOA VDL Mode 2 is not available, POA COMM shows instead of AOA COMM.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Push the transfer switch on RTP-2 to move ACARS or DATA to the STANDBY display and a voice frequency to the ACTIVE display.

1) Make sure that VOICE shows adjacent to the *VHF prompt on the display.

(g) Push the transfer switch on RTP-3 to move ACARS or DATA to the ACTIVE display and a voice frequency to the STANDBY display.

NOTE: Wait for the displays to change in approximately 2 minutes.

1) Make sure that AOA COMM shows adjacent to the *VHF prompt on the display.

NOTE: If the airplane is located in an area where AOA VDL Mode 2 is not available, POA COMM shows instead of AOA COMM.

(h) Push the LSK adjacent to the <RETURN prompt to go back to the ACARS-TECHNICAL MENU page.

SUBTASK 23-27-00-730-018

(6) Do the SATCOM Link Test:

NOTE: This test verifies the interface between the SDU and ACARS CMU.

(a) Select the LSK adjacent to the <LINK STATUS to show the LINK STATUS page.

(b) Select the LSK adjacent to the <*SAT to initiate the SATCOM link test.

(c) Make sure that the SATCOM status changes from SAT IN PROGRESS to LINK TEST SUCCESSFUL and then clears (no text displayed).

(d) Select the LSK adjacent to the <RETURN prompt until the ACARS-APPLICATION MENU page is displayed.

(e) Push the MENU key on the CDU keypad to show the CDU “MENU” page.

SUBTASK 23-27-00-730-008

(7) Do the HF Link Test (optional):

WARNING DO NOT OPERATE THE HF SYSTEM WHILE YOU REFUEL OR DEFUEL THE AIRPLANE. AN EXPLOSION CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE AIRPLANE.

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

CAUTION LEAVE THE HF-SYSTEM IN THE DATA MODE DURING THE HF-LINK-TEST ONLY. IF THE HF-SYSTEM IS IN THE DATA MODE, THEN IT CAN TRANSMIT INDEPENDENTLY OF PERSONNEL CONTROL (PUSH-TO-TALK SWITCH).

(a) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

(b) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

(c) Set the radio tuning panels (RTP) as follows:

1) On the captain's RTP, push the HF-1 switch.

a) If the active display on RTP-1 shows DATA, continue with the test.

b) If the active display shows a HF voice frequency, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> Turn the standby frequency knob on RTP-1 clockwise or counterclockwise until DATA shows in the standby (right) display.

<2> Push the transfer button to transfer DATA to the active display.

NOTE: It can take seven minutes for the selected HF transceiver to establish a link to an ACARS HF ground station. Wait until the *HF prompt appears on the line under the TEST prompt before continuing with the next step.

2) On the first officers RTP, select one of the VHF radios.

3) On RTP-3, select one of the VHF radios.

(d) Push the LSK adjacent to the *HF prompt on the line under the TEST prompt.

1) Make sure that the "*" adjacent to the HF prompt goes away, then shows again after the link test is successful.

2) Make sure that “HF IN PROG” shows on the line above the <RETURN prompt.

3) Make sure that “LINK TEST SUCCESSFUL” shows on the line above the <RETURN prompt.

NOTE: This step can take seven minutes to seven hours to complete. The distance to an ACARS ground station, atmospheric conditions, or sunspots can cause the HF link test to fail.

(e) Set the RTPs as follows:

1) On the captain's RTP, do these steps:

a) Turn the standby frequency knob on RTP-1 clockwise or counterclockwise until a valid voice frequency shows in the standby (right) display.

b) Push the transfer button.

2) On the first officer's RTP, push the HF-2 switch.

a) If the active display on RTP-2 shows DATA, continue with the test.

b) If the active display shows a HF voice frequency, do these steps:

<1> Turn the standby frequency knob on RTP-2 clockwise or counterclockwise until DATA shows in the standby (right) display.

<2> Push the transfer button to transfer DATA to the active display.

NOTE: It can take seven minutes for the selected HF transceiver to establish a link to an ACARS HF ground station. Wait until the *HF prompt appears on the line under the TEST prompt before continuing with the next step.

(f) Push the LSK adjacent to the *HF prompt on the line under the TEST prompt.

1) Make sure that the "*" adjacent to the HF prompt goes away, then shows again after the link test is successful.

2) Make sure that “HF IN PROG” shows on the line above the >RETURN prompt.

3) Make sure that “LINK TEST SUCCESSFUL” shows on the line above the >RETURN prompt.

NOTE: This step can take seven minutes to seven hours to complete. The distance to an ACARS ground station, atmospheric conditions, or sunspots can cause the HF link test to fail.

4) On the first officer's RTP, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Turn the standby frequency knob clockwise or counterclockwise until a valid voice frequency shows in the standby (right) display.

b) Push the transfer button.

(g) On the RTP-3, push the HF-1 switch.

1) If the active display on RTP-3 shows DATA, continue with the test.

2) If the active display shows a HF voice frequency, do these steps:

a) Turn the standby frequency knob on RTP-3 clockwise or counterclockwise until DATA shows in the standby (right) display.

b) Push the transfer button to transfer DATA to the active display.

NOTE: It can take seven minutes for the selected HF transceiver to establish a link to an ACARS HF ground station. Wait until the *HF prompt appears on the line under the TEST prompt before continuing with the next step.

(h) Push the LSK adjacent to the *HF prompt on the line under the TEST prompt.

1) Make sure that the "*" adjacent to the HF prompt goes away, then shows again after the link test is successful.

2) Make sure that “HF IN PROG” shows on the line above the >RETURN prompt.

3) Make sure that “LINK TEST SUCCESSFUL” shows on the line above the >RETURN prompt.

NOTE: This step can take seven minutes to seven hours to complete. The distance to an ACARS ground station, atmospheric conditions, or sunspots can cause the HF link test to fail.

4) On the RTP-3, do these steps:

a) Turn the standby frequency knob clockwise or counterclockwise until a valid voice frequency shows in the standby (right) display.

b) Push the transfer button.

(i) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

(j) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

(k) On CDU-1, push the LSK adjacent to the <RETURN prompt to go back to the ACARS-TECHNICAL MENU page.

SUBTASK 23-27-00-730-009

(8) Do the Peripherals Test:

(a) Push the LSK adjacent to the <PERIPHERALS prompt.

1) Make sure that the ACARS-PERIPHERALS page 1 shows on the display.

2) Make sure that CDU L, CDU R, FMC, FMC BCST, ACMS and PRINTER show PRESENT under the CONNECT line and OK under the STAT line.

(b) Push the NEXT PAGE function key.

1) Make sure that the ACARS-PERIPHERALS page 2 shows on the display.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If the system is installed (and ACARS communication is enabled), make sure that CABIN 1, CABIN 2, SDU 1, SDU 2, HFDR 1 or HFDR 2 shows PRESENT below the CONNECT line and OK below the STAT line.

NOTE: If a system is not installed, or if ACARS communication with that system is not enabled, ABSENT shows below the CONNECT line.

(c) Push the NEXT PAGE function key.

1) Make sure that the ACARS-PERIPHERALS page 3 shows on the display.

2) Make sure that XPDR 1, XPDR 2 and VDR 3 show PRESENT below the CONNECT line and OK below the STAT line.

3) Make sure that VDR 1 and VDR 2 show ABSENT below the CONNECT line.

(d) Push the LSK adjacent to the <RETURN prompt to go back to the ACARS-TECHNICAL MENU page.

(e) Push the LSK adjacent to the <RETURN prompt to go back to the ACARS-APPLICATION MENU page.

SUBTASK 23-27-00-730-010

(9) Do these steps to do the Out-Off-On-In (OOOI) sensor test:

(a) Make sure that the ACARS-AOC MENU page shows on the display.

(b) Push the LSK adjacent to the FLT LOG> prompt.

1) Make sure that the ACARS-FLG LOG-CURR page 1 shows on the display.

(c) Push the LSK adjacent to the SENSORS> prompt.

1) Make sure that the ACARS-SENSORS page shows on the display.

(d) Close all the cabin doors, cargo doors, EE and Section 41 bay doors.

1) Make sure that the FWD DOOR, AFT DOOR, CARGO/EE DOORS, and SERVICE DOORS show CLOSED on the display.

NOTE: The following steps test each OOOI parameter. If the OOOI parameters have been working correctly and the ACARS Communications Management Unit (CMU) was not removed or replaced, these steps are optional.

(e) Open the left forward door.

1) Make sure that the FWD DOOR shows OPEN.

(f) Close the left forward door.

1) Make sure that the FWD DOOR shows CLOSED.

(g) Open the right forward door.

1) Make sure that the SERVICE DOORS show OPEN.

(h) Close the right forward door.

1) Make sure that the SERVICE DOORS show CLOSED.

(i) Open the left aft door.

1) Make sure that the AFT DOOR shows OPEN.

(j) Close the left aft door.

1) Make sure that the AFT DOOR shows CLOSED.

(k) Open the right aft door.

1) Make sure that the SERVICE DOORS show OPEN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Close the right aft door.

1) Make sure that the SERVICE DOORS show CLOSED.

(m) Open the forward cargo compartment door.

1) Make sure that the CARGO/EE DOORS shows OPEN.

(n) Close the forward cargo compartment door.

1) Make sure that the CARGO/EE DOORS shows CLOSED.

(o) Make sure that the PARK BRAKE state is SET.

WARNING MAKE SURE THAT THE AIRPLANE WILL NOT MOVE BEFORE YOU RELEASE THE PARKING BRAKE. IF THE AIRPLANE MOVES, IT CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(p) Release the parking brake.

1) Make sure that the PARK BRAKE state changes from SET to RELEASED.

(q) Set the parking brake.

1) Make sure that the PARK BRAKE state changes from RELEASED to SET.

(r) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

(s) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

1) Make sure that the STRUT state changes from ON GND to AIRBORNE.

(t) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

1) Make sure that the STRUT state changes from AIRBORNE to ON GND.

(u) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

(v) Push the LSK adjacent to the <RETURN prompt to go back to the FLT LOG page.

(w) Push the LSK adjacent to the <RETURN prompt to go back to the ACARS-AOC MENU page.

SUBTASK 23-27-00-730-015

(10) Do these steps to do the MAX Display System (MDS) interface test:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

(b) On the MDS, go to the 31 MDS DISPLAY in the MAINT DATA PGS (TASK 31-65-00-970-801).

(c) Push the NEXT PAGE selection until the DPC DIGITAL INPUTS page shows.

(d) Make sure that the CMU Amber INACTIVE status message shows on the Insert A, pins G01/G02.

(e) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) After the CMU energizes, make sure that the CMU White ACTIVE status message shows on the Insert A, pins G01/G02.

SUBTASK 23-27-00-730-011

(11) If it is necessary, do these steps to change the ACARS fuel units of weight from pounds (LBS) to kilograms (KGS):

(a) Push LSK 6L on the CDU until the ACARS-APPLICATION MENU page shows.

(b) Set LSK 2R (AOC STD) on the CDU.

(c) Set LSK 5R (MISC MENU) on the CDU.

(d) Set LSK 4L (PASSWORD) on the CDU.

(e) Put the time that shows on the bottom of the CDU into the scratchpad of the CDU.

(f) Set LSK 1L (*PASSWORD) on the CDU to show on the ACARS-PARAMETERS page.

NOTE: If "INVALID" shows at LSK 1L, then do the last two steps again.

(g) Set LSK 1L (FUEL UNITS) on the CDU until “KGS” shows (the button toggles between LBS and KGS).

(h) Set LSK 6L until the ACARS-APPLICATION MENU shows.

(i) Set LSK 4R (TECHNICAL) to go back to the ACARS TECHNICAL MENU.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACARS COMMUNICATIONS MANAGEMENT UNIT (CMU) - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A software configuration check for the ACARS Communications Management Unit (CMU).

(2) An installation of the ACARS Communications Management Unit (CMU) software with the Onboard Network System (ONS).

B. For general information for software installation times and data loaders, do this task: On-Airplane Software Installation, TASK 20-15-11-470-801

TASK 23-27-33-700-801

2. ACARS Communications Management Unit (CMU) Software Configuration Check

A. General

(1) This software configuration check makes sure that Aircraft Communications Addressing And Reporting System (ACARS) has the correct software loaded.

(2) A Flight Management Control Display Unit (CDU) is necessary for this procedure.

B. References

Reference	Title
23-27-33-020-801	ACARS Communications Management Unit (CMU) Removal (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. ACARS CMU Software Configuration Check

SUBTASK 23-27-33-840-002

(1) Make sure that you know the correct software part numbers for the ACARS Communications Management Unit (CMU).

NOTE: For ACARS to be an approved installation, the correct software part numbers must be installed.

NOTE: These are the software parts:

Table 201/23-27-33-993-802

Name on CDU	Name of Software Part
CORESW	23 CMU CORE SW
VMDTA	23 CMU VM DATABASE SW
AOC DTA	23 CMU AOC SW

SUBTASK 23-27-33-700-003

(2) Do these steps to verify the software installation for the ACARS CMU:

(a) Push the MENU function key on the CDU.

1) Make sure that the MENU page shows on the CDU.

(b) Push the Line Select Key (LSK) adjacent to the <ACARS prompt.

1) Make sure that the ACARS-APPLICATION MENU page shows on the CDU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Push the LSK adjacent to the TECHNICAL> prompt.

1) Make sure that the ACARS-TECHNICAL MENU page shows on the CDU.

(d) Push the LSK adjacent to the <SYSTEM CONFIG prompt.

1) Make sure that the ACARS-SYSTEM CONFIG page 1 of 2 shows on the CDU.

2) Make sure that the ACARS CMU hardware and software part numbers are correct.

a) If the ACARS software part numbers are not correct, do this task: ACARS Software Installation with Onboard Network System (ONS), TASK 23-27-33-470-802

or replace the ACARS CMU with one that has the correct software loaded (ACARS Communications Management Unit (CMU) Removal, TASK 23-27-33-020-801).

(e) Push the LSK adjacent to the <RETURN prompt to go back to the ACARS-TECHNICAL MENU page.

(f) Push the LSK adjacent to the <RETURN prompt to go back to the ACARS-APPLICATION MENU page.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 23-27-33-860-002

(1) Push the MENU key on the CDU.

——— END OF TASK ———

TASK 23-27-33-470-802

3. ACARS Software Installation with Onboard Network System (ONS)

A. General

(1) This procedure tells you how to install software in the Aircraft Communications Addressing And Reporting System (ACARS) with the Onboard Network System (ONS).

(a) The ACARS CMU must contain these pieces of software:

1) Core software

2) Operational software.

3) Customer Unique Database software.

(b) The ACARS Core software must be installed first.

(c) The ACARS Operational software must be installed before the Customer Unique Database software. If the Operational software is already installed, then you can install only the Customer Unique Database software.

(2) The airplane must be on the ground with the engines shutdown and parking brake set before you can install software.

B. References

Reference	Title
46-13-00-470-801	LRU Software Parts Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. ACARS Software Installation with ONS

SUBTASK 23-27-33-410-008

(1) Make sure that you know the correct software part numbers for the ACARS Communications Management Unit (CMU).

NOTE: For ACARS to be an approved installation, the correct software must be installed.

SUBTASK 23-27-33-410-009

(2) Do these steps to prepare for the software installation:

(a) Make sure that the SYSTEM SELECT switch on the data loader control panel, P61, is set to NORMAL.

(b) Do these steps at the data loader control panel, P61:

1) Set the DATA LOAD SELECTOR switch to L CAPT UPR.

2) Set the SYSTEM SELECT switch to ACARS/CMU.

SUBTASK 23-27-33-470-007

(3) Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: Load CORE SW first. Then load AOC SW or DATABASE.

NOTE: After selecting START LOAD on the MDS or PMD, the data load will not start until you perform the next step (2)(a).

(a) Do these steps to start the data load:

1) Push the LSK adjacent to the “DATALOAD*” prompt on the CDU.

2) Push the LSK adjacent to the “PROCEED WITH DATALOAD” prompt on the CDU.

3) Make sure that the message “SCREEN WILL BLANK WHILE DATALOAD IS IN PROGRESS” appears on the CDU, and then the “MENU” appears without the “ACARS” prompt.

NOTE: While the data load is in progress, status will be displayed on the MDS or PMD.

4) Make sure that the status message "Success" shows on the MDS or PDM.

SUBTASK 23-27-33-410-010

(4) Set the SYSTEM SELECT switch on the data loader control panel, P61, to NORMAL.

(5) Make sure that the <ACARS or <DLNK prompt shows on the CDU.

NOTE: The <ACARS or <DLNK prompt appears on the CDU in 2-4 minutes.

SUBTASK 23-27-33-470-008

(6) Do this task to verify the software was loaded: ACARS Communications Management Unit (CMU) Software Configuration Check, TASK 23-27-33-700-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACARS COMMUNICATIONS MANAGEMENT UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ARINC ACARS Communications Management Unit (CMU)

(2) An installation of the ACARS CMU.

B. The ACARS CMU is located on the E4-1 shelf in the main equipment center.

TASK 23-27-33-020-801

2. ACARS Communications Management Unit (CMU) Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. ACARS Communications Management Unit (CMU) Removal

SUBTASK 23-27-33-860-006

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

SUBTASK 23-27-33-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-27-33-020-003



CAUTION

DO NOT TOUCH THE CONNECTOR PINS OR OTHER CONDUCTORS ON THE CMU. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CMU.

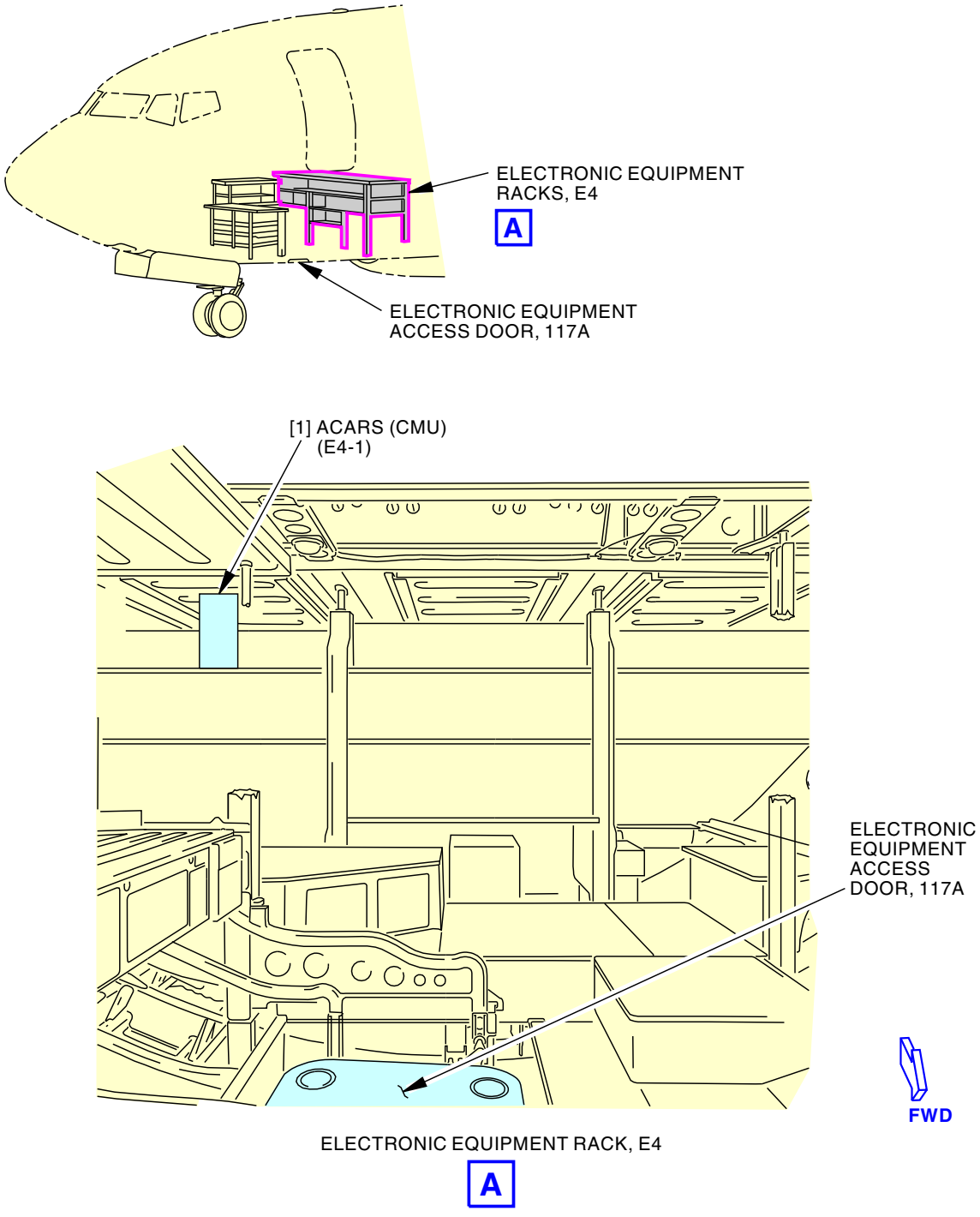
(3) Before touch the ACARS CMU [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 23-27-33-020-004

(4) Remove the ACARS CMU [1] from the shelf, do this task: E/E Box Removal, TASK 20-10-07-000-801.

——— END OF TASK ———

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Aircraft Communications Addressing and Reporting System (ACARS) Communications Management Unit (CMU) Installation
Figure 401/23-27-33-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-27-33-420-801

3. ACARS Communications Management Unit (CMU) Installation

(Figure 401)

A. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-27-00-740-802	ACARS - Operational Test (P/B 501)
23-27-33-700-801	ACARS Communications Management Unit (CMU) Software Configuration Check (P/B 201)
23-27-35-700-801	ACARS Airplane Personality Module (APM) Data Verification Check (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	ACARS CMU	23-27-33-01-100	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. ACARS Communications Management Unit (CMU) Installation

SUBTASK 23-27-33-860-007

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

SUBTASK 23-27-33-420-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS OR OTHER CONDUCTORS ON THE CMU. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CMU.

(2) Before touch the ACARS CMU [1], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-27-33-410-002

(3) Install the ACARS CMU [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-27-33-860-008

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

F. Installation Test

SUBTASK 23-27-33-470-006

(1) Do this task: ACARS Communications Management Unit (CMU) Software Configuration Check, TASK 23-27-33-700-801.

SUBTASK 23-27-33-740-001

(2) Do this task: ACARS - Operational Test, TASK 23-27-00-740-802.

SUBTASK 23-27-33-700-002

(3) Do this task: ACARS Airplane Personality Module (APM) Data Verification Check, TASK 23-27-35-700-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-27-33-410-003

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIRPLANE PERSONALITY MODULE (APM) - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) ACARS APM software configuration check.

(2) ACARS APM software installation.

TASK 23-27-35-700-801

2. ACARS Airplane Personality Module (APM) Data Verification Check

A. General

(1) This data verification check makes sure that the ACARS Airplane Personality Module (APM) contains the correct data values.

(2) A Control Display Unit (CDU) is necessary for this procedure.

B. References

Reference	Title
23-27-35 P/B 401	ACARS AIRPLANE PERSONALITY MODULE - REMOVAL/INSTALLATION
34-61-00-710-801	Flight Management Computer System - Operational Test (P/B 501)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Procedure

SUBTASK 23-27-35-840-001

(1) Make sure that the Flight Management Computer System (FMCS) is serviceable (TASK 34-61-00-710-801).

SUBTASK 23-27-35-730-006

(2) Do these steps on the CDU to check the data in the APM:

(a) Push the MENU button.

(b) Select ACARS.

1) If the ACARS-AOC menu shows, select TECH MENU.

2) If the ACARS-APPLICATION menu shows, select TECHNICAL.

(c) Select PASSWORD.

(d) Enter the UTC time that shows at the bottom of the CDU in the scratchpad.

(e) Select *PASSWORD.

NOTE: The UTC can change while typing it into the scratchpad and cause an INVALID message adjacent to *PASSWORD. If this happens, repeat the last two steps.

(f) Select APM MENU.

1) Make sure that the ACARS-APM MENU shows.

(g) Select IDENT BLK PGM.

1) Make sure that the ACARS-IDENT BLK PGM page 1 shows.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Make sure that the aircraft type adjacent to AC TYPE is correct.

(i) Make sure that the airplane registration number adjacent to REG NUM is correct.

(j) Make sure that the two letter airline identification code adjacent to AIRLINE ID is correct.

(k) Make sure that the three letter ICAO airline identification code adjacent to ICAO AIRLINE ID is correct.

(l) Push the NEXT PAGE button.

1) Make sure that the ACARS-IDENT BLK PGM page 2 shows.

(m) Make sure that the ICAO address adjacent to ICAO ADDR is correct.

(n) Push the NEXT PAGE button.

1) Make sure that the ACARS-IDENT BLK PGM page 3 shows.

(o) Make sure that NO shows adjacent to HFDR 1 INSTALLED and HFDR 2 INSTALLED.

(p) Make sure that YES shows adjacent to VDR INSTALLED.

(q) If one or more of the above parameters are not correct, do this task: ACARS Airplane Personality Module (APM) Software Installation, TASK 23-27-35-700-802, or replace the ACARS APM (PAGEBLOCK 23-27-35/401).

(r) Select RETURN until TECH MENU or TECHNICAL page shows.

——— END OF TASK ———

TASK 23-27-35-700-802

3. ACARS Airplane Personality Module (APM) Software Installation

A. General

(1) This software installation writes data values into the Aircraft Communications Addressing And Reporting System (ACARS) Airplane Personality Module (APM).

(2) A Control Display Unit (CDU) is necessary for this procedure.

B. References

Reference	Title
34-61-00-710-801	Flight Management Computer System - Operational Test (P/B 501)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. ACARS APM Software Installation

SUBTASK 23-27-35-840-002

(1) Make sure that the Flight Management Computer System (FMCS) is serviceable (TASK 34-61-00-710-801).

SUBTASK 23-27-35-730-005

(2) Do these steps on the CDU to write data into the APM:

(a) Push the MENU button.

(b) Select ACARS.

1) If the ACARS-AOC menu shows, select TECH MENU.

2) If the ACARS-APPLICATION menu shows, select TECHNICAL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Select PASSWORD.

(d) Enter the UTC time that shows at the bottom of the CDU into the scratchpad.

(e) Select *PASSWORD.

NOTE: The UTC can change while typing it into the scratchpad and cause an INVALID message adjacent to *PASSWORD. If this happens, repeat the last two steps.

(f) Select PARAMETERS.

(g) Record the six character transponder address (XPDR ADDR), shown at Line Select Key (LSK) 1R, for later use.

- XPDR ADDR: _____.

(h) Select RETURN.

(i) Select APM MENU.

(j) Select IDENT BLK PGM.

(k) Select AC TYPE.

(l) Select BOEING.

(m) Select *737.

1) Make sure that the ACARS-IDENT BLK PGM page 1 shows B737 adjacent to AC TYPE.

NOTE: WRITE* shows at LSK 5R. This will be used later to write the data to the Airplane Personality Module.

(n) Enter the registration number of the airplane as shown on the captain or first officer main instrument panel placard in the scratchpad.

NOTE: If the registration number is less than 7 characters, enter leading decimal points.

NOTE: If the registration number is more than 7 characters, it is necessary to enter the registration number without the dash (-).

Examples of aircraft registration number

DISPLAYED ON PLACARD	ENTER IN SCRATCHPAD
N1239	..N1239
A-2345	.A-2345
N123456	N123456
UP-B3743	UPB3743

(o) Select REG NUM.

1) Make sure that the airplane registration number shown adjacent to REG NUM agrees with the registration number on the captain or first officer main instrument panel placard.

(p) Enter the two letter airline ID into the scratchpad.

(q) Select AIRLINE ID.

1) Make sure that the airline ID shown adjacent to AIRLINE ID is correct.

(r) Enter the three letter ICAO code into the scratchpad.

(s) Select ICAO AIRLINE ID.

1) Make sure that the ICAO airline ID shown adjacent to ICAO AIRLINE ID is correct.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(t) Push the NEXT PAGE button.

1) Make sure that the ACARS-IDENT BLK PGM page 2 shows.

(u) Enter the six character transponder address (XPDR ADDR) of the airplane into the scratchpad.

NOTE: The XPDR ADDR was recorded at an earlier step.

(v) Select ICAO ADDR.

1) Make sure that the ICAO address shown adjacent to ICAO ADDR is correct.

(w) Push the NEXT PAGE button.

1) Make sure that the ACARS-IDENT BLK PGM page 3 shows.

(x) Select HFDR 1 INSTALLED until NO shows.

(y) Select HFDR 2 INSTALLED until NO shows.

(z) Select VDR INSTALLED until YES shows.

(aa) Select WRITE*.

(ab) Select *WRITE.

NOTE: PROGRAMMING shows when the CMU writes to the APM. This can take up to one minute.

(ac) Wait for the CMU to reset or until CONFIRMED shows on the CDU.

(ad) Open this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

(ae) Wait 5 seconds.

(af) Close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

(ag) Wait for ACARS to show on the CDU.

(ah) Do this task: ACARS Airplane Personality Module (APM) Data Verification Check, TASK 23-27-35-700-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACARS AIRPLANE PERSONALITY MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ARINC Communications Addressing and Reporting System (ACARS) airplane personality module (APM).

(2) An installation of the ACARS airplane personality module (APM).

B. The airplane personality module (APM) [1] is located behind the ACARS communications management unit (CMU) , M02127, on the E4-1 shelf in the main equipment center.

TASK 23-27-35-020-801

2. ACARS Airplane Personality Module (APM) Removal

(Figure 401)

A. References

Reference	Title
20-40-12-000-801	ESDS Handling for Printed Circuit Board Removal (P/B 201)
50-11-01-000-801	Forward Cargo Compartment Forward Bulkhead Liner Removal (P/B 401)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door

D. Removal Procedure

SUBTASK 23-27-35-860-003

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-27-35-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 23-27-35-010-002

(3) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 23-27-35-020-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS OR OTHER CONDUCTORS ON THE APM. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE APM.

(4) Before you touch the airplane personality module (APM) [1], do this task: ESDS Handling for Printed Circuit Board Removal, TASK 20-40-12-000-801.

SUBTASK 23-27-35-020-002

(5) Do these steps to remove the airplane personality module (APM) [1]:

NOTE: Access the airplane personality module (APM) [1] from the forward cargo area, through the right pressure relief panel [5] and the wire bundle disconnect bracket [4] on the back of the E4-1 shelf.

(a) Remove the right pressure relief panel [5] (Forward Cargo Compartment Forward Bulkhead Liner Removal, TASK 50-11-01-000-801).

(b) From behind the E4-1 shelf, loosen the four fasteners that hold the E4-1 shelf wire bundle disconnect bracket [4].

(c) Lower the E4-1 shelf wire bundle disconnect bracket [4] for access to the airplane personality module (APM) [1].

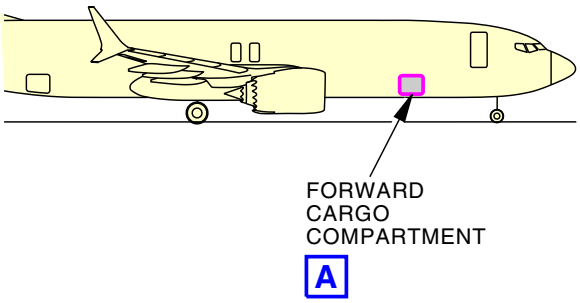
(d) Remove the two screws [2] that hold the airplane personality module (APM) [1] to the APM support bracket [6].

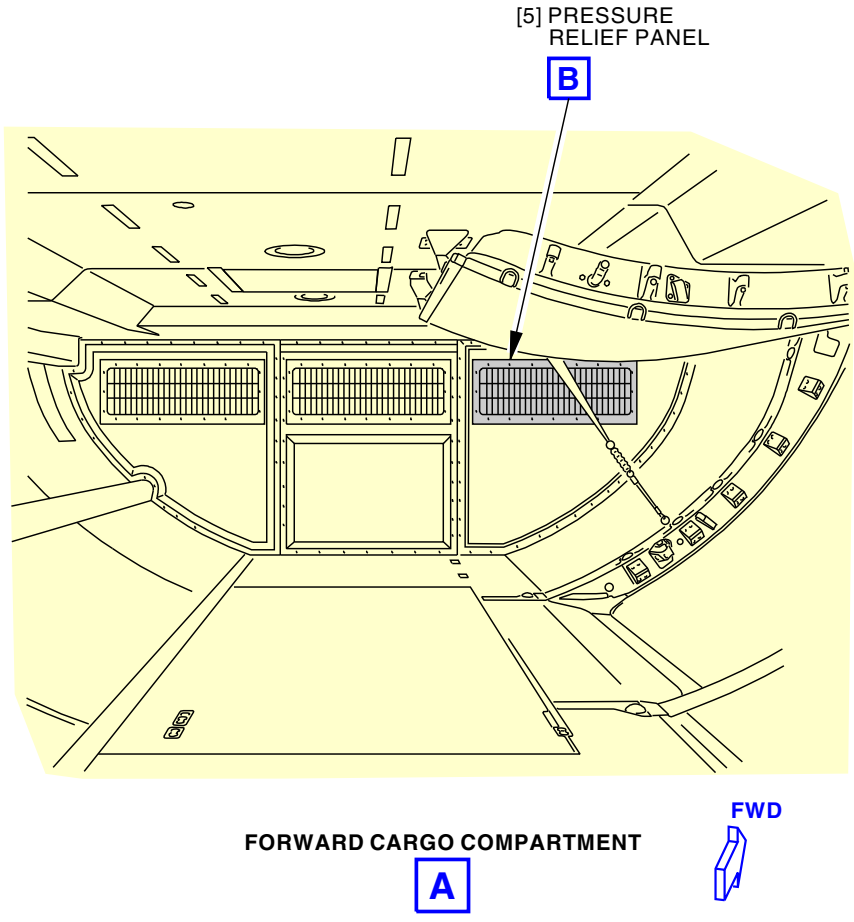
(e) Loosen the two fasteners that hold the connector [3] to the airplane personality module (APM) [1].

(f) Disconnect the connector [3] and remove the airplane personality module (APM) [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

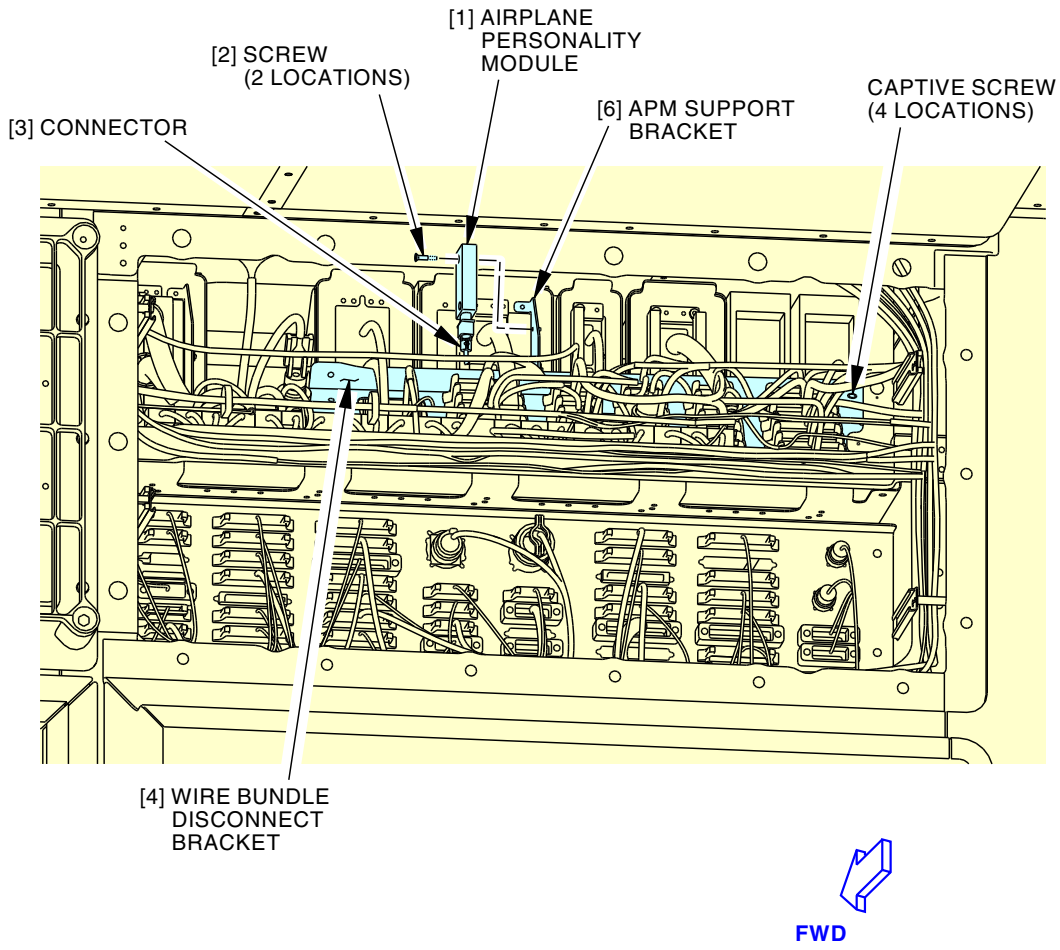




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Aircraft Communications Addressing and Reporting System (ACARS) Airplane Personality Module (APM)
Figure 401/23-27-35-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aircraft Communications Addressing and Reporting System (ACARS) Airplane Personality Module (APM)
Figure 401/23-27-35-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-27-35-420-801

3. ACARS Airplane Personality Module (APM) Installation

(Figure 401)

A. References

Reference	Title
20-40-12-400-801	ESDS Handling for Printed Circuit Board Installation (P/B 201)
23-27-00-740-802	ACARS - Operational Test (P/B 501)
23-27-35-700-801	ACARS Airplane Personality Module (APM) Data Verification Check (P/B 201)
50-11-01-400-801	Forward Cargo Compartment Forward Bulkhead Liner Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Airplane personality module (APM)	23-27-35-05-005	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door

E. Installation Procedure

SUBTASK 23-27-35-860-004

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-27-35-020-003

CAUTION DO NOT TOUCH THE CONNECTOR PINS OR OTHER CONDUCTORS ON THE APM. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE APM.

(2) Before you touch the airplane personality module (APM) [1], do this task: ESDS Handling for Printed Circuit Board Installation, TASK 20-40-12-400-801.

SUBTASK 23-27-35-420-001

(3) Do these steps to install the airplane personality module (APM) [1]:

NOTE: Access to the airplane personality module (APM) [1] is from the forward cargo area, through the right pressure relief panel [5] and the wire bundle disconnect bracket [4] on the back of the E4-1 shelf.

(a) Connect the airplane personality module (APM) [1] to the connector [3].

(b) Tighten the two fasteners that hold the connector [3] to the airplane personality module (APM) [1].

(c) Install the two screws [2] that hold the airplane personality module (APM) [1] to the APM support bracket [6].

(d) Install the E4–1 shelf wire bundle disconnect bracket [4].

1) Make sure the four fasteners that secure the wire bundle disconnect bracket [4] are tightened.

(e) Install the right pressure relief panel [5](TASK 50-11-01-400-801).

SUBTASK 23-27-35-860-005

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	8	C01483	CMU-1 AC

This circuit breaker is inoperative and should remain open:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
E	9	C01500	CMU/ACARS DC (INOP)

F. Installation Test

SUBTASK 23-27-35-700-001

(1) Do this task: ACARS Airplane Personality Module (APM) Data Verification Check, TASK 23-27-35-700-801.

SUBTASK 23-27-35-740-001

(2) Do this task: ACARS - Operational Test, TASK 23-27-00-740-802.

G. Put the Airplane Back to its Initial Condition.

SUBTASK 23-27-35-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-27-35-410-002

(2) Close this access panel:

Number	Name/Location
821	Forward Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SELCAL SYSTEM - MAINTENANCE/PRACTICES

1. General

A. This procedure has these tasks:

(1) SELCAL System Deactivation

(2) SELCAL System Activation

(3) SELCAL Program Switch Module switch settings.

TASK 23-28-00-040-801

2. SELCAL System - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power from the SELCAL System.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-28-00-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

D. SELCAL System - Tryout

NOTE: This tryout is to make sure that the SELCAL system is in a zero energy state.

SUBTASK 23-28-00-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

SUBTASK 23-28-00-700-001

(2) Do these steps to test the SELCAL system:

(a) Set the captain's audio control panel to the VHF 1 communication system.

(b) Set the VHF 1 to an approved test frequency.

(c) Have a SELCAL identified signal transmitted on the VHF 1 test frequency.

NOTE: If a station is not available to transmit from, make a SELCAL signal with a SELCAL ground station encoder. Make sure that the signal is approved and is set to the SELCAL code for the airplane.

(d) Make sure that the VHF 1 light on the SELCAL control panel does not come on.

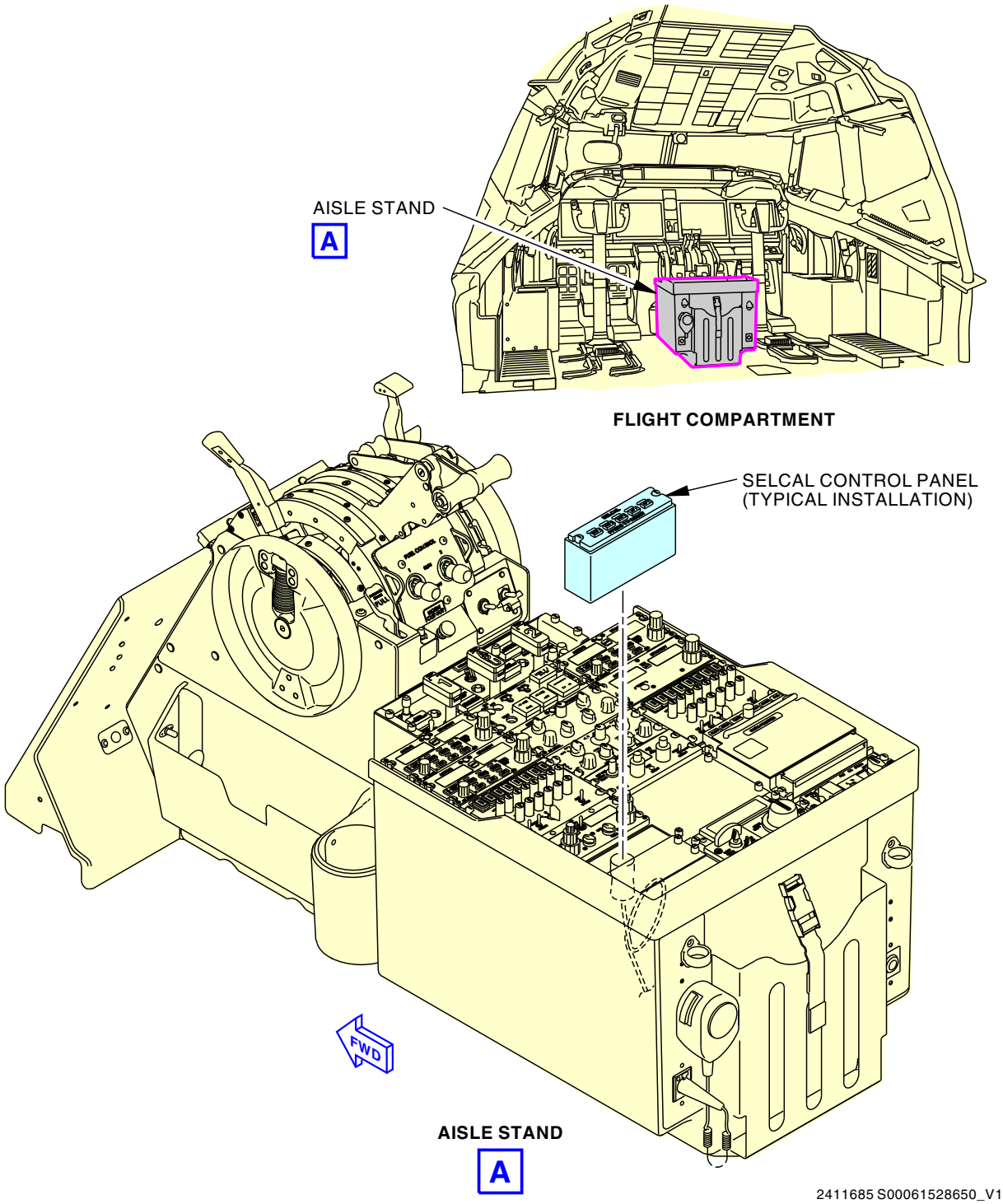
(e) Make sure that the VHF 1 call light on all audio control panels does not come on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Make sure you do not hear a single high-low chime in the flight compartment.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SECAL Control Panel
Figure 201/23-28-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-28-00-440-801

3. SELCAL System - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the SELCAL System.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-28-00-860-003

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

——— END OF TASK ———

TASK 23-28-00-860-801

4. SELCAL Program Switch Module Switch Settings

(Figure 202)

A. General

(1) This task includes the steps to set the DIP switches on the SELCAL Program Switch Module (PSM), M01995. The PSM is located at the rear side of the E4-1 electronics shelf.

(2) This procedure is used to change the SELCAL code (four letter code) when required as a result of an airplane registry change.

B. References

Reference	Title
23-28-00-700-801	SELCAL System - Operational Test (P/B 501)
WDM 23-22-11	Wiring Diagram Manual

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. SELCAL Program Switch Module Switch Settings

SUBTASK 23-28-00-860-004

(1) When necessary, set the SELCAL program switch module switch settings as follows:

NOTE: See WDM 23-22-11 for specific airplane switch settings.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Look at the SELCAL coding table in WDM 23-22-11 and find the airplane you are performing this task on.

(b) Look at the outboard part of the LEFT or RIGHT outboard display units to make sure that the SELCAL code is same.

(c) In the forward cargo compartment, open an access panel to get access the rear side of the E4 electronics rack.

NOTE: The SELCAL PSM, M01995 is on the rear side of the E4-1 electronics rack.

(d) Loosen the two captive screws [4] and washers [5] to remove the security cover [3] from the PSM [1].

(e) Remove the switch seals [2] to get access to the DIP switches [6].

(f) Find the same airplane code in WDM 23-22-11 and read across all columns to find all switch settings.

(g) Move the DIP switches [6] to the ON/OFF position according to the WDM 23-22-11.

- For each switch where you find a “0”, set the switch to ON.
- For each switch where you find a “1”, set the switch to OFF.
- For each switch where you find unused or not connected, set the switch to OFF.

1) If it is necessary to convert the SELCAL code (four letter code) to Binary code, see the SELCAL letter code to binary code conversion table in Figure 202.

2) Make sure that all the DIP switches are set correctly.

(h) Install the two switch seals [2] to cover the DIP switches.

(i) Put the security cover [3] on the PSM [1].

(j) Tighten the two captive screws [4] and washers [5] to attach the security cover [3] on the PSM [1].

(k) Install the access panel for the E4 electronics rack.

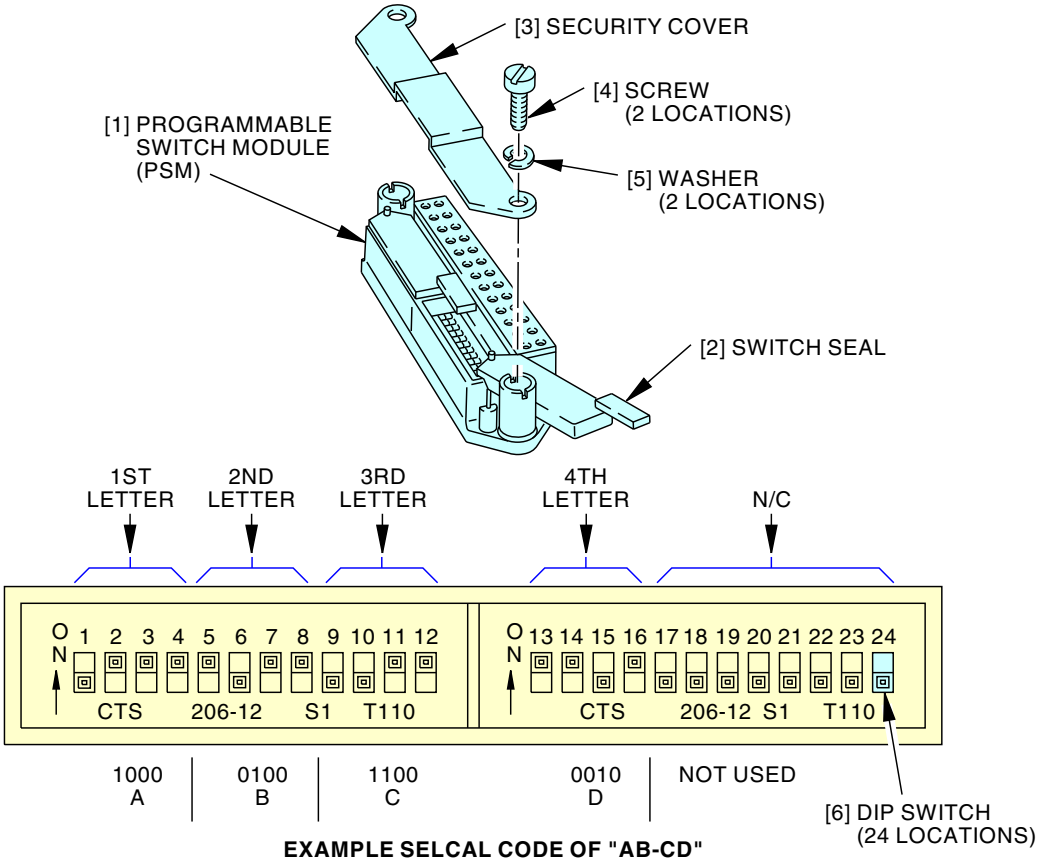
E. Operational Test

SUBTASK 23-28-00-710-001

(1) Do this task: SELCAL System - Operational Test, TASK 23-28-00-700-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EXAMPLE SELCAL CODE OF "AB-CD"

NOTE:

"0"=SWITCH ON

"1"=SWITCH OFF

"NOT UNUSED"=SWITCH OFF

SELCAL LETTER CODE	BINARY CODE
A	1000
B	0100
C	1100
D	0010
E	1010
F	0110
G	1110
H	0001
J	1001
K	0101
L	1101
M	0011
P	1011
Q	0111
R	1111
S	0000

SELCAL LETTER CODE TO BINARY CODE CONVERSION TABLE

2411686 S00061528653_V2

SELCAL Program Switch Module Switch Settings
Figure 202/23-28-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SELCAL - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) An operational test of the Selective Calling (SELCAL) system.

TASK 23-28-00-700-801

2. SELCAL System - Operational Test

A. References

Reference	Title
23-11-00-710-801	HF Communication System - Operational Test (P/B 501)
23-12-00-710-801	VHF Communication System - Operational Test (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1613	Transmitter Tester - Selcal, Ground Equipment Part #: 1000005-004 Supplier: 30242 Part #: 1000005-005 Supplier: 30242 Part #: 90000147 Supplier: 92606 Part #: ARTS-7000 Supplier: 0RDZ5 Part #: AVX-10K-CNS Supplier: 51190 Part #: N1304B-1 Supplier: 30242 Part #: TERC-100-3 Supplier: 3BSK6 Opt Part #: 1000005-001 Supplier: 30242 Opt Part #: 1000005-002 Supplier: 30242 Opt Part #: IFR 4000 Supplier: 51190 Opt Part #: N1304A Supplier: 30242

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. SELCAL System - Operational Test

SUBTASK 23-28-00-860-011

(1) Make sure that the HF communication system is serviceable (TASK 23-11-00-710-801).

SUBTASK 23-28-00-860-012

(2) Make sure that the VHF communication system is serviceable (TASK 23-12-00-710-801).

SUBTASK 23-28-00-710-003

(3) Do the steps that follow for VHF 1:

(a) On the P8-6 captain's audio control panel, push and release the VHF 1 push-button.

(b) On the P8-71 captain's radio tuning panel, set the VHF 1 to an approved test frequency.

(c) Have a SELCAL identified signal transmitted on the VHF 1 test frequency.

NOTE: If a station is not available to transmit from, make a SELCAL signal with a SELCAL ground station encoder. Make sure the signal is approved and the SELCAL Transmitter/Tester, COM-1613 set to the SELCAL code for the airplane.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) On all audio control panels, make sure that the VHF 1 call light comes on.

(e) In the flight compartment, make sure that you hear a single high-low chime.

(f) On the Captain's or the First Officer's control wheel, push the push-to-talk (PTT) switch.

1) Make sure the VHF1 transmitting switch on the ACP panel is selected.

(g) On all audio control panels, make sure that the VHF 1 light goes off.

SUBTASK 23-28-00-710-005

(4) Do the steps that follow for VHF 2:

(a) On the P8-6 captain's audio control panel, push and release the VHF 2 push-button.

(b) On the P8-71 captain's radio tuning panel, set the VHF 2 to an approved test frequency.

(c) Have a SELCAL identified signal transmitted on the VHF 2 test frequency.

NOTE: If a station is not available to transmit from, make a SELCAL signal with a SELCAL ground station encoder. Make sure the signal is approved and the SELCAL Transmitter/Tester, COM-1613 set to the SELCAL code for the airplane.

(d) On all audio control panels, make sure that the VHF 2 call light comes on.

(e) In the flight compartment, make sure that you hear a single high-low chime.

(f) On the Captain's or the First Officer's control wheel, push the push-to-talk (PTT) switch.

1) Make sure the VHF2 transmitting switch on the ACP panel is selected.

(g) On all audio control panels, make sure that the VHF 2 light goes off.

SUBTASK 23-28-00-710-007

(5) Do the steps that follow for VHF 3:

(a) On the P8-6 captain's audio control panel, push and release the VHF 3 push-button.

(b) On the P8-71 captain's radio tuning panel, set the VHF 3 to an approved test frequency.

(c) Have a SELCAL identified signal transmitted on the VHF 3 test frequency.

NOTE: If a station is not available to transmit from, make a SELCAL signal with a SELCAL ground station encoder. Make sure the signal is approved and the SELCAL Transmitter/Tester, COM-1613 set to the SELCAL code for the airplane.

(d) On all audio control panels, make sure that the VHF 3 call light comes on.

(e) In the flight compartment, make sure that you hear a single high-low chime.

(f) On the Captain's or the First Officer's control wheel, push the push-to-talk (PTT) switch.

1) Make sure the VHF3 transmitting switch on the ACP panel is selected.

(g) On all audio control panels, make sure that the VHF 3 light goes off.

SUBTASK 23-28-00-710-009

(6) Do the steps that follow for HF 1:

(a) On the P8-6 captain's audio control panel, set to the HF 1 communication system.

(b) On the P8-71 captain's radio tuning panel, set the HF 1 to an approved test frequency.

(c) Have a SELCAL identified signal transmitted on the HF 1 test frequency.

NOTE: If a station is not available to transmit from, make a SELCAL signal with a SELCAL ground station encoder. Make sure the signal is approved and the SELCAL Transmitter/Tester, COM-1613 set to the SELCAL code for the airplane.

(d) On all audio control panels, make sure that the HF 1 call light comes on.

(e) In the flight compartment, make sure that you hear a single high-low chime.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) On the Captain's or the First Officer's control wheel, push the push-to-talk (PTT) switch.

(g) On all audio control panels, make sure that the HF 1 light goes off.

SUBTASK 23-28-00-710-011

(7) Do the steps that follow for HF 2:

(a) On the P8-6 captain's audio control panel, set to the HF 2 communication system.

(b) On the P8-71 captain's radio tuning panel, set the HF 2 to an approved test frequency.

(c) Have a SELCAL identified signal transmitted on the HF 2 test frequency.

NOTE: If a station is not available to transmit from, make a SELCAL signal with a SELCAL ground station encoder. Make sure the signal is approved and the SELCAL Transmitter/Tester, COM-1613 set to the SELCAL code for the airplane.

(d) On all audio control panels, make sure that the HF 2 call light comes on.

(e) In the flight compartment, make sure that you hear a single high-low chime.

(f) On the Captain's or the First Officer's control wheel, push the push-to-talk (PTT) switch.

(g) On all audio control panels, make sure that the HF 2 light goes off.

SUBTASK 23-28-00-860-009

(8) Set the communication systems back to their usual conditions.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SELCAL DECODER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the selective calling (SELCAL) decoder.

(2) An installation of the SELCAL decoder.

B. The SELCAL decoder is located on the E4-1 electronics shelf in the main equipment center.

TASK 23-28-11-020-801

2. SELCAL Decoder Unit Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Removal Procedure

SUBTASK 23-28-11-010-001

(1) Open this access panel to get access to the main equipment center:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 23-28-11-860-001

(2) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

SUBTASK 23-28-11-860-002

CAUTION

DO NOT TOUCH THE CONNECTOR PINS OR OTHER CONDUCTORS ON THE SELCAL DECODER. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE SELCAL DECODER.

(3) Before you touch the SELCAL decoder [1], do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

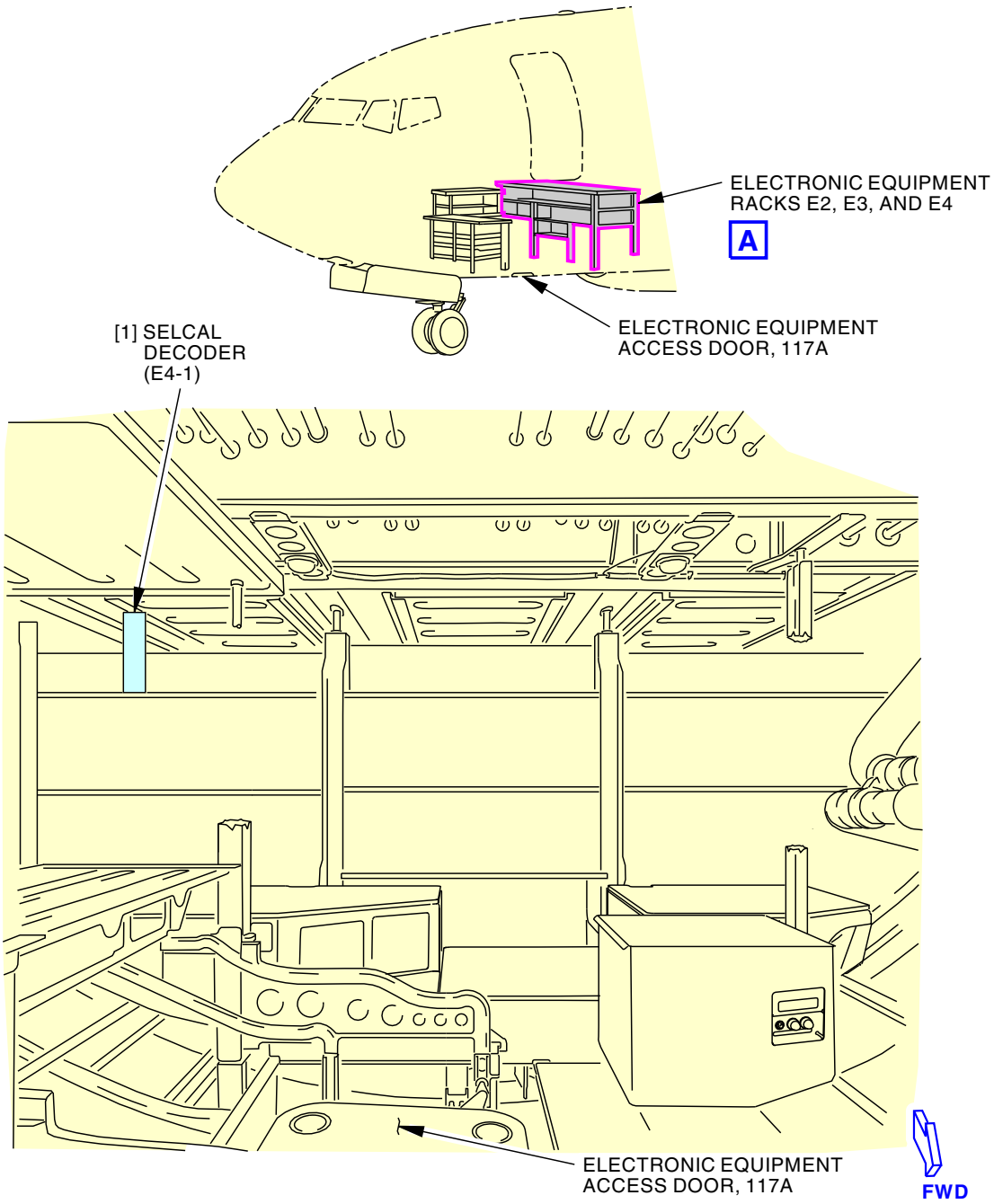
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-28-11-020-001

(4) Remove the SELCAL decoder [1] from the shelf. To remove it, do this task: E/E Box Removal, TASK 20-10-07-000-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRONIC EQUIPMENT RACKS, E2, E3 AND E4

A

2411687 S00061528659_V1

Selective Calling (SELCAL) Decoder Installation
Figure 401/23-28-11-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-28-11-420-801

3. SELCAL Decoder Unit Installation

(Selective Calling (SELCAL) Decoder Installation/Figure 401)

A. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-12-00-710-801	VHF Communication System - Operational Test (P/B 501)
23-28-00-700-801	SELCAL System - Operational Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	SELCAL decoder	23-28-11-01-005	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. SELCAL Decoder Unit Installation

SUBTASK 23-28-11-860-003

(1) Make sure that this circuit breaker is open and has safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

SUBTASK 23-28-11-860-004

CAUTION

DO NOT TOUCH THE CONNECTOR PINS OR OTHER CONDUCTORS ON THE SELCAL DECODER. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE SELCAL DECODER.

(2) Before you touch the SELCAL decoder [1], do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-28-11-420-001

(3) Install the SELCAL decoder [1] on the shelf, do this task: E/E Box Installation, TASK 20-10-07-400-801.

SUBTASK 23-28-11-860-005

(4) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	15	C00058	COMMUNICATIONS SELCAL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Installation Test

SUBTASK 23-28-11-860-007

(1) Make sure that the VHF communication system is serviceable (TASK 23-12-00-710-801).

SUBTASK 23-28-11-860-009

(2) Do the SELCAL system operational test for VHF 1 (TASK 23-28-00-700-801).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-28-11-410-001

(1) Close this access panel for the main equipment center:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PASSENGER ADDRESS SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure does a full test of the passenger address system.

B. This procedure contains:

(1) An operational test of the passenger address system

(2) A system flight mode test of the sound level in the passenger cabin.

TASK 23-31-00-740-801

2. Passenger Address System - Operational Test

A. General

(1) The operational test is a fast check of the system. The test has a number of different parts. All parts of the test follow in a given sequence.

(2) The operational test has these test sections:

(a) Boarding Music Operation Test

(b) Prerecorded Announcement Test

(c) Passenger Address Operation Test

(d) Chime Operation Test

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Passenger Address System Operational Test

SUBTASK 23-31-00-710-006

(1) Do a test of the boarding music operation:

(a) Push the ENT button on the Pre-Recorded Announcement Machine (PRAM).

(b) Push the MUSIC button on the PRAM.

(c) Push a number, '1', '2', '3', or '4', on the PRAM keypad to select a boarding music channel.

(d) Push the START button.

1) Make sure that you can hear music on these speakers:

a) Attendant's speakers

b) Lavatory speakers

c) Passenger Service Unit (PSU) speakers.

(e) Push the up volume button on the PRAM.

1) Make sure that the volume level increases at all of the speakers.

(f) Push the down volume button on the PRAM.

1) Make sure that the volume level decreases at all of the speakers.

(g) Make an announcement from the flight compartment and from each attendant station.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the announcement is heard at all of the speakers.
2) Make sure that the music is not heard.

SUBTASK 23-31-00-710-007

(2) Do a test of the prerecorded announcement operation:
   (a) Push the ANNC button on the PRAM.
   (b) Enter a three-digit number for an announcement.
   (c) Push the START button.
      1) Make sure that you hear an announcement on all of the Passenger Address (PA) speakers.
      2) Make sure that the boarding music is not heard while the prerecorded announcement is heard.
      3) Make sure you hear the boarding music after the announcement stops.
   (d) Make an announcement from the flight compartment and each attendant station while you hear the prerecorded announcement.
      1) Make sure that you can hear the announcement on all of the speakers.
      2) Make sure that you cannot hear the prerecorded announcement while the voice announcement is being made.
   (e) Push the STOP button.

SUBTASK 23-31-00-710-001

(3) Do a test to do a check of the Passenger Address priority system from the aft control stand:
   (a) Make sure that a PA microphone is connected to the left jack on the aft of the control stand.
   (b) Push and hold the Push-To-Talk (PTT) switch on the control stand PA microphone.
      1) Speak into the PA microphone.
         a) Make sure that you can hear the announcement from these speakers:
            <1> Attendant's speakers
            <2> Lavatory speakers
            <3> PSU speakers.
   (c) Release the PTT switch.
   (d) Remove the PA microphone from the jack as applicable.

SUBTASK 23-31-00-710-002

(4) Do a test of the passenger address operation from the from the captain's boom mic:
   (a) Select PA on the captain's audio panel, P8.
   (b) Push and hold the PTT switch on the captain's control wheel.
   (c) Speak into the captain's boom mic.
      1) Make sure that you can hear the announcement from these speakers:
         a) Attendant's speakers
         b) Lavatory speakers
         c) PSU speakers.
   (d) Release the PTT switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-31-00-710-010

(5) Do a test of the passenger address operation from the first officer's boom mic:

(a) Select PA on the first officer’s audio panel, P8.

(b) Push and hold the PTT switch on the first officer's control wheel to MIC.

(c) Speak into the first officer's boom mic.

1) Make sure that you can hear the announcement from these speakers:

a) Attendant’s speakers

b) Lavatory speakers

c) PSU speakers.

(d) Release the PTT switch.

SUBTASK 23-31-00-710-003

(6) Do a test of the passenger address operation from a forward attendant station:

(a) Set the attendant handset to the PA mode.

(b) Push and hold the PTT switch on the captain's control wheel.

(c) Push and hold the PTT button on the attendant handset.

(d) At the same time, speak into the captain's boom mic and have someone speak into the attendant handset.

1) Make sure that you can hear the captain's announcement on these speakers:

a) Attendant's speakers

b) Lavatory speakers

c) PSU speakers.

2) Make sure that you cannot hear the announcement from the attendant handset on the same speakers.

(e) Release the PTT switch on the captain's control wheel.

1) Make sure that you can hear the announcement from the forward attendant handset on these speakers:

a) Aft attendant's speaker

b) Lavatory speakers

c) PSU speakers.

2) Make sure that you cannot hear the announcement on the forward attendant speakers.

(f) Select PA on the Captain's audio panel.

(g) On the captain's audio panel, P8, put the PA volume level to the mid position.

1) Make sure that you can hear the announcement on the captain's headset.

(h) On the captain's audio panel, P8, put the speaker volume level to the mid position.

1) Make sure that you can hear the announcement on the flight deck speaker.

(i) Release the PTT switch on the forward attendant handset.

(j) Put the attendant handset in the handset cradle.

(k) Do this test again using a handset from a different Forward attendant station (if installed).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-31-00-710-004

(7) Do a test of the passenger address operation from an aft attendant station:

(a) Set the attendant handset to the PA mode.

(b) Push and hold the PTT switch on the captain's control wheel.

(c) Push and hold the PTT button on the attendant handset.

(d) At the same time, speak into the captain's boom mic and have someone speak into the attendant handset.

1) Make sure that you can hear the captain's announcement on these speakers:

a) Attendant's speakers

b) Lavatory speakers

c) PSU speakers.

2) Make sure that you cannot hear the announcement from the attendant handset on the same speakers.

(e) Release the PTT switch on the captain's control wheel.

1) Make sure that you can hear the announcement from the aft attendant handset on these speakers:

a) Forward attendant's speaker

b) Lavatory speakers

c) PSU speakers.

2) Make sure that you cannot hear the announcement on the aft attendant speakers.

(f) Select PA on the captain's audio panel, P8.

(g) On the captain's audio panel, P8, put the PA volume level to the mid position.

1) Make sure that you can hear the announcement on the captain's headset.

(h) On the captain's audio panel, P8, put the speaker volume level to the mid position.

1) Make sure that you can hear the announcement on the flight deck speaker.

(i) Release the PTT switch on the attendant handset.

(j) Put the attendant handset in the handset cradle.

(k) Do this test again using a handset from a different Aft attendant station (if installed).

SUBTASK 23-31-00-710-005

(8) Do a test of the chime operation:

(a) Push the attendant call button on a PSU.

1) Make sure that you hear one high tone on the PA speakers.

(b) Push the attendant call button on all other PSUs.

1) Make sure that you hear a high tone each time you push an attendant call button.

(c) Push the attendant call buttons once more to reset them.

1) Make sure that the attendant light goes off.

(d) Push the PILOT call button on each attendant handset.

1) Make sure that you hear a high tone in the flight compartment.

2) Push the RESET button on the attendant handset.

(e) Push the button for the attendant on each attendant handset.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that you hear a high/low tone on the attendant speakers and in the passenger cabin.

(f) Push the ATT call button on the captain's overhead panel, P5.

1) Make sure that you hear a high/low tone on the attendant speakers and in the passenger cabin.

(g) Put the attendant handsets in the handset cradles.

(h) Do these steps to do a check of the Fasten Seat Belt warning function:

1) Make sure that the FASTEN BELTS sign switch on the captain's overhead panel, P5, is in the OFF position.

2) Put the FASTEN BELTS sign switch to the ON position.

a) Make sure that you hear a low tone in the passenger cabin.

3) Put the FASTEN BELTS sign switch to the OFF position.

a) Make sure that you hear a low tone in the passenger cabin.

PTF ALL; AIRPLANES WITH CHIME ONLY NO SMOKING SWITCH

(i) Do these steps to do a check of the chime only function:

1) Make sure that the CHIME ONLY switch on the captain's overhead panel, P5, is in the OFF position.

2) Put the CHIME ONLY switch to the ON position.

a) Make sure that you hear a low tone in the passenger cabin.

3) Put the CHIME ONLY switch to the OFF position.

a) Make sure that you hear a low tone in the passenger cabin.

PTF ALL

——— END OF TASK ———

TASK 23-31-00-730-801

3. Passenger Address System - System Test

A. General

(1) The system test has these test sections:

(a) A ground mode test which checks the sound level of each passenger address speaker.

(b) A flight mode test which makes sure that the sound level of the passenger address speakers increases when the engines are running.

(2) This test makes sure that the sound level of the speakers in the passenger cabin increases by at least 5 dBs when the engines are running.

(3) This test makes sure that the sound level of the speakers in the passenger cabin increases by at least 2 dBs when the Oxygen is running.

(4) This test checks the sound level in the airplane with the engine running relay from the #1 engine, or the #1 and #2 engines, connected to the Passenger Address (PA) amplifier.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1322	Multimeter (Analog / Digital with sufficient internal voltage to measure long cable distances) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: FLUKE 289 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: MODEL 8 MK7 Supplier: 00426
COM-1612	Generator - Random Noise, Audio Output Part #: 543-1 Supplier: 21562
COM-1616	Meter - Sound Level Part #: CEL-240/K1 Supplier: 01RN5 Part #: SE-402-AC3 Supplier: 24575 Opt Part #: 2100-10 Supplier: 90435 Opt Part #: 2400-10 Supplier: 90435 Opt Part #: CEL-254/K1 Supplier: 01RN5

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. System Ground Mode Test

SUBTASK 23-31-00-820-001

(1) Do these steps to calibrate the noise generator:

(a) Connect the multimeter, COM-1322, to the SIGNAL OUTPUT jacks on the random noise generator, COM-1612.

(b) Set the PWR/PTT switch on the noise generator to ON.

(c) Set the OUTPUT switch to the SIG position.

1) Make sure that 1.0 +/- 0.05 volts rms is shown on the voltmeter.

(d) Adjust the SIGNAL ADJUST potentiometer on the noise generator if necessary.

(e) Set the PWR/PTT switch on the noise generator to OFF.

(f) Disconnect the multimeter, COM-1322, from the SIGNAL OUTPUT jacks on the random noise generator, COM-1612.

SUBTASK 23-31-00-480-001

(2) Do these steps to prepare for test:

(a) Connect the random noise generator, COM-1612, to a microphone jack in the flight compartment.

(b) Set the random noise generator, COM-1612, PWR/PTT switch to ON.

(c) Set the PA microphone selector switch on the applicable audio control panel to ON.

1) Make sure that its light comes on.

2) Set the sound level meter, COM-1616, to SLOW and C.

SUBTASK 23-31-00-730-001

(3) Do this test of the sound level of the PA speakers:

(a) Hold the sound level meter, COM-1616, close to the speaker grill at each passenger service unit with an installed speaker and at each lavatory.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do not touch the sound level meter to the speaker grill when you do the sound level test.

2) Make sure that the sound level meter shows a value between 106 dB and 108 dB.

(b) If necessary, adjust the PA GAIN on the front panel of the Remote Electronic Unit (M01353) located on shelf E4-1 in the Electronic Equipment Compartment (EEC).

(c) Hold the sound level meter, COM-1616, close to the speaker grill at any speaker in the overwing area.

1) Do not touch the sound level meter to the speaker grill when you do the sound level test.

2) Make sure that the sound level meter shows a value between 106 dB and 108 dB.

E. System Flight Mode Test

SUBTASK 23-31-00-860-004

(1) Do these steps to prepare for test:

(a) Connect the random noise generator, COM-1612, to a microphone jack in the flight compartment.

(b) Set the random noise generator, COM-1612, PWR/PTT switch to ON.

(c) Measure the sound level of the speakers:

1) Hold the sound level meter, COM-1616, close to the speaker grill at any speaker in the overwing area.

NOTE: Do not touch the speaker grill.

2) Make sure that the value is between 106 db and 108 db.

a) If the value is less than 106 db, do a test of the sound level of the PA speakers.

(d) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	11	C00317	INDICATOR MASTER DIM SECT 5

SUBTASK 23-31-00-730-002

(2) Do a test of engine 1 flight mode:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	3	C01312	ENGINE 1 RUN/PWR

(c) Move the engine 1 start lever to the IDLE position.

(d) Wait at least five minutes.

(e) Measure the sound level of the speakers:

1) Move the sound level meter across the same speaker panel used in the Prepare For Test section.

2) Write the value down.

a) Make sure that the value is at least 5 dBs higher than the first measurement.

(f) Move the engine 1 start lever to the CUTOFF position.

(g) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

(h) Measure the sound level across the speaker.

1) Make sure that the sound level is approximately the same as in the first measurement.

SUBTASK 23-31-00-730-003

(3) Do a test of engine 2 flight mode:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

(b) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR

(c) Move the engine 2 start lever to the IDLE position.

(d) Wait at least five minutes.

(e) Measure the sound level of the speakers:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Move the sound level meter across the same speaker panel used in the Prepare For Test section.

2) Write the value down.

a) Make sure that the value is at least 5 dBs higher than the first measurement.

(f) Move the engine 2 start lever to the CUTOFF position.

(g) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

(h) Measure the sound level across the same speaker.

1) Make sure that the sound level is approximately the same as in the first measurement.

SUBTASK 23-31-00-730-004

(4) Do a test of the oxygen pass system:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	9	C00784	OXYGEN PASS RIGHT
F	10	C00783	OXYGEN PASS LEFT

(b) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	7	C00156	OXYGEN IND
F	8	C00785	OXYGEN MAN CONT

(c) Set the PASS OXYGEN switch on the aft overhead panel, P5, to the ON position.

(d) Measure the sound level of the speakers:

1) Move the sound level meter across the same speaker panel used in the Prepare For Test section.

2) Write the value down.

a) Make sure that the value is at least 2 dBs higher than the first measurement.

(e) Set the PASS OXYGEN switch on the aft overhead panel, P5, to the NORMAL position.

(f) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	9	C00784	OXYGEN PASS RIGHT
F	10	C00783	OXYGEN PASS LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-31-00-860-005

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	11	C00317	INDICATOR MASTER DIM SECT 5

SUBTASK 23-31-00-860-006

(2) Do the steps that follow to put the airplane back to its usual condition:

(a) Set the random noise generator, COM-1612, PWR/PTT switch to OFF.

(b) Disconnect the random noise generator, COM-1612, from the microphone jack in the flight compartment.

1) Remove the test equipment from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PASSENGER ADDRESS AMPLIFIER - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Passenger Address Amplifier Deactivation.

(2) Passenger Address Amplifier Activation.

TASK 23-31-01-040-801

2. Passenger Address Amplifier - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power from the Passenger Address Amplifier.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

SUBTASK 23-31-01-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

E. Passenger Address Amplifier - Tryout

NOTE: This tryout is to make sure that the Passenger Address Amplifier is in a zero energy state.

SUBTASK 23-31-01-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-01-010-001

(2) Open this access panel:

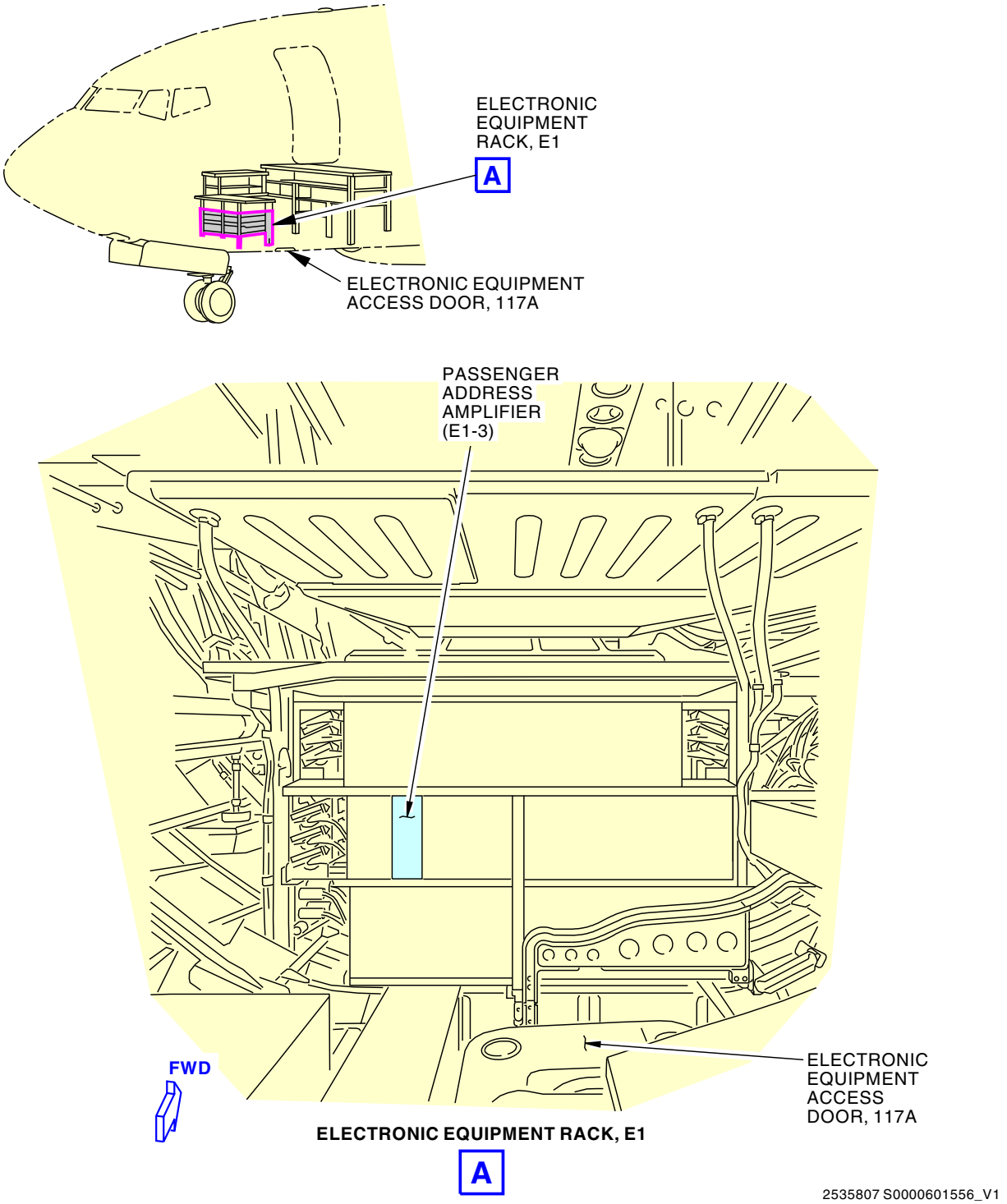
Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 23-31-01-700-001

(3) Make sure the display on the front of the amplifier is blank.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Passenger Address Amplifier
Figure 201/23-31-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-01-440-801

3. Passenger Address Amplifier - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the Passenger Address Amplifier.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

SUBTASK 23-31-01-860-003

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-01-410-001

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PASSENGER ADDRESS AMPLIFIER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Passenger Address (PA) amplifier removal

(2) Passenger Address (PA) amplifier installation.

TASK 23-31-01-000-801

2. Passenger Address (PA) Amplifier Removal

(Figure 401)

A. General

(1) This task is for the removal of the passenger address amplifier.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

D. Prepare for the Removal

SUBTASK 23-31-01-860-009

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

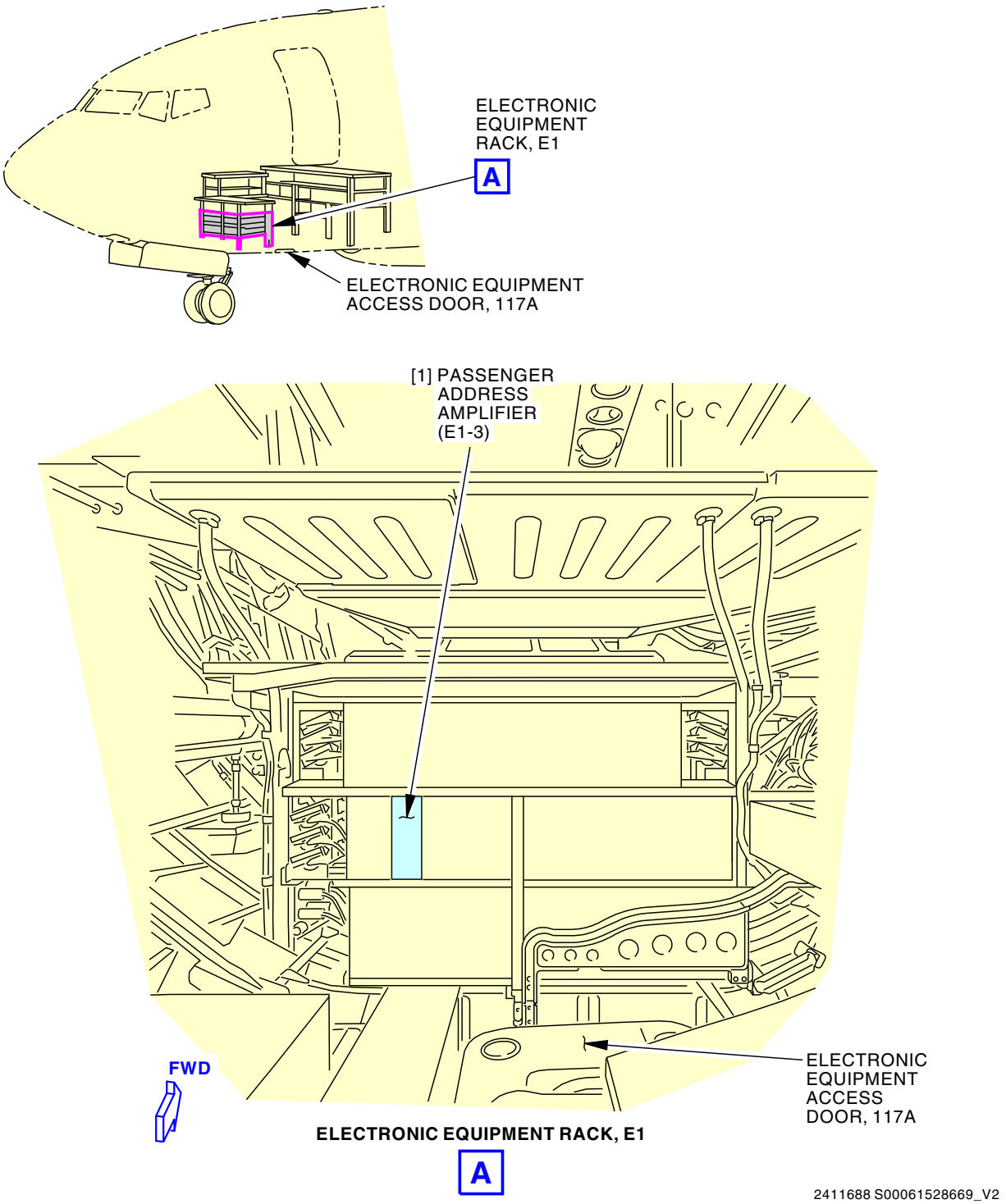
E. Passenger Address (PA) Amplifier Removal

SUBTASK 23-31-01-020-001

(1) Remove the PA amplifier [1] (TASK 20-10-07-000-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Passenger Address Amplifier Installation
Figure 401/23-31-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-01-400-801

3. Passenger Address (PA) Amplifier Installation

(Figure 401)

A. General

(1) This task is for the installation of the passenger address amplifier.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
23-31-00-740-801	Passenger Address System - Operational Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Amplifier	23-31-01-01-010	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

E. Passenger Address (PA) Amplifier Installation

SUBTASK 23-31-01-420-001

(1) Install the PA amplifier [1] (TASK 20-10-07-400-801).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-31-01-860-010

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

G. Installation Test

PTF ALL; AIRPLANES WITH P/N 622-5342-101

SUBTASK 23-31-01-710-001

(1) Do a test of the PA amplifier operation:

(a) Put the OPERATE/TONE switch on the front panel of the PA amplifier to the LEVEL position.

(b) Make sure that the display on the amplifier shows 69.0 to 71.0 VRMS.

NOTE: If the PA amplifier's VRMS output is not within 69-71 VRMS, the output can be adjusted with the Master Gain knob on the front panel.

(c) Put the OPERATE/TONE switch to the TONE position.

(d) Make sure that you can hear a high tone on the passenger address speakers.

(e) Put the OPERATE/TONE switch to the OPERATE position.

PTF ALL

SUBTASK 23-31-01-730-001

(2) Do this task: Passenger Address System - Operational Test, TASK 23-31-00-740-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PASSENGER SERVICE UNIT SPEAKER - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Passenger Service Unit Speaker Deactivation.

(2) Passenger Service Unit Speaker Activation.

TASK 23-31-02-040-801

2. Passenger Service Unit Speaker - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power to the Passenger Service Unit Speaker.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-31-02-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

D. Passenger Service Unit Speaker - Tryout

NOTE: This tryout is to make sure that the Passenger Service Unit Speaker is in a zero energy state.

SUBTASK 23-31-02-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-02-700-001

(2) Do the following to test the PSU speaker:

(a) Lift a handset from an attendant station.

(b) Set the handset to PA.

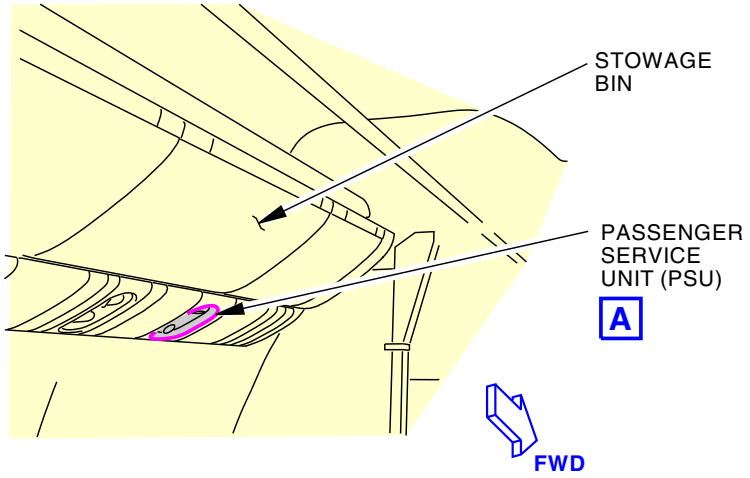
(c) Push the PTT button on the handset.

(d) Make an announcement on the PA system.

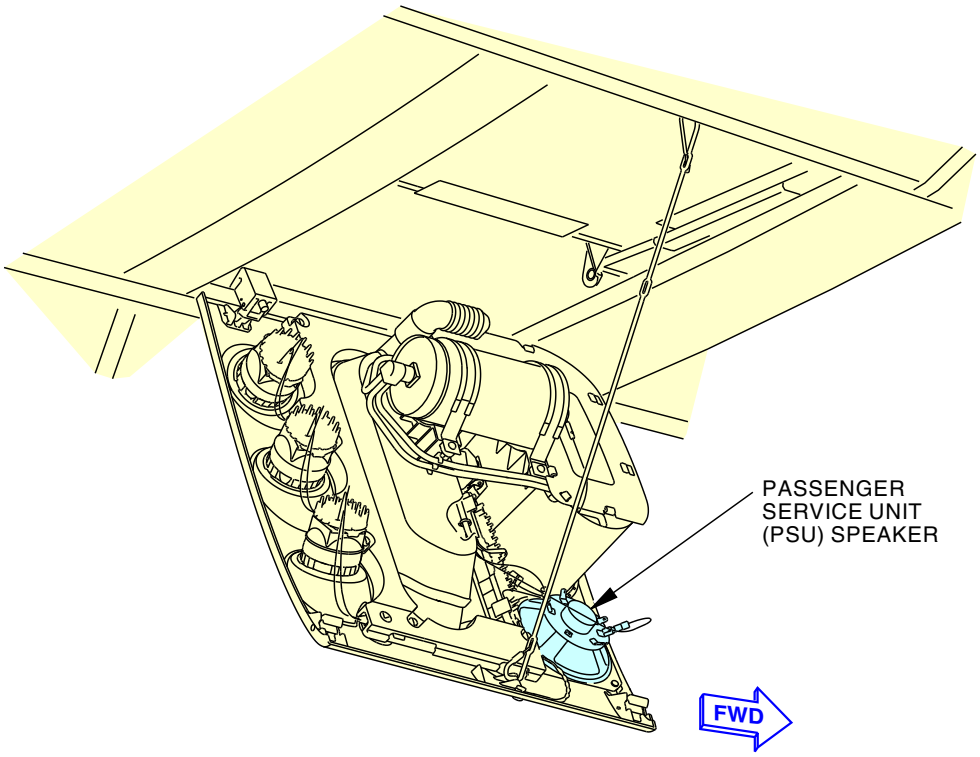
1) Make sure that you cannot hear the announcement on the speaker.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PASSENGER COMPARTMENT



PASSENGER SERVICE UNIT (PSU)
(EXAMPLE)

A

2411689 S00061528677_V1

Passenger Service Unit Speaker
Figure 201/23-31-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-02-440-801

3. Passenger Service Unit Speaker - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the Passenger Service Unit Speaker.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-31-02-860-003

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PASSENGER SERVICE UNIT SPEAKER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the passenger service unit (PSU) speaker

(2) An installation of the PSU speaker.

B. The passenger service units have speakers installed, as necessary, to supply equal sound in the passenger compartment.

TASK 23-31-02-000-801

2. Passenger Service Unit (PSU) Speaker Removal

(Figure 401)

A. Tools/Equipment

Reference	Description
STD-1073	Wire - Standard, 18-22 Gauge, Insulation Removed

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Removal Procedure

SUBTASK 23-31-02-020-002

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-02-020-001

CAUTION MAKE SURE THAT THE RELEASE PIN FOR THE CHEMICAL OXYGEN GENERATOR STAYS ENGAGED IN THE FIRING PIN. IF THE CHEMICAL OXYGEN GENERATOR ACCIDENTALLY OPERATES, IT IS MANDATORY THAT YOU REPLACE IT.

(2) Do these steps to remove the PSU speaker:

(a) Open the PSU.

1) Push a 18-22 gauge insulation removed wire, STD-1073 in the holes of the PSU face panel to open the assembly.

(b) Hold the PSU panel.

1) Push the spring clip with the 18-22 gauge insulation removed wire, STD-1073 to release the assembly.

(c) Slowly lower the PSU panel assembly until the lanyard can hold the PSU.

(d) Disconnect the PSU electrical connector.

(e) Identify the location of the electrical wires on the speaker transformer terminals.

(f) Disconnect the two electrical connectors [2] from the speaker transformer terminals.

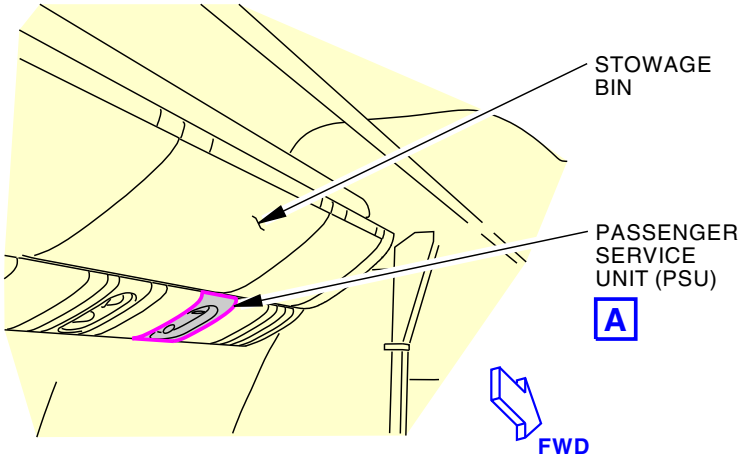
(g) Remove the screws [1] from the speaker [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

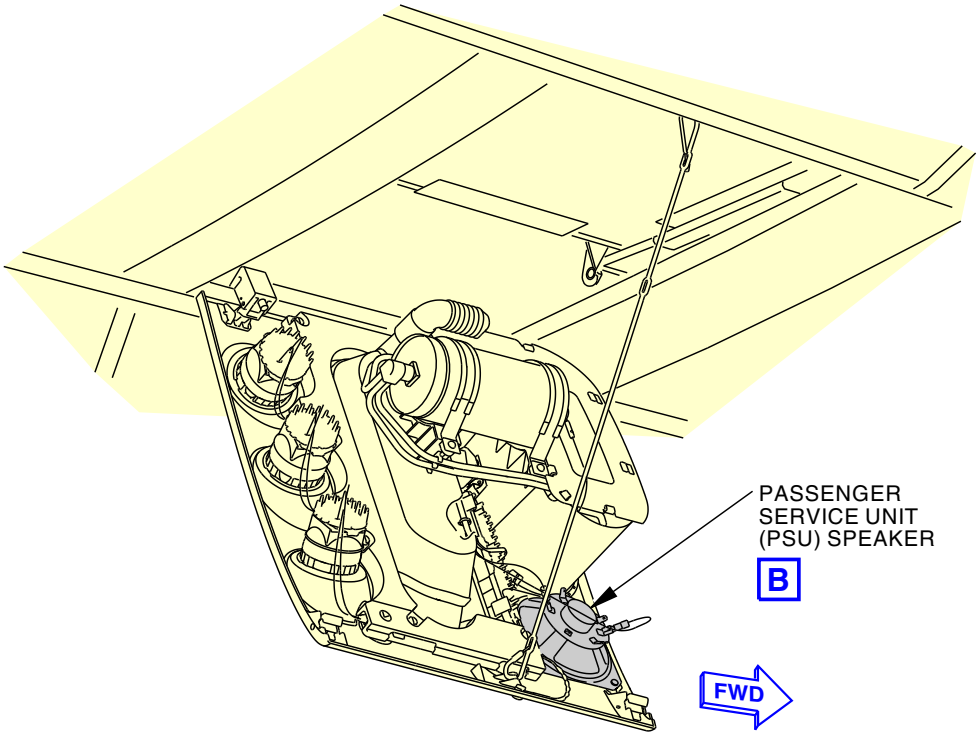
(h) Carefully remove the speaker [3] from the PSU.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PASSENGER COMPARTMENT



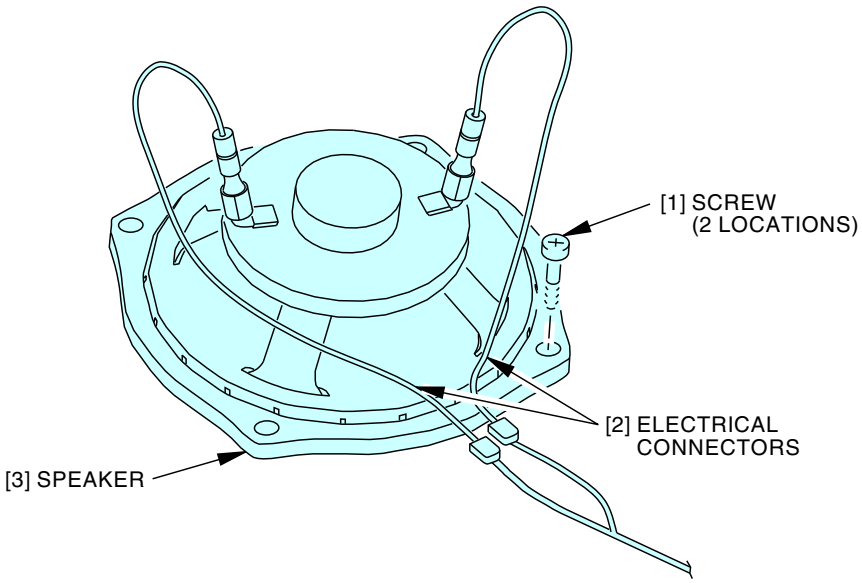
**PASSENGER SERVICE UNIT (PSU)
(EXAMPLE)**

A

2411690 S00061528681_V1

**Passenger Service Unit Speaker Installation
Figure 401/23-31-02-990-802 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**PASSENGER SERVICE UNIT (PSU) SPEAKER
(EXAMPLE)**

B

2411691 S00061528682_V2

**Passenger Service Unit Speaker Installation
Figure 401/23-31-02-990-802 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-02-400-801

3. Passenger Service Unit (PSU) Speaker Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Speaker	23-31-02-02-070	PTF ALL

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Installation Procedure

SUBTASK 23-31-02-860-005

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-02-420-001

CAUTION MAKE SURE THAT THE RELEASE PIN FOR THE CHEMICAL OXYGEN GENERATOR STAYS ENGAGED IN THE FIRING PIN. IF THE CHEMICAL OXYGEN GENERATOR ACCIDENTALLY OPERATES, IT IS MANDATORY THAT YOU REPLACE IT.

(2) Install the PSU speaker:

(a) Install the speaker [3] on the PSU assembly.

(b) Install the screws [1].

(c) Connect the electrical connectors [2] to the speaker transformer terminals.

1) Make sure the electrical connectors are connected to the same terminals that they were removed from.

(d) Connect the PSU electrical connector.

(e) Close the PSU panel assembly.

SUBTASK 23-31-02-420-002

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

D. Installation Test

SUBTASK 23-31-02-710-001

(1) Do a test of the PSU speaker operation:

(a) Lift a handset from an attendant station.

(b) Set the handset to PA.

(c) Push the PTT button on the handset.

(d) Make an announcement on the PA system.

1) Make sure you can hear the announcement on the new PSU speaker.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Put the handset back in the cradle.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT SPEAKER - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Attendant Speaker Deactivation.

(2) Attendant Speaker Activation.

TASK 23-31-03-040-801

2. Attendant Speaker - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power to the Attendant Speaker.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-31-03-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

D. Attendant Speaker - Tryout

NOTE: This tryout is to make sure that the Attendant Speaker is in a zero energy state.

SUBTASK 23-31-03-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-03-700-001

(2) Do the following to test the attendant speaker operation:

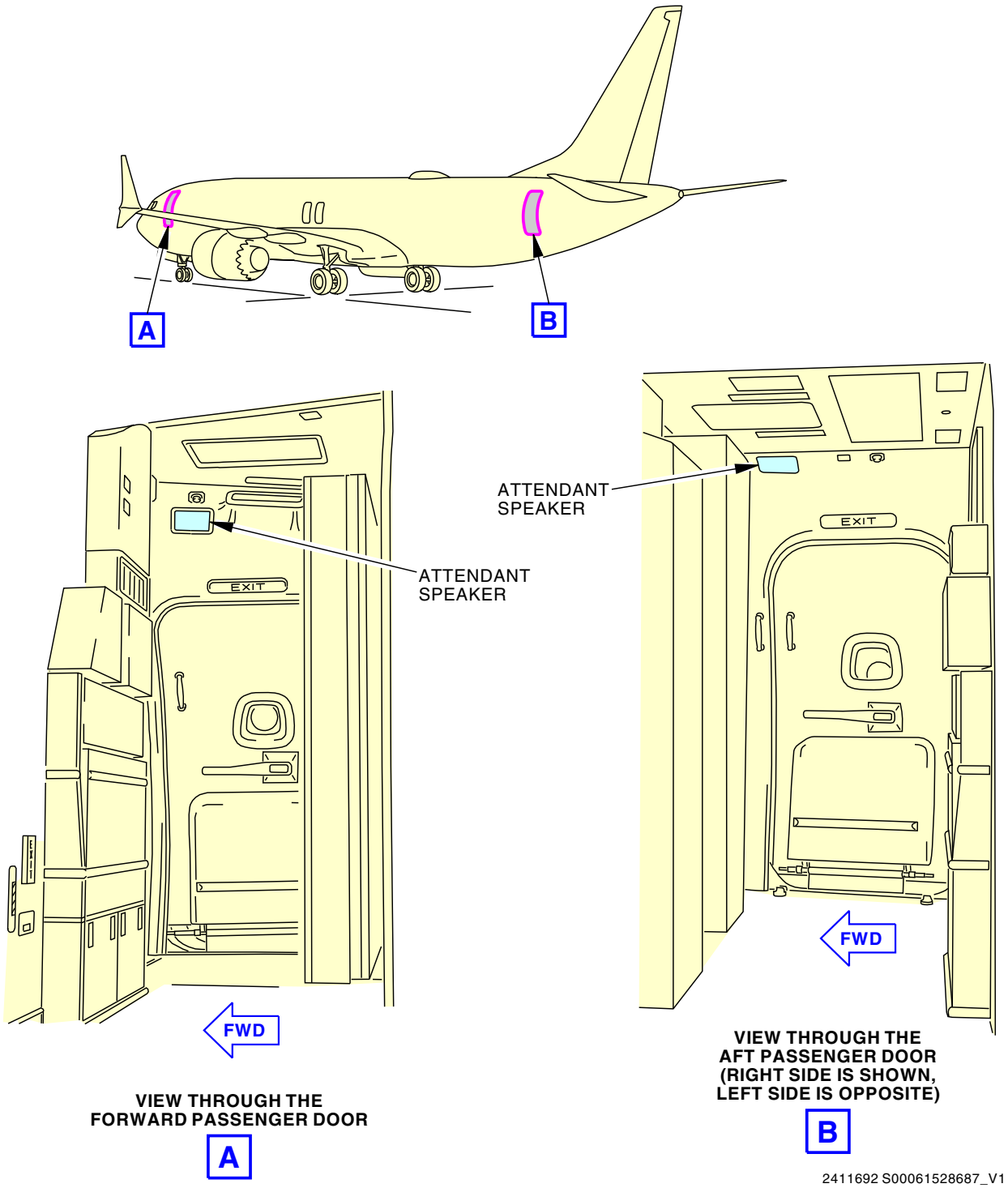
(a) Push the call button on a passenger services unit (PSU).

1) Make sure that you can not hear one high chime from all the attendant speakers.

2) Make sure the PSU call light does not come on.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Attendant Speaker
Figure 201/23-31-03-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-03-440-801

3. Attendant Speaker - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the Attendant Speaker.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-31-03-860-003

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT SPEAKER - REMOVAL/INSTALLATION

1. General

A. This procedure contains:

(1) A removal of the attendant speaker

(2) An installation of the attendant speaker.

B. The attendant speaker is in the header panel over the door. You must remove the door header panel to get access to the speaker.

TASK 23-31-03-000-801

2. Attendant Speaker Removal

(Figure 401)

A. References

Reference	Title
25-21-24-000-801	Aft Door Doorway Header Panel Removal (P/B 401)
25-21-30-020-801	Forward Entry Door Doorway Header Panel Removal (P/B 401)

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Removal Procedure

SUBTASK 23-31-03-860-004

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-03-010-001

(2) For the forward attendant speaker, do this task: Forward Entry Door Doorway Header Panel Removal, TASK 25-21-30-020-801.

SUBTASK 23-31-03-010-004

(3) For the aft attendant speaker, do this task: Aft Door Doorway Header Panel Removal, TASK 25-21-24-000-801.

SUBTASK 23-31-03-020-001

(4) Do these steps to remove the attendant speaker [1]:

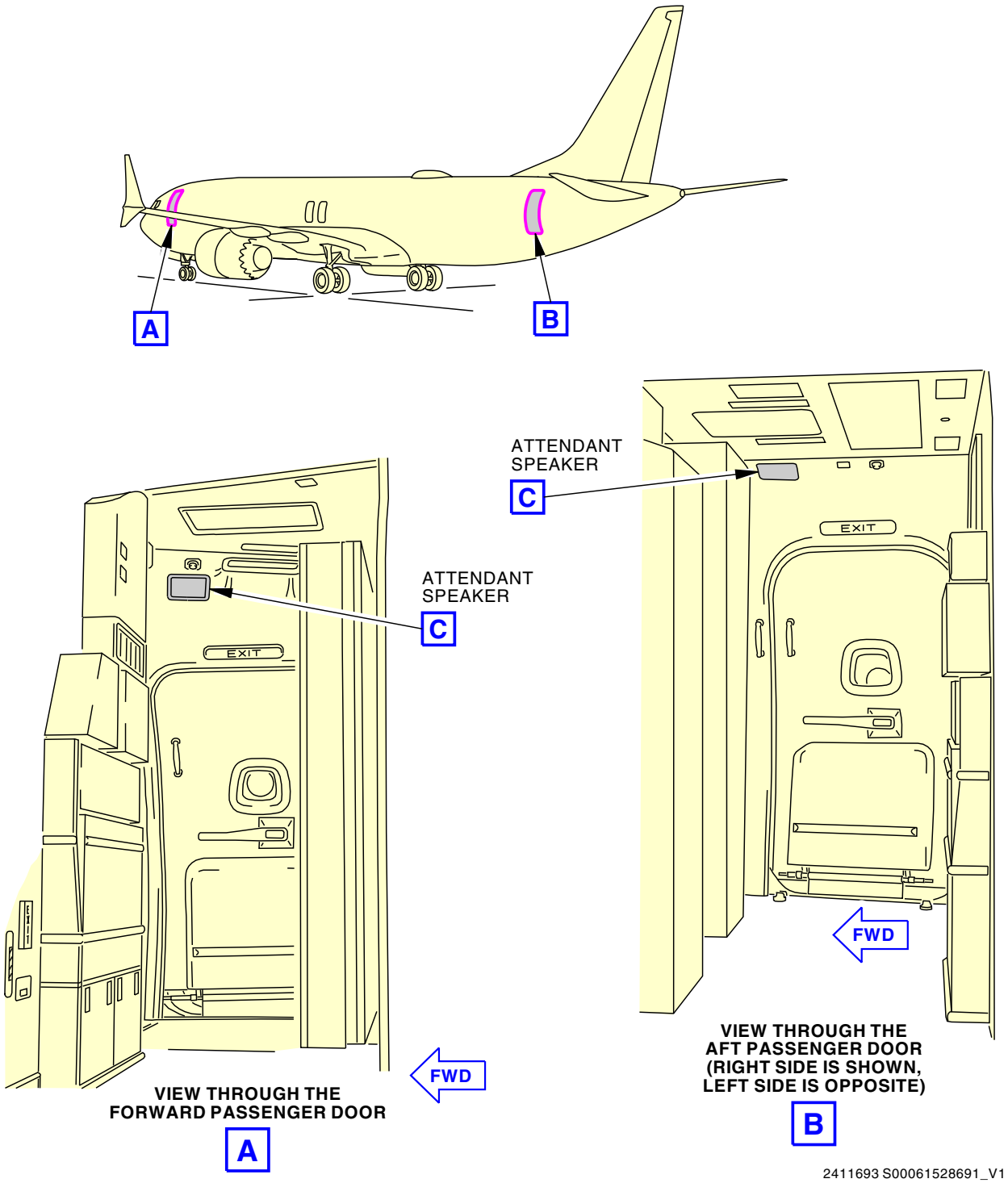
(a) Identify the location of the electrical wires on the speaker transformer terminals.

(b) Disconnect the electrical connectors [2] from the speaker [1].

(c) Remove the screws [3], washers [4] (if applicable), speaker [1], gasket [6], and spacers [5] (if applicable).

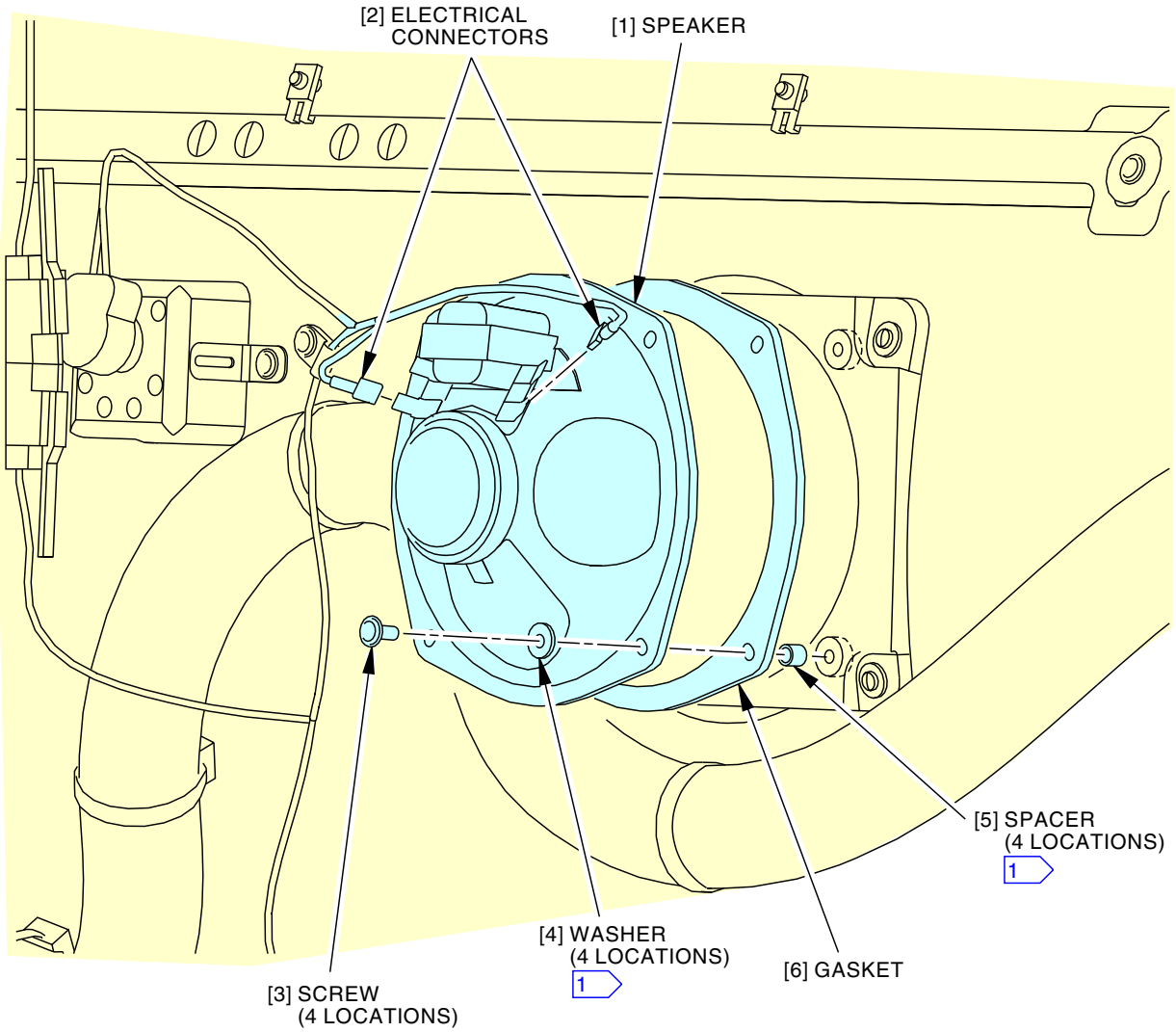
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Attendant Speaker Installation
Figure 401/23-31-03-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Attendant Speaker Installation
Figure 401/23-31-03-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-03-400-801

3. Attendant Speaker Installation

(Figure 401)

A. References

Reference	Title
25-21-24-400-801	Aft Door Doorway Header Panel Installation (P/B 401)
25-21-30-420-801	Forward Entry Door Doorway Header Panel Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Speaker	23-31-03-09A-100	PTF ALL
		23-31-03-09A-125	PTF ALL
		23-31-03-31-100	PTF ALL
		23-31-03-31-125	PTF ALL
		23-31-03-33-085	PTF ALL
		23-31-03-33-110	PTF ALL
		23-31-03-95-080	PTF ALL
		23-31-03-95-100	PTF ALL

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Installation Procedure

SUBTASK 23-31-03-420-001

(1) Do these steps to install the attendant speaker [1]:

(a) Hold the speaker [1] in the correct position.

(b) Install the spacers [5] (if applicable), gasket [6], speaker [1], washers [4] (if applicable), and screws [3].

(c) Connect the two electrical connectors [2] to the speaker [1].

1) Make sure the electrical connectors [2] are connected to the same terminals that they were removed from.

SUBTASK 23-31-03-410-001

(2) For the forward attendant speaker, do this task: Forward Entry Door Doorway Header Panel Installation, TASK 25-21-30-420-801.

SUBTASK 23-31-03-410-002

(3) For the aft attendant speaker, do this task: Aft Door Doorway Header Panel Installation, TASK 25-21-24-400-801.

SUBTASK 23-31-03-860-005

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Installation Test

SUBTASK 23-31-03-710-001

(1) Do a test of the attendant speaker operation:

(a) Push the call button on a passenger services unit (PSU).

1) Make sure that you can hear one high chime from all the attendant speakers.

2) Make sure the PSU call light comes on.

(b) Push the call button on a PSU one more time.

1) Make sure that the PSU call light goes off.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY SPEAKER - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Lavatory Speaker Deactivation.

(2) Lavatory Speaker Activation.

TASK 23-31-05-040-801

2. Lavatory Speaker - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power from the Lavatory Speaker.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-31-05-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

D. Lavatory Speaker - Tryout

NOTE: This tryout is to make sure that the Lavatory Speaker is in a zero energy state.

SUBTASK 23-31-05-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-05-700-001

(2) Do the following to test the lavatory speaker operation:

(a) Lift the PA microphone from at an attendant station.

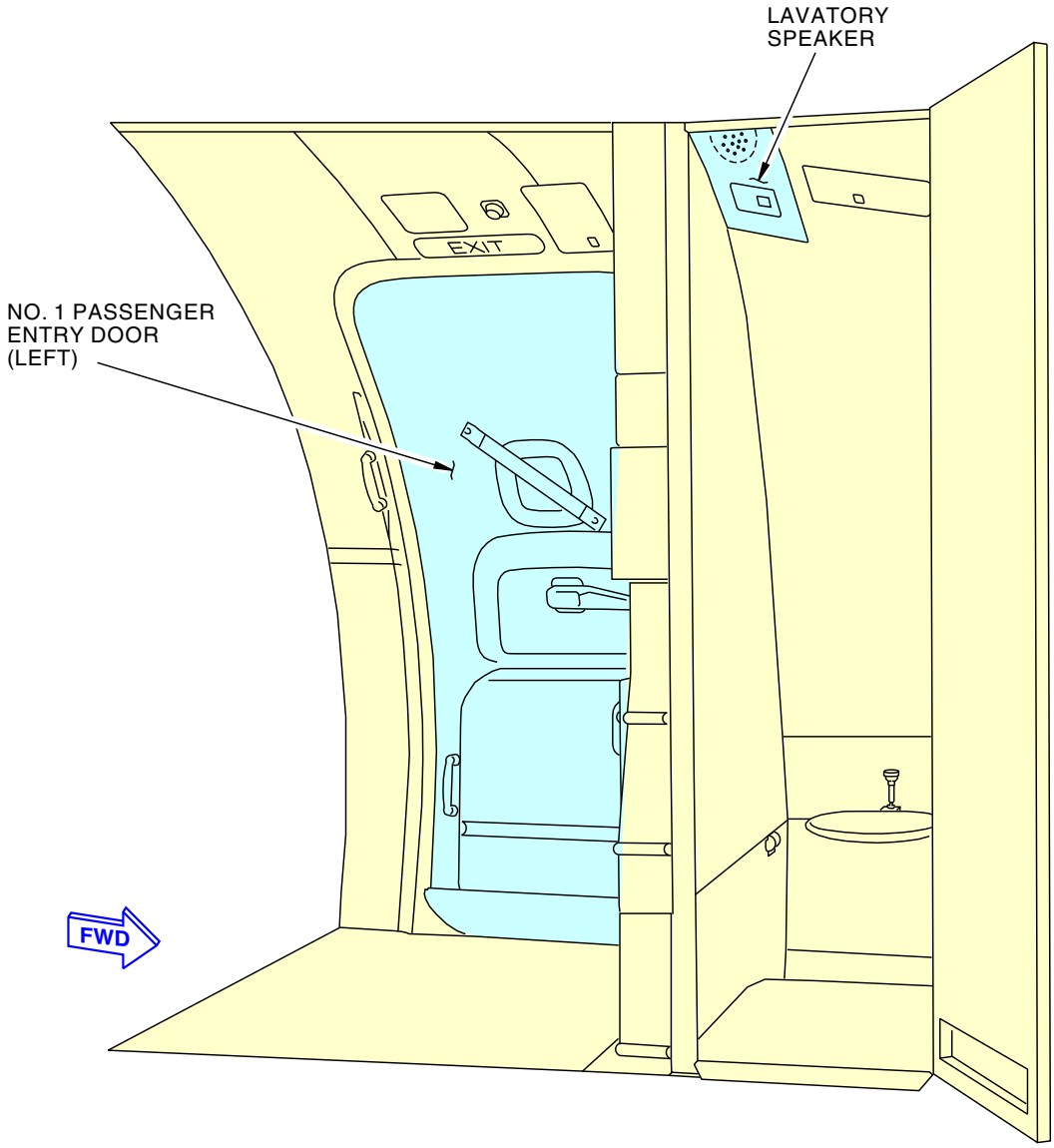
(b) Push the PTT button on the microphone.

(c) Make an announcement on the PA system.

1) Make sure you can not hear the announcement on the speaker.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**PASSENGER COMPARTMENT ENTRYWAY
(TYPICAL)**

A

2411695 S00061528697_V1

**Lavatory Speaker
Figure 201/23-31-05-990-801**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-05-440-801

3. Lavatory Speaker - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the Lavatory Speaker.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-31-05-860-003

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY SPEAKER - REMOVAL/INSTALLATION

1. General

A. This procedure contains:

(1) A removal of the lavatory speaker

(2) An installation of the lavatory speaker.

B. The location of the lavatory speaker is in the passenger service unit (PSU) in the ceiling.

TASK 23-31-05-000-801

2. Lavatory Speaker Removal

(Figure 401)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14266	Screwdriver - Driver Assembly Part #: 29-90-114-10 Supplier: 94222 Part #: 29-90-214-10 Supplier: 94222

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Removal Procedure

SUBTASK 23-31-05-860-004

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

SUBTASK 23-31-05-020-001

(2) Do the steps that follow to remove the forward lavatory speaker [1]:

(a) Open the passenger service unit (PSU).

1) Turn the four quarter turn fasteners [5] on the face of the PSU panel with the driver assembly screwdriver, SPL-14266.

(b) Slowly lower the PSU panel assembly until the lanyards can hold the PSU.

(c) Identify the location of the speaker wires on the speaker transformer terminals.

(d) Disconnect the electrical connectors [6] from the speaker transformer terminals.

(e) Remove the screws [2], washers [3], and spacers [4] from the speaker [1].

(f) Remove the speaker [1].

SUBTASK 23-31-05-420-002

(3) Do the steps that follow to remove the aft lavatory speakers [1]:

(a) Open the lower ceiling panel.

1) Turn the two quarter turn fasteners [5] on the lower ceiling panel to open the assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Slowly lower the ceiling panel assembly until the hinge supports the ceiling panel.

(c) Identify the location of the speaker wires on the speaker transformer terminals.

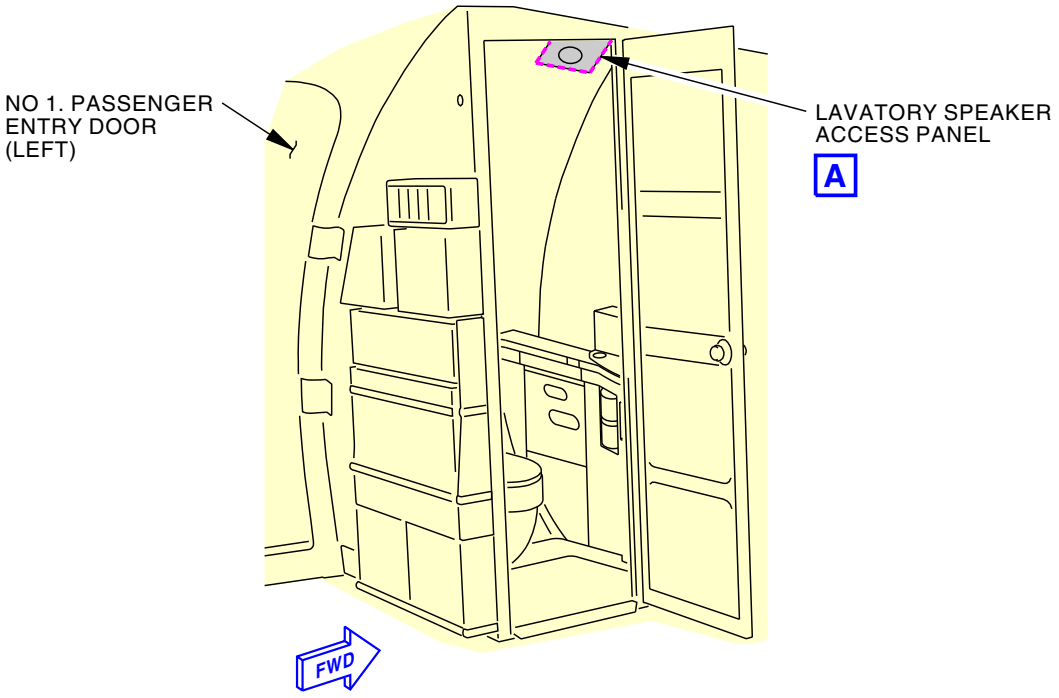
(d) Disconnect theelectrical connectors [6] from the speaker transformer terminals.

(e) Remove the screws [7],washers [3] from the speaker [1].

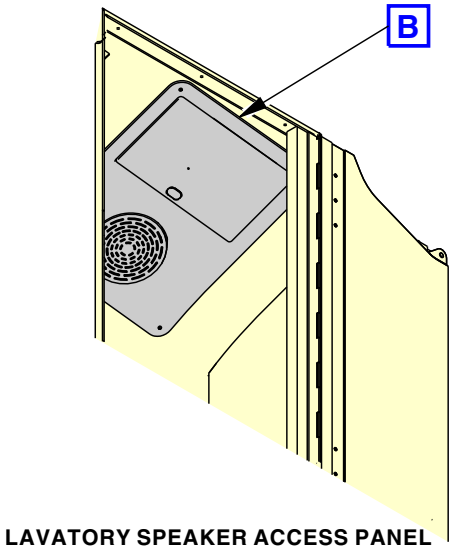
(f) Remove the speaker [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**FORWARD LAVATORY
(TYPICAL)**



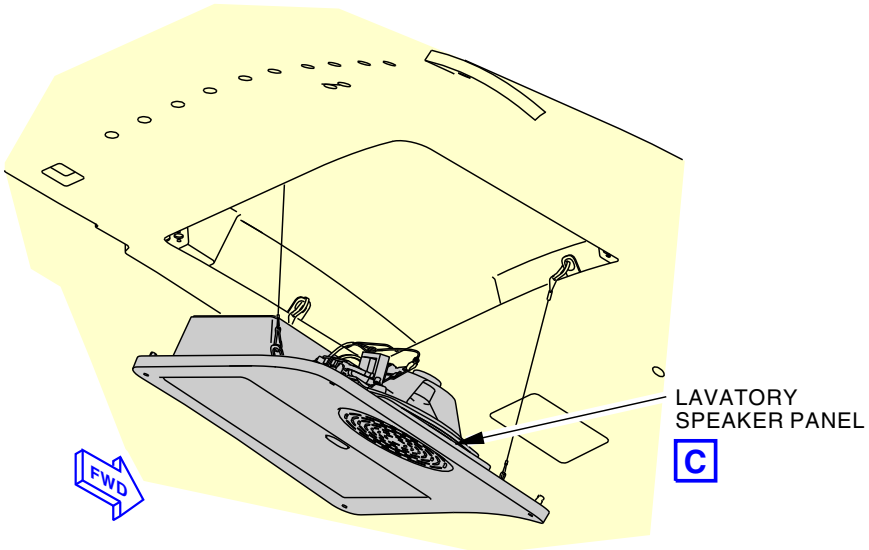
LAVATORY SPEAKER ACCESS PANEL

A

2411696 S00061528701_V1

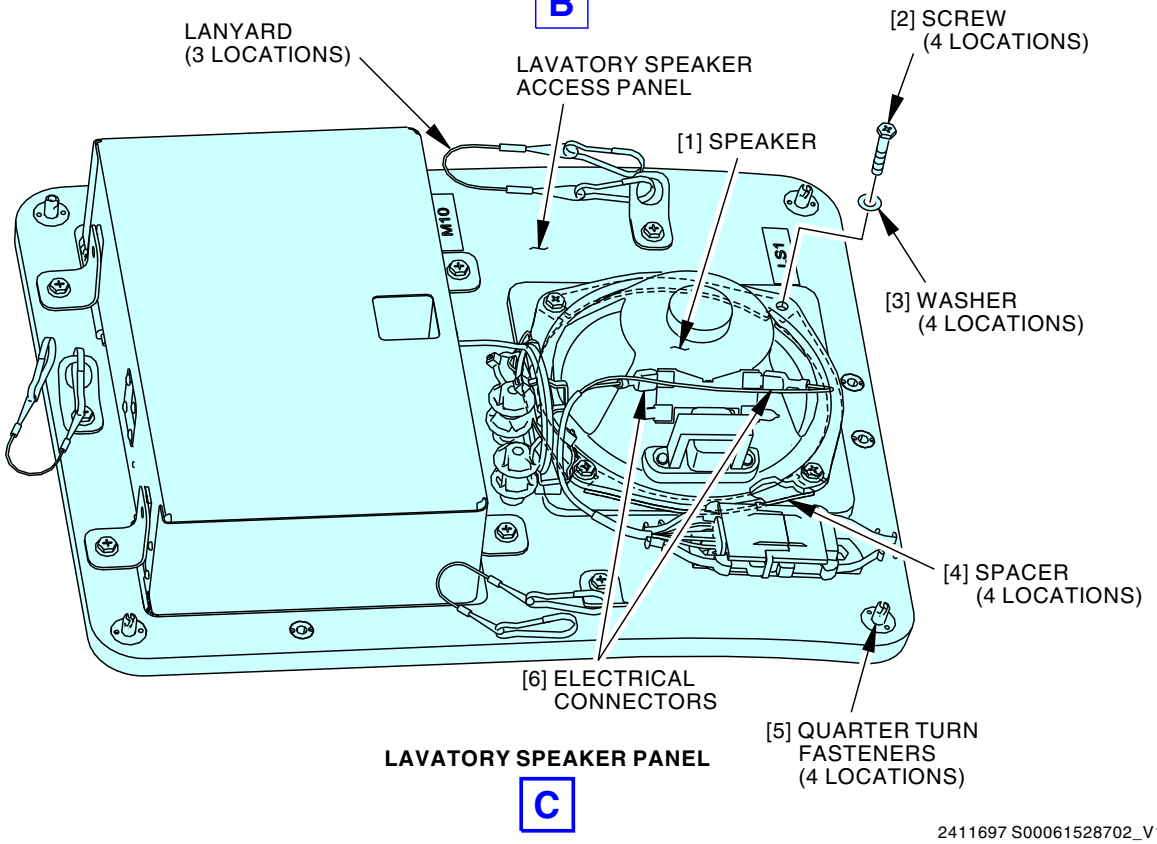
**Lavatory Speaker Installation
Figure 401/23-31-05-990-802 (Sheet 1 of 4)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD LAVATORY SPEAKER ASSEMBLY (EXAMPLE)

B



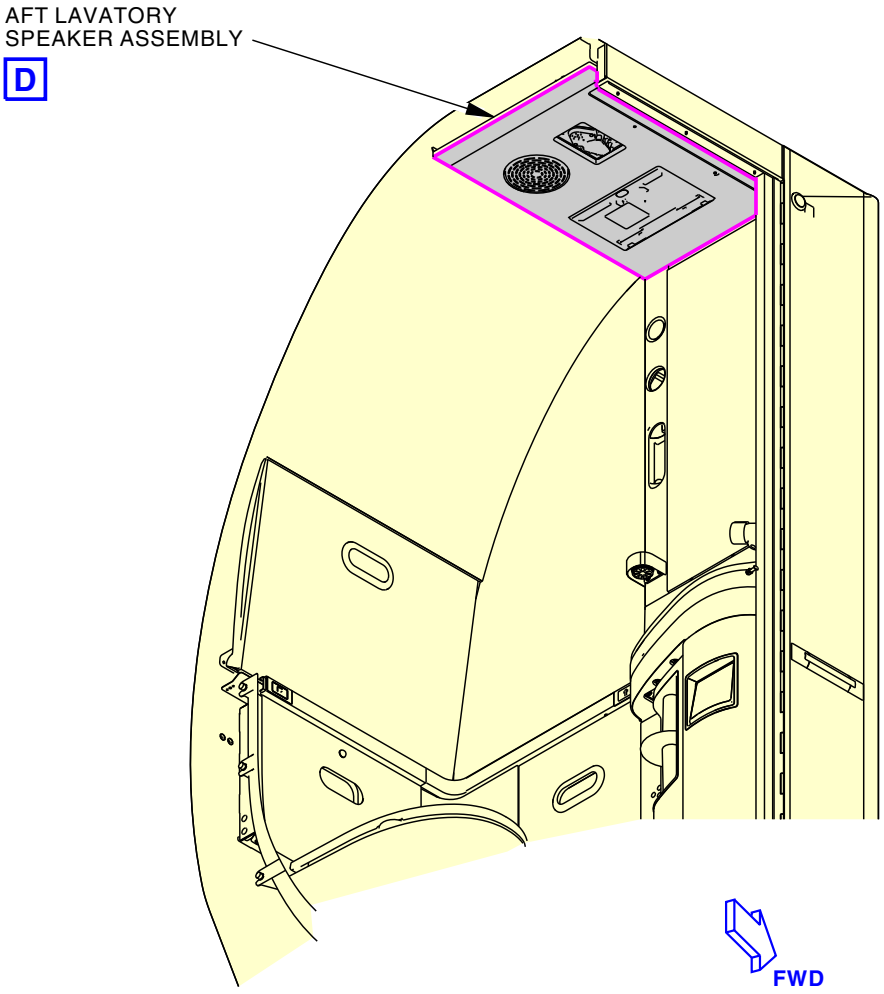
LAVATORY SPEAKER PANEL

C

2411697 S00061528702_V1

Lavatory Speaker Installation
Figure 401/23-31-05-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

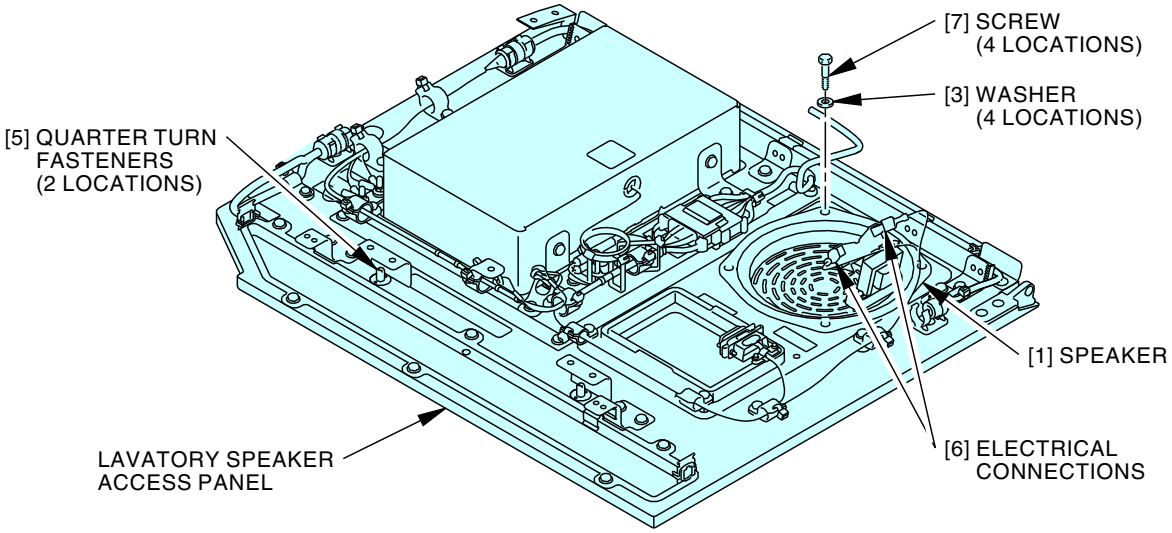


**AFT LAVATORY
(EXAMPLE)**

2540771 S0000602015_V1

**Lavatory Speaker Installation
Figure 401/23-31-05-990-802 (Sheet 3 of 4)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**AFT LAVATORY SPEAKER ASSEMBLY
(EXAMPLE)**

D

2540218 S0000602016_V1

**Lavatory Speaker Installation
Figure 401/23-31-05-990-802 (Sheet 4 of 4)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-05-400-801

3. Lavatory Speaker Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Speaker	23-31-05-01B-050	PTF ALL
		23-31-05-02D-050	PTF ALL
		23-31-05-14D-050	PTF ALL

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Installation Procedure

SUBTASK 23-31-05-420-001

(1) Do the steps that follow to install the lavatory speaker [1]:

(a) Hold the speaker [1] in the correct position.

(b) Make sure the electrical connectors [6] are in the correct position.

(c) Install the screws [2], washers [3], and spacers [4].

(d) Attach the electrical connectors [6] to the speaker [1].

1) Make sure the electrical connectors are connected to the same terminals that they were removed from.

(e) Push the PSU panel assembly up until it closes to the usual position.

SUBTASK 23-31-05-860-006

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

D. Installation Test

SUBTASK 23-31-05-710-001

(1) Do a test of the Lavatory speaker operation:

(a) Lift the PA microphone from at an attendant station.

(b) Push the PTT button on the microphone.

(c) Make an announcement on the PA system.

1) Make sure you can hear the announcement on the new lavatory speaker.

(d) Put the microphone back in its cradle.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRE-RECORDED ANNOUNCEMENT MACHINE (PRAM) - MAINTENANCE PRACTICES

1. General

A. This procedure contains:

(1) Maintenance Modes

(2) Content installation

(3) Pre-Recorded Announcement Machine Deactivation

(4) Pre-Recorded Announcement Machine Activation.

TASK 23-31-07-820-801

2. Pre-Recorded Announcement Machine (PRAM) Maintenance Modes

A. General

(1) You use the PRAM maintenance modes to make adjustments to the announcement playback operation. These are the maintenance modes:

(a) EMERGENCY REPEAT - sets the number of times that the emergency decompression announcement repeats when triggered.

(b) PLAY MODE SETTING - selects single playback mode or sequential playback mode.

(c) MUSIC LEVEL SETTING - sets the boarding music volume level.

(d) ANNC LEVEL SETTING - sets the announcement volume level.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

C. Procedure

SUBTASK 23-31-07-820-001

(1) Do these steps to enter the maintenance mode:

(a) Push and hold the ENT (enter) and then the ANNC (announcement) key.

1) The display screen shows:

PASSWORD

PLEASE INPUT

OK: ENT CANCEL: STOP

(b) Push the number keys to enter the password (7370).

1) Make sure the MAINTENANCE MODE menu screen appears in the display window.

NOTE: If an error message appears in line three and QUIT:STOP appears in line 4, the password entry was incorrect. Press STOP to clear the screen and then press ANNC and ENT to retry the password entry.

(c) Use the UP key or DOWN key to move through the list of maintenance modes.

(d) When the applicable maintenance mode screen is in the display, push the ENT key.

SUBTASK 23-31-07-820-002

(2) Do these steps to do the EMERGENCY REPEAT mode procedure:

(a) Select the EMERGENCY REPEAT mode from the MAINTENANCE MODE menu.

1) The EMERGENCY REPEAT screen shows in the display window.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Line 3 shows the current emergency announcement repeat number.

NOTE: The emergency repeat number may be set from 1 to 16.

(b) Use the "UP" or "DOWN" keys to select the repeat number setting.

(c) Push the ENT key.

(d) The display returns to the MAINTENANCE MODE menu screens.

SUBTASK 23-31-07-820-003

(3) Do these steps to do the MUSIC or ANNC LEVEL SETTING mode procedure.

NOTE: The MUSIC LEVEL and ANNC LEVEL SETTING mode procedures are identical except for the screen title in line 2 of the display.

(a) Select the MUSIC LEVEL SETTING mode or the ANNC LEVEL SETTING mode from the MAINTENANCE MODE menu.

1) The MUSIC LEVEL or ANNC LEVEL SETTING screen shows in the display with the current volume level setting in line 3.

(b) Use the UP key or DOWN key to select the correct volume level setting.

NOTE: The volume level can be set anywhere from -6dB to +6dB in 2dB steps.

(c) When the volume level setting is correct, push the ENT key.

1) The display returns to the MAINTENANCE MODE menu.

SUBTASK 23-31-07-820-004

(4) Do these steps to do the PLAY MODE SETTING mode procedure.

(a) Select the PLAY MODE SETTING mode from the MAINTENANCE MODE menu.

1) The PLAY MODE SETTING screen shows in the display with the current play mode setting in line 2.

(b) Use the "UP" or "DOWN" keys to switch between the play modes.

NOTE: The plays modes are SINGLE or SEQUENTIAL.

(c) When the desired play mode is correct, push the ENT key.

1) The display returns to the MAINTENANCE MODE menu.

SUBTASK 23-31-07-820-005

(5) To exit from the MAINTENANCE MODE push the STOP key.

——— END OF TASK ———

TASK 23-31-07-470-801

3. Pre-Recorded Announcement Machine (PRAM) Content Installation

Figure 201

A. General

(1) A Compact Flash (CF) memory card with the correct announcement and boarding music content is necessary for this task.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Prepare for Installation

SUBTASK 23-31-07-040-003

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

SUBTASK 23-31-07-470-001

(2) Do these steps to install the announcement and boarding music contents:

(a) Remove three screws [6] from PRAM cover [5].

(b) Remove the PRAM cover [5].

(c) Open the access door on the right side of the PRAM [1].

(d) Press the eject button and remove the CF memory card.

CAUTION KEEP THE COMPACT FLASH CARD STRAIGHT WHEN YOU INSTALL IT IN THE PRAM. IF YOU DO NOT INSTALL THE CARD CORRECTLY, IT CAN CAUSE DAMAGE TO THE PRAM UNIT.

(e) Insert the CF memory card with new content into the card slot on the PRAM. Refer toFigure 201 for the correct insertion position.

NOTE: The pre-programmed CF card MUST be left in the PRAM after installation for the unit to function properly.

(f) Carefully slide the new CF card into the CF slot and push it into place.

NOTE: Do not force CF card into slot. Make sure that the card's holes line up with the two rows of pins within the card slot (using inspection mirror or by temporary removal of the unit). Do not permit pin damage/shorting due to improper installation.

(g) Close the access door on the right side of the PRAM.

1) Make sure the door is completely latched.

(h) Install the PRAM cover [5].

(i) Install the three screws [6].

SUBTASK 23-31-07-740-001

(3) Remove the safety tag and close this circuit breaker:

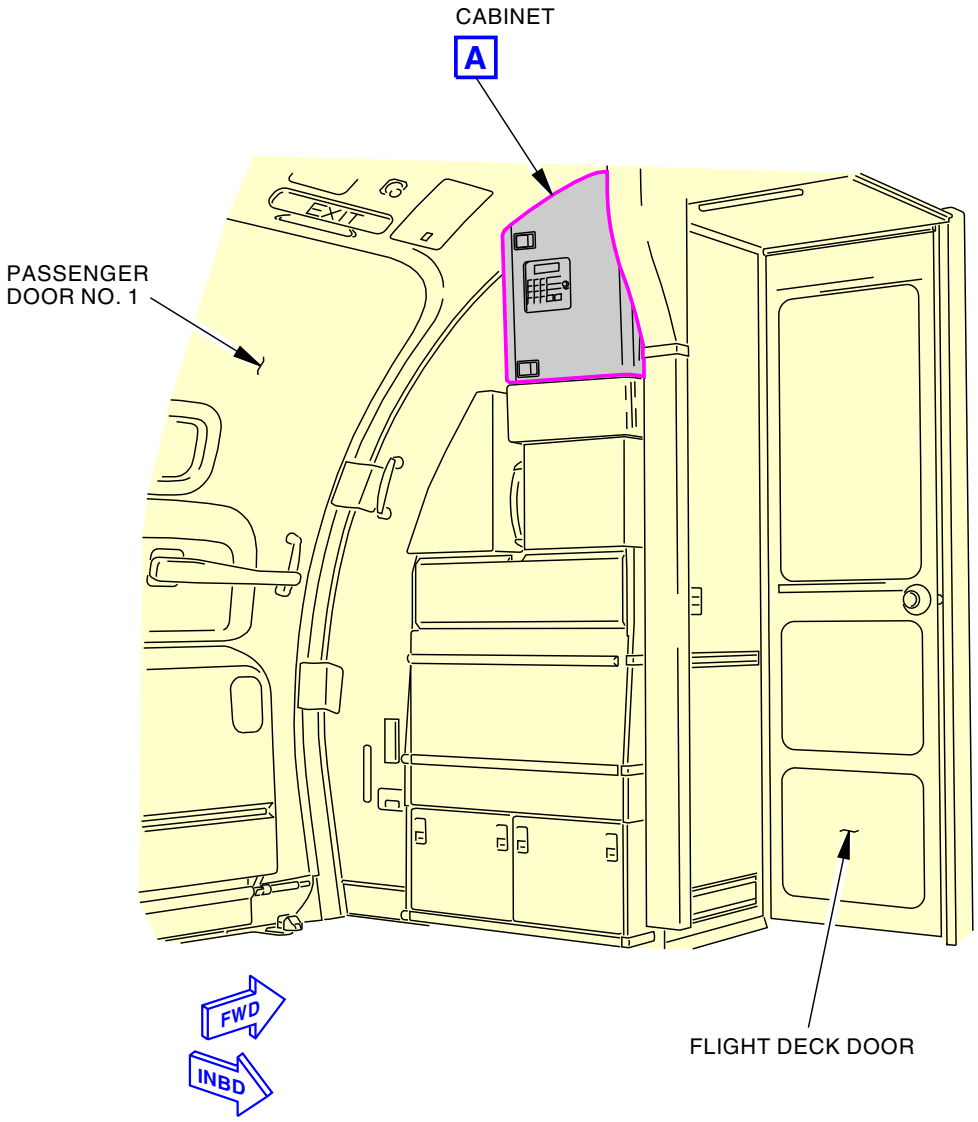
F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

NOTE: Once power is applied to the PRAM, the unit will perform a self BITE test for Audio, LSI, CF and card check. Airline code and card ID is then displayed followed by default screen with ANNC, FLIGHT and MUSIC LED's lit.

END OF TASK

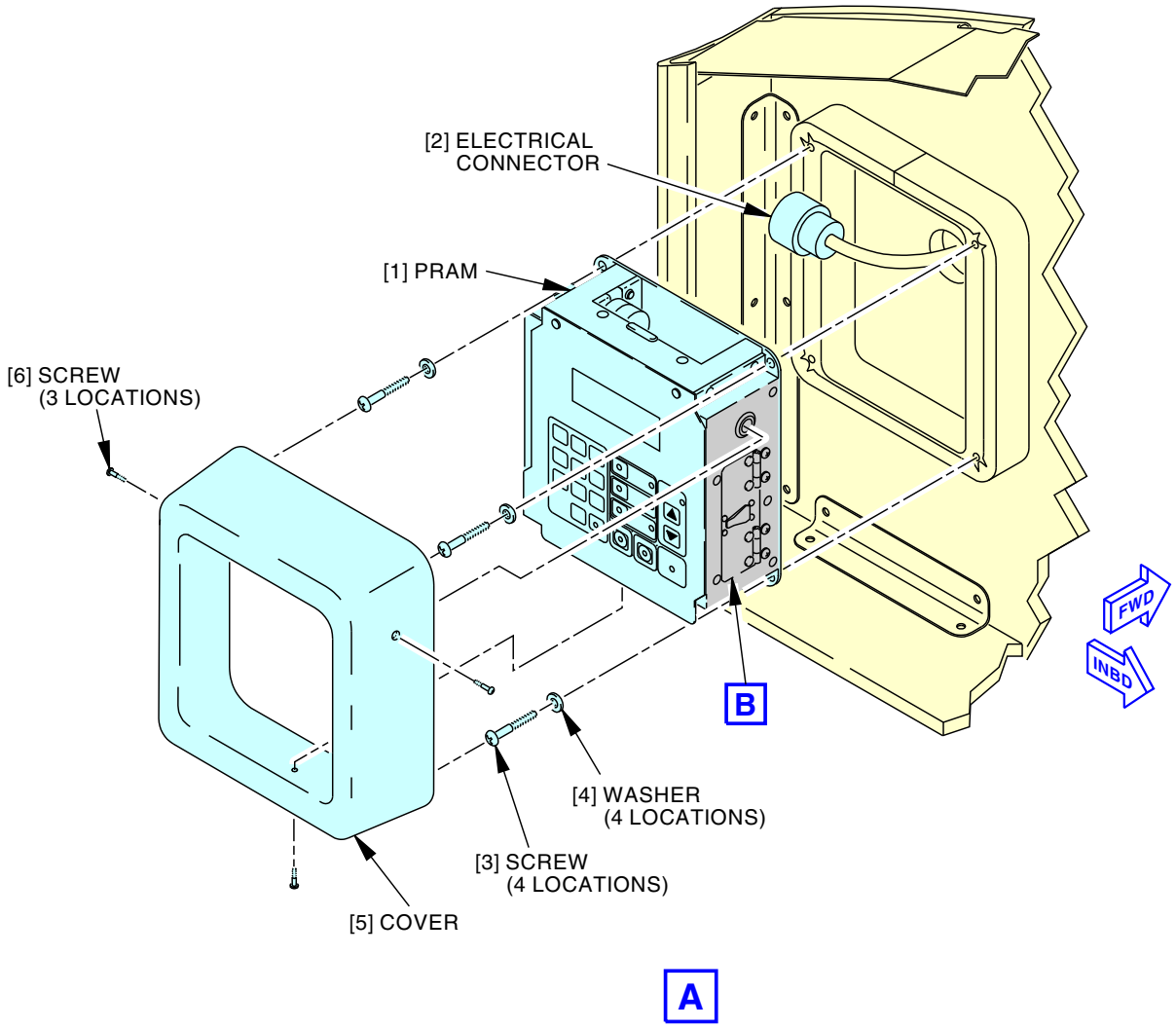
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2578275 S0000620519_V1

Pre-Recorded Announcement Machine (PRAM) Content Installation
Figure 201/23-31-07-990-802 (Sheet 1 of 3)

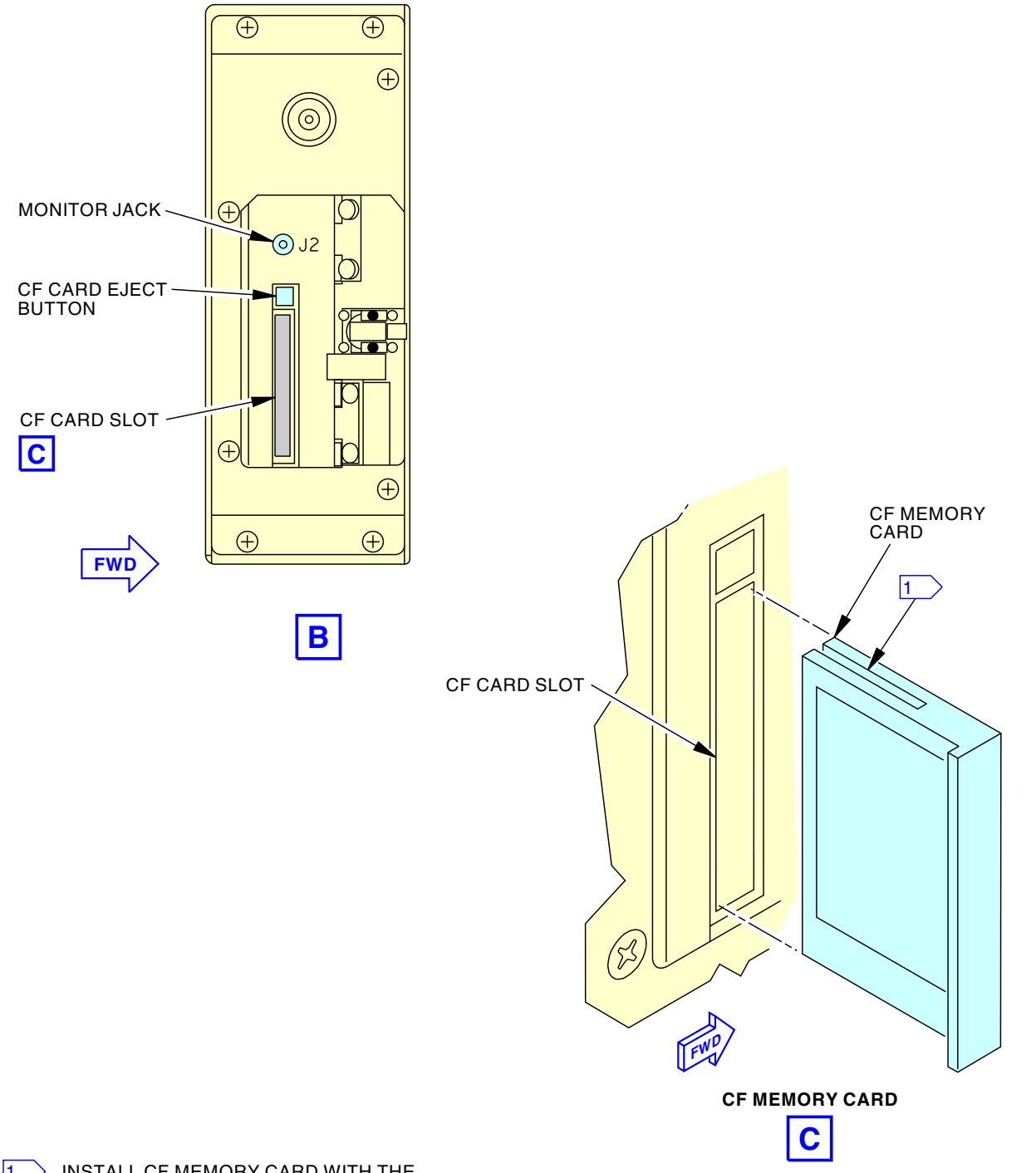
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2578274 S0000620523_V1

Pre-Recorded Announcement Machine (PRAM) Content Installation
Figure 201/23-31-07-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2578271 S0000620527_V1

Pre-Recorded Announcement Machine (PRAM) Content Installation
Figure 201/23-31-07-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-07-040-801

4. Pre-Recorded Announcement Machine (PRAM) - Deactivation

A. General

(1) This procedure removes the electrical power to the Pre-Recorded Announcement Machine (PRAM).

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

C. Procedure

SUBTASK 23-31-07-040-004

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

D. PRAM - Tryout

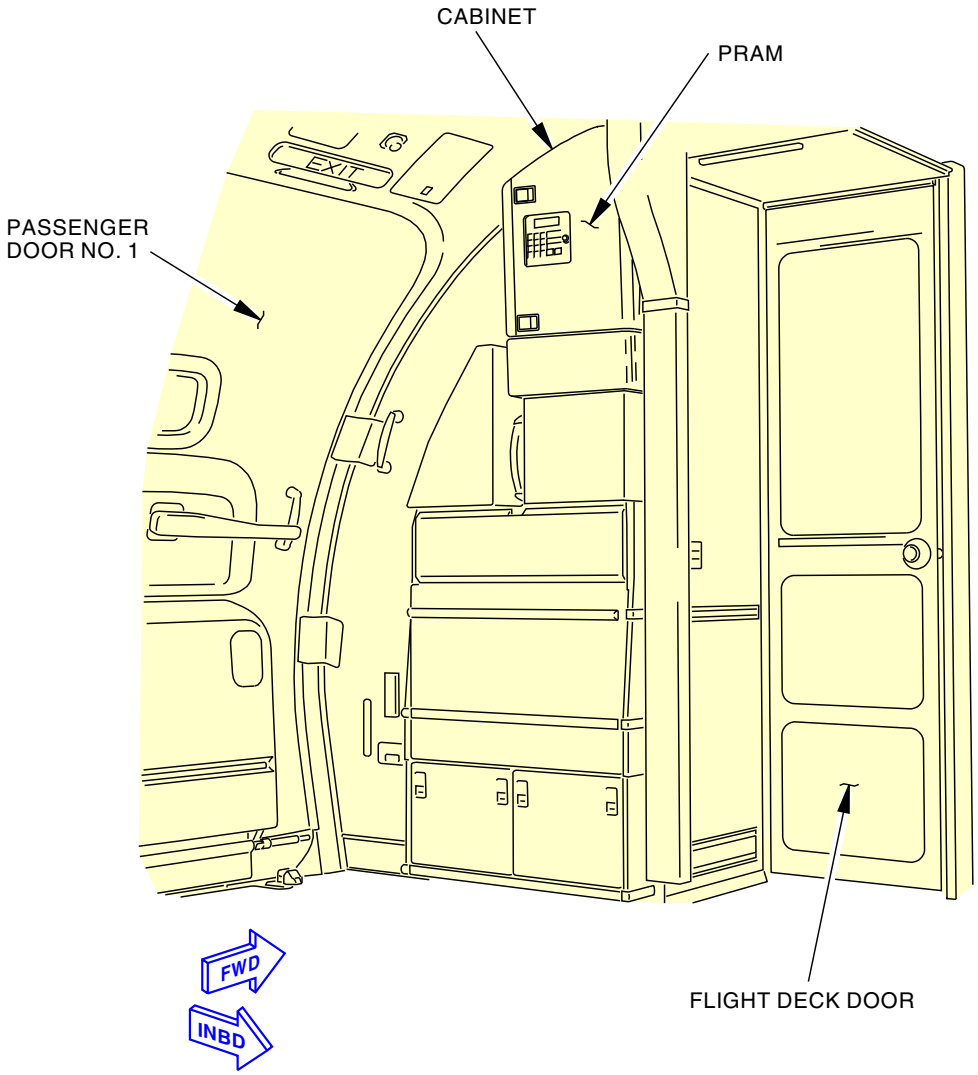
NOTE: This tryout is to make sure the PRAM is in a zero energy state.

SUBTASK 23-31-07-700-001

(1) At the forward attendants station make sure the PRAM is off by viewing the front and that the display is blank.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2578027 S0000620384_V1

Pre-Recorded Announcement Machine Installation
Figure 202/23-31-07-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-07-440-801

5. Pre-Recorded Announcement Machine (PRAM) - Activation

A. General

(1) This procedure adds electrical power to the Pre-Recorded Announcement Machine (PRAM).

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

C. Procedure

SUBTASK 23-31-07-440-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRE-RECORDED ANNOUNCEMENT MACHINE (PRAM) - REMOVAL/INSTALLATION

1. General

A. This procedure contains:

(1) A removal of the PRAM.

(2) An installation of the PRAM.

TASK 23-31-07-000-801

2. Pre-Recorded Announcement Machine (PRAM) Removal

Figure 401

A. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

B. Removal Procedure

SUBTASK 23-31-07-040-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

SUBTASK 23-31-07-000-001

(2) Do these steps to remove the PRAM:

(a) Remove the three screws [6] that attach the cover [5] to the PRAM [1].

(b) Remove the cover [5] from the PRAM [1].

(c) Open the access door on the right side of the PRAM [1].

(d) Press the eject button and remove the CF memory card and retain for installation.

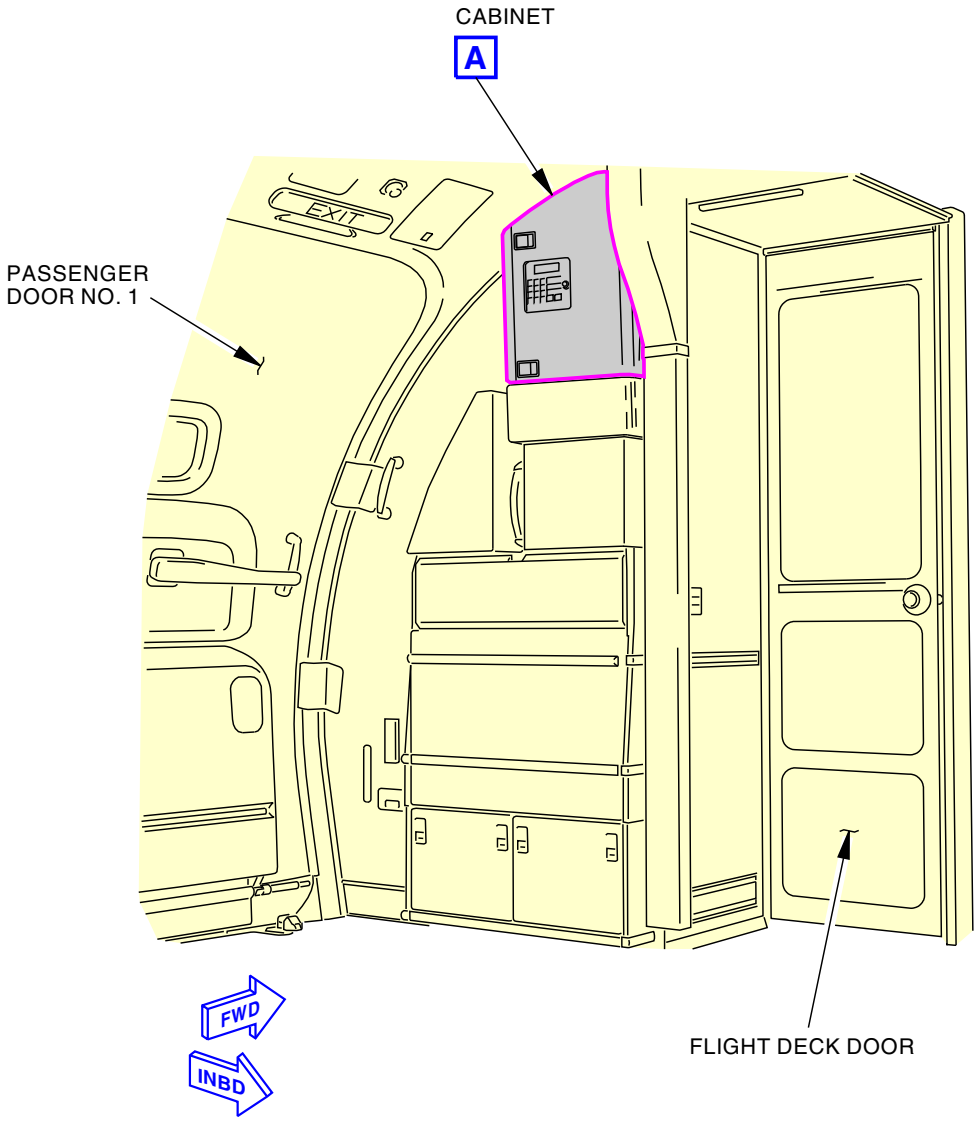
(e) Remove the four screws [3] and washers [4] that attach PRAM to the wall.

(f) Pull the PRAM [1] from the wall.

(g) Remove the electrical connector [2] from the rear of the PRAM [1].

END OF TASK

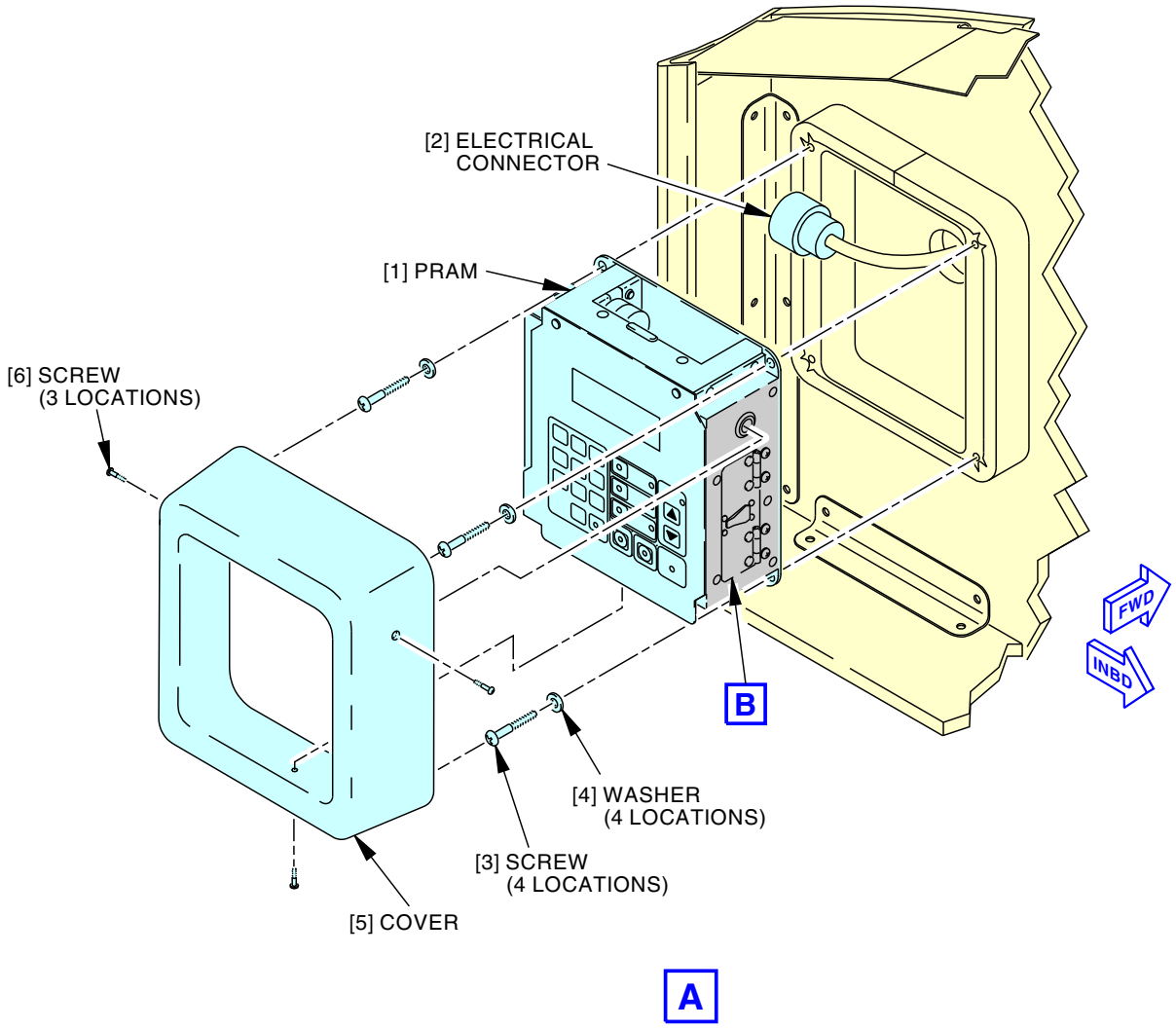
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2578275 S0000620519_V1

Pre-Recorded Announcement Machine (PRAM) Installation
Figure 401/23-31-07-990-801 (Sheet 1 of 3)

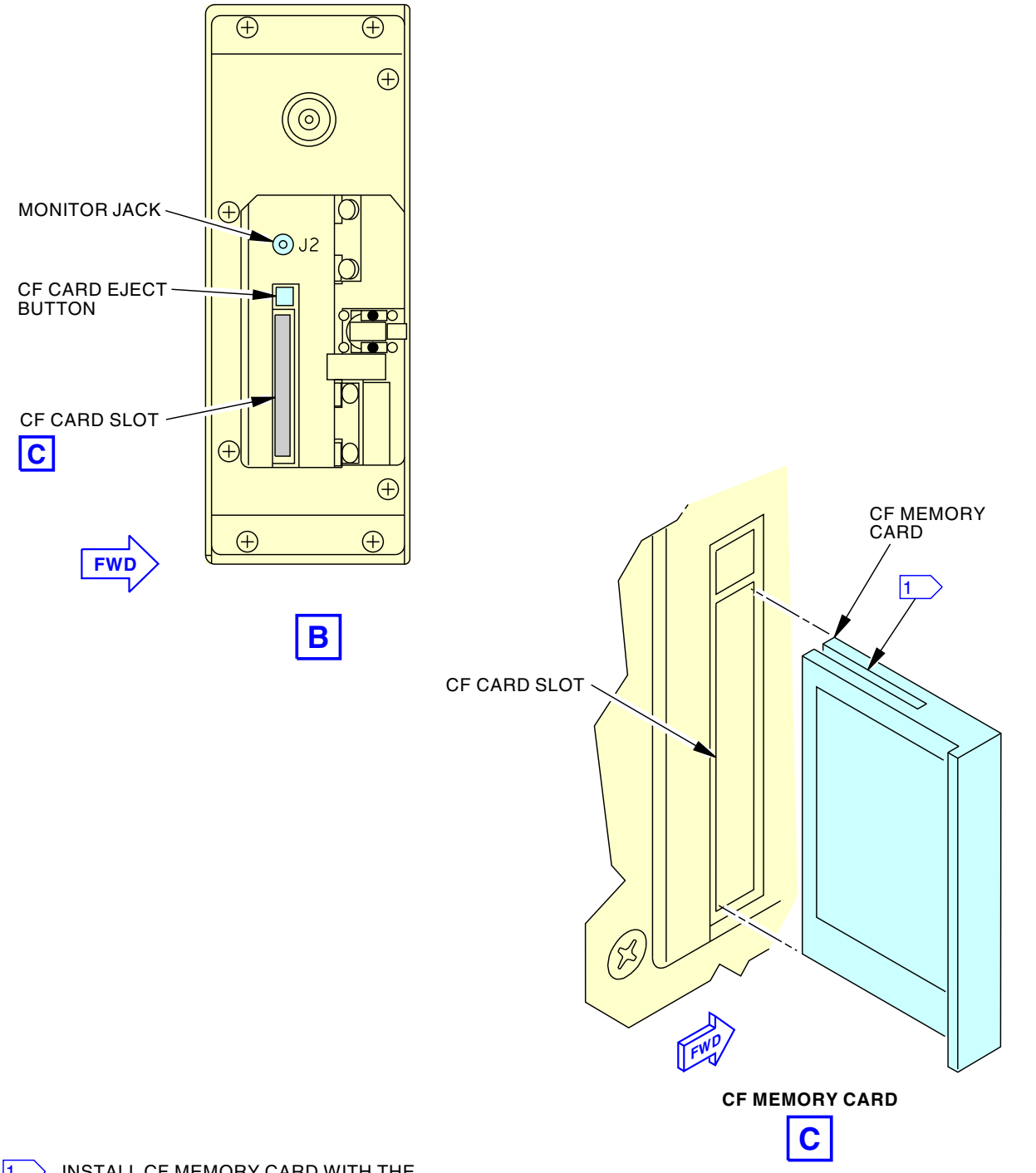
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2578274 S0000620523_V1

Pre-Recorded Announcement Machine (PRAM) Installation
Figure 401/23-31-07-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2578271 S0000620527_V1

Pre-Recorded Announcement Machine (PRAM) Installation
Figure 401/23-31-07-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-31-07-400-801

3. Pre-Recorded Announcement Machine (PRAM) Installation

Figure 401

A. References

Reference	Title
23-31-07-820-801	Pre-Recorded Announcement Machine (PRAM) Maintenance Modes (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	PRAM	23-31-07-01-065	PTF ALL

C. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

D. Procedure

SUBTASK 23-31-07-040-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

SUBTASK 23-31-07-400-001

(2) Do these steps to install the PRAM:

(a) Connect the electrical connector [2] to the rear of the PRAM [1].

(b) Put the PRAM [1] over the bolt holes on the wall.

(c) Install the four washers [4] and screws [3] that attach the PRAM [1] to the wall.

CAUTION KEEP THE COMPACT FLASH CARD STRAIGHT WHEN YOU INSTALL IT IN THE PRAM. IF YOU DO NOT INSTALL THE CARD CORRECTLY, IT CAN CAUSE DAMAGE TO THE PRAM UNIT.

(d) Insert the CF memory card with new content into the card slot on the PRAM. Refer toFigure 401 for the correct insertion position.

NOTE: The pre-programmed CF card MUST be left in the PRAM after installation for the unit to function properly.

(e) Carefully slide the new CF card into the CF slot and push it into place.

NOTE: Do not force CF card into slot. Make sure that the card's holes line up with the two rows of pins within the card slot (using inspection mirror or by temporary removal of the unit). Do not permit pin damage/shorting due to improper installation.

(f) Close the access door on the right side of the PRAM.

1) Make sure the door is completely latched.

(g) Put the PRAM cover [5] over the bolt holes on the PRAM [1].

(h) Install the three screws [6] that attach the cover [5] to the PRAM [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-31-07-440-001

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	9	C00080	ENTERTAINMENT PA TAPE RPDR AC

E. Installation Test Procedure

SUBTASK 23-31-07-750-001

(1) If it is necessary to set the PRAM maintenance modes, do this task: Pre-Recorded Announcement Machine (PRAM) Maintenance Modes, TASK 23-31-07-820-801.

SUBTASK 23-31-07-710-001

(2) Do this test of the PRAM:

(a) Push the ENT button on the PRAM.

(b) Push the MUSIC button.

(c) Push '1' on the PRAM.

(d) Push the START button.

(e) Adjust the volume control if it is necessary.

1) Make sure that you hear clear music on all of the speakers.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SERVICE INTERPHONE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Service Interphone Operational Test

TASK 23-41-00-710-801

2. Service Interphone Operational Test

A. General

(1) This procedure has steps to do a operational test for a Service Interphone system.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2539	Interphone - Boom, Mic/Headset for Ground Crew Part #: 12506G-05 Supplier: 71483 Part #: 12508G-34 Supplier: 71483 Part #: 12515G-12 Supplier: 71483 Opt Part #: 460A Supplier: 39SU9 Opt Part #: 462A Supplier: 39SU9 Opt Part #: H3310 Supplier: 71483 Opt Part #: H3312 Supplier: 71483

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
315	APU Compartment - Left
600	Right Wing

D. Prepare for the Service Interphone Operational Test

SUBTASK 23-41-00-860-002

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-41-00-710-012

(2) Prepare for the Service Interphone Operational Test:

(a) Connect a mic/headset for ground crew boom interphone, COM-2539, to the Captain's, First Officer's, or Observer's boom mic/headset jack on the Audio Control Panel (ACP).

(b) Push the CABIN microphone selector switch on all Audio Control Panels (ACP).

(c) Push the volume control for the CABIN microphone selector switch.

(d) Turn the volume control switch for CABIN to the middle position.

(e) Set the SERVICE INTERPHONE switch, on the P5 overhead panel, to ON.

E. Service Interphone Operational Test

SUBTASK 23-41-00-710-001

(1) Do a communication test between the flight crew and the ground crew at the nose wheel well area:

(a) Connect a mic/headset for ground crew boom interphone, COM-2539, to the service interphone jack.

(b) Push the RADIO - INT switch on the Captain's, First Officer's, or Observer's ACP.

(c) Speak into the Captain's, First Officer's, or Observer's boom microphone.

1) Make sure the ground crew can hear the voice clearly on the headset.

(d) Release the RADIO - INT switch on the Captain's, First Officer's, or Observer's ACP.

(e) Push the PTT switch on the Captain's (First Officer's) glareshield.

1) Speak into the Captain's (First Officer's) boom microphone.

2) Make sure the ground crew can hear the voice clearly on the service interphone headset.

3) Release the PTT switch.

(f) Have the ground crew speak into the boom microphone.

1) Make sure you can hear the voice clearly on the Captain's First Officer's, or Observer's headset.

SUBTASK 23-41-00-710-002

(2) Repeat the communication test between the flight crew and the ground crew with the service interphone jack at the auxiliary power unit.

SUBTASK 23-41-00-710-003

(3) Repeat the communication test between the flight crew and the ground crew with the service interphone jack in the overhead at the aft passenger cabin.

SUBTASK 23-41-00-710-004

(4) Repeat the communication test between the flight crew and the ground crew with the service interphone jack at the refueling station on the right wing.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-41-00-710-005

(5) Repeat the communication test between the flight crew and the ground crew using the service interphone jack at the left wheel well.

SUBTASK 23-41-00-710-006

(6) Repeat the communication test between the flight crew and the ground crew with the service interphone jack at the right wheel well.

SUBTASK 23-41-00-710-007

(7) Repeat the communication test between the flight crew and the ground crew with the service interphone jack at the electrical and electronics compartment.

F. Service Interphone Switch Operational Test

SUBTASK 23-41-00-710-008

(1) Do the test of the SERVICE INTERPHONE switch:

(a) Connect a mic/headset for ground crew boom interphone, COM-2539, to a service interphone jack.

(b) Set the SERVICE INTERPHONE switch to OFF.

(c) Push the RADIO - INT switch on the Captain's, First Officer's, or Observer's ACP.

(d) Speak into the boom microphone.

1) Make sure the ground crew can hear the voice on the service interphone headset.

(e) Release the RADIO - INT switch on the Captain's, First Officer's, or Observer's ACP.

(f) Push the PTT switch on the Captain's (First Officer's) glareshield.

1) Speak into the boom microphone.

2) Make sure the ground crew can hear the voice on the service interphone headset.

3) Release the PTT switch.

(g) Have the ground crew speak into the boom microphone.

1) Make sure you cannot hear the voice on the headset.

SUBTASK 23-41-00-710-009

(2) Repeat the communication test between the flight crew and the ground crew with the First Officer's ACP.

SUBTASK 23-41-00-710-010

(3) Repeat the communication test between the flight crew and the ground crew with the Observer's ACP.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-41-00-020-001

(1) Disconnect the mic/headset for ground crew boom interphone, COM-2539, from the ACP and service interphone boom mic/headset jack.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN INTERPHONE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure contains these tasks:

(1) An Operational Test of the Cabin Interphone system

(2) A Cabin Interphone system / Attendant Panel Installation Test

TASK 23-42-00-710-801

2. Cabin Interphone System - Operational Test

A. General

(1) This task checks to make sure that the Cabin Interphone communication between the flight crew and the attendants operates correctly.

(2) This task requires two persons to complete.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Prepare for the Operational Test

SUBTASK 23-42-00-840-002

(1) Make sure that all handsets are on the hook.

D. Cabin Interphone Operational Test

SUBTASK 23-42-00-710-001

(1) Do the steps that follow for the Cabin Interphone Call Test:

(a) Push the ATTEND call switch on the P5 panel (passenger sign panel).

(b) Make sure that the pink, attendant call light on the FORWARD and AFT EXIT locator signs turn ON.

(c) Make sure that you hear a HI/LO tone in the cabin.

(d) Remove any handset from the holder to reset the light.

(e) Make sure that the attendant call lights turn OFF.

(f) Put the handset back on its holder.

(g) Do the steps that follow for the FORWARD attendant handset:

1) Remove the handset from its holder.

2) Push the button corresponding to “PILOT” on the handset.

3) Make sure that the CABIN call light on the Audio Control Panel (ACP) turns ON.

4) Make sure that you hear a HI tone in the flight deck.

5) Push the “RESET” button on the handset.

6) Push the button corresponding to an “Attendant call” on the handset.

7) Make sure that the attendant call light at the other station turns ON.

8) Make sure that you hear a HI/LO tone in the cabin.

9) Put the handset back on its holder.

10) Make sure that the attendant call light at the other station turns OFF.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Repeat the test for the AFT attendant handset(s).

SUBTASK 23-42-00-710-002

(2) Do the steps that follow for the Cabin Interphone Communication Test:

(a) Push the interphone button on all ACPs.

NOTE: Button may be labeled “CABIN” or “SERVICE”.

(b) Make sure that you get good two way interphone communication between all communication devices: flight deck handset, pilots boom/mic headset, FORWARD and AFT attendant handsets.

——— END OF TASK ———

TASK 23-42-00-700-801

3. Cabin Interphone system - Attendant Panel Installation Test

A. General

(1) This task makes sure that the Cabin Interphone system operates correctly after the ACP installation.

(2) This task requires two persons to complete.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Cabin Interphone / Attendant Panel Installation Test

SUBTASK 23-42-00-720-001

(1) Do the steps that follow to check the Cabin Interphone system after the Attendant Panel installation:

(a) Do the steps that follow for any forward attendant handset:

1) Remove the handset from its holder.
2) Push the button corresponding to an “Attendant call” on the handset.
3) Make sure the attendant call light at the other station turns on.
4) Make sure that you hear a HI/LO tone in the cabin.
5) Make sure that you get a good two way communication between the handset and the other station's attendant handset.
6) Push the “RESET” button on the handset.
7) Push the button corresponding to “PILOT” on the handset.
8) Make sure the CABIN call light on the audio control panel turns on.
9) Make sure you hear a HI tone in the flight deck.
10) Make sure you get a good two way communication between the handset and the pilot's boom microphone.
11) Push the “RESET” button on the handset.
12) Put the handset back on its holder.

(b) Repeat the test for any aft attendant handset.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT HANDSET - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Attendant handset removal

(2) Attendant handset installation

(3) Attendant handset cradle removal

(4) Attendant handset cradle installation

(5) Attendant handset cord removal

(6) Attendant handset cord installation.

TASK 23-42-01-000-801

2. Attendant Handset Removal

(Figure 401)

A. General

(1) This task is for the removal of the attendant handset.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Attendant Handset Removal

SUBTASK 23-42-01-020-001

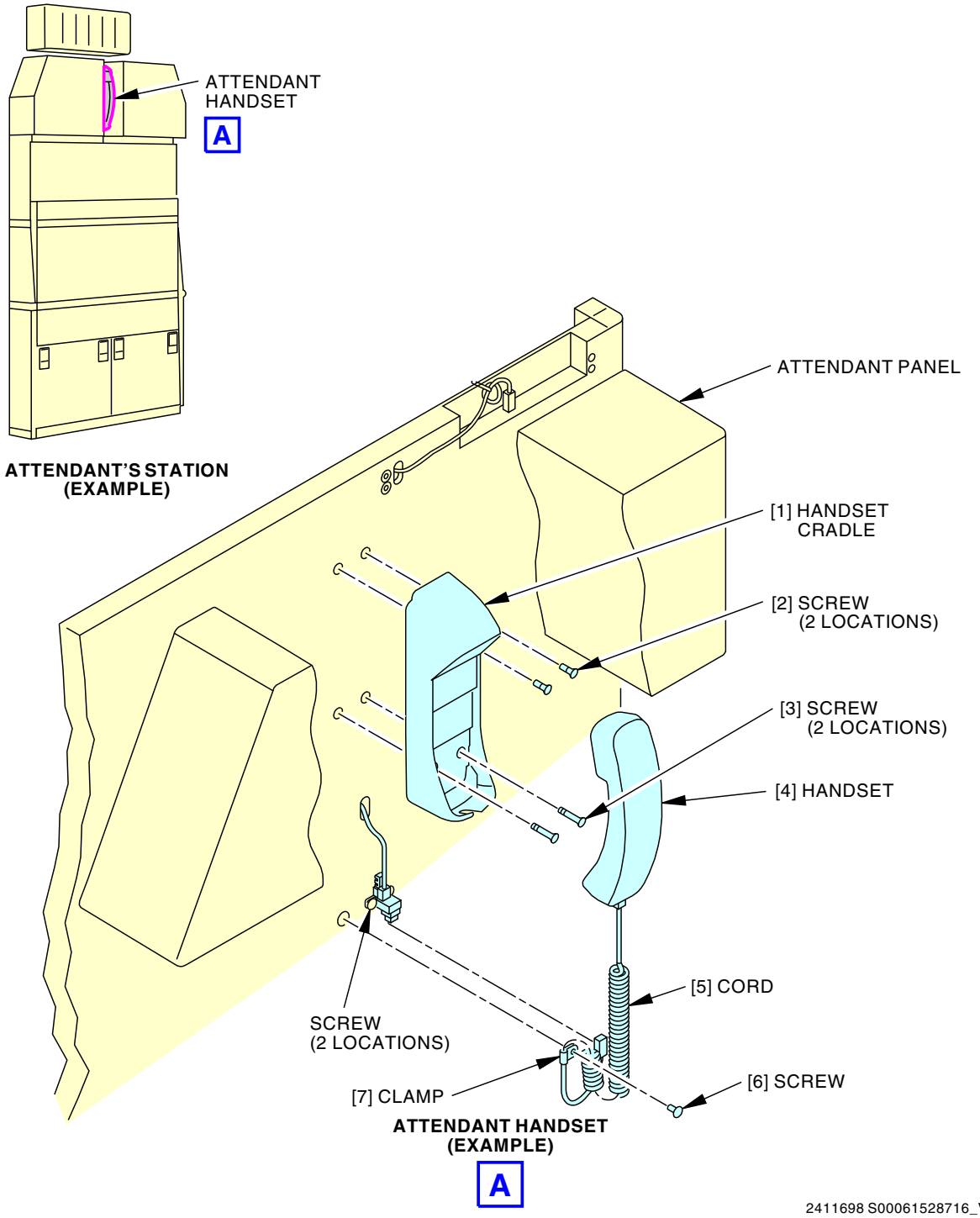
(1) Remove the attendant handset [4] as follows:

(a) Remove the handset [4] from the handset cradle [1].

(b) Disconnect the cord [5] from the handset [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Attendant Handset Installation
Figure 401/23-42-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-42-01-400-803

3. Attendant Handset Installation

(Figure 401)

A. General

(1) This task is for the installation of the attendant handset.

B. References

Reference	Title
23-42-01-700-802	Attendant Handset Installation Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Handset	23-42-01-03-015	PTF ALL
		23-42-01-18-025	PTF ALL
		23-42-01-19-025	PTF ALL

D. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

E. Attendant Handset Installation

SUBTASK 23-42-01-420-003

(1) Install the attendant handset [4] as follows:

(a) Install the marker on the handset if missing.

(b) Connect the cord [5] to the handset [4].

(c) Put the handset [4] on the handset cradle [1].

F. Attendant Handset Installation Test

SUBTASK 23-42-01-720-003

(1) Do this task: Attendant Handset Installation Test, TASK 23-42-01-700-802.

——— END OF TASK ———

TASK 23-42-01-000-803

4. Attendant Handset Cradle Removal

(Figure 401 and Figure 402)

A. General

(1) This task is for the removal of the attendant handset cradle.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Attendant Handset Cradle Removal

SUBTASK 23-42-01-020-003

(1) Remove the handset cradle [1] as follows:

(a) Remove the handset [4] from the handset cradle [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove two screws [2] and two screws [3] from the handset cradle [1].

(c) Remove the handset cradle [1] from the attendant station.

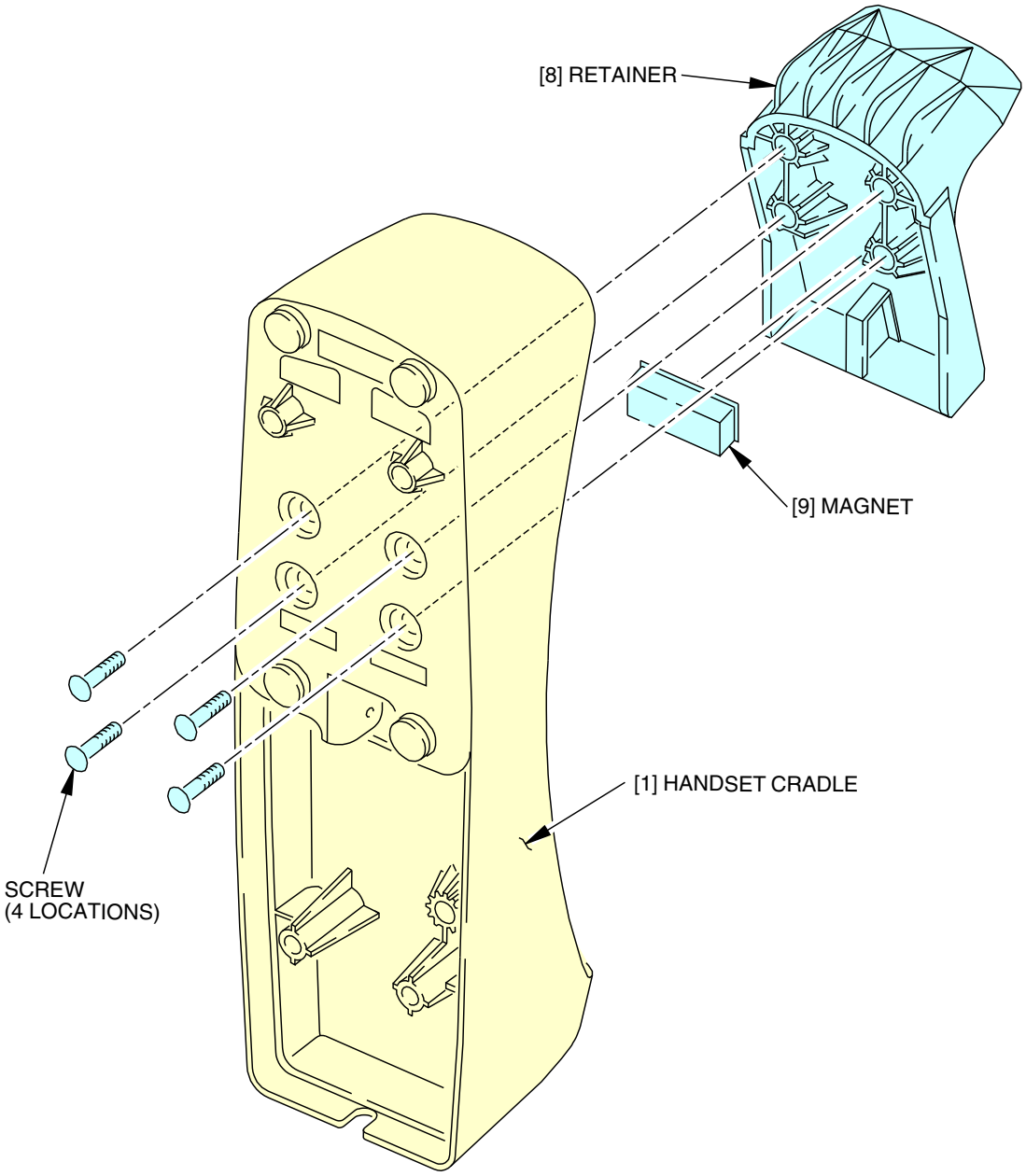
(d) If the magnet [9] is loose inside the handset cradle [1], do these steps:

1) Remove the screws that attach the retainer [8] to the handset cradle [1].

2) Remove the retainer [8] and the magnet [9].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Attendant Handset Cradle Magnet Installation
Figure 402/23-42-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-42-01-400-801

5. Attendant Handset Cradle Installation

(Figure 401 and Figure 402)

A. General

(1) This task is for the installation attendant handset cradle.

B. References

Reference	Title
23-42-01-700-802	Attendant Handset Installation Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
A00188	Adhesive - Urethane, Two Component	BMS5-105 Type V

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Handset cradle	23-42-01-03-035	PTF ALL
		23-42-01-18-040	PTF ALL
		23-42-01-19-040	PTF ALL

E. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

F. Attendant Handset Cradle Installation

SUBTASK 23-42-01-420-004

(1) If it is necessary, do these steps to attach the magnet [9] to the handset cradle [1]:

(a) Use adhesive, A00188, to attach the magnet [9] to the retainer [8].

1) Make sure that the flanged surface of the magnet is centered.

(b) Put the retainer [8] into position on the handset cradle [1].

(c) Install the screws that attach the retainer [8] to the handset cradle [1].

SUBTASK 23-42-01-420-001

(2) Install the handset cradle [1] as follows:

(a) Put the handset cradle [1] in its correct position over the screw holes in the attendant station.

(b) Install the screws [2] and screws [3] to tighten the handset cradle [1] to the attendant station.

(c) Put the handset [4] in the handset cradle [1].

G. Attendant Handset Cradle Installation Test

SUBTASK 23-42-01-700-001

(1) Do this task: Attendant Handset Installation Test, TASK 23-42-01-700-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-42-01-000-802

6. Attendant Handset Cord Removal

(Figure 401)

A. General

(1) This task is for the removal of the attendant handset cord.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Attendant Handset Cord Removal

SUBTASK 23-42-01-020-002

(1) Remove the handset cord [5] as follows:

(a) Disconnect the cord [5] from the handset [4].

(b) Remove the screw [6] from the clamp [7].

(c) Remove the clamp [7].

(d) Pull the handset cord [5] out of the clamp [7].

END OF TASK

TASK 23-42-01-400-802

7. Attendant Handset Cord Installation

(Figure 401)

A. General

(1) This task is for the installation of the attendant handset cord.

B. References

Reference	Title
23-42-01-700-802	Attendant Handset Installation Test (P/B 501)

C. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

D. Attendant Handset Cord Installation

SUBTASK 23-42-01-420-002

(1) Install the handset cord [5] as follows:

(a) Put the handset cord [5] through the clamp [7].

(b) Put the clamp [7] in its location over the screw hole in the attendant's station.

(c) Install the screw [6] to tighten the clamp [7] to the attendant's station.

(d) Connect the cord [5] to the handset [4].

E. Attendant Handset Cord Installation Test

SUBTASK 23-42-01-720-004

(1) Do this task: Attendant Handset Installation Test, TASK 23-42-01-700-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT HANDSET - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Attendant handset installation test.

TASK 23-42-01-700-802

2. Attendant Handset Installation Test

A. General

(1) This task is for the installation test of the attendant handset.

B. Location Zones

Zone	Area
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Attendant Handset Installation Test

SUBTASK 23-42-01-980-001

(1) Lift the handset from the handle cradle.

SUBTASK 23-42-01-720-001

(2) Push the button for the attendant on the handset keypad and check that these occurs:

(a) Make sure that you hear a chime on the PA speakers.

(b) Make sure that the pink attendant call light at the other attendant station comes on.

SUBTASK 23-42-01-720-002

(3) Call another attendant station and make sure that communication goes through.

SUBTASK 23-42-01-980-002

(4) Put the handset in the handset cradle.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT CONTROL PANEL - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Software Loading.

(2) Fault Data Report.

(3) Configuration Check.

(4) Configuration Data Report.

(5) Erasing Mass Storage Area (MSA).

(6) LCD Touch Screen Display Check.

TASK 23-42-03-470-801

2. Attendant Control Panel (ACP) - Software Loading

A. General

(1) This procedure installs Loadable Software Airplane Parts (LSAPs) to the ACP's and the Lighting LRU's.

NOTE: The ACP OPS, CDB and Standard Scenes LDB are required to be installed in the ACP to be fully functional. Optional Custom Scenes LDB may be installed.

NOTE: Software Loading is divided into two phases: Phase 1 data load and Phase 2 data load:

- A Phase 1 data load stages the Loadable Software Airplane Part (LSP) to the forward Attendant Control Panel's (ACP) Mass Storage Area from the ONS. These LSPs are saved onto the forward ACPs memory until ready to be used for a Phase 2 data load.
- A Phase 2 data load transfers the LSPs into the forward ACP (if different) and aft ACP (if different) and to the Lighting LRUs. In order to perform a Phase 2 data load, at least one LSAP type (ACP OPS, CDB, Standard Scenes LDB) needs to be installed from the Current Config or Mass Storage Area of the forward ACP.

(2) Throughout the procedure, the ACP may automatically go into screensaver mode. To exit out of screensaver mode, touch two different corners of the ACP screen sequentially.

B. References

Reference	Title
20-15-11-470-801	On-Airplane Software Installation (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
46-13-00-470-801	LRU Software Parts Installation (P/B 201)

C. Prepare for the Software Load

SUBTASK 23-42-03-860-003

(1) Set the overhead panel P5-13, CAB/UTIL switch to the ON position.

SUBTASK 23-42-03-860-004

(2) Set up the data loader at P61-1 panel located in the flight deck to applicable configuration for loading software to the ACP. Refer to On-Airplane Software Installation, TASK 20-15-11-470-801.

(a) Set the DATA LOAD SELECTOR switch to SINGLE SYS.

(b) Set the SYSTEM SELECT switch to ACP.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-42-03-860-005

(3) Make sure that the forward ACP and aft ACP are ON.

SUBTASK 23-42-03-940-001

(4) Make sure that the airplane is set to ground mode (TASK 32-09-00-860-802).

D. ACP - Software Loading

SUBTASK 23-42-03-470-002

(1) Do these steps to install software:

(a) Push the “Maintenance” button on the ACP main menu column.

(b) Enter password if necessary.

NOTE: Depending on the ACP configuration, you may be required to provide a password.

The default password is '1234'.

(c) Push the “Data Load” button on the Maintenance menu.

NOTE: When you push the “Data Load” button, you will see four buttons in the control window: Phase 1 Data Load, Phase 2 Data Load, Start, and Exit. Data loading instructions will be displayed in the result window.

NOTE: Phase 1 Data Load will load the ACP software from the Onboard Network System (ONS) to the FWD ACP MSA.

Phase 2 Data Load will load software from MSA to FWD ACP. The software will be automatically cross-loaded from FWD ACP to AFT ACP and distributed its data to Lighting LRUs. If AFT ACP and/or Lighting LRUs are not present, the software must be manually cross-loaded from FWD ACP to AFT ACP and distributed its data to Lighting LRUs.

(d) Determine if a Phase 1 data load is required:

1) Press the Phase 2 Data Load button.

2) View the Phase 2 data load screen.

(e) Make sure that at least one of each LSP, for each LSP type (ACP OPS, CDB and Standard Scenes), can be selected.

1) A selectable LSP will have a blue or black background color.

2) Make sure that the correct part number shows in either the Current Config or Mass Storage Area.

3) If the correct part number is not found, then a Phase 1 data load is required.

(f) Press Exit

SUBTASK 23-42-03-470-006

(2) For Phase 1 Data Load, do the steps that follow to load software parts to the forward ACP with the Onboard Network System (ONS):

(a) Push the “Phase 1 Data Load” button on the forward ACP.

(b) Do this task: LRU Software Parts Installation, TASK 46-13-00-470-801.

NOTE: Select the applicable ACP parts (OPS, CDB, Standard Scenes LDB, or Custom Scenes).

NOTE: Only one part can be loaded at a time with ONS. LSAPs may be loaded in any order.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Push the “Exit” button on the ACP after all parts have been successfully loaded with ONS.

SUBTASK 23-42-03-470-007

(3) For Phase 2 Data Load, do the steps that follow:

(a) Push the “Phase 2 Data Load” button.

NOTE: When you select Phase 2 Data Load button, the screen will show the LSAP selection table and control window. The “Current Config” is selected as default. You can select another LSAP to load into the ACP's and the lighting LRU's.

(b) Select the LSAP in the Mass Storage Area or “Current Config”, if not already selected.

1) At least one LSAP from each column will need to be selected in order to begin a Phase 2 data load.

2) The start button in step 2 will stay gray if at least one LSAP type is not selected from each column.

NOTE: If a custom scene is not used, select the "No Custom Scene" button; its back ground will turn blue when selected.

NOTE: The back ground of all the selected part numbers will turn blue.

(c) Push the “Start” button.

NOTE: Once the process has started, the result window will show the current progress of the file transfer. The status bar will show the approximate percent complete as the process executes.

During the Phase 2 data load, the ACPs may reboot. This is normal.

When the Lighting LRUs are being loaded, the lights will display change to a successful addressing color (colored lights will be green or white lights will be amber/warm white) during the process.

SUBTASK 23-42-03-700-001

(4) Do this task: Attendant Control Panel (ACP) - Configuration Check, TASK 23-42-03-700-801.

——— END OF TASK ———

TASK 23-42-03-970-803

3. Attendant Control Panel (ACP) - Fault Data Report

A. General

(1) This task lets you view the system test faults or data load faults using the Attendant Control Panel (ACP).

(2) This task also lets you report or view the system test faults or data load faults using the Onboard Network System (ONS).

(3) Throughout the procedure, the ACP may automatically go into screensaver mode. To exit out of screensaver mode, touch two different corners of the ACP screen sequentially.

B. References

Reference	Title
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
46-13-02-970-803	LRU Reports (P/B 201)

C. Prepare to View or Report

SUBTASK 23-42-03-860-023

(1) Make sure that the overhead panel P5-13, CAB/UTIL switch is set to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-42-03-860-024

(2) Make sure that the forward ACP and aft ACP are ON.

SUBTASK 23-42-03-940-009

(3) Make sure that the airplane is set to ground mode (TASK 32-09-00-860-802).

D. Procedure

SUBTASK 23-42-03-970-004

(1) At the forward ACP, do the steps that follow to view or report the fault data:

(a) In the ACP main menu column, push the Maintenance button.

1) Enter password if necessary.

NOTE: Depending on the ACP configuration, you may be required to provide a password. The default password is “1234”.

(b) In the Maintenance menu column, push the System Test button.

(c) In the Control Window, push the Start Test button.

NOTE: When the system test operates, the System Test and Start Test buttons are active, the other buttons become disabled and the status bar will display the approximate percent complete. When the test completes, the System Test and the Start Test buttons remain active and other buttons will be available for selection.

SUBTASK 23-42-03-970-007

(2) Do these steps to view the System Test Faults:

(a) In the Maintenance menu column, push the Fault Data button.

NOTE: After you push the Fault Data button, four buttons will appear in the control window: System Test Faults, Data Load Faults, Save All and Exit. The result window will show instructions for user to view or save the fault data.

(b) In the Control Window, push the System Test Fault button.

NOTE: If there are System Test Faults found, the ACP will display the number of faults with their maintenance message numbers and corrective actions. If there are no faults, a message will be displayed to indicate that there are no System Test Faults.

SUBTASK 23-42-03-970-008

(3) Do these steps to view the Data Load Faults:

NOTE: These faults are generated during data loading, you will see warning messages about the faults.

(a) Push the Data Load Faults button in the Control Window.

NOTE: The result window will show the faults with their maintenance message numbers and corrective actions. If there are no faults, a message will be displayed to indicate that there are no data load faults.

SUBTASK 23-42-03-970-009

(4) Do these steps to report all ACP fault data using the ONS:

(a) Push the Save All button in the Control Window.

(b) To report all faults for the ACP, do this task: LRU Reports, TASK 46-13-02-970-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-42-03-970-010

(5) To correct the faults, do the actions recommended on the screen or use the maintenance message number and refer to the IFIM and do the applicable procedure(s).

——— END OF TASK ———

TASK 23-42-03-700-801

4. Attendant Control Panel (ACP) - Configuration Check

A. General

(1) The Configuration Check provides part number information for the Attendant Control Panel (ACP)'s and each Lighting LRU. It displays part numbers for hardware and software, including the following software:

- Operational Software (OPS)
- Configuration Database (CDB)
- Standard Scene (LDB)
- Custom Scene (if applicable)

(2) Throughout the procedure, the ACP may automatically go into screensaver mode. To exit out of screensaver mode, touch two different corners of the ACP screen sequentially.

B. References

Reference	Title
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Prepare for the Configuration Check

SUBTASK 23-42-03-860-020

(1) Set the overhead panel P5-13, CAB/UTIL switch to the ON position.

SUBTASK 23-42-03-860-021

(2) Make sure that the forward ACP and aft ACP are ON.

SUBTASK 23-42-03-860-022

(3) Make sure the airplane is set to ground mode (TASK 32-09-00-860-802).

D. ACP Configuration Check

SUBTASK 23-42-03-700-003

(1) Do these steps for a configuration check of the ACP:

(a) Make sure that you know the correct software part numbers for the ACP.

(b) Push the Maintenance button on the ACP.

(c) Enter a password if necessary.

NOTE: Depending on the ACP configuration, you may be required to provide a password.

The default password is '1234'.

(d) Push the “Config Check” Button.

1) The “Configuration Check” screen is displayed.

(e) Push the “Start Check” button in the “Control Window”.

1) Wait for the “Status” progress bar to show 100%.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Make sure that the software part numbers for OPS, CDB, LDB and Custom Scene (if applicable) match the load list or installation plan.

NOTE: Use the scroll bar on the right to see all loaded software parts. Lighting part numbers display first, then the ACP part numbers.

- Firmware Revision: is only for Lighting OPS on Gen 3 lights
- Configuration Database LSAP P/N: is used for ACPs and Lighting LRUs
- Lighting Database (Standard) LSAP P/N: is used for ACPs and Lighting LRUs
- Lighting Database (Custom) LSAP P/N: is used for ACPs and Lighting LRUs
- OPS LSAP P/N: is used for only forward and aft ACP

1) If the software part numbers do not match, software installation is required.

a) Do the Attendant Control Panel (ACP) - Software Loading, TASK 23-42-03-470-801.

b) If the part numbers match, then continue.

——— END OF TASK ———

TASK 23-42-03-970-804

5. Attendant Control Panel (ACP) - Configuration Data Report

A. General

(1) This task lets you view the configuration data for the ACP.

(2) This task also lets you report or view the configuration data using the Onboard Network System (ONS).

(3) Throughout the procedure, the ACP may automatically go into screensaver mode. To exit out of screensaver mode, touch two different corners of the ACP screen sequentially.

B. References

Reference	Title
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
46-13-02-970-803	LRU Reports (P/B 201)

C. Prepare to View or Report

SUBTASK 23-42-03-860-025

(1) Make sure that the overhead panel P5-13, CAB/UTIL switch is set to the ON position.

SUBTASK 23-42-03-860-026

(2) Make sure that the forward ACP and aft ACP display the “Lighting” information screen.

SUBTASK 23-42-03-860-027

(3) Make sure that the airplane is set to ground mode (TASK 32-09-00-860-802).

D. Procedure

SUBTASK 23-42-03-970-005

(1) At the forward ACP, do the steps that follow to view or report the configuration data:

(a) In the ACP main menu column, push the Maintenance button.

1) Enter password if necessary.

NOTE: Depending on the ACP configuration, you may be required to provide a password. The default password is “1234”.

(b) In the Maintenance menu column, push the Config Check button.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) In the Control Window, push the Start Check button.

NOTE: When the check completes, the results will display part number information for the ACP's and each Lighting LRU. It displays part numbers for hardware and software, including the following software:

- Operational Software (OPS)
- Configuration Database (CDB)
- Standard Scene (LDB)
- Custom Scene (if applicable)

SUBTASK 23-42-03-970-006

(2) Do these steps to report all configuration data using the ONS:

(a) Push the Save Config Data button in the Control Window.

(b) To report all configuration data for the ACP, do this task: LRU Reports, TASK 46-13-02-970-803.

——— END OF TASK ———

TASK 23-42-03-070-801

6. Attendant Control Panel (ACP) - Erasing Mass Storage Area (MSA)

A. General

(1) This task has instructions to erase the Mass Storage Area (MSA) at the Forward Attendant Control Panel (ACP).

NOTE: Proceed with caution, by using this function, all LSAPs in MSA will be removed.

(2) Throughout the procedure, the ACP may automatically go into screensaver mode. To exit out of screensaver mode, touch two different corners of the ACP screen sequentially.

B. Erase the Mass Storage Area

SUBTASK 23-42-03-070-001

(1) Do these steps to erase the Mass Storage Area:

(a) Push the Maintenance button in the ACP main menu column.

(b) Enter password if necessary.

NOTE: Depending on the ACP configuration, you may be required to provide a password.

The default password is '1234'.

(c) Push the Data Load button in the Maintenance menu column.

(d) Push the Phase 2 Data Load button in the control window.

(e) Push the Erase MS Area button In the control window.

(f) Push the Start button In the control window.

NOTE: The result window will show the progress of the erase process.

——— END OF TASK ———

TASK 23-42-03-210-801

7. Attendant Control Panel (ACP) - LCD Touch Screen Display Check

A. General

(1) This task determines serviceability of the ACP LCD touch screen displays.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
23-42-03-000-801	Cabin Attendant LCD Control Panel Removal (P/B 401)
23-42-03-400-801	Cabin Attendant LCD Control Panel Installation (P/B 401)

C. LCD Touch Screen Display Check Procedure

SUBTASK 23-42-03-210-001

(1) Make sure that the LCD touch screen display is clear to read.

SUBTASK 23-42-03-210-002

(2) Count the number of defective pixels on the LCD screen of the ACP.

(a) Make sure there are no more than four defective pixels in total with no more than two defective pixels adjacent to each other.

NOTE: Defective is defined as a pixel that does not display the intended color with respect to adjacent pixels.

1) If the LCD touch screen does not meet this criteria, replace the ACP:

a) Do this task: Cabin Attendant LCD Control Panel Removal, TASK 23-42-03-000-801.

b) Do this task: Cabin Attendant LCD Control Panel Installation, TASK 23-42-03-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT LCD CONTROL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cabin attendant Liquid Crystal Display (LCD) control panel

(2) An installation of the cabin attendant LCD control panel.

B. The forward and aft cabin attendant control panels are installed on the aft face of the lavatory wall.

TASK 23-42-03-000-801

2. Cabin Attendant LCD Control Panel Removal

(Figure 401)

A. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 23-42-03-860-013

(1) Turn off emergency lights.

(a) Make sure that electrical power is supplied (TASK 24-22-00-860-801).

(b) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

(c) Set the EMER EXIT LIGHTS switch, on the P5 overhead panel, to the ARMED position.

1) Make sure that the applicable EMERGENCY LIGHTING CONTROL switch, on the attendant panel, is in the NORMAL position.

2) If you will remove electrical power from the airplane after removing the AFT Attendant Control Panel (ACP), set the EMER EXIT LIGHTS switch, on the P5 overhead panel, to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-42-03-860-014

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL
D	19	C01423	VACUUM WASTE

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	6	C01583	ENTERTAINMENT PA SYS BAT
D	4	C00082	COMMUNICATIONS PA AMPL BAT
D	12	C01936	ATTENDANT PANELS

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	21	C00084	INTPH AND WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	18	C01949	CABIN EMER LIGHTS

SUBTASK 23-42-03-010-001

(3) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 23-42-03-860-028

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	3	C00140	WORK AND THSHD LTS
C	9	C00138	WATER QTY IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

These circuit breakers are inoperative and should remain open:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	1	C01350	THRSHOLD LTS (INOP)
E	11	C00413	BRT ENTRY LTS (INOP)

SUBTASK 23-42-03-910-001

CAUTION

OBSERVE SPECIAL HANDLING INSTRUCTIONS FOR ESDS COMPONENT BEFORE INSTALLATION TO PREVENT POSSIBLE DAMAGE.

(5) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

E. Cabin Attendant LCD Control Panel Removal

SUBTASK 23-42-03-020-001

(1) Do these steps to remove the shroud [2]:

(a) Remove the screws [4] from the panel shroud [2].

(b) Pull the shroud [2] from the attendant panel [1].

SUBTASK 23-42-03-020-003

(2) Do these steps to remove the attendant panel [1]:

(a) Remove the screws [3] from the bottom of the attendant panel [1].

(b) Loosen the screws [3] at the top of the attendant panel [1].

CAUTION

BE CAREFUL WHEN YOU REMOVE OR INSTALL THE PANEL. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE ELECTRICAL CABLES AT THE REAR OF THE PANEL.

(c) Lift the attendant panel [1] and pull forward to disconnect the wire bundle [9].

(d) Remove the screw [6], washers [7], and clamps [8] from the back of the attendant panel [1].

(e) Disconnect the electrical conector [5] from the attendant panel [1].

(f) Remove the attendant panel [1].

SUBTASK 23-42-03-860-030

(3) If you will remove electrical power from the airplane after removing the AFT ACP, do these steps:

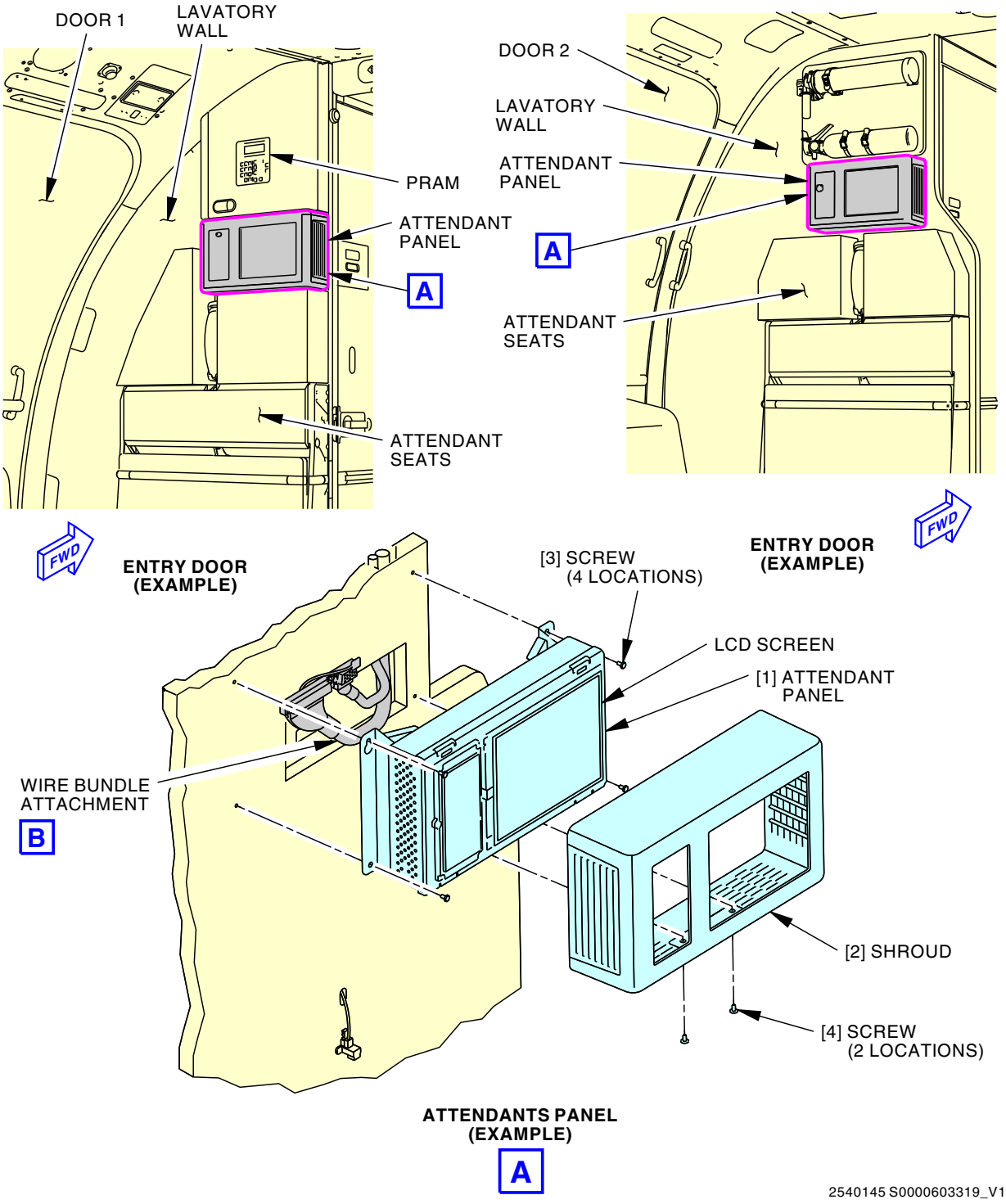
NOTE: Removing the AFT ACP and then removing electrical power will cause the emergency lights to turn on and stay on until they discharge.

(a) Install a jumper wire between the electrical connector J1, pins D17 and E16.

NOTE: The jumper wire will keep the emergency lights turned off.

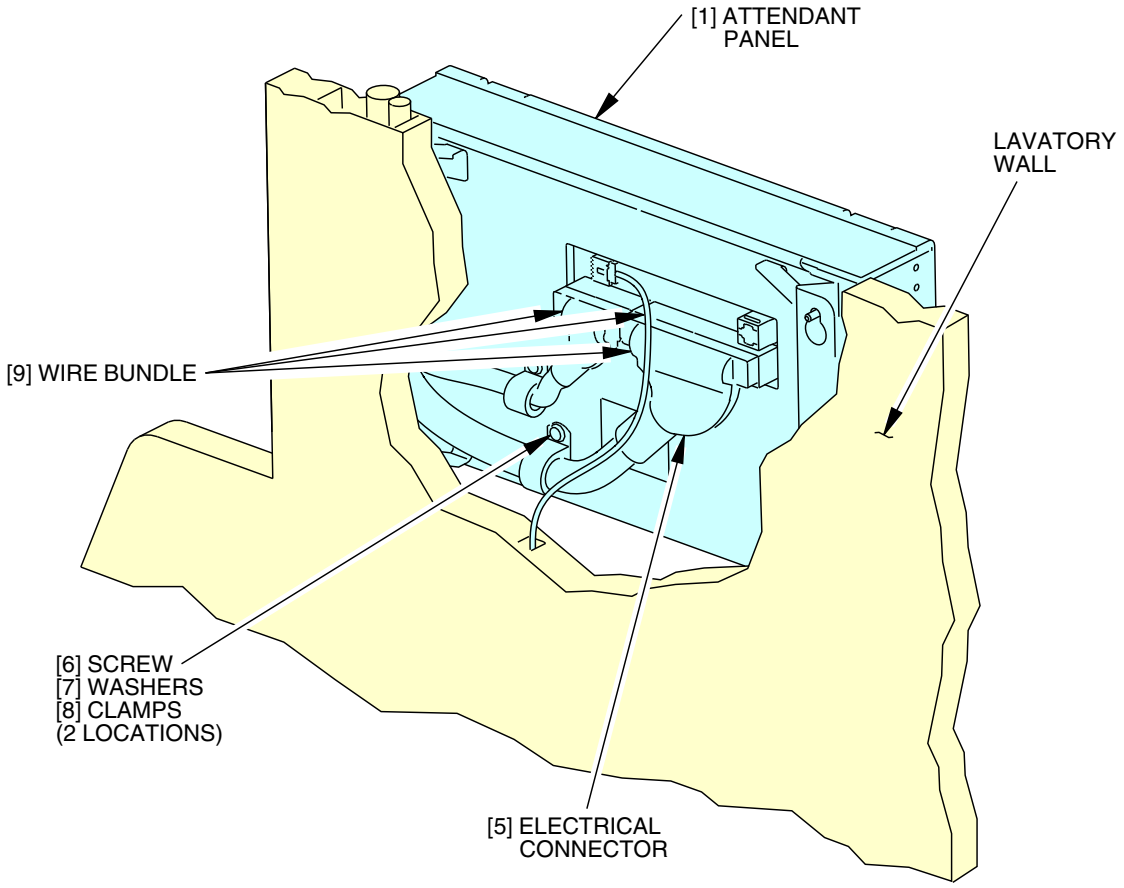
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Attendant LCD Control Panel Installation
Figure 401/23-42-03-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WIRE BUNDLE ATTACHMENT

B

2411915 S00061529455_V2

Cabin Attendant LCD Control Panel Installation
Figure 401/23-42-03-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-42-03-400-801

3. Cabin Attendant LCD Control Panel Installation

(Figure 401)

A. General

(1) This task has instructions to install the cabin attendant control panel (referred to as attendant panel [1]).

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-42-03-470-801	Attendant Control Panel (ACP) - Software Loading (P/B 201)
23-42-03-700-801	Attendant Control Panel (ACP) - Configuration Check (P/B 201)
23-42-03-710-801	Attendant Control Panel - Operational Test (P/B 501)
33-20-00-710-801	Passenger Compartment Lights - Operational Test (P/B 501)
33-51-00-720-801	Emergency Lights Operational Test (P/B 501)
33-51-06-610-801	Power Supply - Charge the Battery Packs (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Attendant panel	23-42-03-01-050	PTF ALL
		23-42-03-50-050	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Installation

SUBTASK 23-42-03-860-015

CAUTION

OBSERVE SPECIAL HANDLING INSTRUCTIONS FOR ESDS COMPONENT BEFORE INSTALLATION TO PREVENT POSSIBLE DAMAGE.

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

G. Cabin Attendant LCD Control Panel Installation

SUBTASK 23-42-03-420-001

(1) Do these steps to install the attendant panel [1]:

(a) If installed, remove the jumper wire from the electrical connector J1 in the AFT attendant panel [1].

(b) Connect the wire bundle [9] and electrical conector [5] to the attendant panel [1].

(c) Install the screws [6], washers [7], and clamps [8] to the attendant panel [1].

(d) Position the attendant panel [1] on the top screws [3] on the wall.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Install the bottom screws [3] to the attendant panel [1].

(f) Tighten the screws [3] attaching the attendant panel [1] to the wall.

SUBTASK 23-42-03-420-006

(2) (3) Do these steps to install the shroud [2]:

(a) Position the shroud [2] on the attendant panel [1].

(b) Install the screws [4] that attach the shroud [2] to the attendant panel [1].

H. Cabin Attendant LCD Control Panel Installation Test

SUBTASK 23-42-03-860-016

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL
D	19	C01423	VACUUM WASTE

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	6	C01583	ENTERTAINMENT PA SYS BAT
D	4	C00082	COMMUNICATIONS PA AMPL BAT
D	12	C01936	ATTENDANT PANELS

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	21	C00084	INTPH AND WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	18	C01949	CABIN EMER LIGHTS

SUBTASK 23-42-03-860-029

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	3	C00140	WORK AND THSHD LTS
C	9	C00138	WATER QTY IND

These circuit breakers are inoperative and should remain open:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	1	C01350	THRSHOLD LTS (INOP)
E	11	C00413	BRT ENTRY LTS (INOP)

SUBTASK 23-42-03-410-001

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

I. Put the Airplane Back to Its Usual Condition

SUBTASK 23-42-03-710-003

(1) Do this task: Emergency Lights Operational Test, TASK 33-51-00-720-801.

SUBTASK 23-42-03-860-017

(2) If it is necessary, do this task: Power Supply - Charge the Battery Packs, TASK 33-51-06-610-801.

SUBTASK 23-42-03-860-018

(3) Do the applicable tests for the light sections that can be controlled from the Attendant Control Panel (ACP) (TASK 33-20-00-710-801):

- LAMP TEST.
- Passenger seating area lights.
- Forward and aft entry area lights.
- Ceiling lights.

SUBTASK 23-42-03-710-004

(4) If a new panel is installed, do this task: Attendant Control Panel (ACP) - Configuration Check, TASK 23-42-03-700-801.

SUBTASK 23-42-03-860-019

(5) If it is necessary, do this task: Attendant Control Panel (ACP) - Software Loading, TASK 23-42-03-470-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-42-03-710-005

(6) Do this task: Attendant Control Panel - Operational Test, TASK 23-42-03-710-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT CONTROL PANEL - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Attendant Control Panel Operational Test

(2) Attendant Control Panel (ACP) BITE Procedure

TASK 23-42-03-710-801

2. Attendant Control Panel - Operational Test

A. General

(1) This task makes sure the Attendant Control Panel (ACP) operates correctly.

(2) The ACP controls and monitors several systems on the airplane. The ACP operational test consists of the tests for all systems that the ACP is associated with.

B. References

Reference	Title
23-42-00-710-801	Cabin Interphone System - Operational Test (P/B 501)

C. Procedure

SUBTASK 23-42-03-700-004

(1) Do the power-up test for the ACP:

NOTE: After power-up, the Over-Temperature Light Emitting Diode (LED) will turn on for 2 seconds and then turn off.

(a) Open and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

(b) Make sure that the LED at the upper left corner of the Integrated Switch Assembly (ISA).

NOTE: The LED should turn on in the first 2 seconds after power-up and then turn off.

NOTE: The ISA is the switch panel located to the left of the ACP's touchscreen.

1) If the LED light (amber) does not turn on in the first 2 seconds after power-up, refer to the IFIM and do the applicable procedure(s).

2) If the LED light continues to stay on after power-up, refer to the IFIM and do the applicable procedure(s).

(c) Make sure that the ACP Touch Screen responds to input.

1) If the ACP Touch Screen does not respond to any inputs, refer to the IFIM and do the applicable procedure(s).

SUBTASK 23-42-03-710-001

(2) Do this task: Cabin Interphone System - Operational Test, TASK 23-42-00-710-801

SUBTASK 23-42-03-710-002

(3) Do the test for the Water/Waste system at the AFT ACP:

(a) Push the ENVIRONMENT menu at the AFT ACP.

1) Make sure that you see the indicators for Potable Water, Vacuum Waste and LAV INOP.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure the Potable Water and Vacuum Waste indicators show the actual system levels.

——— END OF TASK ———

TASK 23-42-03-740-801

3. Attendant Control Panel (ACP) - BITE Procedure

A. General

(1) This procedure provides the system test for the Attendant Control Panel (ACP)s. The system test is performed at the forward ACP.

(a) For the system test, the forward ACP will initiate the Built-In-Test Equipment (BITE) test on both the forward and the aft ACPs. The BITE test will check the ACPs for internal faults and their communication status. The forward ACP will then query the light Line Replaceable Unit (LRU)s for fault information. If there are ACP or lighting faults, the ACP will show the faults at the end of the test.

(2) The procedure also provides the task to view faults that are generated during data loading.

B. System Test Procedure

SUBTASK 23-42-03-730-001

(1) At the forward ACP, do the steps that follow to start the system test:

(a) In the ACP main menu column, push the Maintenance button.

(b) In the Maintenance menu column, push the System Test button.

(c) In the control window, push the Start Test button.

NOTE: When the system test operates, the System Test and Start Test buttons are active, the other buttons become disabled and the status bar will display the approximate percent complete. When the test completes, the System Test and the Start Test buttons remain active and other buttons will be available for selection.

(d) Check the result window for the system test faults.

NOTE: The result window will show the faults with their maintenance message numbers and corrective actions. If there are no faults, a message will be displayed to indicate that there are no system test faults.

(e) To correct the faults, do the actions recommended on the screen or use the maintenance message number and refer to the IFIM and do the applicable procedure(s).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ATTENDANT CONTROL PANEL - CLEANING/PAINTING

1. General

A. This procedure has this task:

(1) Attendant Control Panel (ACP) Touchscreen Cleaning

TASK 23-42-03-100-802

2. Attendant Control Panel (ACP) Touchscreen Cleaning

A. General

(1) This task has instructions to clean the LCD touchscreen.

B. Consumable Materials

Reference	Description	Specification
B50073	Alcohol - Isopropyl	ASTM D 770
G01043	Cloth - Lint-free	

C. LCD Touchscreen Cleaning Procedure

SUBTASK 23-42-03-160-001

(1) Clean the LCD Touchscreen:

(a) Push the Clean screen button in the Display menu.

(b) Clean the LCD Touchscreen assembly with a lint-free cloth, G01043 and alcohol, B50073.

NOTE: Lint-free cloth wipers with alcohol are permitted.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INLINE FILTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of an inline filter.

(2) Installation of an inline filter.

B. The inline filters are installed above the ceiling in the forward and aft sections of the passenger compartment.

TASK 23-42-04-000-801

2. Inline Filter Removal

Figure 401

A. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Inline filter	23-42-04-01-025	PTF ALL
		23-42-04-05-025	PTF ALL

C. Location Zones

Zone	Area
222	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

D. Removal Procedure

SUBTASK 23-42-04-840-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

SUBTASK 23-42-04-010-001

(2) To get access to the inline filter, do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

SUBTASK 23-42-04-840-002

(3) Before you touch the inline filter, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 23-42-04-020-001

(4) Do the steps that follow to remove the inline filter [1]:

(a) Disconnect the electrical connector from the inline filter [1].

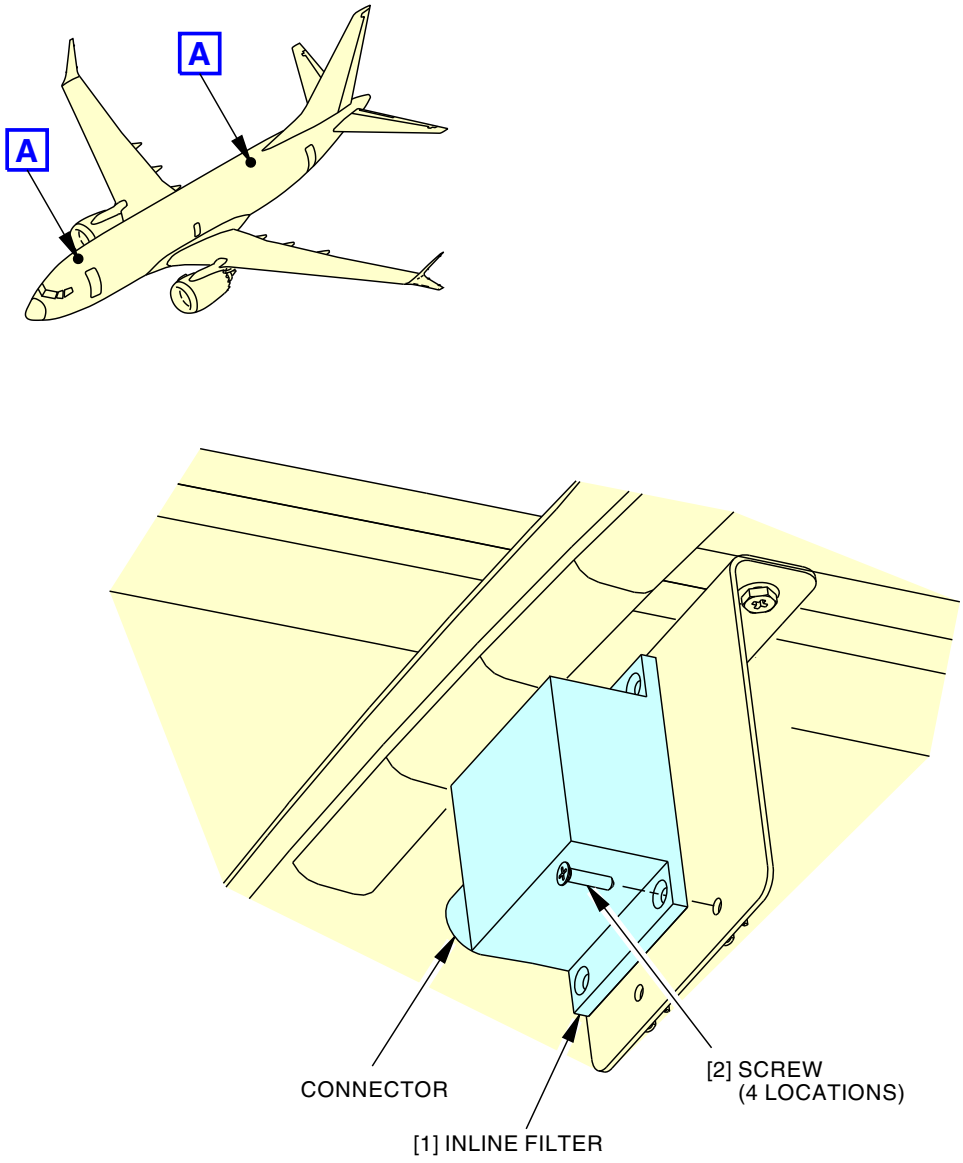
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the four screws [2] that attach the inline filter [1] to the mounting bracket.

(c) Remove the inline filter [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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INLINE FILTER INSTALLATION
Figure 401/23-42-04-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-42-04-400-801

3. Inline Filter Installation

Figure 401

A. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
23-42-00-710-801	Cabin Interphone System - Operational Test (P/B 501)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Inline filter	23-42-04-01-025	PTF ALL
		23-42-04-05-025	PTF ALL

C. Location Zones

Zone	Area
222	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left

D. Installation Procedure

SUBTASK 23-42-04-840-003

(1) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

SUBTASK 23-42-04-840-004

(2) Before you touch the inline filter, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 23-42-04-420-001

(3) Do the steps that follow to install the inline filter [1]:

(a) Attach the inline filter [1] to the mounting bracket with the screws [2] from the removal.

(b) Connect the electrical connector to the inline filter [1].

SUBTASK 23-42-04-410-001

(4) Install the main ceiling panel:

Do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

SUBTASK 23-42-04-840-005

(5) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	12	C01936	ATTENDANT PANELS

E. Installation Test

SUBTASK 23-42-04-700-001

(1) Do voice communication test between the flight compartment and the attendant stations and make sure you can hear the voice clearly. Refer to the task: Cabin Interphone System - Operational Test, TASK 23-42-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT AND GROUND CREW CALL SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Flight and Ground Crew Call System - Operational Test.

TASK 23-43-00-710-801

2. Flight and Ground Crew Call System - Operational Test

A. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Flight and Ground Crew Call System Operational Test

SUBTASK 23-43-00-710-001

(1) Do an operational test of the forward attendant to the aft attendant call:

(a) Do these steps from each forward attendant handset to each aft attendant handset.

1) Push the button for attendant on the handset at the forward attendant's station.

a) Make sure that you hear a chime on the PA speakers.

b) Make sure that the aft pink attendant call light comes on.

2) Push the button for reset on the handset at the aft attendant's station.

a) Make sure that the attendant call light goes off.

3) Speak into the forward attendant's handset.

a) Make sure that you can hear voices on the aft attendant's handset.

4) Speak into the aft attendant's handset.

a) Make sure that you can hear voices on the forward attendant's handset.

SUBTASK 23-43-00-710-002

(2) Do an operational test of the aft attendant to the forward attendant call:

(a) Do these steps from each aft attendant handset to each forward attendant handset.

1) Push the button for attendant on the handset at the aft attendant's station.

a) Make sure that you hear a chime on the PA speakers.

b) Make sure that the pink attendant call light at the forward attendant's station comes on.

2) Push the button for reset on the handset at the forward attendant's station.

a) Make sure that the attendant call light goes off.

SUBTASK 23-43-00-710-003

(3) Do an operational test of the flight crew to the attendant call:

(a) Push the ATTEND button on the P5 Overhead Panel.

1) Make sure that the pink attendant call lights at the forward and aft attendant's stations come on.

2) Make sure that you hear a two-tone chime in the passenger cabin.

(b) Push the button for reset on the handset at the forward attendant's station.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the attendant call light goes off.

SUBTASK 23-43-00-710-004

(4) Do an operational test of the attendant to the flight crew call:

(a) Do these steps from each forward attendant handset and from each aft attendant handset.

1) Push the button for the pilot on the forward or aft attendant's handset.

a) Make sure that you hear a chime in the flight compartment.

b) Make sure that the CABIN interphone call light on each audio control panel comes on.

c) Make sure that the CABIN microphone selector switch is depressed on the audio control panels.

2) Speak into the attendant's handset.

a) Make sure that you can hear voices on the pilot's headset.

3) Speak into the pilot's boom microphone.

a) Make sure that you can hear voices on the attendant's handset.

SUBTASK 23-43-00-710-005

WARNING DO NOT WEAR THE HEADSET OR TOUCH ANY UMBILICAL CONNECTIONS TO THE AIRPLANE DURING ATMOSPHERIC ELECTRICAL DISTURBANCES. A LIGHTNING STRIKE CAN CAUSE DANGEROUS INJURY TO PERSONS.

(5) Do an operational test of the ground crew call horn:

(a) Push and hold the GRD CALL button on the P5 Overhead Panel in the flight compartment.

1) Make sure that you hear the ground crew call horn at the nose wheel well area.

(b) Release the GRD CALL button.

SUBTASK 23-43-00-710-007

(6) Do the operational test of the external power panel, P19:

(a) Push and hold the PILOT CALL button on the external power P19 panel at the nose wheel well area.

1) Make sure that you hear a chime in the flight compartment.

2) Make sure that the flight interphone call light on each audio control panel comes on. The flight interphone is labeled Flt, Int, or Flt Int.

(b) Release the PILOT CALL button.

PTF ALL POST SB 737-24-1236

SUBTASK 23-43-00-710-008

(7) Do an operation test of the nose wheel well interphone box:

WARNING DO NOT WEAR THE HEADSET OR TOUCH ANY UMBILICAL CONNECTIONS TO THE AIRPLANE DURING ATMOSPHERIC ELECTRICAL DISTURBANCES. A LIGHTNING STRIKE CAN CAUSE DANGEROUS INJURY TO PERSONS.

(a) Connect a headset to the interphone box jack.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-24-1236 (Continued)

(b) Push and hold the FLIGHT DECK CALL button on the interphone box.
   1) Make sure that you hear a chime in the flight compartment.
   2) Release the FLIGHT DECK CALL button.

(c) Push and hold the Push-To-Talk (PTT) switch on the interphone box headset.
   1) Speak into the interphone box headset.
   2) Make sure that you hear a voice on the flight compartment.
   3) Release the PTT switch on the interphone box headset.

(d) On the first officer control wheel, set and hold the PTT switch to the INT position.
   1) Speak into the first officer microphone.
   2) Make sure that you hear a voice on the interphone box headset.
   3) Release the PTT switch to the OFF position.

(e) Disconnect the headset from the interphone box jack.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GROUND CREW CALL HORN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Ground crew call horn removal.

(2) Ground crew call horn installation.

B. The ground crew call horn is installed at the nose wheel well area.

TASK 23-43-02-000-801

2. Ground Crew Call Horn Removal

(Figure 401)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left

C. Ground Crew Call Horn Removal

SUBTASK 23-43-02-760-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	17	C01010	ADIRU RIGHT DC

SUBTASK 23-43-02-020-001

(2) Remove the ground crew call horn [1]:

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR BEFORE YOU REMOVE THE GROUND-CREW CALL HORN. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

(b) Remove the bolts [3] and washers [2] from the ground crew call horn [1].

(c) Pull the ground crew call horn [1] out until access to the terminals can be gotten.

(d) Remove the nuts [4] from the ground crew call horn [1] terminals.

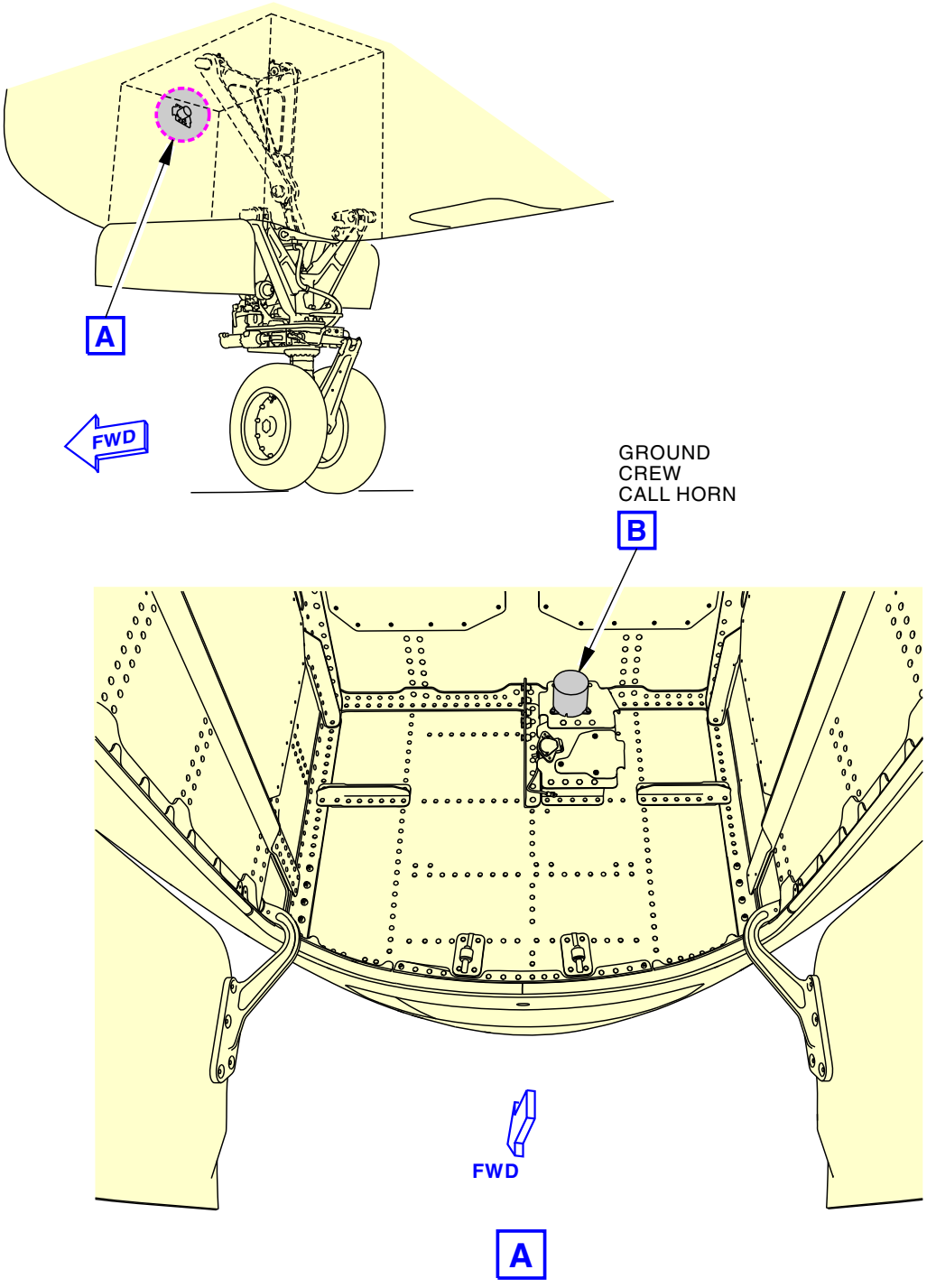
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Disconnect the electrical connectors [5] from the ground crew call horn [1] terminals.

(f) Remove the ground crew call horn [1].

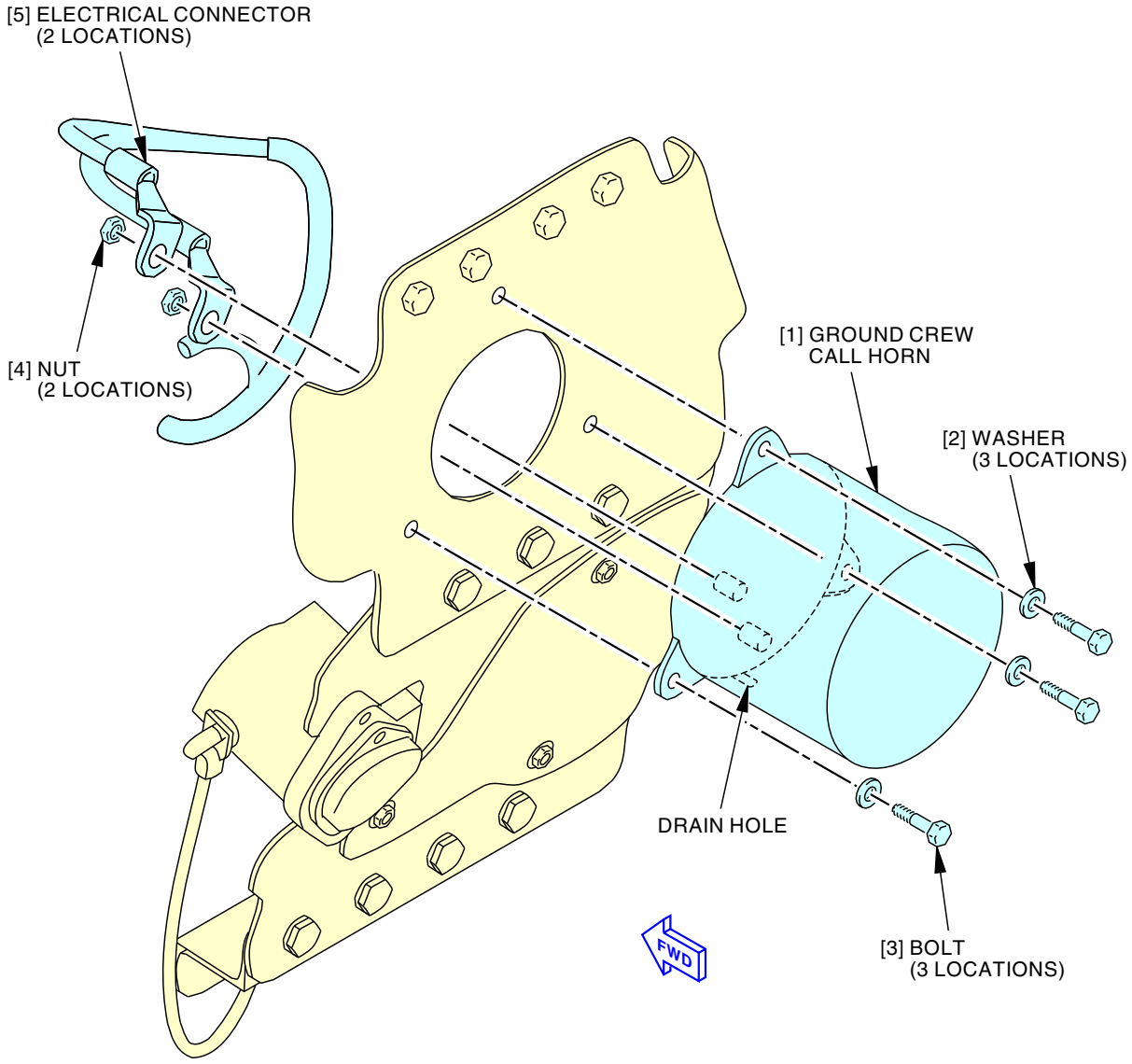
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Crew Call Horn Installation
Figure 401/23-43-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GROUND CREW CALL HORN

B

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Ground Crew Call Horn Installation
Figure 401/23-43-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-43-02-420-801

3. Ground Crew Call Horn Installation

(Figure 401)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Ground crew call horn	23-43-02-01-005	PTF ALL

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left

D. Ground Crew Call Horn Installation

SUBTASK 23-43-02-760-002

(1) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	17	C01010	ADIRU RIGHT DC

SUBTASK 23-43-02-420-001

(2) Install the ground crew call horn [1]:

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR BEFORE YOU REMOVE THE GROUND-CREW CALL HORN. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

(b) Connect the electrical connectors [5] to the ground crew call horn [1] terminals.

NOTE: The ground crew call horn is polarity sensitive and will not operate if the positive connector is not connected to the positive terminal above the +28V label.

(c) Install and tighten the nuts [4] that attach the electrical connectors [5] to the ground crew call horn [1] terminals.

(d) Put the ground crew call horn [1] in its position over the screw holes in the forward area of the nose wheel well.

1) Make sure that the drain hole points down to let the moisture drain.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Install the bolts [3] and washers [2] that attach the ground crew call horn [1] to the airplane.

SUBTASK 23-43-02-760-003

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	17	C01010	ADIRU RIGHT DC

E. Installation Test

SUBTASK 23-43-02-700-001

(1) Do an installation test of the ground crew call horn [1]:

(a) Push the GRD CALL switch momentarily on the P5 pilot overhead panel.

(b) Make sure that the ground crew call horn [1] operates.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE WHEEL WELL INTERPHONE BOX - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the nose wheel well interphone box

(2) An installation of the nose wheel well interphone box.

B. The nose wheel well interphone box is installed at the nose wheel well area.

TASK 23-43-03-000-801

2. Nose Wheel Well Interphone Box Removal

(Figure 401)

A. General

(1) This task is for the removal of the nose wheel well interphone box.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

C. Location Zones

Zone	Area
116	Nose Landing Gear Wheel Well - Right

D. Access Panels

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

E. Prepare for the Removal

SUBTASK 23-43-03-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

SUBTASK 23-43-03-860-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 23-43-03-010-001

(3) Remove this access panel:

(TASK 53-14-01-020-801)

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Interphone Box Removal

SUBTASK 23-43-03-020-001

(1) Remove the interphone box [1]:

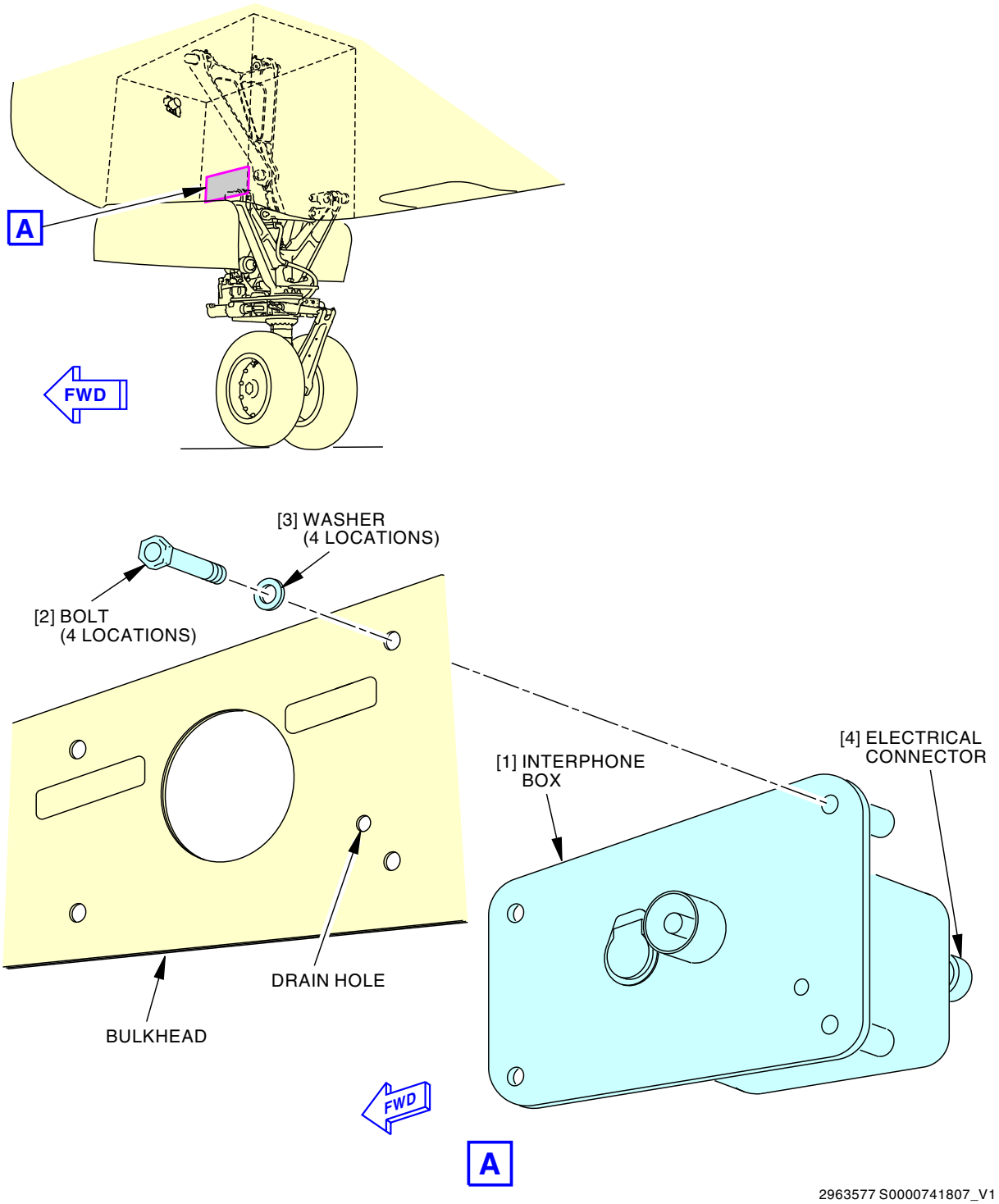
(a) Disconnect the electrical connector [4] from the interphone box [1].

(b) Remove the bolts [2] and washers [3] from the interphone box [1] and bulkhead.

(c) Remove the interphone box [1] from the bulkhead.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Wheel Well Interphone Box Installation
Figure 401/23-43-03-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-43-03-400-801

3. Nose Wheel Well Interphone Box Installation

(Figure 401)

A. General

(1) This task is for the installation of the nose wheel well interphone box.

B. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
23-43-00-710-801	Flight and Ground Crew Call System - Operational Test (P/B 501)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
51-22-00-390-802	Removable Faying (Mated) Surface Seal Application (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50368	Agent - Peelable Parting (Rexco Chemical Company - Partall Coverall Film)	BAC5000

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Interphone box	Not Specified	

E. Location Zones

Zone	Area
116	Nose Landing Gear Wheel Well - Right

F. Access Panels

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

G. Prepare for the Installation

SUBTASK 23-43-03-860-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Interphone Box Installation

SUBTASK 23-43-03-420-001

(1) Install the interphone box [1]:

(a) Use cotton wiper, G00034, moist in solvent, B00148, or solvent, B00666, to clean the mating surfaces of the interphone box [1] and bulkhead.

(b) Apply Rexco Partall Coverall Film peelable parting agent, G50368, to the bulkhead faying surface.

(c) Apply sealant, A00247, between the interphone box [1] and bulkhead for a faying surface seal (TASK 51-22-00-390-802).

1) Make sure that there is no sealant in the drain hole.

2) Make sure that the squeeze-out around the large diameter hole in the bulkhead is sufficient and continuous.

3) Remove excessive sealant after the squeeze out.

(d) Install the interphone box [1] to the bulkhead with the bolts [2] and washers [3].

(e) Apply a fillet seal around the interphone box [1] with sealant, A00247 (TASK 51-22-00-390-806).

(f) Connect the electrical connector [4] to the interphone box [1].

SUBTASK 23-43-03-860-003

(2) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	9	C00073	PASSENGER CABIN CREW CALL

I. Interphone Box Installation Test

SUBTASK 23-43-03-710-001

(1) Do an operation test of the nose wheel well interphone box (TASK 23-43-00-710-801).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 23-43-03-410-001

(1) Install this access panel:

(TASK 53-14-01-420-801)

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

SUBTASK 23-43-03-790-001

(2) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT INTERPHONE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Flight interphone system operational test

(2) Oxygen mask microphone operational test.

TASK 23-51-00-710-801

2. Flight Interphone System Operational Test

A. General

(1) This task is for the test of the flight interphone system.

(2) This task requires two persons to accomplish.

B. References

Reference	Title
35-12-00-440-801	Crew Oxygen System - Activation (P/B 201)
35-12-85-910-801	Crew Oxygen Mask Stowage (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 23-51-00-860-005

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	24	C00085	AUDIO OBS

SUBTASK 23-51-00-860-006

(2) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
B	7	C00629	GND PROX WARN

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	4	C00082	COMMUNICATIONS PA AMPL BAT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	23	C00239	INTERPHONE POWER CAPT DC 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	23	C00083	AUDIO CAPT

SUBTASK 23-51-00-850-001

(3) Set the NORM/ALT switch on the Audio Control Panels (ACP) to the NORM position.

SUBTASK 23-51-00-850-047

(4) Push the INT mic selector on the Captain's Audio Control Panel (ACP) to ON and do these steps:

(a) Make sure that the VHF 2 is available on the First Officer's ACP.

(b) Make sure that the VHF 1 is available on the Observer's ACP.

SUBTASK 23-51-00-860-007

(5) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	23	C00239	INTERPHONE POWER CAPT DC 2

SUBTASK 23-51-00-860-008

(6) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	24	C00240	INTERPHONE POWER CAPT BAT

SUBTASK 23-51-00-850-048

(7) Push the INT mic selector on the Captain's ACP to ON and do these steps:

(a) Make sure that the VHF 2 is available on the First Officer's ACP.

(b) Make sure that the VHF 1 is available on the Observer's ACP.

SUBTASK 23-51-00-860-009

(8) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	23	C00083	AUDIO CAPT

SUBTASK 23-51-00-860-010

(9) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	24	C00085	AUDIO OBS

SUBTASK 23-51-00-850-049

(10) Push the INT mic selector on the Observer's ACP to ON and do these steps:

(a) Make sure that the VHF 1 is available on the Captain's ACP.

(b) Make sure that the VHF 2 is available on the First Officer's ACP.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-00-860-011

(11) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	24	C00240	INTERPHONE POWER CAPT BAT

SUBTASK 23-51-00-860-012

(12) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	23	C00239	INTERPHONE POWER CAPT DC 2

SUBTASK 23-51-00-850-050

(13) Push the INT mic selector on the Observer's ACP to ON and do these steps:

(a) Make sure that the VHF 1 is available on the Captain's ACP.

(b) Make sure that the VHF 2 is available on the First Officer's ACP.

SUBTASK 23-51-00-860-013

(14) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	23	C00239	INTERPHONE POWER CAPT DC 2
D	24	C00085	AUDIO OBS

SUBTASK 23-51-00-860-014

(15) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
D	22	C00086	AUDIO F/O

SUBTASK 23-51-00-850-051

(16) Push the INT mic selector on the First Officer's ACP to ON and do these steps:

(a) Make sure that the VHF 1 is available on the Captain's ACP.

(b) Make sure that the VHF 1 is available on the Observer's ACP.

SUBTASK 23-51-00-860-015

(17) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2

SUBTASK 23-51-00-860-016

(18) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	22	C00561	INTERPHONE POWER F/O BAT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-00-850-052

(19) Push the INT mic selector on the First Officer's ACP to ON and do these steps:

(a) Make sure that the VHF 1 is available on the Captain's ACP.

(b) Make sure that the VHF 1 is available on the Observer's ACP.

SUBTASK 23-51-00-860-017

(20) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-51-00-860-018

(21) Make sure that the GPWS INOP indicator light on the First Officer's Panel is off.

SUBTASK 23-51-00-850-008

(22) Set the switches on the GROUND PROXIMITY MODULE to the NORMAL position.

E. Flight Interphone System Operational Test

SUBTASK 23-51-00-780-001

(1) Make sure that the oxygen system is pressurized (TASK 35-12-00-440-801).

NOTE: The switch in the stowage box will not set if the oxygen system is not pressurized.

SUBTASK 23-51-00-850-010

(2) Set all ACP as follows:

(a) Push the INT microphone selector switch to the ON position.

(b) Make sure that its light comes on.

(c) Push the volume control for the INT microphone selector switch.

(d) Turn the volume control for the INT microphone selector switch clockwise to the middle position.

SUBTASK 23-51-00-850-012

(3) Set the SPKR switch on the Captain's and the First Officer's ACP as follows:

(a) Set the SPKR switch on each ACP to the ON position.

(b) Make sure that its light comes on.

(c) Turn the SPKR volume control switch clockwise to the middle position or to the volume level you are comfortable with.

SUBTASK 23-51-00-850-017

(4) Use a headset for all headphone jacks.

SUBTASK 23-51-00-850-016

(5) Hold the Push-To-Talk (PTT) switch on the Captain's hand microphone.

SUBTASK 23-51-00-860-019

(6) Speak into the Captain's hand microphone and do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that you hear sound in the headphone applicable to the ACP.

NOTE: Use a oxygen mask for a low volume level from the cockpit speaker. Sound from an external source will have low volume from the cockpit speaker.

SUBTASK 23-51-00-860-020

(7) Make sure that the voice level from both Captain's speaker and First Officer's speaker decreases.

SUBTASK 23-51-00-860-021

(8) Hold the PTT switch on the First Officer's hand microphone and do these steps:

(a) Speak into the First Officer's hand microphone.

(b) Make sure that you hear sound in the Captain's Headphone.

(c) Make sure that the voice level from both Captain's speaker and First Officer's speaker decreases.

SUBTASK 23-51-00-860-022

(9) Hold the PTT switch on the Observer's hand microphone and do these steps:

(a) Speak into the Observer's hand microphone.

NOTE: The Observer's hand microphone is on the P61 panel.

(b) Make sure that you hear sound from the Captain's Headphone.

SUBTASK 23-51-00-860-024

(10) On the external power panel, connect the ground cord and headset to the FLT INT Jack.

SUBTASK 23-51-00-850-020

(11) Hold the PTT switch on the ground headset and do these steps:

(a) Speak into the headset microphone.

(b) Make sure that you hear voice on the cockpit speaker.

(c) Release the PTT switch on the ground headset.

SUBTASK 23-51-00-850-021

(12) Hold the PTT switch on the First Officer's Control Wheel to the lower INT position and do these steps:

(a) Speak into the First Officer's microphone.

(b) Make sure that you hear a voice on the ground headset.

(c) Release the PTT switch on the First Officer's Control Wheel to the OFF position.

SUBTASK 23-51-00-850-022

(13) Set the PA Microphone button on all the ACP.

SUBTASK 23-51-00-850-023

(14) Turn the PA volume controls to approximately 1/2 MAX on all the ACP.

SUBTASK 23-51-00-850-024

(15) Hold the PTT switch on the Captain's Control Wheel in the lower INT position and do these steps:

(a) Speak into the Captain's microphone.

(b) Make sure that you hear a voice on the First Officer's headset.

(c) Make sure that there is no PA announcement in the cabin area.

(d) Release the PTT switch on the Captain's Control Wheel to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-00-850-025

(16) Hold the PTT switch on the First Officer's Control Wheel in the lower INT position. and do these steps:

(a) Speak into the First Officer's microphone.

(b) Make sure that there is a voice on the Captain's headset.

(c) Make sure that there is no PA announcement in the cabin area.

(d) Release the PTT switch on the First Officer's Control Wheel to the OFF position.

NOTE: The microphone for the oxygen mask removed from the stowage box is on.

SUBTASK 23-51-00-020-001

(17) Remove the oxygen masks from the stowage boxes

SUBTASK 23-51-00-710-004

(18) Test the Captain/First Officer oxygen mask microphone as follows:

NOTE: Do a test of the Captains's then the First Officer's oxygen mask microphone.

(a) Hold the PTT switch on the applicable control wheel in the MIC position.

(b) Speak into the oxygen mask microphone.

(c) Make sure that you hear voice from the cockpit speaker.

(d) Release the PTT switch.

SUBTASK 23-51-00-850-027

(19) Test the PTT switches on the glareshield as follows:

NOTE: Do a test of the Captains's then the First Officer's PTT switch.

(a) Hold the PTT glareshield switch.

(b) Speak into the oxygen mask microphone.

(c) Make sure that you hear a voice from the cockpit speaker.

(d) Make sure that there is a voice from the cabin speakers.

(e) Release the PTT glareshield switch.

SUBTASK 23-51-00-710-005

(20) Test the Observer's mask microphone as follows:

(a) Set the RADIO-INT switch to the RADIO position at the Observer's ACP.

(b) Speak into the Observer's oxygen mask microphone.

(c) Make sure that you hear voice from the cockpit speaker.

(d) Release the RADIO-INT switch.

SUBTASK 23-51-00-420-001

(21) Put the oxygen masks back into the stowage boxes (TASK 35-12-85-910-801).

F. ALT/NORM Switch Test

SUBTASK 23-51-00-850-041

(1) Set the Captain's ALT/NORM switch to the ALT position on the ACP and do this step:

(a) Make sure that the VHF 1 switch light comes on.

SUBTASK 23-51-00-850-042

(2) Set the Observer's ALT/NORM switch to the ALT position on the ACP and do this step:

(a) Make sure that the VHF 1 switch light comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-00-850-043

(3) Set the First Officer's ALT/NORM switch to the ALT position on the ACP and do this step:

(a) Make sure that the VHF 2 switch light comes on.

SUBTASK 23-51-00-850-044

(4) Make sure that the ALT/NORM switch on the ACP is in the NORM position.

G. AURAL Warning Signal to Headsets Test

SUBTASK 23-51-00-850-045

(1) Push the GPWS SYS TEST BUTTON for approximately 2 seconds on the GROUND PROXIMITY MODULE and do this step:

NOTE: The self test switch pushed for longer than 2 seconds will cancel the self test.

(a) Make sure that there are AURAL warning sounds from the headsets.

——— END OF TASK ———

TASK 23-51-00-710-802

3. Oxygen Mask Microphone Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the operational test of the oxygen masks microphones.

B. References

Reference	Title
35-12-85-910-801	Crew Oxygen Mask Stowage (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Oxygen Mask Microphone Operational Test

SUBTASK 23-51-00-210-001

(1) Make sure that the flight crew oxygen system is pressurized.

SUBTASK 23-51-00-210-002

(2) Make sure that the SERVICE INTERPHONE switch on the aft overhead panel is in the OFF position.

SUBTASK 23-51-00-860-025

(3) Set all Audio Control Panels (ACP) as follows:

(a) Push the PA MIC SELECTOR switch to the ON position.

(b) Make sure that the MIC SELECTOR lamp comes on.

(c) Push the volume control for the INT microphone selector switch.

(d) Turn the volume control for the INT microphone selector switch clockwise to the middle position.

SUBTASK 23-51-00-860-026

(4) Set the Captain's and First Officer's ACP as follows:

(a) Push the SPKR volume control switch on each ACP to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that its light comes on.

(c) Turn the SPKR volume control switches clockwise to the middle position or to the desired volume level.

SUBTASK 23-51-00-010-001

(5) Remove oxygen masks from the stowage boxes.

SUBTASK 23-51-00-710-001

(6) Test the Captain/First Officer oxygen mask microphone as follows:

NOTE: Do a test of the Captains's then the First Officer's oxygen mask microphone.

(a) Hold the Push-To-Talk (PTT) switch on the applicable control wheel in the MIC position.

(b) Speak into the oxygen mask microphone.

(c) Make sure that you hear voice from the cockpit speaker.

(d) Make sure that there is a voice from the cabin speakers.

(e) Release the PTT switch.

SUBTASK 23-51-00-710-003

(7) Test the PTT switches on the glareshield as follows:

NOTE: Do a test of the Captains's then the First Officer's PTT switch.

(a) Hold the PTT glareshield switch.

(b) Speak into the oxygen mask microphone.

(c) Make sure that you hear a voice from the cockpit speaker.

(d) Make sure that there is a voice from the cabin speakers.

(e) Release the PTT glareshield switch.

SUBTASK 23-51-00-710-002

(8) Test the first observer's mask microphone as follows:

(a) Set the RADIO-INT switch to the RADIO position at the observer's Audio Control Panel (ACP).

(b) Speak into the first observer's oxygen mask microphone.

(c) Make sure that there is voice from the cockpit speaker.

(d) Release the RADIO-INT switch.

SUBTASK 23-51-00-410-001

(9) Put the oxygen masks back into the stowage boxes (TASK 35-12-85-910-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

REMOTE ELECTRONICS UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Remote Electronics Unit Removal

(2) Remote Electronics Unit Installation.

B. The remote electronics unit is on the E4-1 shelf in the electrical/electronic compartment.

TASK 23-51-01-000-801

2. Remote Electronics Unit Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the remote electronics unit [1].

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Remote Electronics Unit Removal

SUBTASK 23-51-01-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	6	C01583	ENTERTAINMENT PA SYS BAT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-01-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 23-51-01-860-002

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Before the remote electronics unit [1] is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

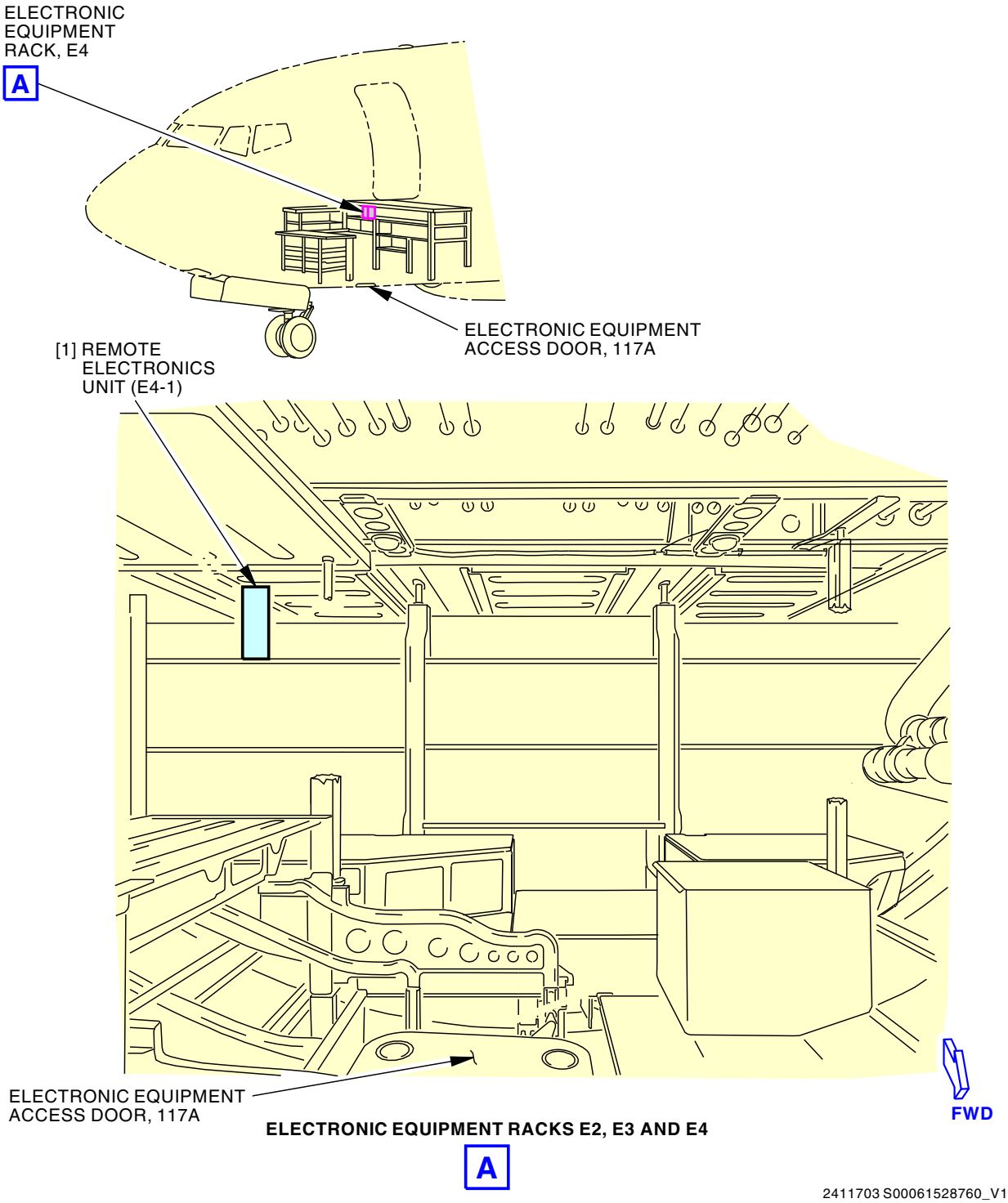
F. Remote Electronics Unit Removal

SUBTASK 23-51-01-020-001

(1) Remove the remote electronics unit [1] (TASK 20-10-07-000-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Remote Electronics Unit Installation
Figure 401/23-51-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-51-01-400-801

3. Remote Electronics Unit Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the remote electronics unit [1].

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Remote electronics unit	23-51-01-01-010	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Remote Electronic Unit Installation

SUBTASK 23-51-01-860-003

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	6	C01583	ENTERTAINMENT PA SYS BAT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-51-01-860-004

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(2) Before the remote electronics unit [1] is touched, do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Remote Electronics Unit Installation

SUBTASK 23-51-01-420-001

(1) Install the remote electronics unit [1] (TASK 20-10-07-400-801).

SUBTASK 23-51-01-200-001

(2) If there is an AURAL WARN MUTE switch on the front panel of the remote electronics unit [1], adjust the switch to the applicable airline standard as follows:

NOTE: If no AURAL WARN MUTE switch is present, the REU is preset.

(a) Vertical Position - Aural warnings are delivered to the flight deck headsets and speakers.

(b) Horizontal Position - Aural warnings are muted from the flight deck headsets and are only heard through the flight deck speakers.

H. Remote Electronics Unit Installation Test

SUBTASK 23-51-01-860-005

(1) Close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	6	C01583	ENTERTAINMENT PA SYS BAT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-51-01-710-001

(2) Do the operational test for the remote electronics unit [1]:

(a) Set the MIC button on the audio control panel's to the ON position.

(b) Make sure that the MIC buttons come on for the audio control panel's.

(c) Turn the volume controls to approximately 1/2 MAX on the audio control panel's.

(d) Set the SPKR volume control to the ON position.

(e) Turn the SPKR volume control on the Captain’s Audio Control Panel to approximately 1/2 MAX.

(f) Turn the SPKR volume control on the First Officer’s Audio Control Panel to approximately 1/2 MAX.

(g) Make sure that you hear an announcement between the Captain, First Officer and Observer when you use Hand Microphones, Headphones, and Boom Mic headsets.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 23-51-01-860-006

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AUDIO CONTROL PANEL - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Audio Control Panel (ACP) mic select switch lamp replacement.

TASK 23-51-02-300-801

2. Audio Control Panel - Mic Select Switch Lamp Replacement

(Figure 201)

A. General

(1) This task is to replace the mic selector switch lamps when the lamps do not come on and the switch is still functional.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Mic Selector Switch Lamp Replacement

SUBTASK 23-51-02-040-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 23-51-02-300-001

(2) Get access to the thumbnail slots on the perimeter of the switch bezel.

SUBTASK 23-51-02-300-002

CAUTION

DO NOT USE SHARP TOOLS TO PULL THE SWITCH BEZEL. IF YOU DO NOT OBEY, DAMAGE TO THE SEAL THAT IS AROUND THE SWITCH CAN OCCUR.

(3) Carefully pull the switch forward, away from the edgelit panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-02-300-003

(4) Pivot the switch bezel to the side to gain access to the lamps, then slide the lamps out of the switch body.

SUBTASK 23-51-02-300-004

(5) Insert new replacement lamps all the way into the socket in the back of the switch body.

SUBTASK 23-51-02-300-005

(6) Pivot the switch body back into normal position and gently push on the switch cap until the switch body is fully seated.

SUBTASK 23-51-02-440-002

(7) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

F/O Electrical System Panel, P6-3

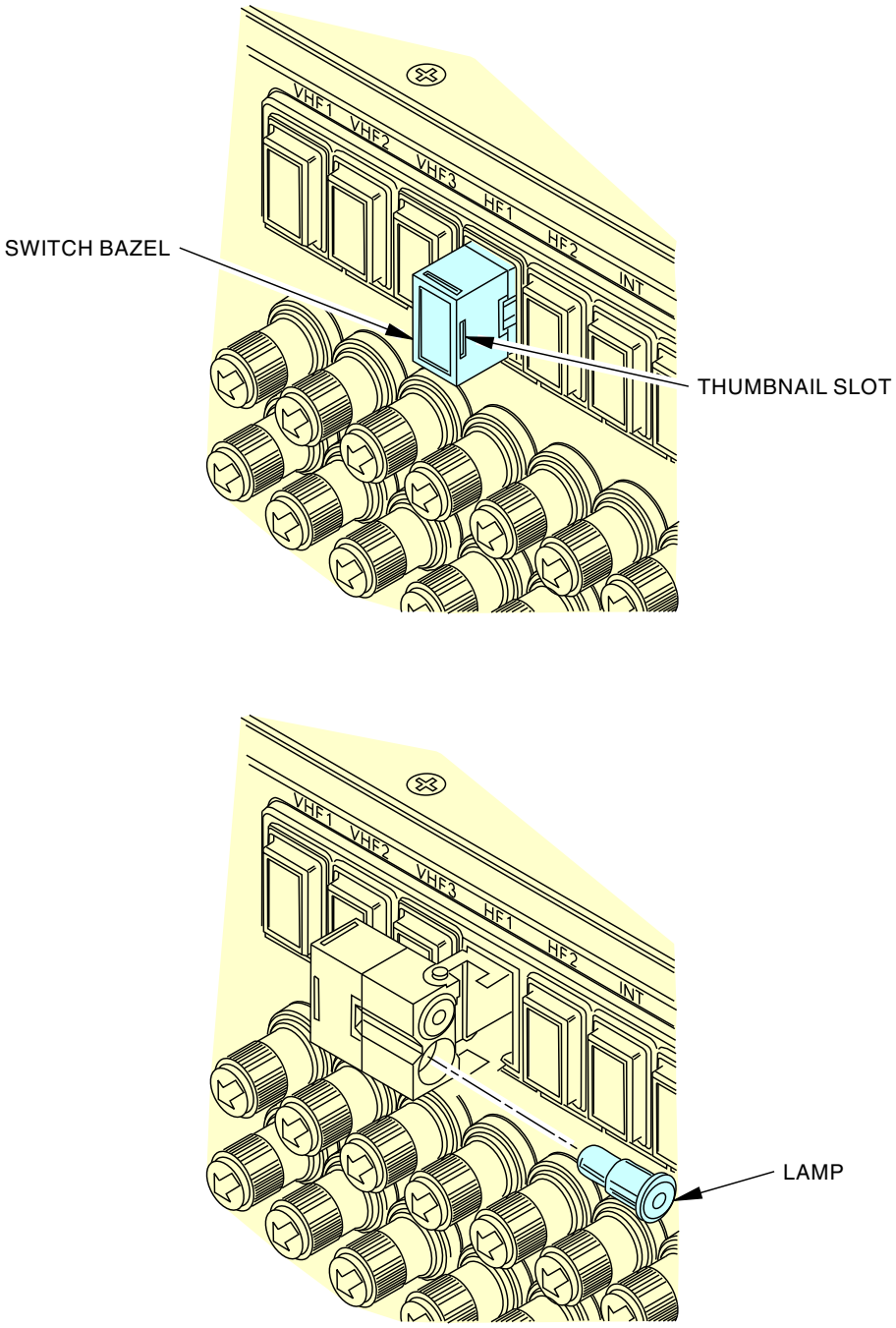
Row	Col	Number	Name
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 23-51-02-700-001

(8) Push the mic select switch and make sure that the light comes on.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Audio Control Panel Mic Select Switch
Figure 201/23-51-02-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AUDIO CONTROL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Audio Control Panel Removal

(2) Audio Control Panel Installation.

TASK 23-51-02-000-801

2. Audio Control Panel Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the audio control panel [1].

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Audio Control Panel Removal

SUBTASK 23-51-02-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	10	C00284	PANEL & INSTR ELEX PANEL
D	8	C00701	EMER PANEL LTG
F	10	C00390	OVERHEAD 5V SECONDARY
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

D. Audio Control Panel Removal

SUBTASK 23-51-02-020-001

(1) Loosen the four quarter-turn fasteners [2] on the front panel of the audio control panel [1].

SUBTASK 23-51-02-020-002

(2) Lift the audio control panel [1] to get access to the electrical connector [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-02-020-003

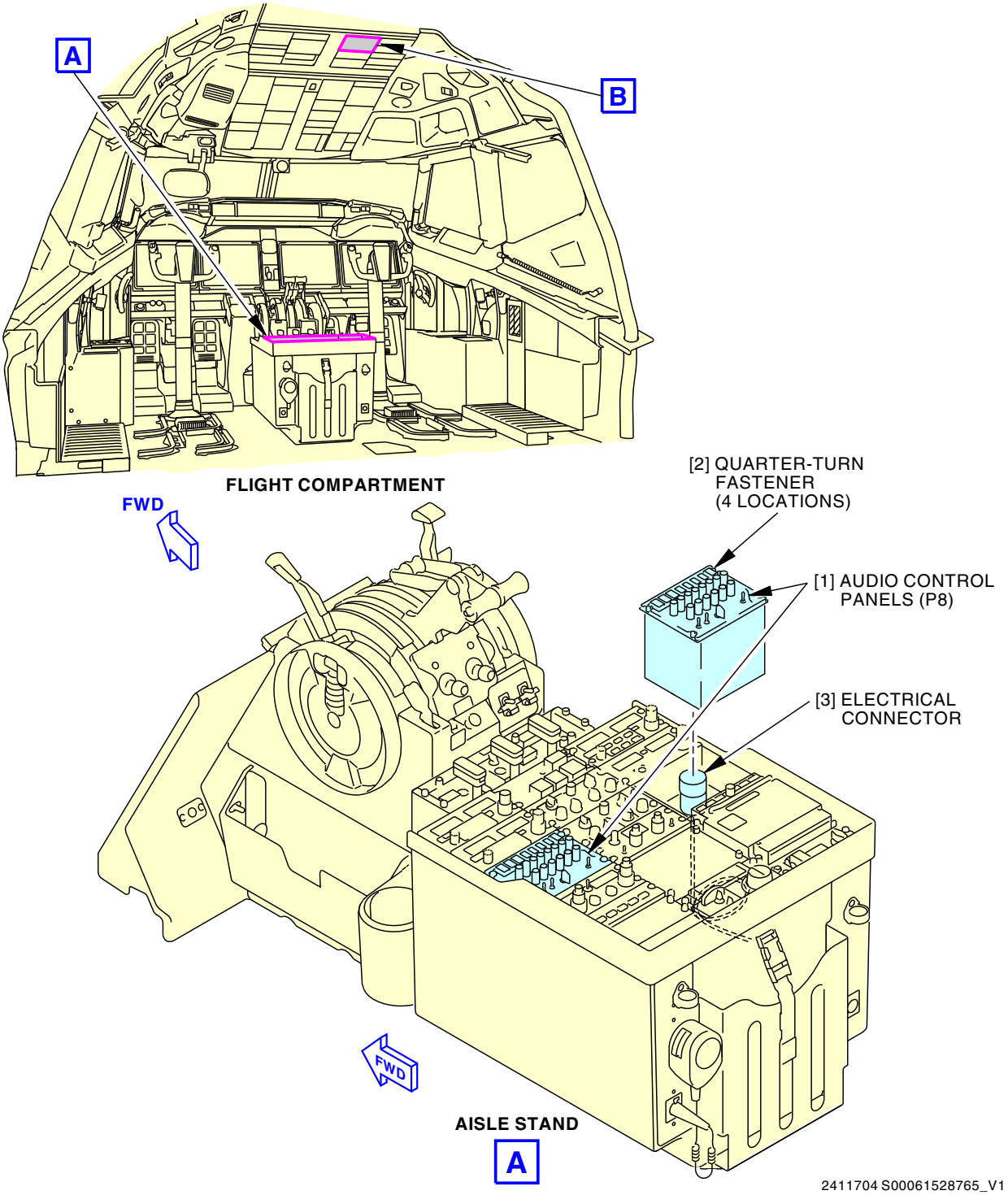
(3) Disconnect the electrical connector [3] from the rear of the audio control panel [1].

SUBTASK 23-51-02-020-004

(4) Remove the audio control panel [1].

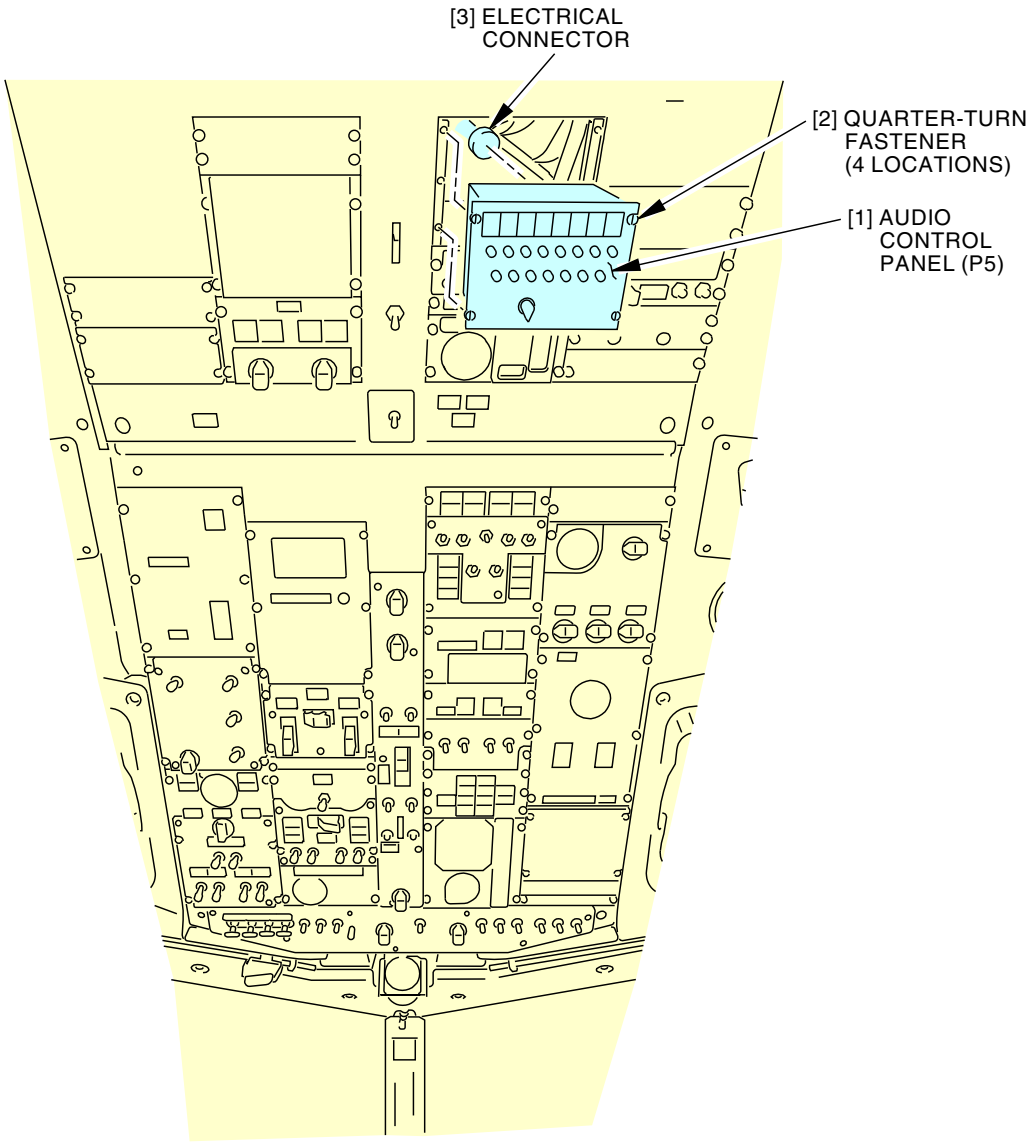
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Audio Control Panel Installation
Figure 401/23-51-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OVERHEAD PANEL
(VIEW IN THE FORWARD DIRECTION)

B

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Audio Control Panel Installation
Figure 401/23-51-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-51-02-400-801

3. Audio Control Panel Installation

(Figure 401)

A. General

(1) This procedure has steps to install the audio control panel [1].

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Audio Control Panel Installation

SUBTASK 23-51-02-420-001

(1) Connect the electrical connector [3] in the rear of the audio control panel [1].

SUBTASK 23-51-02-420-002

(2) Put the audio control panel [1] into its position.

SUBTASK 23-51-02-420-003

(3) Tighten the four quarter-turn fasteners [2] on the front panel of the audio control panel [1].

D. Audio Control Panel Installation Test

SUBTASK 23-51-02-860-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	10	C00284	PANEL & INSTR ELEX PANEL
D	8	C00701	EMER PANEL LTG
F	10	C00390	OVERHEAD 5V SECONDARY
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

SUBTASK 23-51-02-710-001

(2) Do an operational test of the Audio Control Panel (ACP):

(a) Push the INT microphone selector switch on the audio control panels [1].

(b) Turn the INT volume controls to approximately 1/2 MAX on all the audio control panels [1].

(c) Make sure that there is sound from all three audio control panels [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Set the RADIO-INT Push-To-Talk (PTT) switch on the audio control panel [1] to the INT position.

NOTE: There can be feedback caused by this audio coupling between the headphones and the area microphone. The feedback is intermittent and does not affect the test results.

1) If there is background noise caused by the audio feedback from the hot microphone, cover the oxygen mask microphone.

(e) Speak into the pilot's boom microphone.

NOTE: For the observer's ACP test when the boom microphone is not installed, you can connect a boom microphone at the observer's boom microphone jack and use it for the test. You can also use the oxygen mask microphone for the test.

1) Make sure that the voice can be heard from the other pilot's headset.

2) Make sure that the voice cannot be heard on the PA.

(f) Make sure that there is sound from the cockpit speaker.

(g) Set the RADIO-INT PTT switch on the audio control panel [1] to the middle switch position.

(h) Push the PILOT button on the attendant handset.

1) Make sure that the CABIN microphone light comes ON.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT INTERPHONE SPEAKER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Flight Interphone Speaker Removal

(2) Flight Interphone Speaker Installation

B. The flight interphone speaker is in the forward ceiling panel in the flight compartment.

TASK 23-51-03-000-801

2. Flight Interphone Speaker Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the flight interphone speaker (referred to as the speaker [1] in this task).

B. References

Reference	Title
25-11-21-000-801	Flight Compartment Forward Ceiling Panel Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. Prepare for the Flight Interphone Speaker Removal

SUBTASK 23-51-03-760-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-51-03-010-001

(2) Remove the ceiling panel (TASK 25-11-21-000-801).

E. Flight Interphone Speaker Removal

SUBTASK 23-51-03-020-001

(1) Remove the four screws [3] from the speaker [1].

SUBTASK 23-51-03-020-002

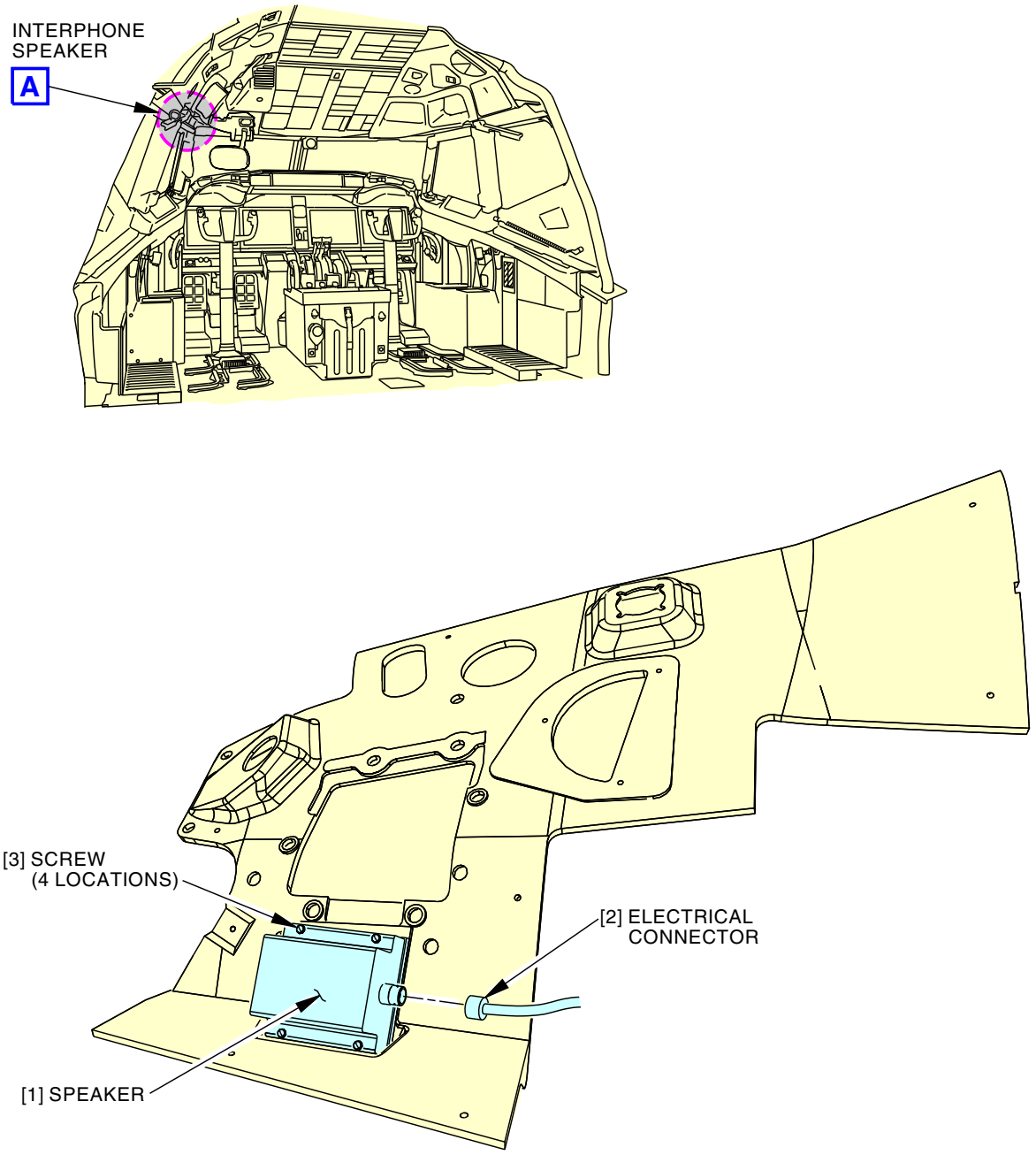
(2) Disconnect the electrical connector [2] from the speaker [1].

SUBTASK 23-51-03-020-003

(3) Remove the speaker [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT INTERPHONE SPEAKER

A

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Flight Interphone Speaker Installation
Figure 401/23-51-03-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-51-03-400-801

3. Flight Interphone Speaker Installation

(Figure 401)

A. General

(1) This procedure has steps to install a flight interphone speaker (referred to as the speaker [1] in this task).

B. References

Reference	Title
25-11-21-400-801	Flight Compartment Forward Ceiling Panel Installation (P/B 201)
35-12-00-440-801	Crew Oxygen System - Activation (P/B 201)
35-12-85-000-802	Oxygen Mask/Regulator Removal (P/B 401)
35-12-85-400-802	Oxygen Mask/Regulator Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Speaker	23-51-03-01-010	PTF ALL
		23-51-03-02-030	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left

E. Flight Interphone Speaker Installation

SUBTASK 23-51-03-420-001

(1) Put the speaker [1] in its correct position in the forward overhead panel.

SUBTASK 23-51-03-420-002

(2) Connect the electrical connector [2] to the speaker [1].

SUBTASK 23-51-03-420-003

(3) Install the four screws [3] to the speaker [1].

F. Flight Interphone Installation Test

SUBTASK 23-51-03-860-001

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-51-03-440-001

(2) Make sure that the oxygen system is pressurized (TASK 35-12-00-440-801).

SUBTASK 23-51-03-020-004

(3) Remove the oxygen masks from the stowage box (TASK 35-12-85-000-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-03-860-003

(4) Make sure that there is sound from the speaker [1]:

(a) Set the INT volume control to ON.

(b) Push the INT mic selector button ON.

1) Turn the INT volume controls to approximately 1/2 MAX.

(c) Set the SPKR switch to ON.

1) Turn the SPKR volume control to approximately 1/2 MAX.

(d) On the external power panel, connect the ground cord and headset to the FLT INT Jack.

1) Hold the Push-To-Talk (PTT) switch on the ground headset.

2) Speak into the ground headset microphone.

3) Make sure that there is sound from the cockpit speaker.

4) Release the PTT switch on the ground headset.

(e) Hold the PTT switch on the control wheel to the MIC position

1) Speak into the oxygen mask microphone.

2) Make sure that there is sound from the cockpit speaker.

3) Release the PTT switch on the control wheel.

(f) Hold the PTT glareshield switch:

1) Speak into the oxygen mask microphone.

2) Make sure that there is sound from the cockpit speaker.

3) Release the PTT glareshield switch.

(g) Set the RADIO-INT switch to the RADIO position.

1) Speak into the oxygen mask microphone.

2) Make sure that there is sound from the cockpit speaker.

3) Release the RADIO-INT switch.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-51-03-410-001

(1) Install the ceiling panel (TASK 25-11-21-400-801).

SUBTASK 23-51-03-420-004

(2) Put the oxygen masks in the stowage box (TASK 35-12-85-400-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PUSH TO TALK CONTROL WHEEL SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Push To Talk (PTT) Control Wheel Switch Removal

(2) Push To Talk (PTT) Control Wheel Switch Installation

B. The PTT switch is installed on the control wheel.

TASK 23-51-04-000-801

2. Push-To-Talk (PTT) Control Wheel Switch Removal

(Figure 401)

A. General

(1) This procedure has steps to remove a PTT switch assembly [1].

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the PTT Control Wheel Switch Removal

SUBTASK 23-51-04-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. PTT Control Wheel Switch Removal

SUBTASK 23-51-04-020-001

(1) Remove the two mounting screws [2] from the control wheel.

SUBTASK 23-51-04-020-002

(2) Remove the PTT switch assembly [1] to get access to the wires.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-04-020-003

(3) Disconnect the lugs from the interphone switch [3] in the PTT switch assembly [1].

- S519 - Captain
- S520 - First Officer

(a) Make sure to identify each wire for correct installation on the switch.

NOTE: Correct wire installation is important for correct operation.

SUBTASK 23-51-04-020-004

(4) Disconnect the lugs from the microphone switch [6] in the PTT switch assembly [1].

- S161 - Captain
- S214 - First Officer

(a) Make sure to identify each wire for correct installation on the switch.

NOTE: Correct wire installation is important for correct operation.

SUBTASK 23-51-04-020-005

(5) Remove the locking screw [5] from the clamp [4].

SUBTASK 23-51-04-020-006

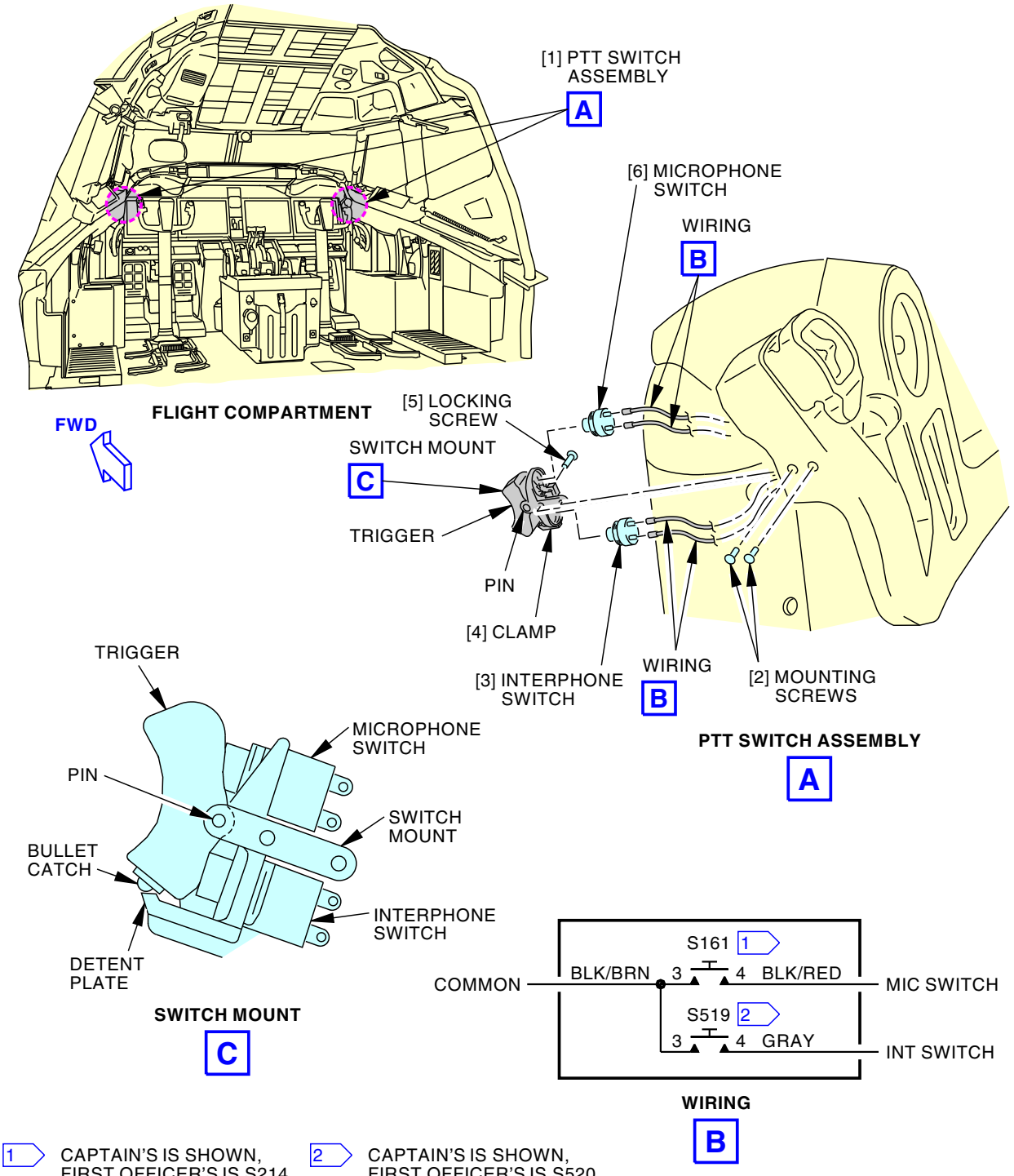
(6) Turn the PTT switch assembly [1] counterclockwise.

SUBTASK 23-51-04-020-007

(7) Remove the PTT switch assembly [1] from the clamp [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PTT Control Wheel Switch Installation
Figure 401/23-51-04-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-51-04-400-801

3. Push To Talk (PTT) Control Wheel Switch Installation

(Figure 401)

A. General

(1) This procedure has steps to install a PTT switch assembly [1].

B. References

Reference	Title
35-12-00-440-801	Crew Oxygen System - Activation (P/B 201)
35-12-85-000-802	Oxygen Mask/Regulator Removal (P/B 401)
35-12-85-400-802	Oxygen Mask/Regulator Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. PTT Control Wheel Switch Installation

SUBTASK 23-51-04-420-001

(1) Install the interphone switch [3] into the clamp [4].

SUBTASK 23-51-04-420-002

(2) Install the microphone switch [6] into the clamp [4].

SUBTASK 23-51-04-820-001

(3) Turn the PTT switch assembly [1] clockwise.

SUBTASK 23-51-04-820-002

(4) Adjust the microphone switch [6] in the PTT switch assembly [1].

SUBTASK 23-51-04-820-003

(5) Adjust the interphone switch [3] in the PTT switch assembly [1].

SUBTASK 23-51-04-420-003

(6) Install the locking screw [5] in the clamp [4].

SUBTASK 23-51-04-820-004

(7) Push the trigger to the INT position and then release the trigger.

(a) Make sure that the trigger stays in the detent position.

SUBTASK 23-51-04-020-008

(8) Loosen the locking screw [5] on the clamp [4].

SUBTASK 23-51-04-820-005

(9) Push the trigger to the OFF position.

SUBTASK 23-51-04-820-006

(10) Push the trigger to the MIC position and then release the trigger.

(a) Make sure that the trigger returns to the center (OFF) position.

SUBTASK 23-51-04-420-004

(11) Connect the lugs to the interphone switch [3] in the PTT switch assembly [1].

- S519 - Captain
- S520 - First Officer

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that you connect the identified wires on the terminals.

NOTE: Correct wire installation is important for correct operation.

NOTE: Connections for terminals 3 and 4 are interchangeable.

SUBTASK 23-51-04-420-005

(12) Connect the lugs to the microphone switch [6] in the PTT switch assembly [1].

- S161 - Captain
- S214 - First Officer

(a) Make sure that you connect the identified wires on the terminals.

NOTE: Correct wire installation is important for correct operation.

NOTE: Connections for terminals 3 and 4 are interchangeable.

SUBTASK 23-51-04-420-006

(13) Tighten the locking screw [5] on the PTT switch assembly [1].

SUBTASK 23-51-04-420-007

(14) Install the two mounting screws [2] on the control wheel.

E. PTT Control Wheel Switch Installation Test

SUBTASK 23-51-04-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-51-04-440-001

(2) Make sure that the oxygen system is pressurized (TASK 35-12-00-440-801).

NOTE: The switch in the stowage box will not set if the oxygen system is not pressurized.

SUBTASK 23-51-04-850-001

(3) Push the INT microphone selector switch on the pilot's Audio Control Panel (ACP) to on.

SUBTASK 23-51-04-850-010

(4) Turn the volume control for the INT microphone selector switch clockwise to the middle position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-51-04-850-004

(5) Set the SPKR switch on the Audio Control Panels (ACP) to ON.

SUBTASK 23-51-04-850-005

(6) Turn the SPKR volume control on the ACP clockwise to the middle position or to the volume level you are comfortable with.

SUBTASK 23-51-04-850-006

(7) Hold the Push-To-Talk (PTT) switch on the control wheel in the lower INT position.

(a) Speak into the microphone.

(b) Make sure that there is a voice on the headset.

(c) Make sure that there is no PA announcement in the cabin area.

(d) Release the PTT switch.

(e) Make sure that the switch stays in the INT position.

(f) Push the PTT switch to the center (OFF) position.

SUBTASK 23-51-04-020-009

(8) Remove the oxygen masks from the stowage box (TASK 35-12-85-000-802).

SUBTASK 23-51-04-850-007

(9) Hold the PTT switch on the control wheel in the MIC position.

(a) Speak into the oxygen mask microphone.

(b) Make sure that there is a voice from the cockpit speaker.

(c) Make sure that there is no voice from the cabin speakers.

(d) Release the PTT switch.

(e) Make sure that the switch returns to the center (OFF) position.

SUBTASK 23-51-04-420-008

(10) Put the oxygen masks in the stowage box (TASK 35-12-85-400-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PUSH TO TALK CONTROL WHEEL SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) An adjustment of the catchball.

B. Do this procedure when the PTT switch is not in the INT locked position.

TASK 23-51-04-720-801

2. Catchball Adjustment

(Figure 501)

A. General

(1) This task has the steps to adjust the catchball on the control wheel.

B. References

Reference	Title
23-51-04-000-801	Push-To-Talk (PTT) Control Wheel Switch Removal (P/B 401)
23-51-04-400-801	Push To Talk (PTT) Control Wheel Switch Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00270	Compound - Threadlocking, Low-strength - Loctite 222	
A00648	Compound - Retaining - Loctite 242	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Catchball Adjustment

SUBTASK 23-51-04-760-001

(1) Do these steps when the Push-To-Talk (PTT) switch does not lock in the INT position:

(a) Remove the PTT switch assembly (TASK 23-51-04-000-801).

(b) Do the steps to adjust the ball catch:

NOTE: The assembly consists of three small parts that will spring apart when released.

1) Remove the ball catch assembly from the trigger.

2) Apply a thin layer of Loctite 222 compound, A00270, or Loctite 242 compound, A00648, to 1 to 3 full threads centered on thread length of ball catch sleeve.

3) Assemble the catchsleeve, ball, and spring in the trigger.

4) Push the trigger to the INT position.

5) Turn the ball catch clockwise until the ball no longer makes contact with the switch mount detent.

6) Push the trigger to the INT position.

7) Turn the bullet catch counterclockwise until the ball just makes contact with the switch mount detent.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

8) Turn the bullet catch counterclockwise 1/4 turn.

9) Do not extend the bullet catch further than 0.078 in. (1.981 mm) from the trigger.

NOTE: If the PTT switch will not stay in the INT position when the bullet catch is fully extended, adjust the PTT switch (Figure 501).

10) Push the trigger to the bottom (INT) position.

11) Make sure that the trigger stays in the locked INT position.

(c) Install the PTT switch assembly (TASK 23-51-04-400-801):

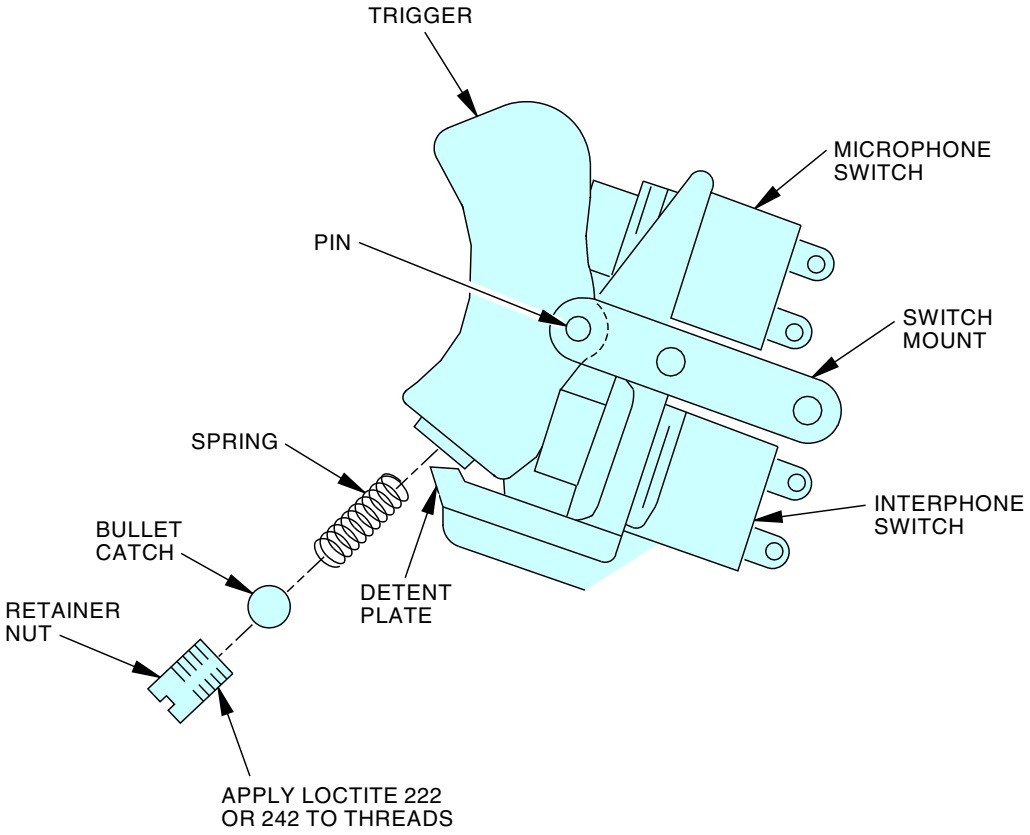
SUBTASK 23-51-04-850-008

(2) Push the trigger to the bottom (INT) position.

(a) Make sure that the trigger stays in the locked INT position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PTT SWITCH ASSEMBLY

2411708 S00061528779_V1

Control Wheel PTT Switch Bullet Catch Assembly
Figure 501/23-51-04-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GLARESHIELD PUSH TO TALK SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the glareshield PTT switch

(2) An installation of the glareshield PTT switch.

TASK 23-51-05-000-801

2. Glareshield Push To Talk (PTT) Switch Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the glareshield Push-To-Talk (PTT) switch.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 23-51-05-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

D. Glareshield PTT Switch Removal

SUBTASK 23-51-05-020-003

(1) Remove the panel [6] as follows:

(a) Remove the screw [9].

(b) Release the quarter turn fastener [12] and quarter turn fasteners [13] or remove the screw [10] and screws [11].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If it is necessary, make sure that you record the location of the different screws for the installation.

(d) Remove the panel [6].

SUBTASK 23-51-05-020-007

(2) Remove the glareshield PTT switch [1] on the glareshield as follows:

(a) Remove the screws [3] from the switch plate [2] on the glareshield.

(b) Carefully pull the switch plate [2] from the glareshield panel.

(c) Remove the lock nut [4] and lock washer [7] from the glareshield PTT switch [1].

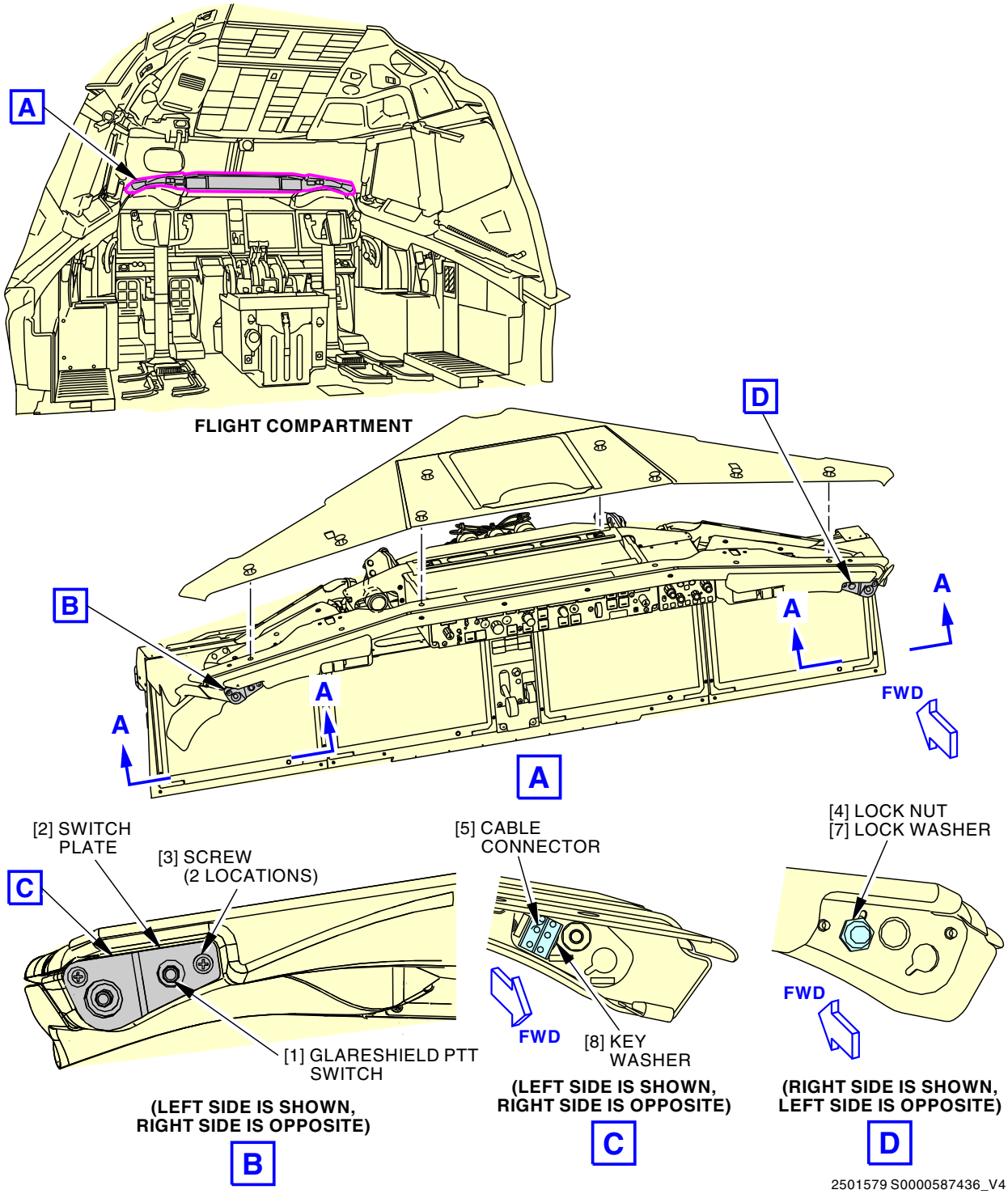
(d) Remove the glareshield PTT switch [1] from the glareshield.

(e) Remove the key washer [8] from the glareshield PTT switch [1].

(f) Disconnect the cable connector [5] from the glareshield PTT switch [1].

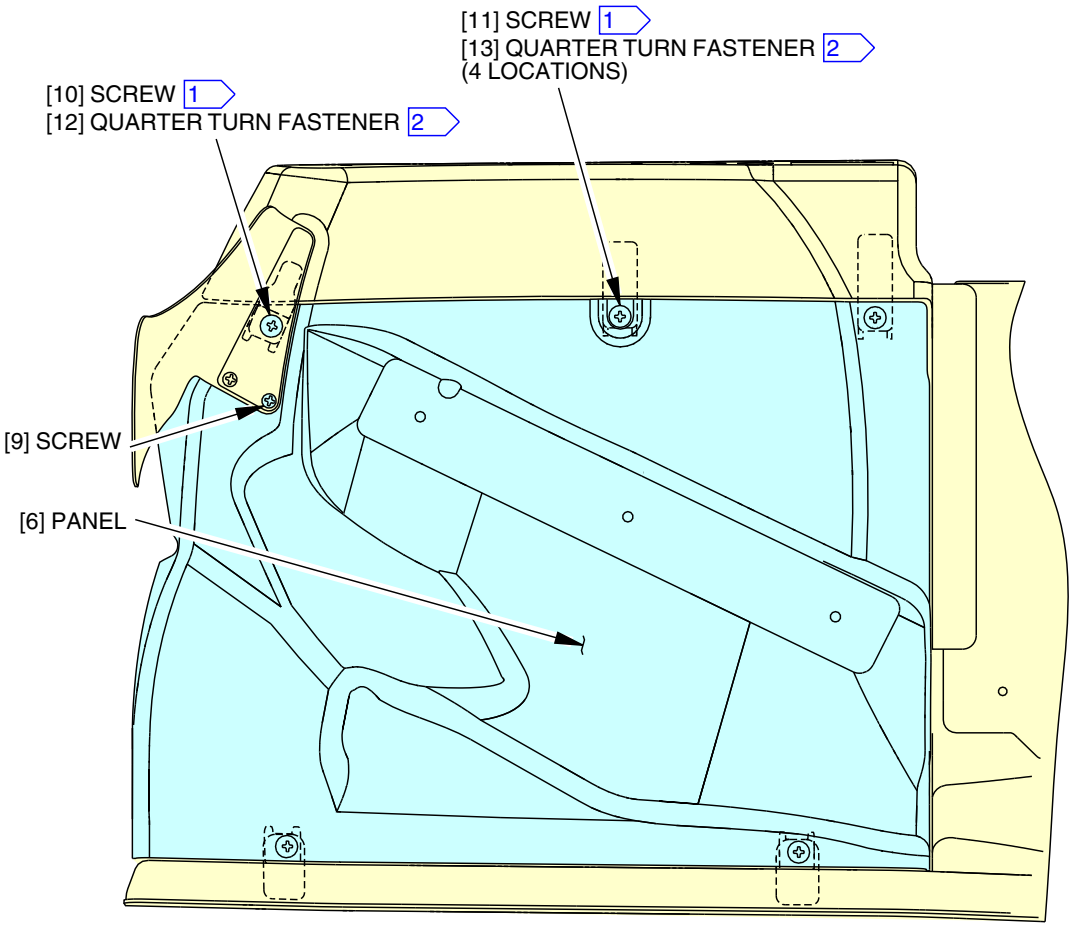
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Glareshield PTT Switch Installation
Figure 401/23-51-05-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A-A

1 PANEL INSTALLATION WITH SCREWS
2 PANEL INSTALLATION WITH QUARTER TURN FASTENERS

2864312 S0000675094_V2

Glareshield PTT Switch Installation
Figure 401/23-51-05-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-51-05-400-801

3. Glareshield Push To Talk (PTT) Switch Installation

(Figure 401)

A. General

(1) This procedure has steps to install the glareshield Push-To-Talk (PTT) switch.

B. References

Reference	Title
35-12-00-440-801	Crew Oxygen System - Activation (P/B 201)
35-12-85-000-802	Oxygen Mask/Regulator Removal (P/B 401)
35-12-85-400-802	Oxygen Mask/Regulator Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Glareshield PTT switch	23-51-05-01-010	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Glareshield PTT Switch Installation

SUBTASK 23-51-05-420-001

(1) Install the glareshield PTT switch [1] on the glareshield as follows:

(a) Connect the cable connector [5] to the glareshield PTT switch [1].

(b) Place the key washer [8] on the glareshield PTT switch [1].

(c) Install the glareshield PTT switch [1] on the glareshield.

(d) Install the lock washer [7] and lock nut [4] to the glareshield PTT switch [1].

1) Make sure that the lock nut [4] is tight.

(e) Place the switch plate [2] in its position on the glareshield.

(f) Tighten the screws [3] that attach the switch plate [2] to the glareshield.

SUBTASK 23-51-05-410-001

(2) Install the panel [6] as follows:

(a) Put the panel [6] in its position.

(b) Engage the quarter turn fastener [12] and quarter turn fasteners [13] or install the screw [10] and screws [11].

(c) Install the screw [9].

F. Glareshield PTT Switch Installation Test

SUBTASK 23-51-05-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	11	C00165	COMMUNICATIONS VHF 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	3	C00166	COMMUNICATIONS VHF 2

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	21	C00560	INTERPHONE POWER F/O DC 2
C	22	C00561	INTERPHONE POWER F/O BAT
C	23	C00239	INTERPHONE POWER CAPT DC 2
C	24	C00240	INTERPHONE POWER CAPT BAT
D	21	C00084	INTPH AND WARN
D	22	C00086	AUDIO F/O
D	23	C00083	AUDIO CAPT
D	24	C00085	AUDIO OBS

SUBTASK 23-51-05-440-001

(2) Make sure that the oxygen system is pressurized (TASK 35-12-00-440-801).

NOTE: If the oxygen system is not pressurized, the switch in the stowage box will not set.

SUBTASK 23-51-05-850-001

(3) Set the FLT INT volume control to the ON position.

NOTE: The volume control is automatic for the microphone.

SUBTASK 23-51-05-850-002

(4) Set the FLT INT MIC button, on the audio control panel's, to the ON position.

SUBTASK 23-51-05-850-003

(5) Turn the FLT INT volume controls to approximately 1/2 MAX, on the audio control panel's.

SUBTASK 23-51-05-850-004

(6) Set the SPKR switch, on the audio control panel's, to the ON position.

SUBTASK 23-51-05-850-005

(7) Turn the SPKR volume control, on the audio control panel's, to approximately 1/2 MAX.

SUBTASK 23-51-05-020-006

(8) Remove the oxygen masks from the stowage box (TASK 35-12-85-000-802).

SUBTASK 23-51-05-850-006

(9) Hold the glareshield PTT switch [1].

(a) Speak into the oxygen mask microphone.

(b) Make sure that there is a voice from the cockpit speaker.

(c) Make sure that there is no voice from the cabin speakers.

(d) Release the glareshield PTT switch [1].

SUBTASK 23-51-05-420-005

(10) Put the oxygen masks in the stowage box (TASK 35-12-85-400-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STATIC DISCHARGERS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Static Discharger Removal

(2) Static Discharger Installation

(3) Static Discharger Base with Screws Removal

(4) Static Discharger Base with Screws Installation

(5) Static Discharger Base with Rivets Removal

(6) Static Discharger Base with Rivets Installation

(7) Static Discharger Base with Adapter Plate Assembly Removal

(8) Static Discharger Base with Adapter Plate Assembly Installation

TASK 23-61-00-000-801

2. Static Discharger Removal

(Figure 401)

A. General

(1) The static dischargers [1] are installed on the edge of the airplane wing and tail.

(2) The static discharger [1] is held by a set screw [2] on its discharger base [3].

(3) The discharger base [3] is attached to the airplane surface.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Static Discharger Removal

SUBTASK 23-61-00-040-001

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from the hydraulic systems for the applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802
- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802

SUBTASK 23-61-00-040-002

(2) Remove the electrical power (TASK 24-22-00-860-802).

E. Static Discharger Removal

SUBTASK 23-61-00-010-001

(1) Get access to the static discharger [1].

SUBTASK 23-61-00-020-001

(2) Loosen the set screw [2].

SUBTASK 23-61-00-020-002

(3) Remove the static discharger [1] from the discharger base [3].

F. Put the Airplane Back to Its Usual Condition

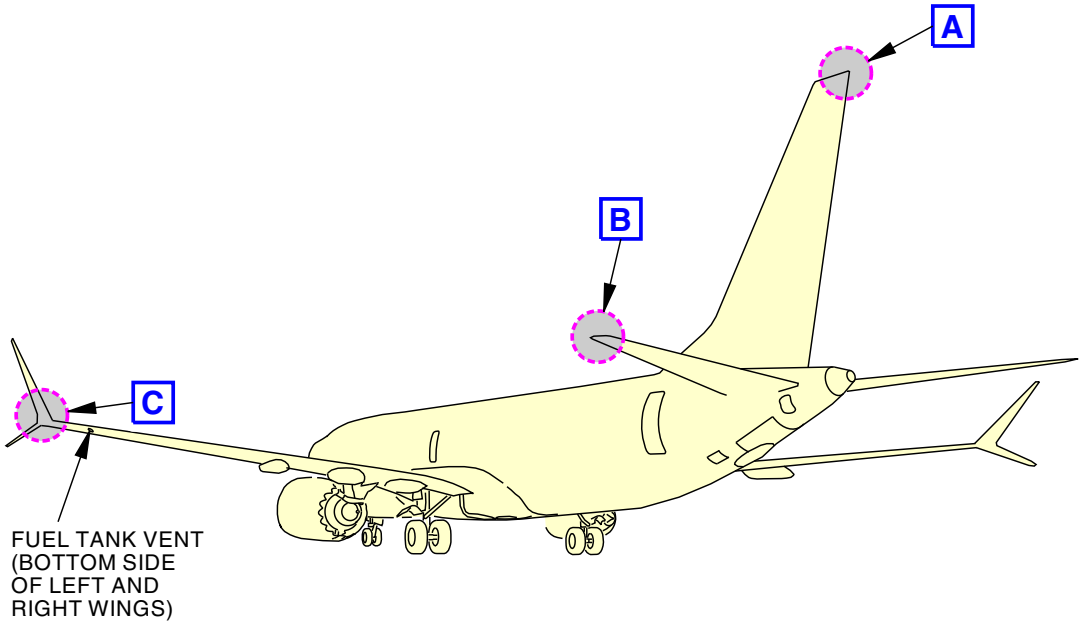
SUBTASK 23-61-00-440-001

(1) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

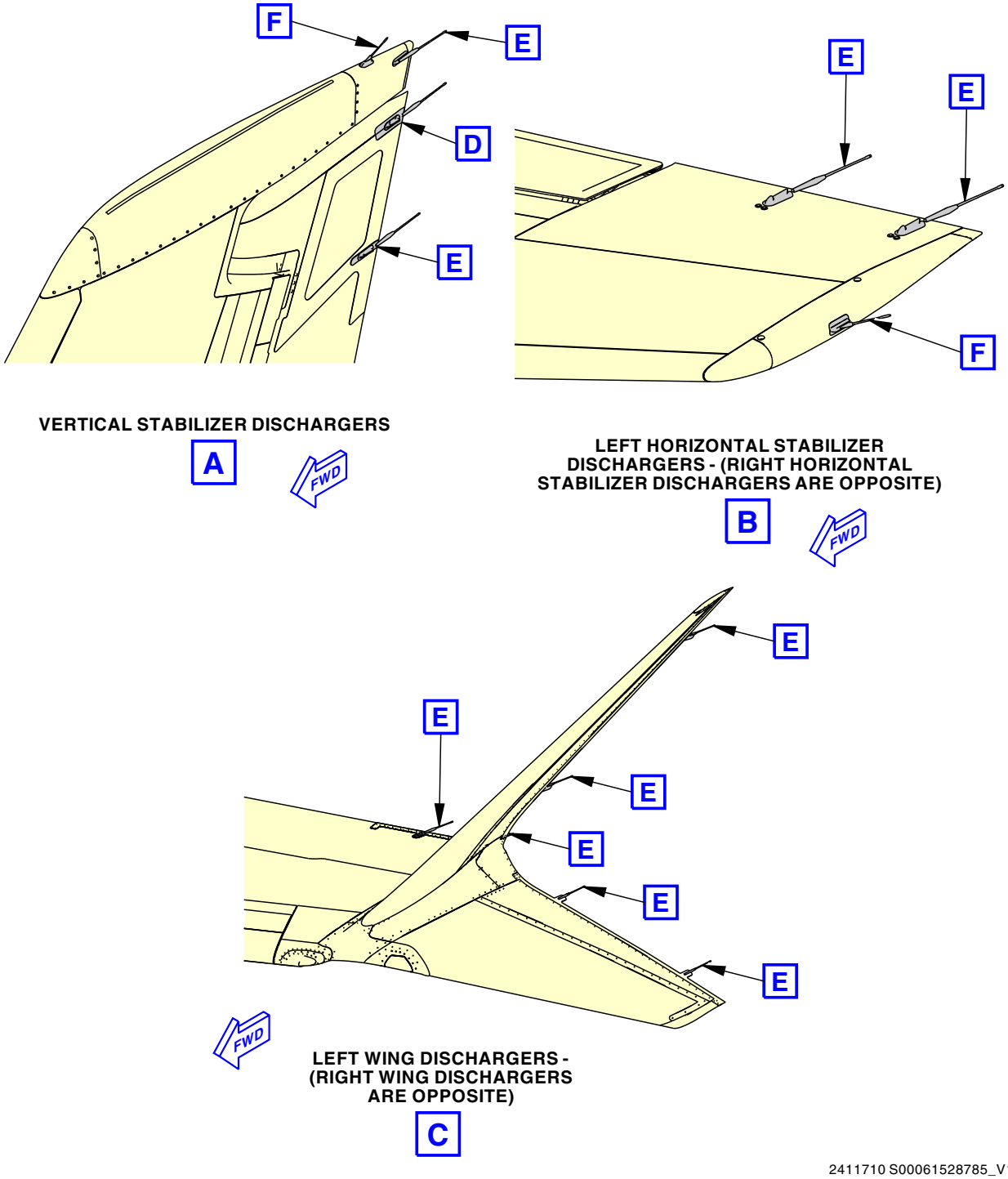


STATIC DISCHARGER LOCATIONS

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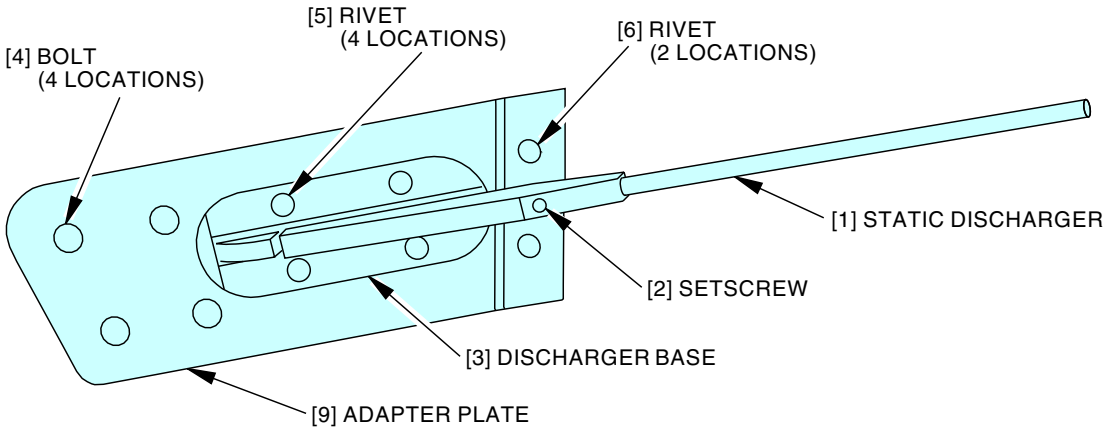
Static Discharger Installation
Figure 401/23-61-00-990-804 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



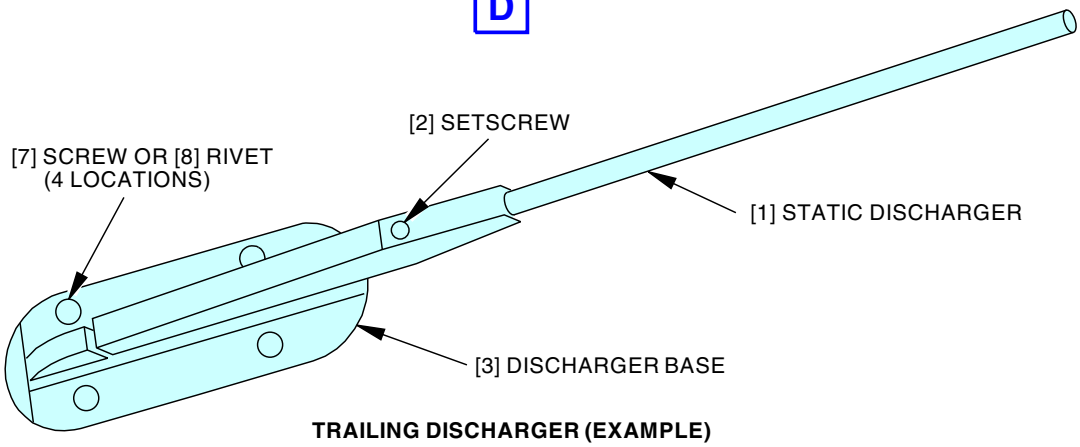
Static Discharger Installation
Figure 401/23-61-00-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



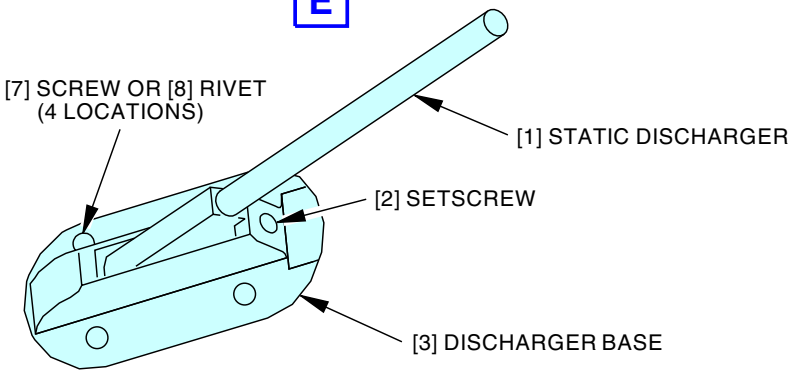
TRAILING DISCHARGER WITH ADAPTER PLATE (EXAMPLE)

D



TRAILING DISCHARGER (EXAMPLE)

E



TIP DISCHARGER (EXAMPLE)

F

2411711 S00061528786_V1

Static Discharger Installation
Figure 401/23-61-00-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-61-00-400-801

3. Static Discharger Installation

(Figure 401)

A. General

(1) The static dischargers [1] are installed on the edge of the airplane wing and tail.

(2) The discharger is held by a set screw [2] on its base.

(3) The discharger base [3] is attached to the airplane surface.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1587	Wrench - Torque, 30 in-lbs, +/-5% accuracy Part #: 5718A42 Supplier: 3A054 Opt Part #: TE3FUA Supplier: 55719

D. Consumable Materials

Reference	Description	Specification
A00648	Compound - Retaining - Loctite 242	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)
B00130	Alcohol - Isopropyl	TT-I-735
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
B01026	Solvent - FCC-55	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Static discharger	23-61-00-02-110	PTF ALL
		23-61-00-02-210	PTF ALL
		23-61-00-02-305	PTF ALL
		23-61-00-02-355	PTF ALL

F. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

G. Prepare for the Static Discharger Installation

SUBTASK 23-61-00-040-003

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from the hydraulic systems for the applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802
- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802

SUBTASK 23-61-00-040-004

(2) Remove the electrical power (TASK 24-22-00-860-802).

SUBTASK 23-61-00-210-001

(3) Inspect the static discharger [1].

(a) If the static discharger [1] is damaged, replace the static discharger [1].

H. Static Discharger Installation

SUBTASK 23-61-00-420-001

(1) Install the set screw [2] into the static discharger [1].

(a) Use cotton wiper, G00034, moist in alcohol, B00130, or solvent, B00148, or solvent, B00184, or solvent, B00666, or FCC-55 solvent, B01026, to clean the mating surfaces of the static discharger [1] and discharger base [3].

CAUTION DO NOT USE THIS MATERIAL WITH PLASTIC-BASED MATERIALS. IT WILL NOT BOND CORRECTLY. THE STATIC DISCHARGER WILL COME OFF. DAMAGE TO EQUIPMENT WILL OCCUR.

(b) Apply a thin layer of Loctite 242 compound, A00648 to the set screw [2] threads.

NOTE: Static dischargers without NY-LOC insert in the set screw and static dischargers that are not plastic based material.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do not use Loctite with the plastic-based material static discharger or with NY-LOC insert in the setscrew.

NOTE: Loctite reacts to the plastic materials that will cause eventual failure of the bond and result in separation of the static discharger from the airplane.

(d) Install the set screw [2] on the static discharger [1].

(e) Adjust the set screw [2] until it makes a continuous surface with the static discharger [1] casting.

SUBTASK 23-61-00-420-002

(2) Install the static discharger [1] into the discharger base [3].

(a) Hold the static discharger [1] in the correct position and tighten the set screw [2] to this value:

1) Tighten the set screw [2] to 6 in-lb (0.68 N·m) -9 in-lb (1.02 N·m) using a torque wrench (0-30 lbs), COM-1587.

(b) Make sure that the static discharger [1] is attached to its base correctly.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-440-002

(1) If it is necessary, put the hydraulic systems back to the usual condition:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

——— END OF TASK ———

TASK 23-61-00-000-802

4. Static Discharger Base with Screws Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the static discharger base [3] with screws.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-1053	Knife - Putty, Broad Blade

D. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

E. Prepare for the Static Discharger with Screws Removal

SUBTASK 23-61-00-040-005

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from hydraulic systems for the applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802

SUBTASK 23-61-00-040-006

(2) Remove the electrical power (TASK 24-22-00-860-802).

F. Static Discharger with Screws Removal

SUBTASK 23-61-00-020-003

(1) Remove the discharger base [3]:

(a) Loosen the set screw [2] from the discharger base [3].

(b) Remove the static discharger [1] from the discharger base [3].

(c) Make a mark with a grease pencil larger than the area of the discharger base [3] be cleaned.

1) Make sure that the line is parallel to the airflow.

(d) Remove the screws [7] from the discharger base [3].

1) Remove the weather/aerodynamic fillet seal around the edge of the discharger base [3] with a sealant removal tool, COM-2481.

2) Keep the screws [7] for installation.

CAUTION

BE CAREFUL WITH A PUTTY KNIFE, DAMAGE TO THE AIRCRAFT SURFACE CAN OCCUR.

3) Put a broad blade putty knife, STD-1053 below the edge of the discharger base [3].

4) Remove the discharger base [3] from the surface.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-440-003

(1) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

——— END OF TASK ———

TASK 23-61-00-400-802

5. Static Discharger Base with Screws Installation

(Figure 401)

A. General

(1) This procedure has steps to install the static discharger base [3] with screws.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
51-21-99-300-802	Decorative Exterior Paint System Repair (P/B 701)
51-22-00-390-806	Fillet Seal Application (P/B 201)
51-22-00-390-808	Fastener Seal Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-739	Respirator - Half-Mask, Air Purifying (with filters/cartridges)
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00102	Abrasive - Aluminum Oxide Coated Cloth	
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
B01026	Solvent - FCC-55	
B01054	Solvent - Methyl Ethyl Ketone and sec-Butyl Alcohol Blend - (MEK:secButyl Alcohol - 42:58 Percent)	BAC5750, ASTM D740/ASTM D1007
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Static discharger	23-61-00-02-110	PTF ALL
		23-61-00-02-210	PTF ALL
		23-61-00-02-305	PTF ALL
		23-61-00-02-355	PTF ALL

F. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

G. Prepare for the Static Discharger with Screws Installation

SUBTASK 23-61-00-040-007

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from hydraulic systems for applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802
- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802

SUBTASK 23-61-00-040-008

(2) Remove the electrical power (TASK 24-22-00-860-802).

SUBTASK 23-61-00-110-001

(3) Clean the surface for the installation of the discharger base [3]:

(a) Remove sealant with a sealant removal tool, COM-2481.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(b) Clean the area to be bonded with the solvent, B00148, solvent, B00666, solvent, B00184, FCC-55 solvent, B01026, or solvent, B01054.

1) Wear a half-mask respirator, STD-739 to clean with solvents.

2) Apply the solvent with a cotton wiper, G00034.

(c) Remove the solvent with cotton wiper, G00034 before it dries.

1) Make sure that you do not touch the surface.

(d) Rub the surface with the abrasive cloth, B00102 (600 grit) (TASK 20-10-34-120-801).

NOTE: Change the sandpaper frequently.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that you have a smooth surface.

2) Make sure that the surface is not shiny.

(e) Clean with cotton wiper, G00034.

NOTE: Do not use solvent.

H. Static Discharger with Screws Installation

SUBTASK 23-61-00-420-003

(1) Install the static discharger base [3] to the aircraft surface:

(a) Remove the metal discharger base [3] from the plastic container.

1) Make sure not to touch the bottom surface of the discharger base [3] which touches the airplane surface.

(b) Rub the discharger base [3] with a abrasive cloth, B00102.

1) Make sure that the discharger base [3] is installed in less than 5 minutes.

NOTE: Do not rub a plated discharger base with sandpaper. A plated discharger base has a green dot for identification. Put a pair of gloves on to touch the bases.

2) Make sure that you have a smooth surface.

NOTE: Change the sandpaper frequently.

3) Make sure that the airplane surface is not shiny.

(c) Attach the discharger base [3] to the airplane surface.

1) Make sure that you install the base in the same direction as the airflow.

(d) Install the screws [7] to attach the discharger base [3].

(e) Apply sealant, A00247 to the screws [7] (TASK 51-22-00-390-808).

(f) Apply sealant, A00247 to the edge of the discharger base [3] (TASK 51-22-00-390-806).

(g) Make the sealant smooth with a hardwood or plastic fillet smoothing spatula, STD-810.

(h) Make sure that you do not see cracks between the discharger base [3] and the airplane surface.

(i) Remove the sealant, A00247 with a cotton wiper, G00034 which is moist with solvent, B00062 or solvent, B00184.

SUBTASK 23-61-00-210-002

(2) Examine the installation of the discharger base [3].

(a) Make sure that there are no cracks in the sealant, A00247.

(b) If you find a crack, fill the crack with the sealant, A00247.

I. Static Discharger with Screws Installation Test

SUBTASK 23-61-00-760-007

WARNING OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Check the electrical bond between the discharger base [3] and airplane surface:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Use a intrinsically safe approved bonding meter, COM-1550 to measure the electrical bond between the discharger base [3] and the airplane surface (SWPM 20-20-00).

NOTE: This resistance value is for a new installed static discharger base and a new bond. The resistance value is different for an in-service bond and the discharger base.

The resistance value of a new discharger base [3] must not be more than:

NOTE: A resistance of more than these value can cause airplane equipment damage if a lightning strike occurs.

1) 0.01 ohms for discharger base [3] installed on metal panels.
2) 0.10 ohms for discharger base [3] installed on an aluminum flame spray on top of the fiberglass laminate or the composite panel.

NOTE: This includes discharger bases installed on aluminum-covered fabric.

3) 1.0 ohms for discharger base [3] installed on composite panels without a cover of the aluminum flame spray.

SUBTASK 23-61-00-420-004

(2) Install the static discharger [1] into the discharger base [3], do this task: Static Discharger Installation, TASK 23-61-00-400-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-370-001

(1) Paint the area around the discharger base [3] (TASK 51-21-99-300-802).

SUBTASK 23-61-00-440-004

(2) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

——— END OF TASK ———

TASK 23-61-00-000-803

6. Static Discharger Base with Rivets Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the static discharger base [3] base with rivets.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-1053	Knife - Putty, Broad Blade

D. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Static Discharger Base with Rivets Removal

SUBTASK 23-61-00-040-009

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from the hydraulic systems for the applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802
- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802

SUBTASK 23-61-00-040-010

(2) Remove the electrical power (TASK 24-22-00-860-802).

F. Static Discharger Base with Rivets Removal

SUBTASK 23-61-00-020-004

(1) Remove the discharger base [3] with rivets:

(a) Loosen the set screw [2].

(b) Remove the static discharger [1] from the base.

(c) Make a mark with a grease pencil larger than the area of the discharger base [3].

1) Make sure that the line is parallel to the airflow.

(d) Remove the discharger base [3].

1) Remove the weather/aerodyanamic fillet seal around the edge of the discharger base [3] with a sealant removal tool, COM-2481.

(e) Remove the rivets [8] from the discharger base [3] (SRM SUBJECT 51-40-02).

CAUTION BE CAREFUL WITH A PUTTY KNIFE, DAMAGE TO THE AIRCRAFT SURFACE CAN OCCUR.

(f) Put a broad blade putty knife, STD-1053 below the edge of the discharger base [3].

(g) Remove the discharger base [3] from the airplane surface.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-440-005

(1) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-61-00-400-803

7. Static Discharger Base with Rivets Installation

(Figure 401)

A. General

(1) This procedure has steps to install the discharger base [3] with rivets.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
51-21-99-300-802	Decorative Exterior Paint System Repair (P/B 701)
51-22-00-390-806	Fillet Seal Application (P/B 201)
51-22-00-390-808	Fastener Seal Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-739	Respirator - Half-Mask, Air Purifying (with filters/cartridges)
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00102	Abrasive - Aluminum Oxide Coated Cloth	
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
B01026	Solvent - FCC-55	
B01054	Solvent - Methyl Ethyl Ketone and sec-Butyl Alcohol Blend - (MEK:secButyl Alcohol - 42:58 Percent)	BAC5750, ASTM D740/ASTM D1007
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Static discharger	23-61-00-02-110	PTF ALL
		23-61-00-02-210	PTF ALL
		23-61-00-02-305	PTF ALL
		23-61-00-02-355	PTF ALL

F. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

G. Prepare for the Static Discharger Base with Rivets Installation

SUBTASK 23-61-00-040-011

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from the hydraulic systems for the applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802
- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802

SUBTASK 23-61-00-040-012

(2) Remove the electrical power (TASK 24-22-00-860-802).

SUBTASK 23-61-00-110-002

(3) Clean the surface for the installation of the discharger base [3]:

(a) Remove sealant with a sealant removal tool, COM-2481.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(b) Clean the area to be bonded with solvent, B00148, solvent, B00666, solvent, B00184, FCC-55 solvent, B01026, or solvent, B01054.

1) Wear a half-mask respirator, STD-739 to clean with solvents.

2) Apply the solvent with a cotton wiper, G00034.

(c) Remove the solvent with cotton wiper, G00034 before it dries.

1) Make sure that you do not touch the cleaned surface.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Rub the surface with the abrasive cloth, B00102 (600 grit) 1183 (TASK 20-10-34-120-801).

NOTE: Change the sandpaper frequently.

1) Make sure that you have a smooth surface.

2) Make sure that the surface is not shiny.

(e) Clean with cotton wiper, G00034.

NOTE: Do not use solvent.

H. Static Discharger Base with Rivets Installation

SUBTASK 23-61-00-420-005

(1) Install the discharger base [3] with rivets:

(a) Remove the metal discharger base [3] from the plastic container.

1) Make sure not to touch the bottom surface of the discharger base [3] which touches the airplane surface.

(b) Rub the discharger base [3] with a abrasive cloth, B00102.

1) Make sure that the discharger base [3] is installed in less than 5 minutes.

NOTE: Do not rub a plated discharger base with sandpaper. A plated discharger base has a green dot for identification. Put a pair of gloves on to touch the bases.

2) Make sure that you have a smooth surface.

NOTE: Change the sandpaper frequently.

3) Make sure that the airplane surface is not shiny.

(c) Attach the discharger base [3] to the airplane surface.

1) Make sure that you install the base in the same direction as the airflow.

(d) Install the rivets [8] to the discharger base [3] (SRM SUBJECT 51-40-02).

(e) Apply sealant, A00247 to the rivets [8] (Fastener Seal Application, TASK 51-22-00-390-808).

(f) Apply sealant, A00247 to the edge of the discharger base [3] (TASK 51-22-00-390-806).

(g) Make the sealant smooth with a hardwood or plastic fillet smoothing spatula, STD-810.

(h) Make sure that you do not see cracks between the discharger base [3] and the airplane surface.

(i) Remove the sealant, A00247 with a cotton wiper, G00034 which is moist with solvent, B00062 or solvent, B00184.

SUBTASK 23-61-00-210-003

(2) Examine the installation of the discharger base [3]:

(a) Make sure that there are no cracks in the sealant, A00247.

(b) If you find a crack, fill the crack with the sealant, A00247.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Static Discharger Base with Rivets Installation Test

SUBTASK 23-61-00-760-008

WARNING OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Check the electrical bond between the discharger base [3] and airplane surface:

(a) Use a intrinsically safe approved bonding meter, COM-1550 to measure the electrical bond between the discharger base [3] and the airplane surface (SWPM 20-20-00).

NOTE: This resistance value is for a new installed static discharger base and a new bond. The resistance value is different for an in-service bond and the discharger base.

The resistance value of a new discharger base [3] must not be more than:

NOTE: A resistance of more than these value can cause airplane equipment damage if a lightning strike occurs.

1) 0.01 ohms for discharger base [3] installed on metal panels.

2) 0.10 ohms for discharger base [3] installed on an aluminum flame spray on top of the fiberglass laminate or the composite panel.

NOTE: This includes discharger bases installed on aluminum-covered fabric.

3) 1.0 ohms for discharger base [3] installed on composite panels without a cover of the aluminum flame spray.

SUBTASK 23-61-00-420-006

(2) Install the static discharger [1] into the discharger base [3], do this task: Static Discharger Installation, TASK 23-61-00-400-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-370-002

(1) Paint the area around the discharger base [3] (TASK 51-21-99-300-802).

SUBTASK 23-61-00-440-006

(2) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

END OF TASK

TASK 23-61-00-000-804

8. Static Discharger Base with Adapter Plate Assembly Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the static discharger base [3] with adapter plate assembly.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-1053	Knife - Putty, Broad Blade

D. Location Zones

Zone	Area
300	Empennage
500	Left Wing

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
600	Right Wing

E. Prepare for the Static Discharger Base with Adapter Plate Assembly Removal

SUBTASK 23-61-00-040-013

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from the hydraulic systems for the applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802
- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802

SUBTASK 23-61-00-040-014

(2) Remove the electrical power (TASK 24-22-00-860-802).

F. Static Discharger Base with Adapter Plate Assembly Removal

SUBTASK 23-61-00-020-005

(1) Remove the static discharger base [3] with adapter plate [9] assembly:

(a) Loosen the set screw [2].

(b) Remove the static discharger [1] from the discharger base [3].

(c) Make a mark with a grease pencil larger than the area of the discharger base [3] to be cleaned.

1) Make sure that the line is parallel to the airflow.

(d) Remove the adapter plate [9] assembly:

1) Remove the weather/aerodynamic fillet seal around the edge of the adapter plate [9] with a sealant removal tool, COM-2481.

2) Remove the bolts [4] and rivets [6] from the adapter plate [9] (SRM SUBJECT 51-40-02).

CAUTION BE CAREFUL WITH A PUTTY KNIFE, DAMAGE TO THE AIRCRAFT SURFACE CAN OCCUR.

(e) Put a broad blade putty knife, STD-1053 below the edge of the adapter plate [9].

(f) Remove the adapter plate [9] from the airplane surface.

(g) Remove the discharger base [3] from the adapter plate [9].

1) Remove the rivets [5] from the discharger base [3] (SRM SUBJECT 51-40-02).

2) Remove the discharger base [3] from the adapter plate [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-440-007

(1) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

——— END OF TASK ———

TASK 23-61-00-400-804

9. Static Discharger Base with Adapter Plate Assembly Installation

(Figure 401)

A. General

(1) This procedure has steps to install the static discharger base [3] with adapter plate assembly.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
51-21-99-300-802	Decorative Exterior Paint System Repair (P/B 701)
51-22-00-390-806	Fillet Seal Application (P/B 201)
51-22-00-390-808	Fastener Seal Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-739	Respirator - Half-Mask, Air Purifying (with filters/cartridges)
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00102	Abrasive - Aluminum Oxide Coated Cloth	
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
B01026	Solvent - FCC-55	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
B01054	Solvent - Methyl Ethyl Ketone and sec-Butyl Alcohol Blend - (MEK:secButyl Alcohol - 42:58 Percent)	BAC5750, ASTM D740/ASTM D1007
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Static discharger	23-61-00-02-110	PTF ALL
		23-61-00-02-210	PTF ALL
		23-61-00-02-305	PTF ALL
		23-61-00-02-355	PTF ALL

F. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

G. Prepare for the Static Discharger Base with Adapter Plate Assembly Installation

SUBTASK 23-61-00-040-015

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from the hydraulic systems for the applicable flight control surfaces, do these tasks:

- Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801
- Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802
- Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 23-61-00-040-016

(2) Remove the electrical power (TASK 24-22-00-860-802).

SUBTASK 23-61-00-110-003

(3) Clean the surface for the installation of the discharger base [3]:

(a) Remove sealant with a sealant removal tool, COM-2481.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(b) Clean the adapter plate [9] with solvent, B00148, solvent, B00666, solvent, B00184, FCC-55 solvent, B01026, or solvent, B01054.

1) Wear a half-mask respirator, STD-739 to clean with solvent.

2) Apply the solvent with a cotton wiper, G00034.

(c) Remove the solvent with cotton wiper, G00034 before it dries.

1) Do not touch the cleaned surface.

(d) Rub the adapter plate [9] with the abrasive cloth, B00102 (600 grit) (TASK 20-10-34-120-801).

NOTE: Change the sandpaper frequently.

1) Make sure that you have a smooth surface.

2) Make sure that the surface is not shiny.

(e) Clean with cotton wiper, G00034.

NOTE: Do not use solvent.

H. Static Discharger Base with Adapter Plate Assembly Installation

SUBTASK 23-61-00-420-007

(1) Install the discharger base [3]:

(a) Remove the metal discharger base [3] from the plastic container.

1) Do not to touch the bottom surface of the discharger base [3] which touches the adapter plate [9].

(b) Rub the discharger base [3] with a abrasive cloth, B00102.

1) Make sure that the discharger base [3] is installed in less than 5 minutes.

NOTE: Do not rub a plated discharger base with sandpaper. A plated discharger base has a green dot for identification. Put a pair of gloves on to touch the bases.

2) Make sure that you have a smooth surface.

NOTE: Change the sandpaper frequently.

3) Make sure that the adapter plate [9] is not shiny.

(c) Attach the discharger base [3] to the adapter plate [9].

(d) Install the rivets [5] which attaches the discharger base [3] to the adapter plate [9] (SRM SUBJECT 51-40-02).

(e) Apply sealant, A00247 to the rivets [5] (TASK 51-22-00-390-808).

(f) Apply sealant, A00247 to the edge of the discharger base [3] (TASK 51-22-00-390-806).

(g) Make the sealant smooth with a hardwood or plastic fillet smoothing spatula, STD-810.

(h) Make sure that you do not see cracks between the discharger base [3] and the adapter plate [9] surface.

(i) Remove the unwanted sealant, A00247 with a cotton wiper, G00034 which is moist with solvent, B00062 or solvent, B00184.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-61-00-420-008

(2) Install the adapter plate [9] assembly:

(a) Rub the airplane surface with abrasive cloth, B00102.

1) Make sure that you have a smooth surface.

NOTE: Change the sandpaper frequently.

2) Make sure that the airplane surface is not shiny.

(b) Attach the adapter plate [9] assembly to the airplane surface.

1) Make sure that you install the adapter plate [9] assembly in the same direction as the airflow.

(c) Install the bolts [4] that attaches to the adapter plate [9] assembly.

(d) Install the rivets [6] that attach to the adapter plate [9] assembly (SRM SUBJECT 51-40-02).

(e) Apply sealant, A00247 to the rivets [6] and bolts [4] (TASK 51-22-00-390-808).

(f) Apply sealant, A00247 to the edge of the adapter plate [9] (TASK 51-22-00-390-806).

(g) Make sure that you do not see cracks between the adapter plate [9] and the airplane surface.

(h) Remove the sealant, A00247 with a cotton wiper, G00034 which is moist with solvent, B00148.

SUBTASK 23-61-00-210-004

(3) Examine the discharger base [3]:

(a) Make sure that there are no cracks in the sealant, A00247.

(b) If you find a crack, fill the crack with the sealant, A00247.

I. Static Discharger Base with Adapter Plate Assembly Installation Test

SUBTASK 23-61-00-760-009

WARNING OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Check the electrical bond between the discharger base [3] and aircraft surface:

(a) Use a intrinsically safe approved bonding meter, COM-1550 to measure the electrical bond between the discharger base [3] and the airplane surface (SWPM 20-20-00).

NOTE: This resistance value for a new static discharger base and a new bond is different for an in-service bond and the discharger base.

(b) Make sure that the resistance value of a new discharger base [3] must not be more than:

NOTE: A resistance of more than these value can cause airplane equipment damage if a lightning strike occurs.

1) 0.01 ohms for discharger base [3] installed on metal panels.

2) 0.10 ohms for discharger base [3] installed on an aluminum flame spray on top of the fiberglass laminate or the composite panel.

NOTE: This includes discharger bases installed on aluminum-covered fabric.

3) 1.0 ohms for discharger base [3] installed on composite panels without a cover of the aluminum flame spray.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-61-00-420-009

(2) Install the static discharger [1] into the discharger base [3], do this task: Static Discharger Installation, TASK 23-61-00-400-801.

SUBTASK 23-61-00-370-003

(3) Paint the area around the discharger base [3] (TASK 51-21-99-300-802).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-440-008

(1) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

- Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802
- Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802
- Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STATIC DISCHARGER - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Static Discharger Inspection

(2) Static Discharger Resistance Measurement

B. Do a regular check of the static dischargers to prevent static interference and bad radio operation.

TASK 23-61-00-210-801

2. Static Discharger Inspection

(Figure 601)

A. General

(1) This task has steps to inspect static dischargers.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

D. Prepare for the Static Discharger Inspection

SUBTASK 23-61-00-860-001

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove the hydraulic pressure and power from the hydraulic systems for the applicable flight control surfaces:

- For the aileron hydraulic systems, do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.
- For the rudder and standby hydraulic systems, do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- For the elevator hydraulic systems, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 23-61-00-860-002

(2) Remove the electrical power (TASK 24-22-00-860-802).

E. Static Discharger Inspection

SUBTASK 23-61-00-210-005

(1) Make sure that at least 19 static dischargers are installed correctly in the locations that follow:

NOTE: There are 22 static dischargers installed on the airplane.

(a) Make sure that at least two static dischargers are installed on each horizontal stabilizer.

1) Make sure that one is in the tip position or in the outermost trailing position.

(b) Make sure that at least two static dischargers are installed on the vertical stabilizer.

1) Make sure that one of the two must be on the tip or top-most position.

2) Where there are only 2 dischargers on the vertical, one of the two must be in the tip position or in the uppermost trailing position.

(c) The 2 dischargers on the aircraft wings, inboard from the winglet, may be missing.

(d) Make sure that 4 static dischargers are installed on each winglet.

SUBTASK 23-61-00-210-006

(2) Make sure that there is no lightning damage.

NOTE: This damage will be seen as a burned or a rough area on the black conductive surfaces.

(a) You could also find damage as pits on the discharger base.

(b) If it is necessary, replace the discharger.

SUBTASK 23-61-00-210-007

(3) Do a visual check for erosion of the discharger surface.

NOTE: The leading edge erosion must not extend more than a third the width of the discharger.

(a) Replace all dischargers that show damage.

(b) Replace all dischargers that show degradation of shrink tube (orange or black).

NOTE: The shrink tube protects the conductive paint that connects the tip to the base.

F. Put the Airplane Back to Its Usual Condition

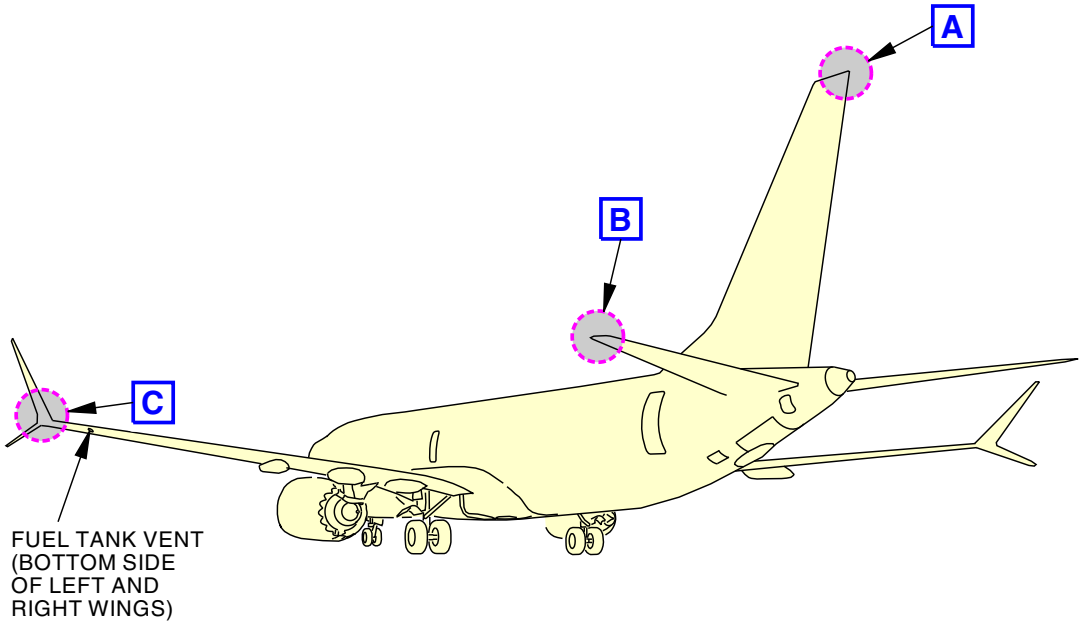
SUBTASK 23-61-00-010-002

(1) If it is necessary, put the applicable hydraulic systems back to the usual condition:

- For the aileron hydraulic systems, do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.
- For the rudder and standby hydraulic systems, do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.
- For the elevator hydraulic systems, do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

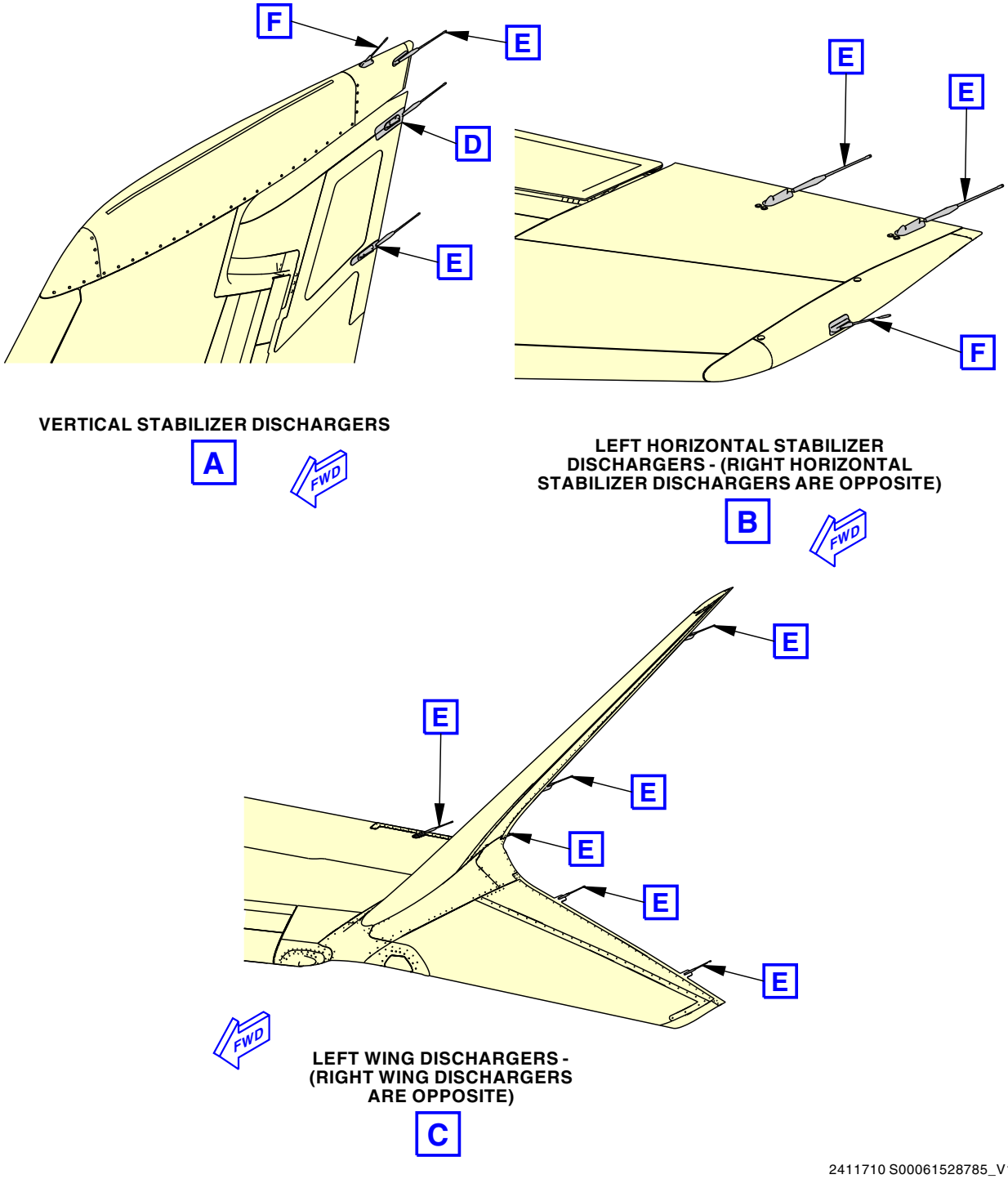


STATIC DISCHARGER LOCATIONS

2411709 S00061528784_V1

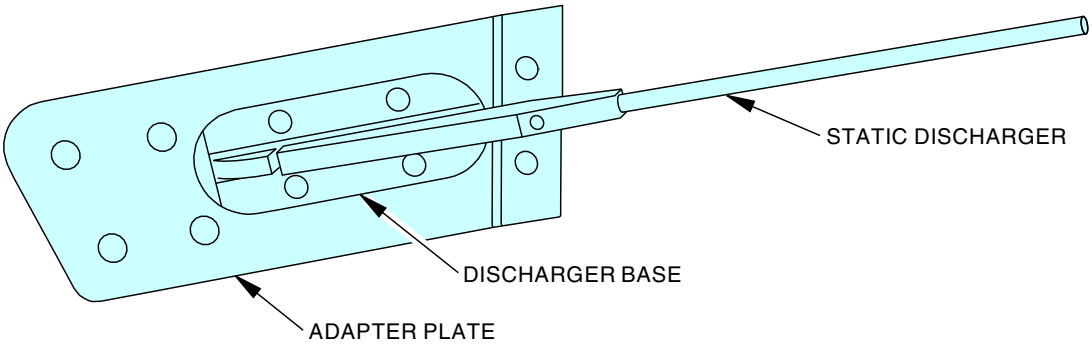
Static Discharger Inspection
Figure 601/23-61-00-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



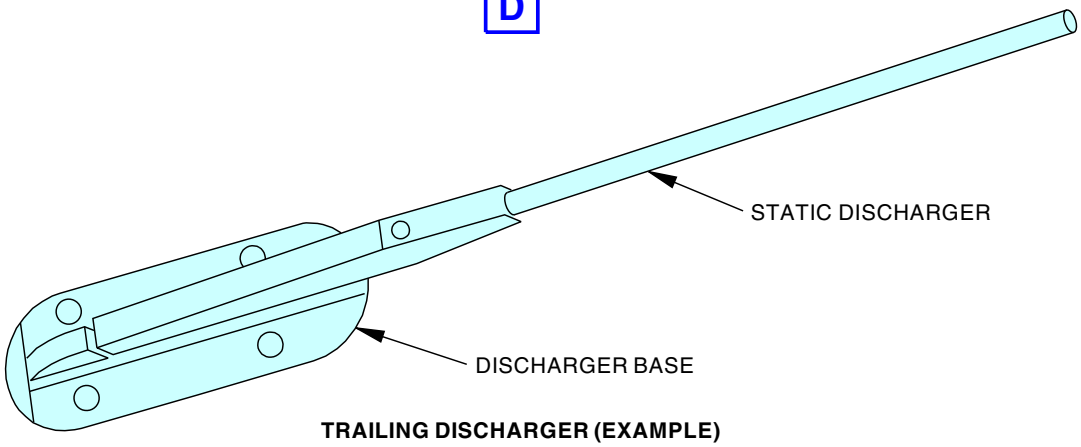
Static Discharger Inspection
Figure 601/23-61-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



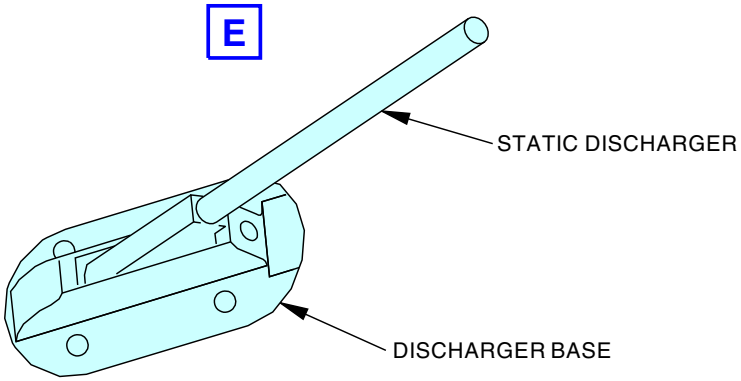
TRAILING DISCHARGER WITH ADAPTER PLATE (EXAMPLE)

D



TRAILING DISCHARGER (EXAMPLE)

E



TIP DISCHARGER (EXAMPLE)

F

2411714 S00061528798_V1

Static Discharger Inspection
Figure 601/23-61-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-61-00-760-801

3. Static Discharger Resistance Measurement

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
23-61-00-000-801	Static Discharger Removal (P/B 401)
23-61-00-000-802	Static Discharger Base with Screws Removal (P/B 401)
23-61-00-000-803	Static Discharger Base with Rivets Removal (P/B 401)
23-61-00-000-804	Static Discharger Base with Adapter Plate Assembly Removal (P/B 401)
23-61-00-400-801	Static Discharger Installation (P/B 401)
23-61-00-400-802	Static Discharger Base with Screws Installation (P/B 401)
23-61-00-400-803	Static Discharger Base with Rivets Installation (P/B 401)
23-61-00-400-804	Static Discharger Base with Adapter Plate Assembly Installation (P/B 401)
24-22-00-860-802	Remove Electrical Power (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
51-21-99-300-802	Decorative Exterior Paint System Repair (P/B 701)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-6457	Meter - Insulation (Range: 1-1,000 VDC or equivalent, select meter per test requirements) Part #: 1864-9700 Supplier: 62015 Part #: 1865PLUS Supplier: 62015 Part #: 1865PLUSCE Supplier: 62015 Part #: 2471F Supplier: 21844 Opt Part #: 1865-00-CE Supplier: 62015

C. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
300	Empennage
500	Left Wing
600	Right Wing

E. Prepare for the Static Discharger Resistance Measurement

SUBTASK 23-61-00-860-003

WARNING MAKE SURE TO REMOVE ALL PRESSURE FROM THE HYDRAULIC SYSTEM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Remove pressure and power from the hydraulic systems for the applicable flight control surfaces, do these tasks:

(a) Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

(b) Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

(c) Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 23-61-00-860-004

(2) Remove the electrical power (TASK 24-22-00-860-802).

SUBTASK 23-61-00-760-010

WARNING OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Check for fuel vapors:

NOTE: If you must do a resistance check for a static discharger that is in a five foot diameter area around a fuel tank vent, open fuel tank doors or open fuel tank access panels, then do the Alternative Discharger Resistance Test procedures.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Use these precautions for possible fuel vapors when you use a megohmmeter to measure the discharger resistance.

(a) Use a insulation meter, COM-6457 or equivalent meter with a 500 VDC test voltage and a maximum 5 milliampere short circuit current.

(b) Do not use a megohmmeter at these locations:

1) A 5.000 ft (1.524 m) diameter, vertical column areas adjacent to or below:

- a wing fuel tank vent (from the vent to the ground).
- open fuel tank doors (from the door to the ground).
- open fuel tank access panels (from the panel to the ground).

2) Zero to 18 in. (457 mm) above the ground in the area around the airplane.

(c) Make sure that:

1) Area is well ventilated.

2) Metal workstands are grounded.

3) Megohmmeter is plugged into a grounded receptacle.

NOTE: It is not necessary to connect shock-resistant portable megohmmeters to a grounded receptacle.

4) Megohmmeter is insulated from metal work stand.

F. Static Discharger Resistance Measurement Test

NOTE: If you must do a resistance check for a static discharger that is in a five foot diameter area around a fuel tank vent, open fuel tank doors or open fuel tank access panels, then do the Alternative Discharger Resistance Test procedures below.

SUBTASK 23-61-00-760-011

WARNING OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do a resistance check of static dischargers:

NOTE: Dayton-Granger and other static dischargers not supplied by Chelton.

(a) Use a megohmmeter to measure the resistance between the discharger tip and the discharger base:

(b) Set the megohmmeter to 500 VDC test voltage.

(c) Make a good surface bond with the end of the static discharger:

NOTE: A good connection between the discharger end and the megohmmeter is necessary for a correct resistance value.

1) Use water to make a cotton wiper, G00034 wet.

2) Make sure that the full end of the discharger touches the wet cotton wiper, G00034.

(d) Put one lead of the megohmmeter on the discharger base.

(e) Put the other lead of the megohmmeter on the wet cotton wiper, G00034.

(f) Measure the resistance value.

(g) Make sure that the resistance value is in a range of 6-100 megohms for dischargers.

NOTE: If the measured value is too high, add water to the wet material and then measure the resistance value again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) For a discharger that does not agree with the correct resistance value, replace the static discharger.

These are the tasks:

- Static Discharger Removal, TASK 23-61-00-000-801
- Static Discharger Installation, TASK 23-61-00-400-801

SUBTASK 23-61-00-760-012

WARNING OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do the resistance check of static dischargers:

NOTE: Chelton supplied static dischargers only.

(a) Use a megohmmeter to measure the resistance between the discharger tip and the discharger base:

(b) Set the megohmmeter to 500 VDC test voltage.

NOTE: A good electrical bond between the end of the discharger tip and the megohmmeter is necessary for a correct measure of resistance.

(c) To make a good connection with the end of the discharger tip, use a wet material.

(d) Put the wet material on the END of the discharger tip.

NOTE: DO NOT WRAP the wet material around the discharger tip of Chelton static dischargers. This can give incorrect resistance values that cause unnecessary removals of serviceable static dischargers. Put the wet material between the tip of the discharger and the megohmmeter probe

(e) Put one megohmmeter probe on the base and one probe on the wet material at the end of the discharger tip. .

NOTE: See On Wing Discharger Resistance Test Detail I (Figure 602)

1) Make sure that the measured resistance is between 6-100 megohms.

2) If the resistance is more than 6-100 megohms, remove the wet material and put megohmmeter probe directly on the tip of the discharger core material.

(f) Add more water if it is necessary and measure again.

1) Use the lowest measured value.

(g) For a discharger that does not agree with the correct resistance value, replace the static discharger.

These are the tasks:

- Static Discharger Removal, TASK 23-61-00-000-801
- Static Discharger Installation, TASK 23-61-00-400-801

G. Alternative Static Discharger Resistance Test

SUBTASK 23-61-00-020-006

(1) Remove the static discharger (TASK 23-61-00-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-61-00-760-013

WARNING

OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do a resistance check of discharger tip and discharger shank:

NOTE: Dayton-Granger and other static dischargers not supplied by Chelton.

(a) Use a megohmmeter to measure the resistance between the discharger tip and the discharger shank:

(b) Set the megohmmeter to 500 VDC test voltage.

NOTE: A good electrical bond between the end of the discharger tip and the megohmmeter is necessary for a correct measure of resistance.

(c) To make a good connection with the end of the discharger tip, use a wet paper towel, cotton cloth or a sponge.

(d) Put the wet towel on the end of the discharger tip.

(e) Put the megohmmeter connectors on the shank of the static discharger and on the wet towel at the end of the tip.

1) Make sure that the resistance is between 6-100 megohms.

2) If the resistance is high, make sure that the meter lead and the end of the discharger tip touch the wet towel.

(f) Add more water if it is necessary and measure again.

1) Use the lowest measured value.

(g) For a discharger that does not agree with the correct resistance value, replace the static discharger.

SUBTASK 23-61-00-760-014

WARNING

OBEY THE INSTRUCTIONS TO USE A MEGOHMETER, FAILURE TO OBEY THE INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Do a resistance check between discharger tip and discharger shank:

NOTE: Chelton supplied static discharger only.

(a) Use a megohmmeter to measure the resistance between the discharger tip and the discharger shank:

(b) Set the megohmmeter to 500 V dc (volts direct current) test voltage.

NOTE: A good electrical bond between the end of the discharger tip and the megohmmeter is necessary for a correct measure of resistance.

(c) To make a good connection with the end of the discharger tip, use a wet material.

(d) Put the wet material on the END of the discharger tip.

NOTE: DO NOT WRAP the wet material around the discharger tip of Chelton static dischargers. This can give incorrect resistance values that cause unnecessary removals of serviceable static dischargers. Put the wet material between the tip of the discharger and the megohmmeter probe.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Put one megohmmeter probe on the shank of the static discharger and one probe on the edge of the wet material at the end of the discharger tip.

NOTE: See Alternate (Off Wing) Discharger Resistance Test Detail II (Figure 602).

1) Make sure that the measured resistance is between 6-100 megohms.

2) If the resistance is more than 6-100 megohms, remove the wet material and put megohmmeter probe directly on the tip of the discharger core material.

(f) Add more water if it is necessary and measure again.

1) Use the lowest measured value.

(g) For a discharger that does not agree with the correct resistance value, replace the static discharger.

SUBTASK 23-61-00-420-010

(4) Install the Static Discharger (TASK 23-61-00-400-801).

H. Resistance Test for a Static Discharger

SUBTASK 23-61-00-760-015

WARNING OBEY THE INSTRUCTIONS TO USE A BONDING METER, FAILURE TO OBEY INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do the resistance test of static dischargers:

(a) Do a resistance test of static dischargers with a metallic shank.

1) Use a intrinsically safe approved bonding meter, COM-1550, (SWPM 20-20-00) to measure the resistance between the static discharger shank and the static discharger base.

2) Make sure that the resistance is not more than 1 ohm.

NOTE: A resistance of more than these values can cause damage if a lightning strike occurs.

(b) Do a resistance test of static dischargers with a plastic shank.

1) Use a intrinsically safe approved bonding meter, COM-1550, (SWPM 20-20-00) to measure the resistance between the set screw in the static discharger shank and the static discharger base.

2) Make sure that the resistance is not more than 1 ohm.

NOTE: A resistance of more than these values can cause damage if a lightning strike occurs.

I. Resistance Test for a Static Discharger Base

SUBTASK 23-61-00-760-016

WARNING OBEY THE INSTRUCTIONS TO USE A BONDING METER, FAILURE TO OBEY INSTRUCTIONS WILL CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do a electrical bond check between static discharger bases and airplane surface:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Method 1 - Measure the bond resistance from the discharger base to its mounting surface.

NOTE: It may be necessary to scratch the paint on the mounting surface to get a good connection.

1) Use a intrinsically safe approved bonding meter, COM-1550 to measure the bond between the static discharger base and the airplane surface (SWPM 20-20-00).

NOTE: This resistance value is for an in-service discharger. There is a different value for a new installed discharger base.

2) Put one lead on the static discharger base.

3) Put the other lead on the airplane surface.

4) Make sure that the two leads touch bare metal.

5) If it is necessary, remove a small quantity of paint or use a sharp probe to go into the paint.

6) Make sure the resistance value for an in-service static discharger base is not more than:

NOTE: A resistance of more than these values can cause damage if a lightning strike occurs. Take the resistance measurement between the discharger base and the discharger mounting surface.

a) 0.05 ohms for the discharger base installed on the metal panels.

b) 0.5 ohms for the discharger base installed on an aluminum flame spray on top of the fiberglass laminate or the composite panels.

c) 0.5 ohms for the discharger base installed on an aluminum-covered fabric.

d) 5.0 ohms for the discharger base installed on a composite panels that do not have an aluminum layer.

(b) Method 2 - Measure the bond resistance between two adjacent discharger bases.

1) Use a intrinsically safe approved bonding meter, COM-1550 to measure the bond between a static discharger base and an adjacent static discharger base (SWPM 20-20-00).

NOTE: This resistance value is for an in-service discharger. There is a different value for a new installed discharger base.

2) Make sure that the resistance between the two adjacent static discharger bases is not more than:

NOTE: A resistance of more than these values can cause damage if a lightning strike occurs. Take the resistance measurement between two discharger bases.

a) 0.05 ohms for the discharger base installed on the metal panels.

b) 0.5 ohm for the discharger base installed on an aluminum flame spray on top of the fiberglass laminate or the composite panels.

c) 0.5 ohm for the discharger base installed on an aluminum-covered fabric.

d) 5.0 ohms for the discharger base installed on a composite panels that do not have an aluminum layer.

3) If the resistance shows more than the resistance values in Method 2, then do the procedure in Method 1 to find the defective static discharger base.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) For a static discharger base that shows more than the resistance value in Method 1, replace the static discharger base:

1) For a static discharger base attached with screws, do these tasks:

a) Static Discharger Base with Screws Removal, TASK 23-61-00-000-802.

b) Static Discharger Base with Screws Installation, TASK 23-61-00-400-802.

2) For a static discharger base attached with rivets, do these tasks:

a) Static Discharger Base with Rivets Removal, TASK 23-61-00-000-803.

b) Static Discharger Base with Rivets Installation, TASK 23-61-00-400-803.

3) For a static discharger adapter plate assembly attached with rivets and screws, do these tasks:

a) Static Discharger Base with Adapter Plate Assembly Removal, TASK 23-61-00-000-804.

b) Static Discharger Base with Adapter Plate Assembly Installation, TASK 23-61-00-400-804.

SUBTASK 23-61-00-370-004

(2) Paint the area where the paint was removed for the surface resistance test, do this task:

(a) Decorative Exterior Paint System Repair, TASK 51-21-99-300-802.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 23-61-00-010-003

(1) If it is necessary, put the hydraulic systems back to the usual condition, do these tasks:

(a) Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

(b) Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

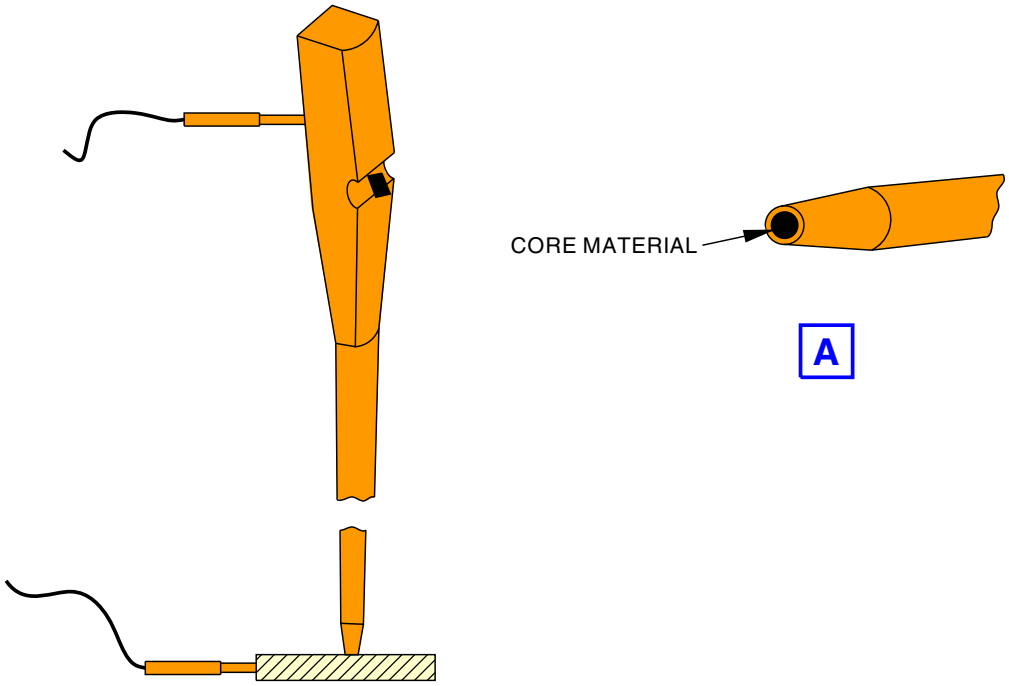
(c) Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ON WING DISCHARGER RESISTANCE TEST



ALTERNATE (OFF WING) DISCHARGER RESISTANCE TEST

NOTE:

TEST PROCEDURES FOR THE TRAILING EDGE AND WING TIP DISCHARGERS ARE THE SAME.

2411715 S00061528800_V1

Chelton Static Discharger Resistance Test
Figure 602/23-61-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VOICE RECORDER SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Voice Recorder System Deactivation

(2) Voice Recorder System Activation

(3) Make a copy of the Cockpit Voice Recorder (CVR) data with the Hand Held Multipurpose Interface (HHMPI)

(4) Make a copy of the CVR data with the Extended Handheld Download Unit (EHHDLU)

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X OR 2100-1925-2X

(5) Make a copy of the Cockpit Voice Recorder (CVR) data with the Data Acquisition Ruggedized Tool (DART).

PTF ALL

TASK 23-71-00-040-801

2. Voice Recorder System Deactivation

A. General

(1) This task disconnects the electrical power to the Cockpit Voice Recorder (CVR) system.

B. Tools/Equipment

Reference	Description
STD-1390	Headphone - 600 Ohm, with 1/4 Inch Mono Audio Plug

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Voice Recorder System Deactivation

SUBTASK 23-71-00-860-001

(1) Deactivate the CVR as follows:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 23-71-00-210-001

(2) Verify that the electrical power to the voice recorder system is off as follows:

(a) Set the CVR AUTO/ON switch (P5) to the ON position.

(b) Connect a headphone, STD-1390 to the voice recorder control panel at the overhead panel, P5.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Push and hold the TEST switch on the voice recorder control panel for one second.

1) Make sure that you do not hear a tone in the headphone.

2) Make sure that the STATUS light does not come on.

——— END OF TASK ———

TASK 23-71-00-440-801

3. Voice Recorder System Activation

A. General

(1) This procedure connects the electrical power to the Cockpit Voice Recorder (CVR) system.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Voice Recorder System Activation

SUBTASK 23-71-00-860-003

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 23-71-00-710-011

(2) Push and hold the TEST switch on the voice recorder control panel for one second.

(a) Make sure that the STATUS light comes on in five seconds or less.

——— END OF TASK ———

TASK 23-71-00-970-801

4. Make a Copy of the CVR Data with the HHMPI

A. General

(1) This task gives the steps on how to use the Hand Held Multipurpose Interface (HHMPI) to make a copy of the Cockpit Voice Recorder (CVR) data.

(2) The copied data then can be analyzed at a different location by the applicable airline personnel.

B. References

Reference	Title
23-71-11-000-801	Voice Recorder Removal (P/B 401)
23-71-11-400-801	Voice Recorder Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-13694	Unit - Interface, Hand-Held Multi-Purpose (HHMPI) Part #: FDS40-0301 Supplier: Z7C70 Opt Part #: FDS400-301 Supplier: Z7C70
COM-14627	Adapter - CVR Rack, ARINC-757 Part #: FDS40-0247 Supplier: Z7C70 Opt Part #: FDS400-247 Supplier: Z7C70
COM-14992	Cable - Interface, HHMPI (For Honeywell HRF5-V or C) Part #: FDS40-0274 Supplier: Z7C70

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

E. Prepare to Make a Copy of the CVR Data

SUBTASK 23-71-00-480-002

(1) Make sure that the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694 is OFF.

SUBTASK 23-71-00-480-003

(2) If it is necessary, install the removable media into the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694.

NOTE: The downloaded CVR data can be stored on its internal memory, a USB flash drive, a compact flash card, SD card, or an optional PCMCIA card.

SUBTASK 23-71-00-480-013

(3) If it is necessary, do these steps to prepare the CVR:

NOTE: These steps are not necessary if audio data is not necessary.

(a) Do this task: Voice Recorder Removal, TASK 23-71-11-000-801.

(b) Connect the ARINC-757 CVR rack adapter, COM-14627, to the CVR equipment rack and the rear of the CVR.

(c) Activate the CVR system (TASK 23-71-00-440-801).

SUBTASK 23-71-00-860-025

(4) Set the voice recorder switch panel AUTO/ON switch to ON.

SUBTASK 23-71-00-480-008

(5) Connect the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694, as follows:

(a) Connect the HHMPI interface cable (For Honeywell HRF5-V or C), COM-14992, to the CVR front connector.

(b) Connect the HHMPI interface cable (For Honeywell HRF5-V or C), COM-14992, to the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694.

SUBTASK 23-71-00-480-019

(6) Turn on the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694, as follows:

(a) Press CANCEL or OK for 2-3 seconds until the unit beeps.

NOTE: The interface cable is detected automatically. A blue aircraft icon will show in the status bar of the display to indicate correct connection to the CVR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Make a Copy of the CVR Data

SUBTASK 23-71-00-700-005

(1) Make a copy of the CVR data with the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694, as follows:

(a) If it is necessary, select the aircraft type, model and tail number.

(b) If it is necessary, select the memory storage device.

NOTE: The option to download to the HHMPI Internal Memory will not appear if an SD card is connected. Remove the SD storage card if access to the internal memory is required.

(c) Select DOWNLOAD OPERATIONS from the main menu.

(d) Select one of the CVR download options.

(e) Push the OK button when the download has completed.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-00-080-009

(1) Turn off the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694, as follows:

(a) Press CANCEL until the CONFIRM POWER OFF alert is on the display.

(b) Press OK to confirm.

SUBTASK 23-71-00-080-004

(2) Disconnect the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694, as follows:

(a) Disconnect the HHMPI interface cable (For Honeywell HRF5-V or C), COM-14992, from the Hand-Held Multi-Purpose Interface Unit (HHMPI), COM-13694.

(b) Disconnect the HHMPI interface cable (For Honeywell HRF5-V or C), COM-14992, from the CVR front connector.

SUBTASK 23-71-00-080-007

(3) If it is necessary, do these steps to re-install the CVR:

(a) Deactivate the CVR system (TASK 23-71-00-040-801).

(b) Disconnect the ARINC-757 CVR rack adapter, COM-14627 from the CVR equipment rack and the rear of the CVR.

(c) Do this task: Voice Recorder Installation, TASK 23-71-11-400-801.

SUBTASK 23-71-00-860-031

(4) Set the voice recorder switch panel AUTO/ON switch to AUTO.

——— END OF TASK ———

TASK 23-71-00-970-802

5. Make a Copy of the CVR Data with the EHHDLU

A. General

(1) This task gives the steps on how to use the Extended Handheld Download Unit (EHHDLU) to make a copy of the Cockpit Voice Recorder (CVR) data.

(2) The copied data then can be analyzed at a different location by the applicable airline personnel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) Use download unit, COM-20206 (P/N 69001074-080) for this task.

NOTE: P/N 69001074-081 is an acceptable equivalent to the P/N 69001074-080.

NOTE: P/N 69001074-060 is an acceptable equivalent to the P/N 69001074-080.

(4) Rack adapter FDS40-0247 can be used if the download unit kit does not include a rack adapter.

B. References

Reference	Title
23-71-11-000-801	Voice Recorder Removal (P/B 401)
23-71-11-400-801	Voice Recorder Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-20206	Download Unit - Recorder Part #: 69001074-090 Supplier: 97896 Opt Part #: 69001074-080 Supplier: 97896

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

E. Prepare to Make a Copy of the CVR Data

SUBTASK 23-71-00-480-021

(1) Do this task: Voice Recorder Removal, TASK 23-71-11-000-801.

SUBTASK 23-71-00-480-023

(2) Do these steps to connect the download unit, COM-20206:

(a) Connect the USB-C adapter to the handheld unit.

(b) Connect the download cable to the USB-C adapter.

(c) Connect the download cable to the CVR front connector.

SUBTASK 23-71-00-480-026

(3) Do these steps to install the CVR with the rack adapter:

(a) Connect the rack adapter to the CVR equipment rack and the rear of the CVR.

(b) Activate the CVR system (TASK 23-71-00-440-801).

SUBTASK 23-71-00-860-035

(4) Set the voice recorder switch panel AUTO/ON switch to the ON position.

SUBTASK 23-71-00-480-022

(5) Push the power button on the handheld unit to turn it on.

SUBTASK 23-71-00-480-024

(6) If it is necessary, move one finger across the screen to unlock the handheld unit.

F. Network Settings

SUBTASK 23-71-00-480-025

(1) Do these steps to prepare the network settings on the handheld unit:

(a) Move one finger down from the top of the screen.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) At the bottom of the screen, select the configuration icon.

(c) Select Network & internet.

(d) Select Ethernet.

(e) Enter Netmask: 255.255.255.0.

(f) Enter IP Address: 192.168.27.1.

(g) Select OK.

(h) Select the home button to go back to the main menu.

G. Make a Copy of the CVR Data

SUBTASK 23-71-00-700-017

(1) Do these steps to make a copy the CVR data with the handheld unit:

(a) Select the eHHDLU application.

(b) Make sure that the Download screen is displayed.

(c) Select HFR5-V SSCVR from the Recorder Type list.

(d) Select DOWNLOAD.

(e) If the security notification is shown, select ALLOW.

(f) Select the folder location to save the file.

1) If it is necessary, make a new folder.

a) Select the three vertical dots in the upper right corner.

b) Select New Folder

c) Enter the folder name.

(g) If it is necessary, change the file name.

(h) Select SAVE to start the download.

(i) Wait for the download to finish.

1) Do not disconnect any cables from the CVR.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-00-080-013

(1) Do these steps to turn off the handheld unit:

(a) Press and hold the power button.

(b) Select Power off.

SUBTASK 23-71-00-080-016

(2) Do these steps to remove the CVR and the rack adapter:

(a) Deactivate the CVR system (TASK 23-71-00-040-801).

(b) Disconnect the rack adapter from the CVR equipment rack and the rear of the CVR.

SUBTASK 23-71-00-080-014

(3) Do these steps to disconnect the download unit, COM-20206:

(a) Disconnect the download cable from the CVR front connector.

(b) Disconnect the download cable from the USB-C adapter.

(c) Disconnect the USB-C adapter from the handheld unit.

SUBTASK 23-71-00-080-015

(4) Do this task: Voice Recorder Installation, TASK 23-71-11-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-71-00-860-034

(5) Make sure that the voice recorder switch panel AUTO/ON switch is set to the AUTO position.

——— END OF TASK ———

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X OR 2100-1925-2X

TASK 23-71-00-970-804

6. Make a copy of the Cockpit Voice Recorder (CVR) data with the Data Acquisition Ruggedized Tool (DART)

A. General

(1) This task gives the instructions on how to use the Data Acquisition Ruggedized Tool (DART) to make a copy of the Cockpit Voice Recorder (CVR) data.

(2) The copied data then can be analyzed at a different location by the applicable airline personnel.

B. References

Reference	Title
23-71-11-000-801	Voice Recorder Removal (P/B 401)
23-71-11-400-801	Voice Recorder Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-23709	Ruggedized Tool - Data Acquisition (DART) Part #: KGB750-001 Supplier: 73ZG6
COM-23711	Cable - Interface, DART (For Honeywell HFR5) Part #: KGB750-108 Supplier: 73ZG6
COM-23909	Connector - Power Part #: KGB750-121 Supplier: 73ZG6

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

E. Prepare to Make a Copy of the CVR Data

SUBTASK 23-71-00-020-002

(1) If the audio data is necessary, do these steps:

NOTE: These steps are not necessary if audio data is not necessary.

(a) Do this task: Voice Recorder Removal, TASK 23-71-11-000-801.

(b) Connect power connector, COM-23909, to the CVR equipment rack and rear of the CVR.

SUBTASK 23-71-00-040-001

(2) Do this task: Voice Recorder System Deactivation, TASK 23-71-00-040-801.

SUBTASK 23-71-00-970-001

(3) Make sure that data acquisition ruggedized tool, COM-23709, is turned ON.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X

SUBTASK 23-71-00-970-002

(4) Connect DART interface cable, COM-23711, to the CVR front connector and J2 connector on data acquisition ruggedized tool, COM-23709.

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X OR 2100-1925-2X

SUBTASK 23-71-00-970-004

(5) Make sure that DETECTED CABLE field, on the DART screen, indicates the attached cable.

SUBTASK 23-71-00-440-002

(6) Do this task: Voice Recorder System Activation, TASK 23-71-00-440-801.

SUBTASK 23-71-00-970-005

(7) Make sure that RECORDER field, on the DART screen, shows CONNECTED.

SUBTASK 23-71-00-970-006

(8) Make sure that function buttons, F1, F2, and F3, on the DART screen, are red in color.

F. Make a Copy of the CVR Data

SUBTASK 23-71-00-970-007

(1) Push the F2 button to open the DOWNLOAD RECORDER screen on the DART screen.

SUBTASK 23-71-00-970-008

(2) Select Download Type.

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X

(a) Use the Up and Down arrow buttons to select CVR Narrow Data or CVR Wide Data.

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X OR 2100-1925-2X

(b) Push the Enter button.

SUBTASK 23-71-00-970-009

(3) Select the applicable download option:

- For the DOWNLOAD ALL option, push F2 button.
- For the DOWNLOAD X MINUTES option (if applicable), push F3 button.
- For the DOWNLOAD SINCE LAST option (if applicable), push F4 button.

SUBTASK 23-71-00-970-010

(4) If the DOWNLOAD X MINUTES option is selected, do these steps:

(a) Use the Up and Down arrow buttons to specify the desired download minutes.

(b) Push the Enter button.

(c) Push the F4 button to select START DOWNLOAD.

SUBTASK 23-71-00-970-011

(5) Enter the aircraft tail number.

(a) Use the Up, Down, Left, and Right arrow buttons to navigate around the keyboard.

(b) Push the Enter button to add the selected alphanumeric digit to the aircraft tail number text entry area.

(c) Push F4 button to select OK.

SUBTASK 23-71-00-970-012

(6) Enter the Recorder Part Number and Recorder Serial Number.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X OR 2100-1925-2X (Continued)

(a) Use F2 and F3 buttons to navigate between different fields on the DOWNLOAD INFORMATION screen.

(b) Use the Up, Down, Left, and Right arrow buttons to navigate around the keyboard.

(c) Push the Enter button to add the selected alphanumeric digit to the Recorder Part Number or Recorder Serial Number text entry area.

(d) Push F4 button to select OK.

SUBTASK 23-71-00-970-013

(7) Push F3 to advance to the Reason For Download options.

SUBTASK 23-71-00-970-014

(8) Select the applicable Reason For Download option.

(a) Use the Up and Down arrow buttons to select the Reason For Download.

(b) Push the Enter button.

SUBTASK 23-71-00-970-015

(9) Push the F4 button to select START DOWNLOAD.

(a) Make sure that the DART screen shows the progress bar at the center of the screen.

NOTE: When download is complete, the file name displays at the top of the UPLOADED DATA screen and OK displays in the file name.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-00-040-002

(1) Do this task: Voice Recorder System Deactivation, TASK 23-71-00-040-801.

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X

SUBTASK 23-71-00-970-016

(2) Disconnect DART interface cable, COM-23711, from the CVR front connector and J2 connector on data acquisition ruggedized tool, COM-23709.

PTF ALL; AIRPLANES WITH COCKPIT VOICE RECORDER P/N 980-6032-00X OR 2100-1925-2X

SUBTASK 23-71-00-970-018

(3) If it is removed, do these steps to install the CVR:

(a) Disconnect power connector, COM-23909, from the CVR equipment rack and rear of the CVR.

(b) Do this task: Voice Recorder Installation, TASK 23-71-11-400-801.

SUBTASK 23-71-00-440-003

(4) Do this task: Voice Recorder System Activation, TASK 23-71-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VOICE RECORDER SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks for the voice recorder system:

(1) Voice Recorder System - Operational Test

(2) Voice Recorder System - Audio Fidelity Test

(3) Voice Recorder System - System Test

TASK 23-71-00-710-801

2. Voice Recorder System - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test will perform an operational test of the voice recorder system.

B. References

Reference	Title
23-27-00-740-802	ACARS - Operational Test (P/B 501)

C. Tools/Equipment

Reference	Description
STD-1390	Headphone - 600 Ohm, with 1/4 Inch Mono Audio Plug

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 23-71-00-860-005

(1) Make sure that the Aircraft Communications Addressing And Reporting System (ACARS) system is serviceable (TASK 23-27-00-740-802).

SUBTASK 23-71-00-860-006

(2) Set the airplane's parking brake.

(a) Pull the parking brake lever, located on the control stand.

(b) Make sure that the parking brake light comes on.

SUBTASK 23-71-00-420-002

(3) Connect a headphone, STD-1390, to the voice recorder control panel at the P5 pilot's overhead panel.

SUBTASK 23-71-00-860-007

(4) Set the volume control switches, on the Audio Control Panels (ACP), to the OFF position.

SUBTASK 23-71-00-860-017

(5) Set the CVR AUTO/ON switch (P5) to the ON position.

(a) Make sure that the CVR AUTO/ON switch stays in the ON position.

SUBTASK 23-71-00-710-002

(6) Make sure that you can hear the flight deck conversation in the headphone.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Operational Test

SUBTASK 23-71-00-700-001

(1) Do the Operational Test as follows:

NOTE: When you push the TEST switch on the voice recorder control panel, a test is performed on the Cockpit Voice Recorder (CVR), Datalink, and Recorder Independent Power Supply (RIPS).

(a) Push and hold the TEST switch on the voice recorder control panel for two seconds.

(b) In five seconds or less, make sure that:

1) You hear a test tone (800 Hz) in the headphone.

2) The STATUS light comes on.

a) A “PASS” Test is when the STATUS light is steady and bright.

b) A “FAIL” is when the STATUS light blinks, flickers, or dims. Refer to the IFIM and do the applicable procedure(s).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-00-080-001

(1) Disconnect the headphone from the voice recorder's control panel.

SUBTASK 23-71-00-860-018

(2) Set the CVR AUTO/ON switch (P5) back to the AUTO position.

SUBTASK 23-71-00-860-036

(3) If it is necessary, release the parking brake.

——— END OF TASK ———

TASK 23-71-00-730-801

3. Voice Recorder System - Audio Fidelity Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure contains a task for the four-channel microphone test of the voice recorder system using the flight crew's microphones and the area microphone.

B. References

Reference	Title
23-51-00-710-801	Flight Interphone System Operational Test (P/B 501)

C. Tools/Equipment

Reference	Description
STD-1390	Headphone - 600 Ohm, with 1/4 Inch Mono Audio Plug
STD-13898	Headset with Boom Mic and two-prong plug (For use on Flight Deck during CVR testing)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Test

SUBTASK 23-71-00-860-019

(1) Set the CVR AUTO/ON switch (P5) to the ON position.

(a) Make sure that the CVR AUTO/ON switch stays in the ON position.

SUBTASK 23-71-00-760-002

(2) Make sure that the Flight Interphone System is serviceable (TASK 23-51-00-710-801).

SUBTASK 23-71-00-480-001

(3) Connect a headphone, STD-1390 to the voice recorder control panel at the pilot's overhead panel, P5.

SUBTASK 23-71-00-860-009

(4) Set the volume control switches on the audio control panels (ACP) to the off position.

F. Audio Fidelity Test

SUBTASK 23-71-00-710-003

(1) Do an operational test for the voice recorder system:

NOTE: When you push the TEST switch on the voice recorder control panel, a test is performed on the CVR and the Recorder Independent Power Supply (RIPS).

(a) Push and hold the TEST switch on the voice recorder control panel for two seconds.

(b) In five seconds or less, make sure that:

1) You hear a tone in the headphone.

2) The STATUS light comes on.

SUBTASK 23-71-00-860-010

(2) Put a cover on the area microphone for the voice recorder system.

SUBTASK 23-71-00-710-004

(3) Do a test for the voice recorder microphone at the captain's position:

(a) Connect a on-board boom microphone headset, STD-13898 to the captain's jack panel.

(b) If the captain's ACP has a BOOM microphone switch, then make sure it is set to the BOOM position.

(c) Speak into the boom microphone.

1) Make sure that you hear your voice through the headphone at the voice recorder control panel.

NOTE: When you test the digital or solid state voice recorder you will hear your voice in the headset as you speak. When you test the analog or tape based voice recorder you will hear your voice in the headset after approximately a 0.5 second delay.

(d) Disconnect the boom microphone from the captain's jack panel.

SUBTASK 23-71-00-710-005

(4) Do a test for the voice recorder microphone at the first officer's position:

(a) Connect a boom microphone to the first officer's jack panel.

(b) If the first officer's ACP has a BOOM microphone switch, then make sure it is set to the BOOM position.

(c) Speak into the boom microphone.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that you hear your voice through the headphone at the voice recorder control panel.

NOTE: When you test the digital or solid state voice recorder you will hear your voice in the headset as you speak. When you test the analog or tape based voice recorder you will hear your voice in the headset after approximately a 0.5 second delay.

(d) Disconnect the boom microphone from the first officer's jack panel.

SUBTASK 23-71-00-710-006

(5) Do a test for the voice recorder microphone at the first observer's position:

(a) Connect a boom microphone or a hand mic, as applicable, to the first observer's jack panel.

(b) If the first observer's ACP has a BOOM microphone switch, then make sure it is set to the BOOM position.

(c) Speak into the boom microphone or hand mic.

1) Make sure that you hear your voice through the headphone at the voice recorder control panel.

NOTE: When you test the digital or solid state voice recorder you will hear your voice in the headset as you speak. When you test the analog or tape based voice recorder you will hear your voice in the headset after approximately a 0.5 second delay.

(d) Disconnect the boom microphone or hand mic from the first observer's jack panel.

SUBTASK 23-71-00-700-002

(6) Remove the cover from the area microphone for the voice recorder system.

SUBTASK 23-71-00-710-007

(7) Do a test for the area microphone of the voice recorder system:

(a) Speak in an usual voice to the area microphone.

NOTE: Keep 3 to 4 feet (1 meter) away from the area microphone for the voice recorder.

1) Make sure that you hear your voice through the headphone at the voice control panel.

NOTE: When you test the digital or solid state voice recorder you will hear your voice in the headset as you speak. When you test the analog or tape based voice recorder you will hear your voice in the headset after approximately a 0.5 second delay.

NOTE: There can be feedback caused by the audio coupling between the headphones and the area mic. The feedback is a nuisance only and does not affect the test results.

SUBTASK 23-71-00-860-011

(8) Do these steps to erase the memory for the voice recorder:

(a) Set the airplane's parking brake.

(b) Push and hold the ERASE switch on the voice recorder control panel for three seconds.

1) Make sure that you hear a tone in the headphone in six seconds or less.

G. Put the Airplane Back to the Usual Condition

SUBTASK 23-71-00-020-001

(1) Disconnect the headphone from the voice recorder control panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-71-00-860-020

(2) Set the CVR AUTO/ON switch (P5) to the AUTO position.

——— END OF TASK ———

TASK 23-71-00-730-802

4. Voice Recorder System Test

A. General

(1) This procedure contains the steps to test the voice recorder system.

(2) This system test includes these tests:

(a) Operational test

(b) Audio fidelity test

(c) Cockpit Voice Recorder (CVR) engine relay test

(d) Bulk erasure test.

B. References

Reference	Title
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
23-51-00-710-801	Flight Interphone System Operational Test (P/B 501)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1390	Headphone - 600 Ohm, with 1/4 Inch Mono Audio Plug

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 23-71-00-860-013

(1) Make sure that the flight interphone system is serviceable (TASK 23-51-00-710-801).

SUBTASK 23-71-00-860-014

(2) Connect a headphone, STD-1390, to the voice recorder's control panel at the pilot's overhead panel, P5.

SUBTASK 23-71-00-860-015

(3) Set the volume control switches on the Audio Control Panels (ACP) to the off position.

F. Voice Recorder System Test

SUBTASK 23-71-00-710-008

(1) Do this task: Voice Recorder System - Operational Test, TASK 23-71-00-710-801.

SUBTASK 23-71-00-710-009

(2) Do this task: Voice Recorder System - Audio Fidelity Test, TASK 23-71-00-730-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-71-00-860-028

(3) Set the CVR AUTO/ON switch (P5) to the ON position as follows:

(a) Make sure that the CVR AUTO/ON switch stays in the ON position.

SUBTASK 23-71-00-730-001

(4) Do a test for the CVR Engine Relay as follows:

(a) Set the CVR AUTO/ON switch to the AUTO position.

(b) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(c) Set the left engine start lever to the IDLE position and do these steps:

1) Make sure that the right engine start lever is in the CUTOFF position.

2) Wait approximately five minutes.

NOTE: The CDS will toggle the engine running relay five minutes after the start lever is placed in the IDLE position.

3) Make sure that you can hear the flight deck conversation in the headphone, STD-1390.

(d) Push and hold the TEST switch on the voice recorder control panel for two seconds.

(e) In five seconds or less, make sure that:

1) You hear a tone in the headphone, STD-1390.

2) The STATUS light comes on.

(f) Set the left engine start lever to the CUTOFF position.

1) Make sure that you cannot hear the flight deck conversation in the headphone, STD-1390.

(g) Set the right engine start lever to the IDLE position.

1) Wait approximately five minutes.

NOTE: The CDS will toggle the engine running relay five minutes after the start lever is placed in the IDLE position.

2) Make sure that you can hear the flight deck conversation in the headphone, STD-1390.

(h) Push and hold the TEST switch on the voice recorder control panel for two seconds.

(i) In five seconds or less, make sure that:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) You hear a tone in the headphone, STD-1390.

2) The STATUS light comes on.

(j) Set the right engine start lever to the CUTOFF position.

1) Make sure that you cannot hear the flight deck conversation in the headphone, STD-1390.

(k) Set the CVR AUTO/ON switch to the ON position (P5 panel).

NOTE: Make sure that the CVR AUTO/ON switch stays in the ON position.

(l) Make sure that you can hear the flight deck conversation in the headphone, STD-1390.

SUBTASK 23-71-00-710-010

(5) Do a test for the Bulk Erasure of the voice recorder as follows:

(a) Make sure that the airplane wheels have chocks in place (TASK 10-11-05-500-801 or TASK 10-11-05-500-802).

(b) Make sure that you connect a headphone, STD-1390, to the voice recorder's control panel.

WARNING MAKE SURE LANDING GEAR WHEELS ARE CHOCKED, BEFORE PARKING BRAKE LEVER IS RELEASED. INJURIES TO PERSONNEL AND DAMAGE TO THE AIRCRAFT CAN OCCUR.

(c) Release the parking brake.

(d) Push and hold the ERASE switch on the voice recorder control panel for three seconds.

1) Make sure that you do not hear a tone in the headphone, STD-1390, in six seconds or less.

(e) Set the parking brake.

(f) Push and hold the ERASE switch on the voice recorder control panel for three seconds.

1) Make sure that you hear a tone in the headphone, STD-1390, in six seconds or less.

WARNING MAKE SURE THAT YOU INSTALL CHOCKS AT THE WHEELS. THE BRAKES WILL RELEASE WHEN YOU PUT THE AIRPLANE IN AIR MODE. THIS CAN CAUSE THE AIRPLANE TO MOVE SUDDENLY. IF YOU DO NOT INSTALL THE CHOCKS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(g) Do the PSEU BITE procedure to put the No. 1 air/ground system in air mode (TASK 32-09-00-860-801).

(h) Push and hold the ERASE switch on the voice recorder control panel for three seconds.

1) Make sure that you do not hear a tone in the headphone, STD-1390, in six seconds or less.

(i) Put the airplane back to the ground mode (TASK 32-09-00-860-802).

(j) Push and hold the ERASE switch on the voice recorder control panel for three seconds.

1) Make sure that you hear a tone in the headphone, STD-1390, in six seconds or less.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-71-00-860-021

(6) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

G. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-00-860-016

(1) Disconnect the headphone, STD-1390, from the voice recorder's control panel.

SUBTASK 23-71-00-860-022

(2) Set the CVR AUTO/ON switch (P5) back to the AUTO position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VOICE RECORDER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Voice Recorder Removal

(2) Voice Recorder Installation.

B. The voice recorder is installed in the voice recorder rack in the aft cargo compartment.

TASK 23-71-11-000-801

2. Voice Recorder Removal

(Figure 401)

A. General

(1) This procedure has steps to remove a voice recorder.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
23-71-21-000-801	Underwater Locator Beacon Removal (P/B 401)

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
822	Aft Cargo Door

E. Prepare for the Voice Recorder Removal

SUBTASK 23-71-11-010-001

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 23-71-11-860-001

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 23-71-11-010-002

(3) Open the access panel on the E6 rack to get access to the voice recorder [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-71-11-020-002

(4) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

F. Voice Recorder Removal

SUBTASK 23-71-11-020-003

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

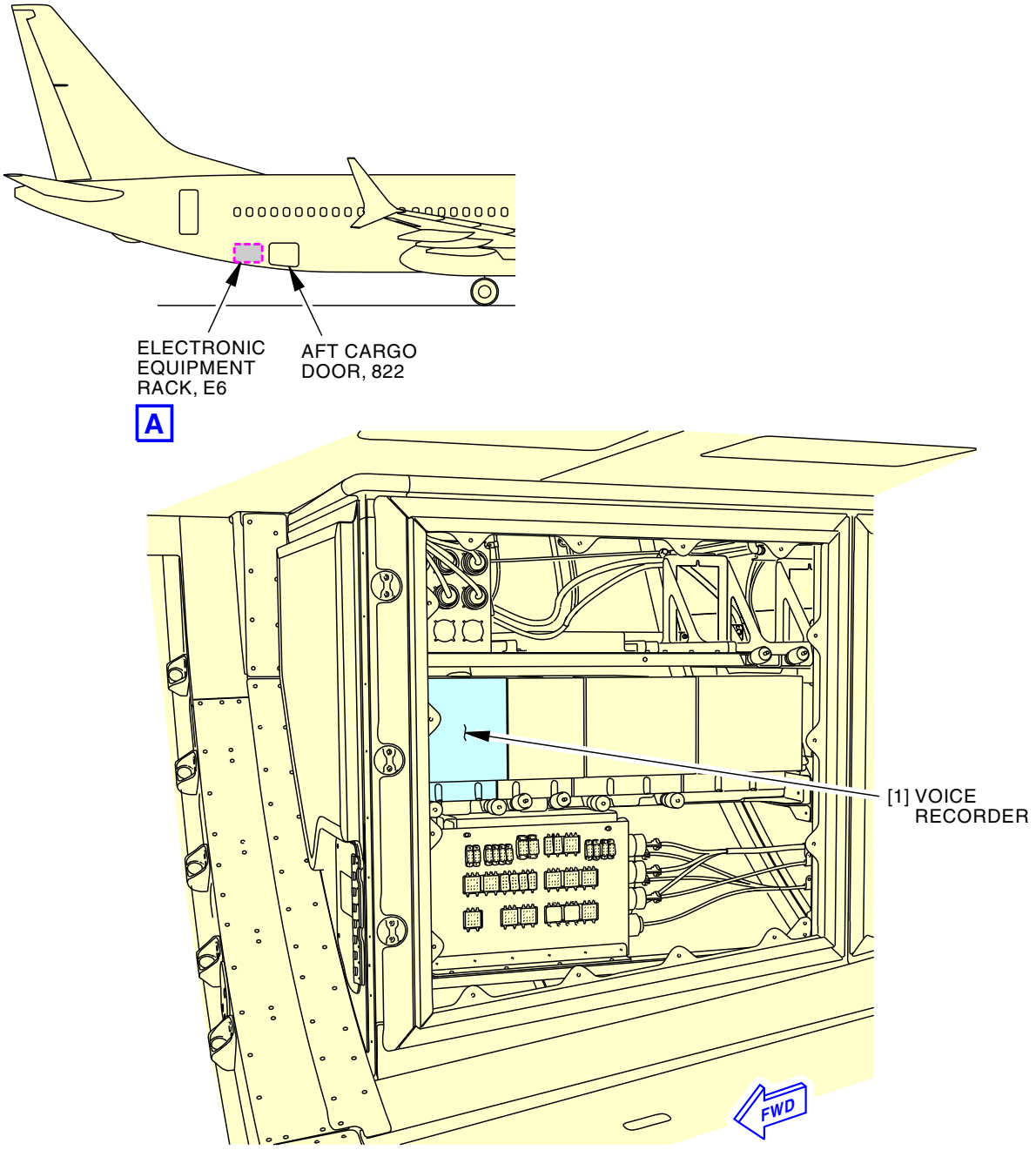
(1) Remove the voice recorder [1], (TASK 20-10-07-000-801).

SUBTASK 23-71-11-020-004

(2) If it is necessary, do this task: Underwater Locator Beacon Removal, TASK 23-71-21-000-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRONIC EQUIPMENT RACK, E6

A

2411717 S00061528814_V3

Voice Recorder Installation
Figure 401/23-71-11-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-71-11-400-801

3. Voice Recorder Installation

(Figure 401)

A. General

(1) This procedure has steps to install a voice recorder.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-71-00-710-801	Voice Recorder System - Operational Test (P/B 501)
23-71-21-400-801	Underwater Locator Beacon Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Voice recorder	23-71-11-01-010	PTF ALL

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

E. Access Panels

Number	Name/Location
822	Aft Cargo Door

F. Prepare for the Voice Recorder Installation

SUBTASK 23-71-11-420-001

(1) If the ULB is removed, do this task: Underwater Locator Beacon Installation, TASK 23-71-21-400-801.

SUBTASK 23-71-11-860-002

(2) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 23-71-11-420-002

(3) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Voice Recorder Installation

SUBTASK 23-71-11-420-003

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(1) Install the voice recorder [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

SUBTASK 23-71-11-410-001

(2) Close the access door on the E6 rack.

SUBTASK 23-71-11-420-004

(3) Connect the connector, D13185 to the Recorder Independent Power Supply (RIPS).

SUBTASK 23-71-11-410-002

(4) Close the lowered ceiling panel in the aft passenger compartment.

H. Voice Recorder Installation Test

SUBTASK 23-71-11-860-003

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 23-71-11-710-001

(2) Do the operational test for the voice recorder (TASK 23-71-00-710-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-11-410-003

(1) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VOICE RECORDER CONTROL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the voice recorder control panel

(2) An installation of the voice recorder control panel.

B. The voice recorder control panel is installed in the pilot's overhead panel P5.

TASK 23-71-12-000-801

2. Voice Recorder Control Panel Removal

(Figure 401)

A. General

(1) This task gives instruction to remove voice recorder control panel.

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 23-71-12-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

D. Voice Recorder Control Panel Removal

SUBTASK 23-71-12-020-002

(1) Loosen the quarter-turn fasteners [2] on the voice recorder control panel [1].

SUBTASK 23-71-12-020-003

(2) Remove the voice recorder control panel [1] to get access to the electrical connectors [3].

SUBTASK 23-71-12-020-004

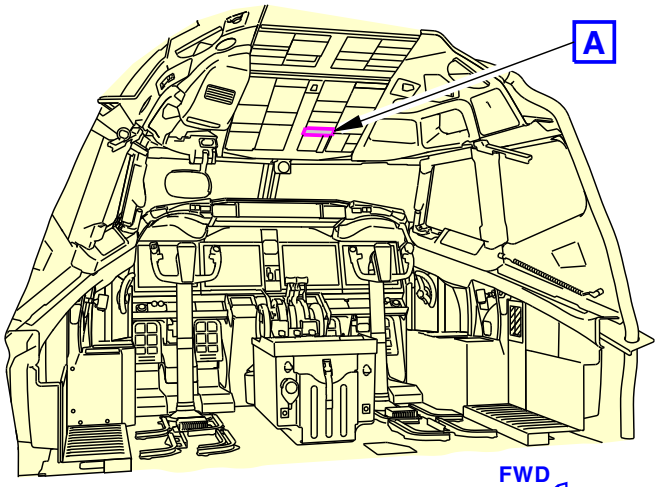
(3) Disconnect the electrical connector [3] from the voice recorder control panel [1].

SUBTASK 23-71-12-020-005

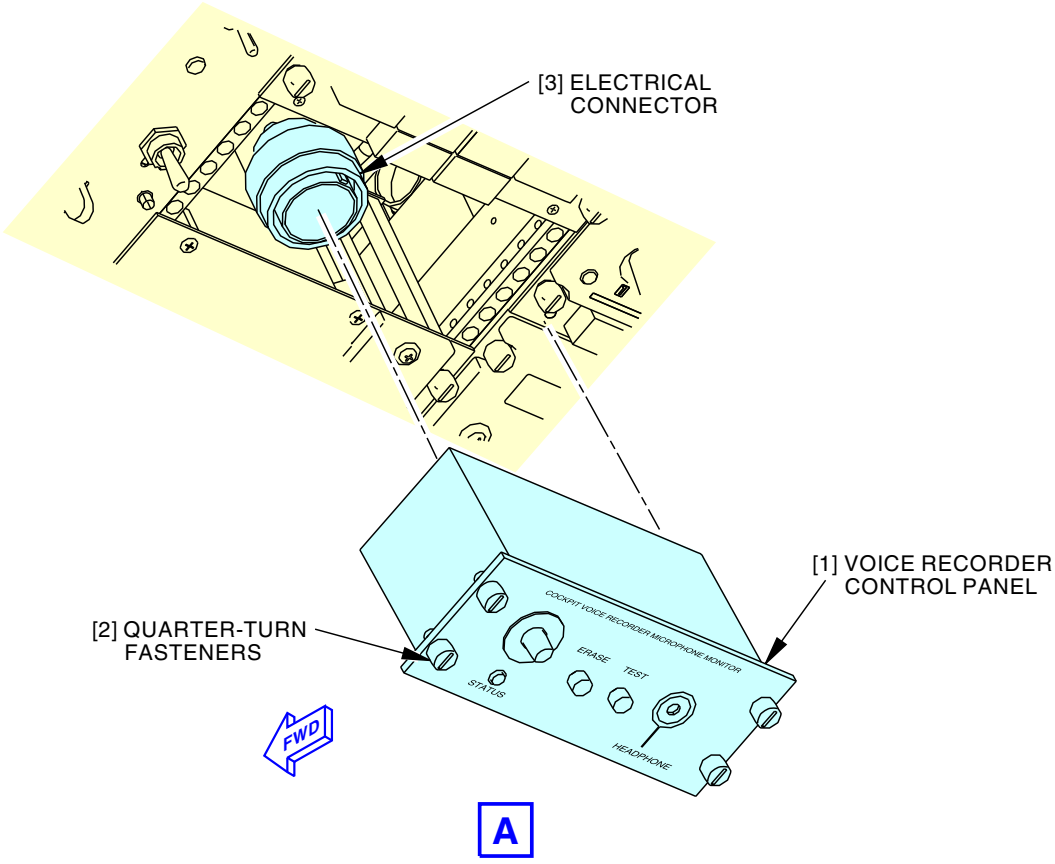
(4) Put a protective cover on the electrical connector [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



2411719 S00061528820_V3

Voice Recorder Control Panel Installation
Figure 401/23-71-12-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-71-12-400-801

3. Voice Recorder Control Panel Installation

(Figure 401)

A. General

(1) This task gives instruction to install the voice recorder control panel.

B. References

Reference	Title
23-71-00-710-801	Voice Recorder System - Operational Test (P/B 501)

C. Tools/Equipment

Reference	Description
STD-1390	Headphone - 600 Ohm, with 1/4 Inch Mono Audio Plug

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Voice recorder control panel	23-71-12-01-050	PTF ALL
		31-11-94-01-255	PTF ALL

E. Location Zones

Zone	Area
212	Flight Compartment - Right

F. Voice Recorder Control Panel Installation

SUBTASK 23-71-12-420-001

(1) Remove the protective cover from the electrical connector [3].

SUBTASK 23-71-12-210-001

(2) Examine the electrical connectors for bent or broken pins, dirt, and damage.

SUBTASK 23-71-12-420-002

(3) Connect the electrical connector [3] to the voice recorder control panel [1].

SUBTASK 23-71-12-420-003

(4) Install the voice recorder control panel [1].

SUBTASK 23-71-12-420-004

(5) Tighten the quarter-turn fasteners [2] on the voice recorder control panel [1].

G. Voice Recorder Control Panel Installation Test

SUBTASK 23-71-12-860-003

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 23-71-12-710-001

(2) Do the operational test of the voice recorder (TASK 23-71-00-710-801).

SUBTASK 23-71-12-860-004

(3) Make sure that the airplane's parking brake is set.

SUBTASK 23-71-12-860-005

(4) Make sure that a headphone is connected to the voice recorder control panel [1], at P5.

SUBTASK 23-71-12-860-006

(5) Set the voice recorder Auto/On switch to the ON position.

SUBTASK 23-71-12-730-001

(6) Push and hold the ERASE switch on the voice recorder control panel [1] for three seconds.

(a) Make sure that you hear a tone in the headphone, STD-1390, in six seconds or less.

SUBTASK 23-71-12-020-006

(7) Disconnect the headphone from the voice recorder control panel [1].

SUBTASK 23-71-12-860-007

(8) Set the CVR AUTO/ON switch (P5) back to the AUTO position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VOICE RECORDER SWITCH PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Voice Recorder Switch Panel Removal

(2) Voice Recorder Switch Panel Installation.

TASK 23-71-13-000-801

2. Voice Recorder Switch Panel Removal

(Figure 401)

A. General

(1) This task has instructions to remove the voice recorder switch panel.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for Removal

SUBTASK 23-71-13-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

D. Voice Recorder Switch Removal

SUBTASK 23-71-13-020-002

(1) Remove the voice recorder switch panel [1]:

(a) Loosen the quarter-turn fasteners [2].

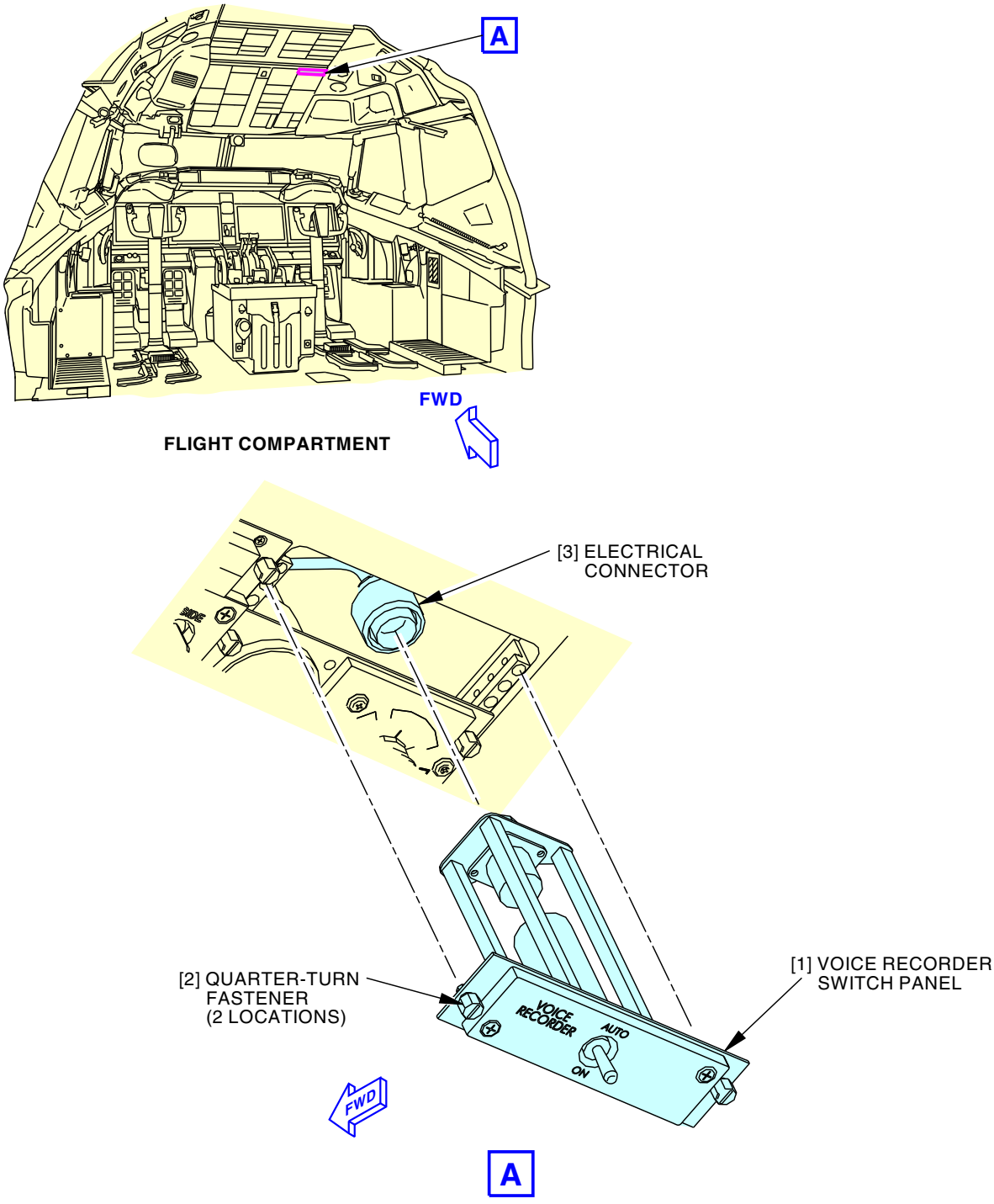
(b) Pull the voice recorder switch panel [1].

(c) Disconnect the electrical connector [3].

(d) Put the protective cover on the electrical connector [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Voice Recorder Switch Panel Installation
Figure 401/23-71-13-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-71-13-400-801

3. Voice Recorder Switch Panel Installation

(Figure 401)

A. General

(1) This task has instructions to install the voice recorder switch panel.

B. Tools/Equipment

Reference	Description
STD-1390	Headphone - 600 Ohm, with 1/4 Inch Mono Audio Plug

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Voice Recorder Switch Panel Installation

SUBTASK 23-71-13-420-001

(1) Install the voice recorder switch panel [1]:

(a) Remove the protective cover from the electrical connector [3].

(b) Connect the electrical connector [3].

(c) Put the voice recorder switch panel [1] into position.

(d) Tighten the quarter-turn fasteners [2].

E. Voice Recorder Switch Panel Installation Test

SUBTASK 23-71-13-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

SUBTASK 23-71-13-420-003

(2) Connect a headphone, STD-1390 to the HEADPHONE jack on the voice recorder control panel.

SUBTASK 23-71-13-710-001

(3) Do these steps to test the voice recorder switch panel [1]:

(a) Set the VOICE RECORDER AUTO/ON switch to the ON position.

(b) Make sure that the switch is latched in the ON position.

(c) Make sure that a conversation is heard in the headphone.

(d) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

(e) Make sure that the VOICE RECORDER AUTO/ON switch is in the AUTO position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY

(g) Make sure that a conversation is not heard in the headphone.

SUBTASK 23-71-13-020-003

(4) Disconnect the headphone, STD-1390 from the HEADPHONE jack on the voice recorder control panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

UNDERWATER LOCATOR BEACON - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Dukane Underwater Locator Beacon Battery - Replacement

(2) L-3 Underwater Beacon Battery - Replacement

B. The underwater locator beacon (ULB) is attached to the front of the voice recorder. The voice recorder is located in the voice recorder rack at the aft cargo compartment.

TASK 23-71-21-960-801

2. Dukane Underwater Locator Beacon Battery - Replacement

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure contains the steps to replace the Dukane Underwater Locator Beacon (ULB) battery.

B. References

Reference	Title
23-71-21-000-801	Underwater Locator Beacon Removal (P/B 401)
23-71-21-400-801	Underwater Locator Beacon Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1619	Wrench - Spanner, Underwater Locator Beacon Part #: 810-2005 Supplier: 94970 Part #: 810-2007/KVS Supplier: 94970 Part #: 810-325 Supplier: 94970 Part #: B362-09111 Supplier: 61257 Opt Part #: 008407 Supplier: 26858 Opt Part #: B362-04180A Supplier: 26858
COM-15071	Wrench - Spanner Part #: 810-325 Supplier: 94970 Part #: 9809800007 Supplier: 06141
COM-15496	Clamp - Vise Part #: 810-546 Supplier: 94970 Part #: 9809800002 Supplier: 06141
STD-1066	Hose - Radiator, Split, 1-1/4 Inch Diameter, 5 Inch Length

D. Consumable Materials

Reference	Description	Specification
D50082	Lubricant - 810-346	
G51422	Kit - Lithium Battery, 810-2042/K	
G51671	Kit - Lithium Battery, 810-2050K	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Dukane Underwater Locator Beacon Battery Removal

SUBTASK 23-71-21-010-001

(1) Remove the ULB, do this task: Underwater Locator Beacon Removal, TASK 23-71-21-000-801.

SUBTASK 23-71-21-800-001

WARNING DO NOT REMOVE THE BATTERY FROM THE UNDERWATER LOCATOR BEACON. THE BATTERY HAS DANGEROUS MATERIALS AND CAN CAUSE SERIOUS INJURY TO PERSONNEL.

(2) If you have a DK100/90 or DK130/90 ULB, send it to the manufacturer for servicing.

SUBTASK 23-71-21-020-014

(3) If you have a DK120/90 or DK140/90 ULB, remove the battery [28] as follows:

CAUTION DO NOT PUT THE UNDERWATER LOCATOR BEACON IN A VISE. DAMAGE TO THE UNDERWATER LOCATOR BEACON CAN OCCUR.

(a) Hold the ULB [4] with a radiator hose - 1-1/4 Inch Diameter, 5 Inch Length, STD-1066, or vise clamp, COM-15496.

(b) Use the spanner wrench, COM-1619, or spanner wrench, COM-15071, to remove the end cap [25] that is identified BATTERY ACCESS.

(c) Remove the rubber shock cushion [27] from the battery end if it is not removed with the end cap [25].

(d) Remove and discard the o-ring [26] from the end cap [25].

(e) Hit the ULB [4] lightly to remove the battery [28].

SUBTASK 23-71-21-020-015

(4) If you have a DK290 ULB, remove the batteries [28] as follows:

CAUTION DO NOT CLAMP THE BEACON IN A VISE, UNLESS YOU USE VISE CLAMP COM-15496. IF YOU DO NOT, IT CAN CAUSE DAMAGE.

(a) Hold the ULB [4] with a radiator hose - 1-1/4 Inch Diameter, 5 Inch Length, STD-1066, or vise clamp, COM-15496.

(b) Use the spanner wrench, COM-1619, or spanner wrench, COM-15071, to remove the end caps [25].

CAUTION DO NOT USE A STEEL SCREWDRIVER OR SHARP TOOL. THIS WILL PREVENT DAMAGE TO THE O-RING GROOVE.

(c) Remove and discard the o-rings [26] from the end caps [25].

(d) Hit the ULB [4] lightly to remove the batteries [28].

F. Dukane Underwater Locator Beacon Battery Installation

SUBTASK 23-71-21-420-014

(1) If you have a DK120/90 or DK140/90 ULB, install the new battery [28] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the new battery is the same as the battery code on the ULB label. See the table below.

BATTERY CODE	REQUIRED BATTERY KIT
F	810-2042/K

NOTE: The Dukane 810-2042/K battery is a 7 year lithium battery used in the Dukane model DK120/90 and DK140/90 ULB.

NOTE: The battery, o-ring and lubricant are provided in the battery replacement kit, G51422.

CAUTION DO NOT REVERSE THE POLARITY OF THE UNDERWATER LOCATOR BEACON BATTERY. DAMAGE TO THE UNDERWATER BEACON WILL OCCUR.

(b) Put the new battery [28] in the ULB [4] with the end identified INSERT THIS END in first.

CAUTION DO NOT GET DIRT OR UNWANTED MATERIALS NEAR THE O-RING. DAMAGE TO THE O-RING AND THREADS CAN CAUSE DAMAGE TO THE UNDERWATER LOCATOR BEACON.

(c) Clean the threads and the o-ring groove in the ULB [4].

(d) Apply a thin layer of lubricant, D50082, to the new o-ring [26], o-ring groove, and threads.

(e) Install the new o-ring [26] on the end cap [25].

(f) Put the rubber shock cushion [27] smoothly on the end cap [25].

(g) Hold the ULB [4] with a radiator hose - 1-1/4 Inch Diameter, 5 Inch Length, STD-1066, or vise clamp, COM-15496.

(h) Put the end cap [25] into the ULB [4].

(i) Hand tighten the end cap [25] with a spanner wrench, COM-1619, or spanner wrench, COM-15071, until the cap flange touches the ULB [4].

(j) Put a new battery replacement label [23] on the ULB [4].

(k) On the battery replacement label [23] write the next scheduled replacement date for the new battery [28].

NOTE: The battery replacement label [23] is blank so you can write in a replacement date based on your maintenance schedule.

SUBTASK 23-71-21-420-015

(2) If you have a DK290 ULB, install the new batteries [28] as follows:

(a) Make sure that the new batteries are the same as the battery code on the ULB label.

NOTE: The Dukane 810-2050K batteries are 6 year lithium batteries used in the Dukane model DK290 ULB.

NOTE: The battery, o-ring and lubricant are provided in the battery replacement kit, G51671.

(b) Clean the threads and the o-ring grooves in the ULB [4] and end caps [25].

(c) Apply a thin layer of lubricant, D50082, to the new o-rings [26], o-ring grooves, and threads.

(d) Install the newo-rings [26] on the end caps [25].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) For each battery [28], do these steps:

NOTE: There are two batteries [28], one at each end of the ULB [4].

CAUTION

DO NOT REVERSE THE POLARITY OF THE UNDERWATER LOCATOR BEACON BATTERY. DAMAGE TO THE UNDERWATER BEACON WILL OCCUR.

1) Put the new battery [28] in the ULB [4].
   a) Make sure that the bullseye pattern side of the battery [28] is inserted first into the ULB [4].
2) Hold the ULB [4] with a radiator hose - 1-1/4 Inch Diameter, 5 Inch Length, STD-1066, or vise clamp, COM-15496.
3) Put the end cap [25] into the ULB [4].
4) Hand tighten the end cap [25] with a spanner wrench, COM-1619, or spanner wrench, COM-15071, until the cap flange touches the ULB [4] or has a clearance of 0.003 in. (0.076 mm) or less.

(f) Put a new battery replacement label [23] on the ULB [4].

(g) On the battery replacement label [23], write the next scheduled replacement date for the new batteries [28].

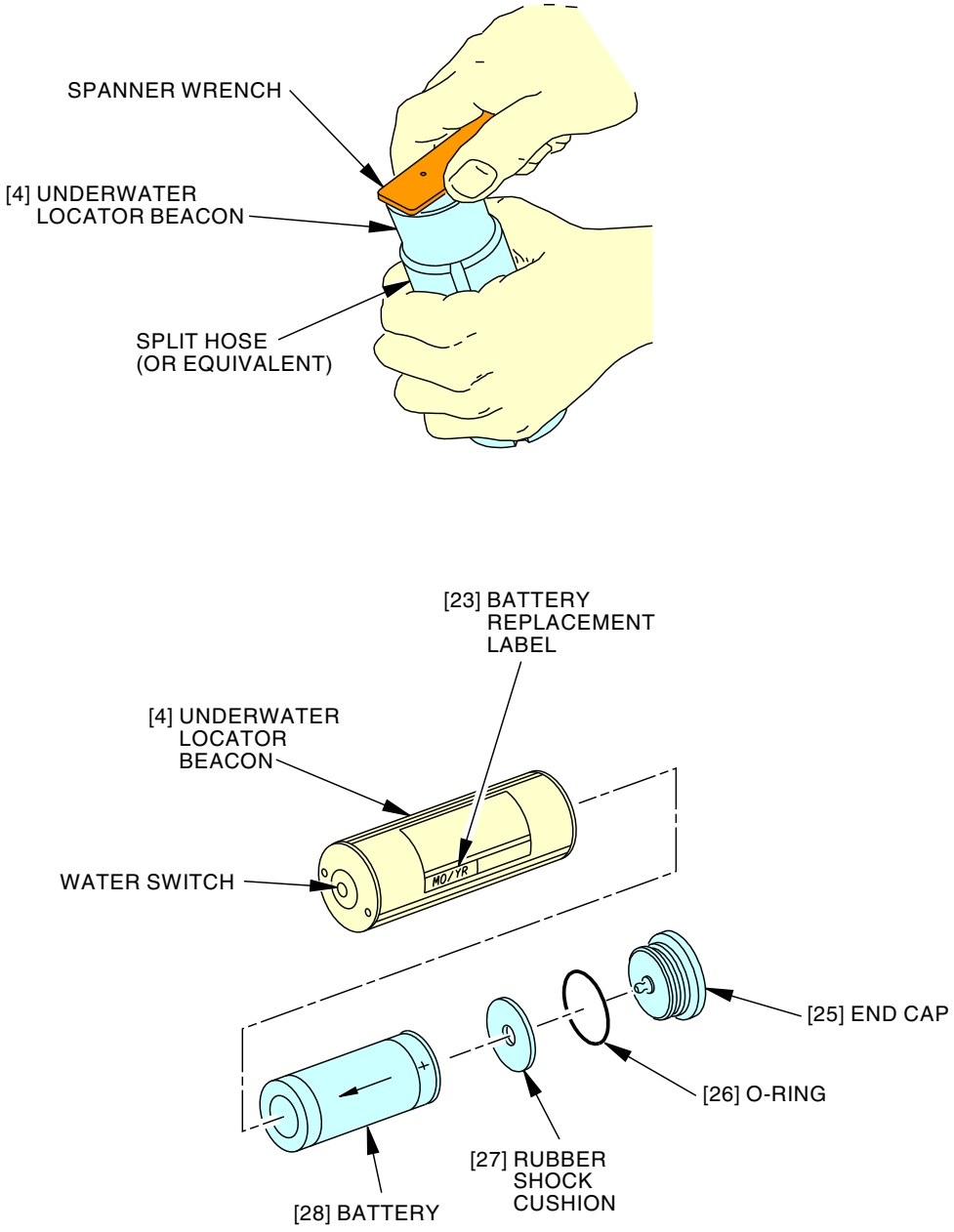
NOTE: The battery replacement label [23] is blank so you can write in a replacement date based on your maintenance schedule.

SUBTASK 23-71-21-410-001

(3) Install the ULB, do this task: Underwater Locator Beacon Installation, TASK 23-71-21-400-801.

——— END OF TASK ———

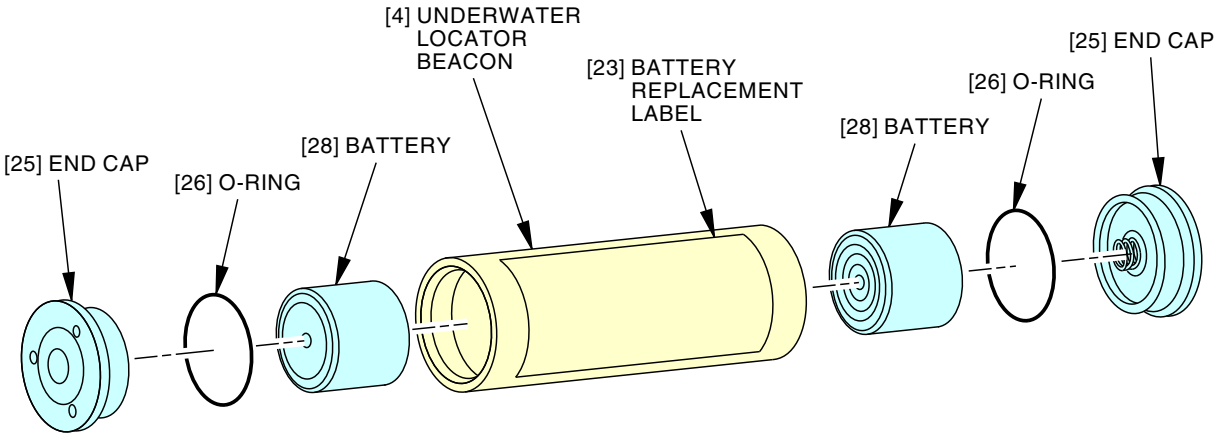
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2411722 S00061528828_V2

Underwater Locator Beacon Battery Replacement
Figure 201/23-71-21-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2954530 S0000732223_V2

Underwater Locator Beacon Battery Replacement
Figure 201/23-71-21-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-71-21-960-803

3. L-3 Underwater Locator Beacon Battery - Replacement

(Figure 202)

A. General

(1) This procedure contains the steps to replace the L-3 Underwater Locator Beacon (ULB) battery.

(a) Use a Battery Replacement Kit, P/N 147E5542−00, to replace the ULB battery.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-15071	Wrench - Spanner Part #: 810-325 Supplier: 94970 Part #: 9809800007 Supplier: 06141
COM-15496	Clamp - Vise Part #: 810-546 Supplier: 94970 Part #: 9809800002 Supplier: 06141

C. Consumable Materials

Reference	Description	Specification
D50082	Lubricant - 810-346	

D. L-3 Underwater Locator Beacon Battery Removal

SUBTASK 23-71-21-000-001

CAUTION REMOVE ALL OF THE CONTAMINATION FROM THE THREADS AND THE O-RING GROOVES. CONTAMINATION CAN CAUSE DAMAGE TO THE THREADS. THREAD DAMAGE CAN CAUSE LEAKS.

(1) Remove the battery [28] from the L-3 ULB [4].

(a) Secure the L-3 ULB [4] with vise clamp, COM-15496.

CAUTION DO NOT GET DIRT OR UNWANTED MATERIALS NEAR THE O-RING. DAMAGE TO THE O-RING AND THREADS CAN CAUSE DAMAGE TO THE UNDERWATER LOCATOR BEACON.

(b) Use the spanner wrench, COM-15071, to remove the battery end cap [25] with two wrench holes by unscrewing counterclockwise.

NOTE: The breakaway torque is usually high, so the spanner wrench should be held firmly in contact with the battery end cap in order to prevent damage to wrench holes.

(c) Remove the rubber shock cushion [27] from the battery end if it is not removed with the battery end cap [25].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

DO NOT USE A STEEL SCREWDRIVER OR SHARP TOOL. THIS WILL PREVENT DAMAGE TO THE O-RING GROOVE.

(d) Remove the o-ring [26] from the end cap [25].

(e) Remove the battery [28].

1) Make sure that the rubber shock cushion [27] remains in the battery end cap [25].

E. L-3 Underwater Locator Beacon Battery Installation

SUBTASK 23-71-21-420-013

CAUTION

DO NOT INSTALL A BATTERY OTHER THAN THAT FROM THE BATTERY REPLACEMENT KIT. IF YOU DO NOT FOLLOW THIS CAUTION, IT CAN CAUSE DAMAGE.

CAUTION

DO NOT INSTALL A 90-DAY BATTERY IN A 30-DAY BEACON OR A 30-DAY BATTERY IN A 90-DAY BEACON. IF YOU DO, IT CAN CAUSE DAMAGE.

(1) Make sure that the new battery is same as the battery on the ULB battery replacement label [23] (Table 201).

Table 201/23-71-21-993-802 Beacon Battery Replacement Kits

Description	Kit Part Number
30-Day Battery	147E030500
90-Day Battery	147E554200

SUBTASK 23-71-21-400-001

CAUTION

REMOVE ALL OF THE CONTAMINATION FROM THE THREADS AND THE O-RING GROOVES. CONTAMINATION CAN CAUSE DAMAGE TO THE THREADS. THREAD DAMAGE CAN CAUSE LEAKS.

CAUTION

MAKE SURE THAT YOU INSTALL THE BATTERY CORRECTLY. INCORRECT INSTALLATION OF THE BATTERY WILL CAUSE PERMANENT DAMAGE TO THE BEACON.

CAUTION

DO NOT CLAMP THE BEACON IN A VISE, UNLESS YOU USE VISE CLAMP COM-15496. IF YOU DO NOT, IT CAN CAUSE DAMAGE.

(2) Install the L-3 ULB battery [28]:

(a) Apply a thin coating of lubricant, D50082, to new o-ring [26], o-ring groove and threads on end cap [25].

(b) Carefully install the new o-ring [26] on the battery end cap [25].

(c) Install new battery [28].

1) Make sure that the end marked “INSERT THIS END FIRST” goes in first.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Position rubber shock cushion [27] around contact spring and on the inside of the battery end cap [25].

(e) Install the battery end cap [25] and tighten it until the cover flange contacts the body or leaves less than a 0.003 in. (0.076 mm) gap.

1) Hold the L-3 ULB in the vise clamp, COM-15496, use only hand force to tighten.

(f) Clean the L-3 ULB [4] exterior of excess lubricant.

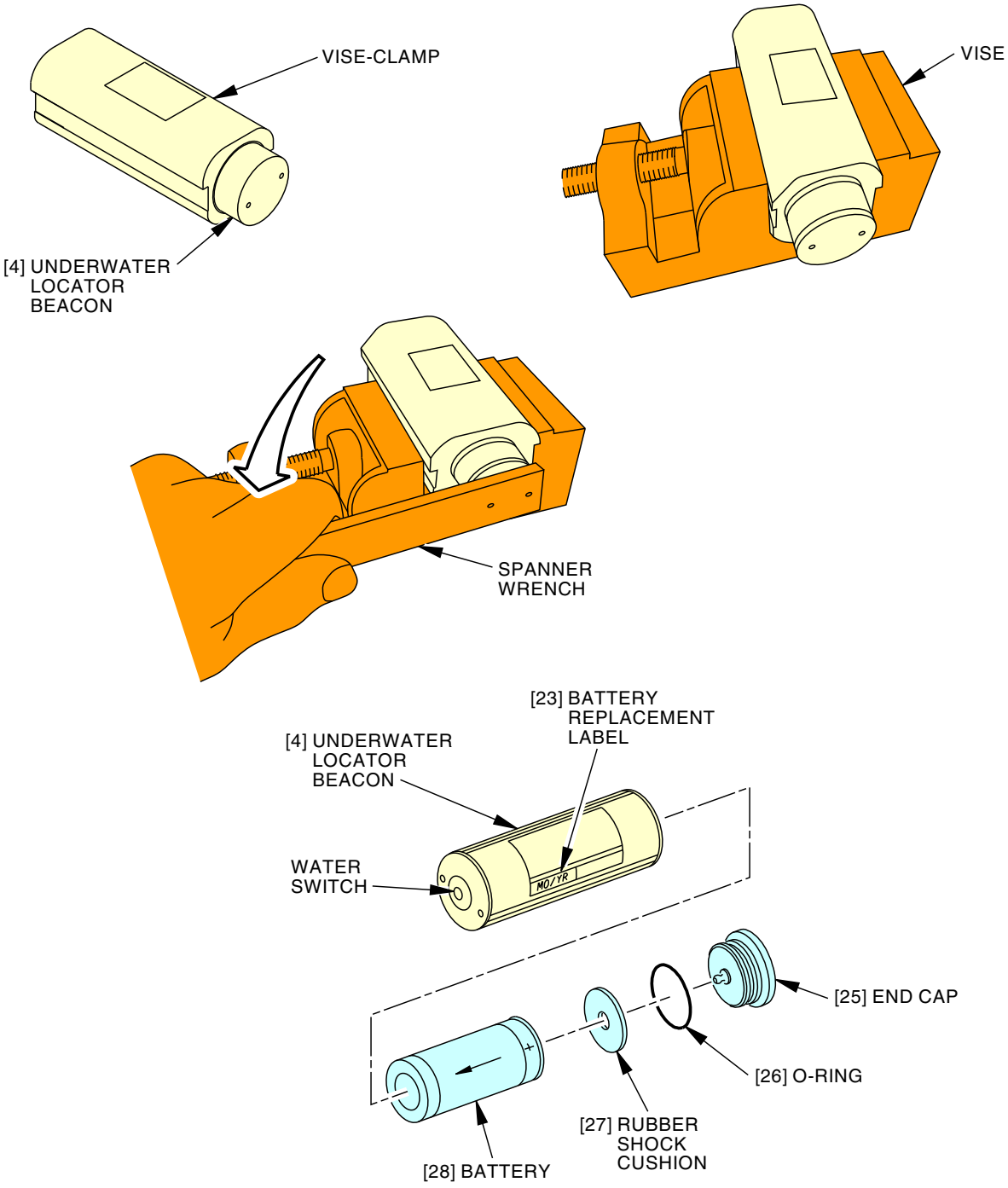
(g) Replace the battery replacement label [23] on the outside of the L-3 ULB [4] case, with the new label that is supplied with the battery.

1) Locate the battery replacement label [23] on the L-3 ULB [4] body in the space provided.

a) Do not put label on the water switch end of the beacon.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2797221 S0000638845_V1

Underwater Locator Beacon Battery Replacement
Figure 202/23-71-21-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

UNDERWATER LOCATOR BEACON - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ULB

(2) An installation of the ULB.

TASK 23-71-21-000-801

2. Underwater Locator Beacon Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to remove the Underwater Locator Beacon (ULB) from the cockpit voice recorder.

(2) The ULB only has a battery as the power source.

B. References

Reference	Title
23-71-11-000-801	Voice Recorder Removal (P/B 401)

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

D. Underwater Locator Beacon Removal

SUBTASK 23-71-21-020-003

(1) If it is necessary, do this task: Voice Recorder Removal, TASK 23-71-11-000-801.

SUBTASK 23-71-21-020-004

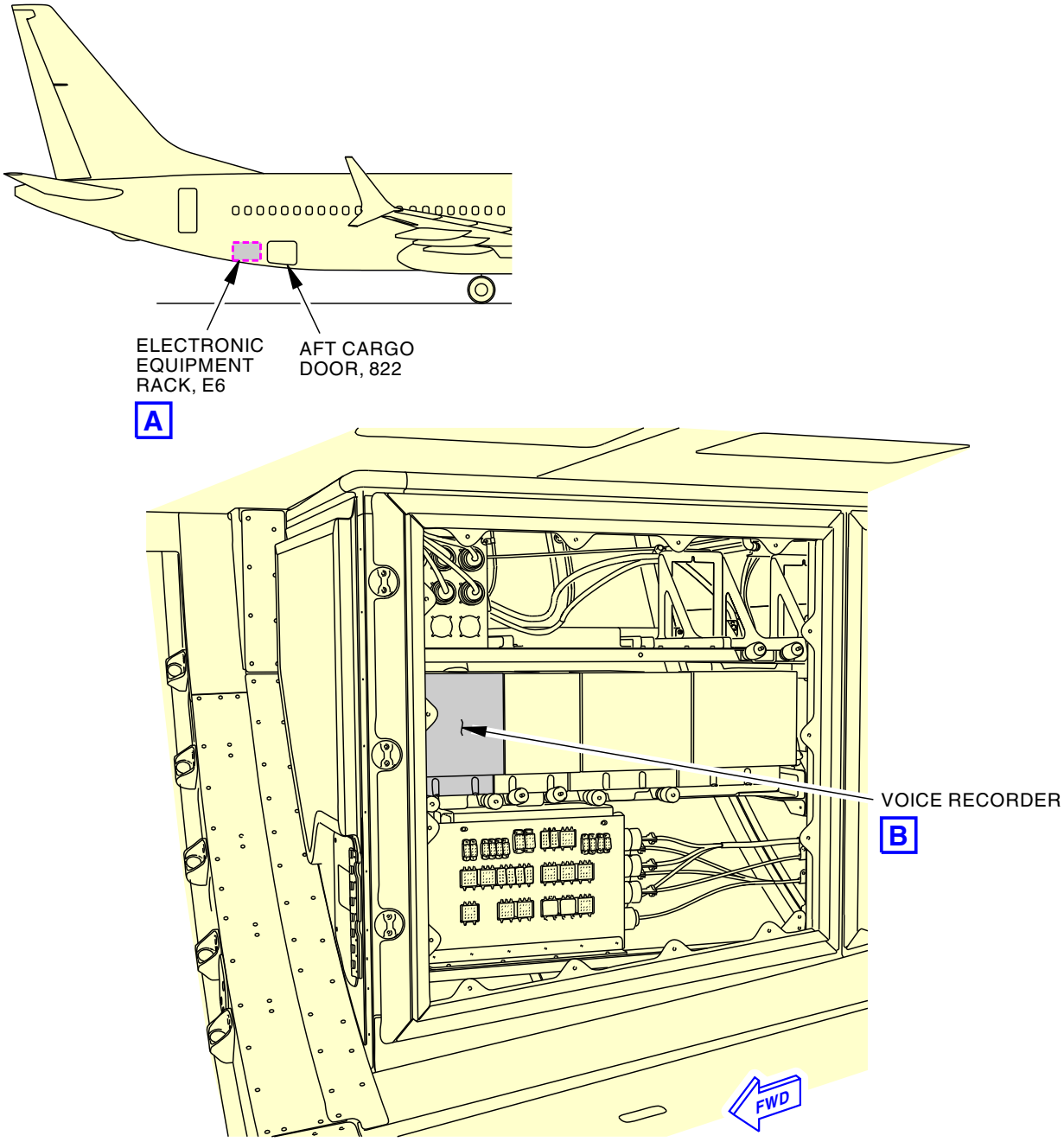
(2) Remove the ULB [1] from the cockpit voice recorder:

(a) Remove the screws [3] that attach the ULB [1] to the supports.

(b) Remove the ULB [1] from the supports.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



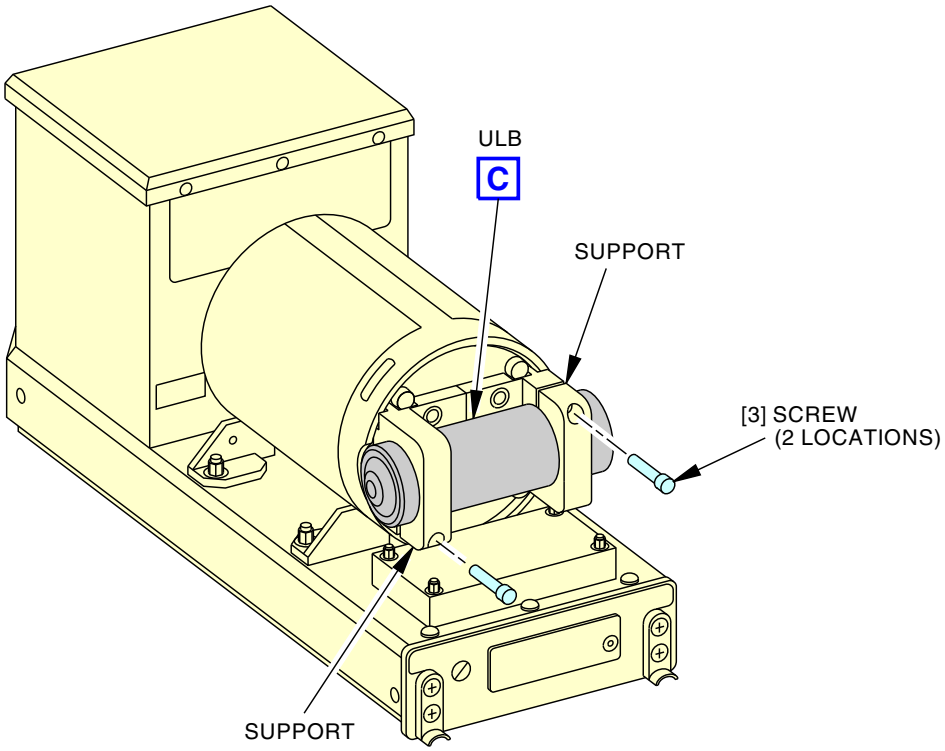
ELECTRONIC EQUIPMENT RACK, E6

A

2411720 S00061528825_V2

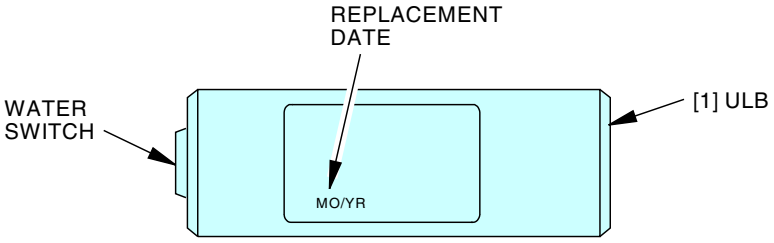
Underwater Locator Beacon Installation
Figure 401/23-71-21-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RECORDER
(EXAMPLE)

B



UNDERWATER LOCATOR BEACON

C

2983565 S0000761070_V1

Underwater Locator Beacon Installation
Figure 401/23-71-21-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-71-21-400-801

3. Underwater Locator Beacon Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to install the Underwater Locator Beacon (ULB) to the cockpit voice recorder.

(2) The ULB only has a battery as the power source.

B. References

Reference	Title
23-71-11-400-801	Voice Recorder Installation (P/B 401)
23-71-21-730-801	Underwater Locator Beacon Test with a 42A12 Test Set (P/B 501)
23-71-21-730-802	Underwater Locator Beacon Test with a PL1 Test Set (P/B 501)
23-71-21-730-803	Underwater Locator Beacon Test with a PL3 Test Set (P/B 501)
23-71-21-730-806	Underwater Locator Beacon Test with a TS200 Test Set (P/B 501)
23-71-21-730-807	Underwater Locator Beacon Test with a L-3AR Test Set (P/B 501)
23-71-21-730-808	Underwater Locator Beacon Test with a TS500 Test Set (P/B 501)

C. Consumable Materials

Reference	Description	Specification
B00541	Cleaner - General Purpose Household Detergent	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	ULB	23-71-21-01-040	PTF ALL

E. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

F. Underwater Locator Beacon Pre-Installation Test

SUBTASK 23-71-21-730-001

(1) If the ULB is repaired or replaced, do the ULB test with applicable test set (Underwater Locator Beacon Test with a 42A12 Test Set, TASK 23-71-21-730-801 or Underwater Locator Beacon Test with a PL1 Test Set, TASK 23-71-21-730-802 or Underwater Locator Beacon Test with a PL3 Test Set, TASK 23-71-21-730-803 or Underwater Locator Beacon Test with a TS200 Test Set, TASK 23-71-21-730-806 or Underwater Locator Beacon Test with a L-3AR Test Set, TASK 23-71-21-730-807 or Underwater Locator Beacon Test with a TS500 Test Set, TASK 23-71-21-730-808).

G. Underwater Locator Beacon Installation

SUBTASK 23-71-21-420-003

(1) Install the ULB [1] on the cockpit voice recorder:

(a) Make sure that the water switch of the ULB [1] has no grease or dirt.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Clean the water switch with a weak general purpose household detergent cleaner, B00541.

(b) Install the ULB [1] into the supports.

(c) Make sure that you can read the replacement date on the ULB [1].

(d) Install the screws [3].

(e) Tighten the screws [3] to 13 in-lb (1.47 N·m) - 20 in-lb (2.26 N·m).

SUBTASK 23-71-21-420-004

(2) If it is necessary, do this task: Voice Recorder Installation, TASK 23-71-11-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

UNDERWATER LOCATOR BEACON - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Underwater Locator Beacon Test with a 42A12 Test Set

(2) Underwater Locator Beacon Test with a PL1 Test Set

(3) Underwater Locator Beacon Test with a PL3 Test Set

(4) Underwater Locator Beacon Test with a TS200 Test Set

(5) Underwater Locator Beacon Test with a L-3AR Test Set

(6) Underwater Locator Beacon Test with a TS500 Test Set

(7) Underwater Locator Beacon Operational Test with a TS500 Test Set

(8) Underwater Locator Beacon Voltage Test with a TS500 Test Set

TASK 23-71-21-730-801

2. Underwater Locator Beacon Test with a 42A12 Test Set

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to do a test of the Underwater Locator Beacon (ULB) with a 42A12 Test Set.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-10768	Test Set - 42A12, Underwater Locator Beacon (ULB) Opt Part #: 42A12-1 Supplier: 94970
STD-11018	Voltmeter - High Impedance (Minimum 10Mohm Input Impedance)

C. Battery Test

SUBTASK 23-71-21-760-001

(1) Do a check of the battery voltage:

(a) Make sure that the case and water switch are clean and dry.

(b) Place the negative lead of the high impedance voltmeter, STD-11018, on the water switch pin and the positive of the meter on the beacon case or mounting kit.

(c) Measure the battery voltage.

1) Refer to the ULB label for the minimum permitted range of the beacon battery voltage:

a) Code: F - 2.97 Volts

b) P/N: DK290 - 2.75 Volts.

D. Operation Test

SUBTASK 23-71-21-730-002

(1) If you have a 42A12 ULB Test Set, COM-10768, do this test of the ULB:

NOTE: 42A12 can do a test for all ULBs.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put the 42A12 ULB Test Set, COM-10768 as close as possible to the ULB.

(b) Set the GAIN control switch on the 42A12 ULB Test Set, COM-10768 to the maximum clockwise position.

NOTE: A background noise is heard. If you do not hear noise from the test set, replace the test set battery.

(c) Set the TUNING control switch to the middle position.

(d) Make sure that the 42A12 ULB Test Set, COM-10768 operates correctly.

1) Rub your thumb and fingers together in front of the microphone to make sure that it operates.

NOTE: This will produce a rushing noise from the speaker.

a) Make sure that you hear sounds through the speaker.

(e) Use tape to attach a piece of wire, a shorting tab, or other conductive material to the ULB case and to the center of the water switch.

NOTE: This will make a short circuit from the center of the water switch to the outer part of the ULB.

(f) Set the GAIN control switch to a comfortable listening level.

(g) Point the microphone of the test set towards the water switch end of the beacon for best results.

1) Make sure you hear a pulse tone.

(h) Remove the wire, shorting tab, or other conducting material from the ULB case and the center of the water switch.

(i) Set the GAIN control switch to the OFF position.

(j) Make sure that the water switch on the ULB has no grease or dirt.

(k) If necessary, do the steps that follow:

1) Clean the switch with water and detergent.

2) Dry the switch with a clean cloth.

END OF TASK

TASK 23-71-21-730-802

3. Underwater Locator Beacon Test with a PL1 Test Set

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to do a test of the Underwater Locator Beacon with a PL1 Test Set.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-10771	Test Set - PL1, Underwater Locator Beacon (ULB) Opt Part #: PL1 Supplier: 94970

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
G00270	Tape - Scotch Flatback Masking 250	ASTM D6123 (Supersedes A-A-883)

D. Procedure

SUBTASK 23-71-21-730-003

(1) If you have a PL1 ULB test set, COM-10771, do this test of the underwater locator beacon (ULB):

NOTE: PL1 can only do a test for the DK100–90 and DK120–90 ULB.

(a) Use Scotch Flatback Masking Tape 250, G00270 to attach a piece of wire or other conductive material to the ULB case and the center of the water switch.

NOTE: This will make a short circuit from the center of the water switch to the outer part of the ULB.

(b) Put the end of the PL1 ULB test set, COM-10771 against the ULB, approximately one inch from the water switch.

(c) Push and hold the operation switch on the PL1 ULB test set, COM-10771.

1) Make sure that the BEACON ACTIVE WHEN FLASHING light flashes.

2) Remove the piece of wire or other conductive material from the ULB case and the center of the water switch.

3) Make sure that the BEACON ACTIVE WHEN FLASHING light does not flash.

(d) Release the operation switch on the PL1 ULB test set, COM-10771.

(e) Remove the PL1 ULB test set, COM-10771.

(f) Make sure that the water switch end of the ULB has no grease or dirt.

(g) If necessary, do the steps that follow:

1) Clean the switch with water and detergent.

2) Dry the switch with a clean cloth.

——— END OF TASK ———

TASK 23-71-21-730-803

4. Underwater Locator Beacon Test with a PL3 Test Set

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to do a test of the Underwater Locator Beacon with a PL3 Test Set.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-10772	Test Set - PL3, Underwater Locator Beacon (ULB) Opt Part #: PL3 Supplier: 94970

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Procedure

SUBTASK 23-71-21-730-004

(1) If you have a PL3 ULB test set, COM-10772, do this test of the underwater locator beacon (ULB):

NOTE: PL3 can only do a test for the DK100–90 and DK120–90 ULBs.

(a) Push and hold the PL3 ULB test set, COM-10772 against the ULB water switch.

1) Make sure that you hear a pulse sound.

2) Make sure that you see the LED light comes on and off.

(b) Remove the PL3 ULB test set, COM-10772.

(c) Make sure that the water switch end of the ULB has no grease or dirt.

(d) If necessary, do the steps that follow:

1) Clean the switch with water and detergent.

2) Dry the switch with a clean cloth.

——— END OF TASK ———

TASK 23-71-21-730-806

5. Underwater Locator Beacon Test with a TS200 Test Set

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to do a test of the Underwater Locator Beacon with a TS200 Test Set.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-10770	Test Set - TS200, Underwater Locator Beacon (ULB) Opt Part #: TS200 Supplier: 94970

C. Procedure

SUBTASK 23-71-21-730-007

(1) If you have a TS200 ULB test set, COM-10770, do this test of the Underwater Locator Beacon (ULB):

NOTE: TS200 can do a test for all ULBs except Honeywell ULB DK290-11. Use Test Set TS500 for Honeywell ULB DK290-11.

(a) Attach the test probe clip of the TS200 ULB test set, COM-10770, to the beacon in its mount.

(b) Put the tip of the probe on the silver pad of the water switch at the end of the beacon.

1) The LCD display will show the battery voltage of the beacon.

(c) Refer to the applicable battery code for the minimum permitted range of the beacon battery voltage:

NOTE: Examine the battery replacement label to find the battery code.

1) Code F – 2.97 Volts

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Push the red button on the TS200 ULB test set, COM-10770.

1) The beacon starts and you hear a pinging noise from the TS200 ULB test set, COM-10770.

(e) Remove the test probe clip of the TS200 ULB test set, COM-10770.

(f) Replace the ULB if necessary.

(g) Make sure that the water switch end of the ULB has no grease or dirt.

(h) If necessary, do the steps that follow:

1) Clean the switch with water and detergent.

2) Dry the switch with a clean cloth.

——— END OF TASK ———

TASK 23-71-21-730-807

6. Underwater Locator Beacon Test with a L-3AR Test Set

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to do a test of the Underwater Locator Beacon with a L-3AR Test Set.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-15070	Test Set - 17TES0015, Underwater Locator Beacon (ULB) Part #: 17TES0015 Supplier: 06141

C. Procedure

SUBTASK 23-71-21-730-008

(1) If you have a underwater locator beacon test set, COM-15070, do this test of the underwater locator beacon (ULB):

NOTE: L-3AR ULB test set can only do a test for the L-3 266–E5542–00 ULBs.

(a) Attach the test probe clip of the test set to the beacon case.

(b) Place the tip of the probe in contact with the metal pad at the end of the water switch.

1) The display will show the battery voltage of the beacon.

2) The minimum permitted voltage of the battery is 2.97 Volts.

(c) Press and hold the red button on the test set.

1) The beacon starts and you hear a pinging noise from the test set.

(d) Remove the test probe of the test set.

(e) Replace the ULB if necessary.

(f) Make sure that the water switch end of the ULB has no grease or dirt.

(g) If necessary, do the steps that follow:

1) Clean the switch with water and detergent.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Dry the switch with a clean cloth.

——— END OF TASK ———

TASK 23-71-21-730-808

7. Underwater Locator Beacon Test with a TS500 Test Set

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to do a test of the Underwater Locator Beacon (ULB) with a TS500 Test Set.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-19623	Test Set - TS500, Underwater Locator Beacon (ULB) Part #: TS500 Supplier: 94970

C. Underwater Locator Beacon Test with a TS500 Test Set

SUBTASK 23-71-21-730-009

(1) Testing for the ULB battery voltage:

(a) Attach the test probe clip of the TS500 underwater locator beacon test set, COM-19623 to the beacon in its mount.

(b) Put the tip of the probe on the silver pad of the water switch at the end of the beacon.

1) The LCD display will show the battery voltage of the beacon.

(c) Refer to the ULB label for the minimum permitted range of the beacon battery voltage:

1) Code: F - 2.97 Volts

2) P/N: DK290 - 2.75 Volts

(2) Testing for the operation of the ULB:

(a) Keep the probes connected to the ULB.

NOTE: The test set display will remain blank during this test.

(b) Push the red button on the TS500 underwater locator beacon test set, COM-19623.

1) The beacon starts and you hear a pinging noise from the TS500 underwater locator beacon test set, COM-19623.

2) The TS500 underwater locator beacon test set, COM-19623 LED indicator will light up red every time the ULB pings.

(c) Remove the test probe clip of the TS500 underwater locator beacon test set, COM-19623.

(3) If voltage or operation test does not pass, replace the ULB if necessary.

(a) Make sure that the water switch end of the ULB has no grease or dirt.

(b) If necessary, do the steps that follow:

1) Clean the switch with water and detergent.

2) Dry the switch with a clean cloth.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-71-21-710-801

8. Underwater Locator Beacon Operational Test with a TS500 Test Set

A. General

(1) This procedure has steps to do the operational test of the Underwater Locator Beacon (ULB) with a TS500 Test Set.

B. References

Reference	Title
23-71-21-000-801	Underwater Locator Beacon Removal (P/B 401)
23-71-21-400-801	Underwater Locator Beacon Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-19623	Test Set - TS500, Underwater Locator Beacon (ULB) Part #: TS500 Supplier: 94970

D. Prepare for the ULB Operation Test

SUBTASK 23-71-21-020-012

(1) If it is necessary, remove the ULB (TASK 23-71-21-000-801).

SUBTASK 23-71-21-480-001

(2) Set the TS500 underwater locator beacon test set, COM-19623, power switch to ON.

E. ULB Operation Test

SUBTASK 23-71-21-710-001

(1) Do the operational test of the ULB:

(a) Hold the TS500 underwater locator beacon test set, COM-19623, test probes onto the ULB water switch caps.

NOTE: The polarity is shown on the ULB label.

(b) Push the red button on the TS500 underwater locator beacon test set, COM-19623.

NOTE: The test set display will remain blank during this test.

1) Make sure that you hear the beep from the TS500 underwater locator beacon test set, COM-19623.

NOTE: The test set red LED indicator will come on with every beep.

(c) Remove the test probes.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-21-080-001

(1) Set the TS500 underwater locator beacon test set, COM-19623, power switch to OFF.

SUBTASK 23-71-21-420-011

(2) If removed, install the ULB (TASK 23-71-21-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-71-21-720-801

9. Underwater Locator Beacon Voltage Test with a TS500 Test Set

A. General

(1) This procedure has steps to do the voltage test of the Underwater Locator Beacon (ULB) with a TS500 Test Set.

B. References

Reference	Title
23-71-21-000-801	Underwater Locator Beacon Removal (P/B 401)
23-71-21-400-801	Underwater Locator Beacon Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-19623	Test Set - TS500, Underwater Locator Beacon (ULB) Part #: TS500 Supplier: 94970

D. Prepare for the ULB Voltage Test

SUBTASK 23-71-21-020-013

(1) If it is necessary, remove the ULB (TASK 23-71-21-000-801).

SUBTASK 23-71-21-480-002

(2) Set the TS500 underwater locator beacon test set, COM-19623, power switch to ON.

E. ULB Voltage Test

SUBTASK 23-71-21-720-001

(1) Do the voltage test of the ULB:

(a) Refer to the ULB label to get the minimum required voltage:

1) Code: F - 2.97 Volts

2) P/N: DK290 - 2.75 Volts

(b) Hold the TS500 underwater locator beacon test set, COM-19623, test probes onto the ULB water switch caps.

NOTE: The polarity is shown on the ULB label.

1) Make sure that the voltage shown on the TS500 underwater locator beacon test set, COM-19623, display is not below the minimum requirement.

(c) Remove the test probes.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-21-080-002

(1) Set the TS500 underwater locator beacon test set, COM-19623, power switch to OFF.

SUBTASK 23-71-21-420-012

(2) If removed, install the ULB (TASK 23-71-21-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

UNDERWATER LOCATOR BEACON - CLEANING/PAINTING

1. General

A. This procedure has the task to clean the underwater locator beacon.

TASK 23-71-21-100-801

2. Underwater Locator Beacon Cleaning

A. General

(1) This procedure has steps to clean the Underwater Locator Beacon (ULB).

B. References

Reference	Title
23-71-21-000-801	Underwater Locator Beacon Removal (P/B 401)
23-71-21-400-801	Underwater Locator Beacon Installation (P/B 401)

C. Prepare to clean the ULB

SUBTASK 23-71-21-020-011

(1) If it is necessary, remove the ULB (TASK 23-71-21-000-801).

D. Clean the ULB

SUBTASK 23-71-21-100-001

(1) Clean the ULB switch caps as follows:

(a) Clean the switch caps with a soft cloth with water and detergent.

(b) Dry the switch caps with a clean cloth.

(c) Make sure that the switch caps do not have grease or dirt.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 23-71-21-420-010

(1) If removed, install the ULB (TASK 23-71-21-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT DECK ENTRY VIDEO SURVEILLANCE SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Video Surveillance System - Deactivation.

(2) Video Surveillance System - Activation.

TASK 23-75-00-040-801

2. Video Surveillance System - Deactivation

A. General

(1) This procedure removes the electrical power to the Video Surveillance System.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Procedure

SUBTASK 23-75-00-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	2	C04093	SURVEILLANCE CAMERA DC

D. Video Surveillance System - Tryout

SUBTASK 23-75-00-211-001

NOTE: This tryout is to make sure the Video Surveillance system is in a zero energy state.

(1) Do the following steps to verify the Video Surveillance System is off:

(a) Select a camera on the Camera Control Panel.

NOTE: The Camera Control Panel is in the flight compartment on the aisle stand P8 panel.

(b) View the MAX Display System.

(c) If the Video Surveillance System is off there will be no display from any of the cameras.

——— END OF TASK ———

TASK 23-75-00-440-801

3. Video Surveillance System - Activation

A. General

(1) This procedure adds electrical power to the Video Surveillance System.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Procedure

SUBTASK 23-75-00-860-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	2	C04093	SURVEILLANCE CAMERA DC

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT DECK ENTRY VIDEO SURVEILLANCE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has one task:

(1) Flight Deck Entry Video Surveillance System - System Test.

TASK 23-75-00-730-801

2. Flight Deck Entry Video Surveillance System - System Test

A. References

Reference	Title
31-65-00 P/B 501	MAX DISPLAY SYSTEM - ADJUSTMENT/TEST

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Procedure

SUBTASK 23-75-00-860-003

(1) Make sure that the P5-13 IFE/PASS SEAT switch in the Flight Deck is in the ON position.

SUBTASK 23-75-00-860-004

(2) Make sure that the MAX Display System (MDS) is serviceable (PAGEBLOCK 31-65-00/501).

D. Test Procedure

SUBTASK 23-75-00-710-001

(1) Push the “DSPL” button on the Camera Control Panel (CCP).

NOTE: The CCP is on the aisle stand, P8 panel in the flight compartment.

(a) Make sure that a video image appears on the captain's or the first officer's inboard display unit.

SUBTASK 23-75-00-710-006

(2) Rotate the “CAM SEL” knob on the CCP to the position “R”.

(a) Make sure that the right camera (Main camera) image shows on the captain's or the first officer's inboard display unit.

NOTE: The camera image will show the Door 1 Right area.

SUBTASK 23-75-00-710-007

(3) Rotate the “CAM SEL” knob on the CCP to the position “C”.

(a) Make sure that the center camera (Secondary camera) image shows on the captain's or the first officer's inboard display unit.

NOTE: The camera image will show the area for the aft of the flight deck door and the Door 1 Left area.

SUBTASK 23-75-00-710-005

(4) Set the P5-13 IFE/PASS SEAT switch to OFF.

(a) Make sure that the surveillance camera images no longer show on the captain's or the first officer's inboard display unit.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VIDEO CAMERAS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the video camera

(2) An installation of the video camera.

B. The camera assemblies are installed in the ceiling above the flight compartment, and the aft of the flight deck security door.

TASK 23-75-01-000-801

2. Video Camera Removal

(Figure 401)

A. General

(1) This task is for the removal of these cameras:

(a) Main camera on the forward galley doorway header 1R.

(b) Secondary camera on the back-side of the forward entry ceiling panel.

NOTE: On the camera Control Panel (CP), position R is main camera and position C is secondary camera.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)
25-21-34-000-801	Forward Galley Door Doorway Header Panel Removal (P/B 401)
25-21-71-000-806	Forward Entry Ceiling Panel Removal (P/B 401)

C. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

D. Prepare for the Removal

SUBTASK 23-75-01-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	2	C04093	SURVEILLANCE CAMERA DC

SUBTASK 23-75-01-010-001

(2) Get access to the applicable camera:

(a) To get access to main camera, do this task: Forward Galley Door Doorway Header Panel Removal, TASK 25-21-34-000-801.

(b) To get access to secondary camera, do this task: Forward Entry Ceiling Panel Removal, TASK 25-21-71-000-806.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Video Camera Removal

SUBTASK 23-75-01-860-002

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

CAUTION DO NOT PULL, TWIST, OR BEND THE CAMERA CABLE MORE THAN THE MINIMUM RADIUS OF 1 INCH (25 MM). IF YOU BEND, PULL, OR TWIST THE CABLE TOO MUCH, DAMAGE TO THE CABLE CAN OCCUR.

(1) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 23-75-01-020-007

(2) For the main camera with the shroud mounted on the outside of the panel, do these steps:

(a) Remove the screws [33] and washers [34] that hold the shroud [32] to the panel.

(b) Remove the screws [35] and washers [36] that hold the main camera [30] to the shroud [32].

SUBTASK 23-75-01-020-008

(3) For the secondary camera with the shroud mounted on the outside of the panel, do these steps:

(a) Remove the screws [39] and washers [40] that hold the shroud [38] to the camera mounting bracket [41].

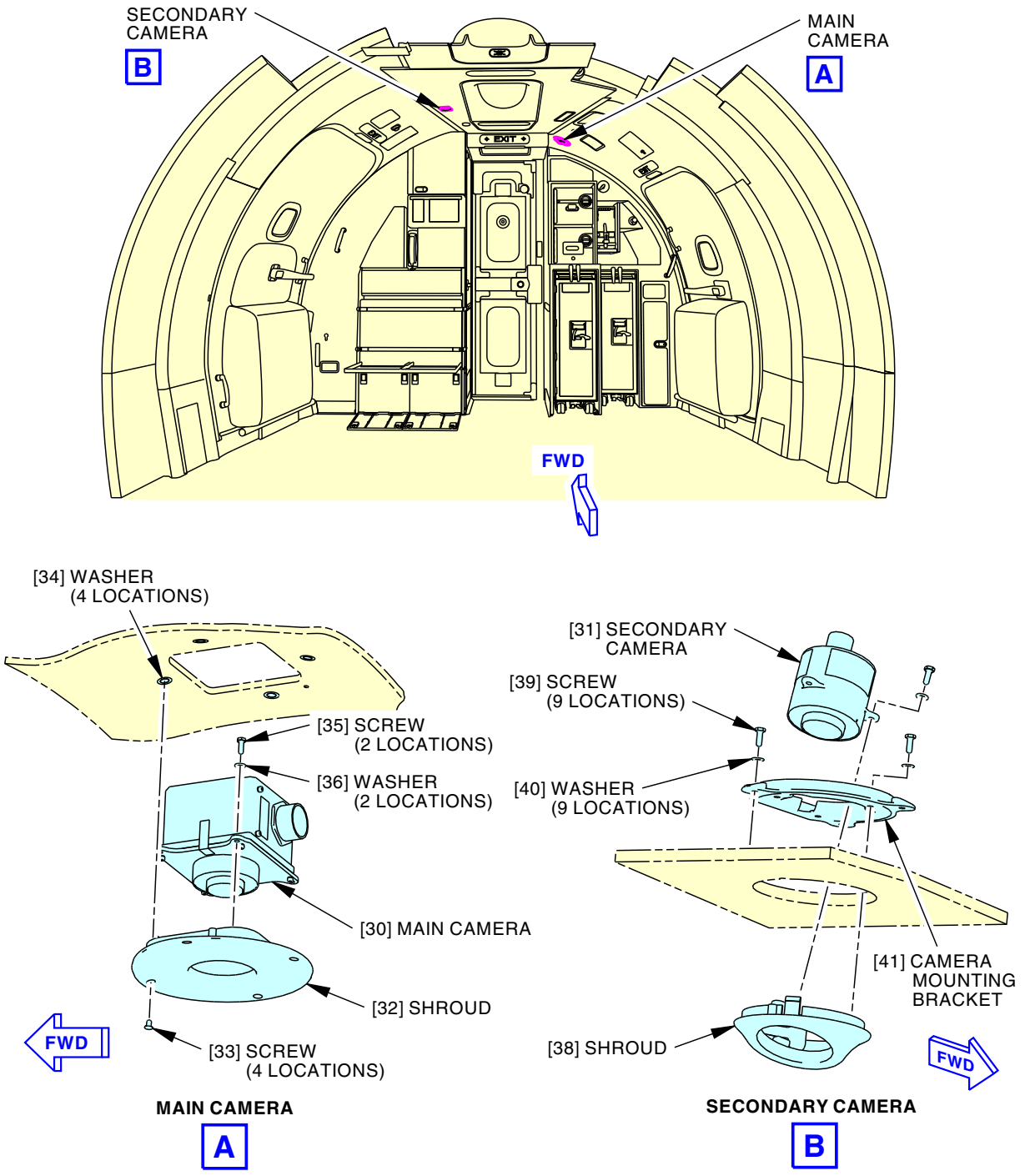
(b) Remove the screws [39] and washers [40] that hold the secondary camera [31] to the shroud [38].

SUBTASK 23-75-01-480-001

(4) Install a conductive dust cap and connector cover on the video camera electrical connector (TASK 20-40-12-400-803).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Video Camera Installation
Figure 401/23-75-01-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-75-01-400-801

3. Video Camera Installation

(Figure 401)

A. General

(1) This task is for the installation of the video camera assemblies.

B. References

Reference	Title
20-40-12-000-803	Conductive Dust Cap and Connector Cover Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
23-75-00-730-801	Flight Deck Entry Video Surveillance System - System Test (P/B 501)
25-21-34-400-801	Forward Galley Door Doorway Header Panel Installation (P/B 401)
25-21-71-400-806	Forward Entry Ceiling Panel Installation (P/B 401)

C. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

D. Video Camera Installation

SUBTASK 23-75-01-860-004

CAUTION DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

CAUTION DO NOT PULL, TWIST, OR BEND THE CAMERA CABLE MORE THAN THE MINIMUM RADIUS OF 1 INCH (25 MM). IF YOU BEND, PULL, OR TWIST THE CABLE TOO MUCH, DAMAGE TO THE CABLE CAN OCCUR.

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 23-75-01-080-001

(2) Remove the protective cover from the video camera electrical connector (TASK 20-40-12-000-803).

SUBTASK 23-75-01-420-008

(3) For the main camera, M3382, with the shroud mounted on the outside of the panel, do these steps:

(a) Attach the main camera [30] to shroud [32] with screws [35] and washers [36].

(b) Attach the shroud [32] to the doorway header panel with screws [33] and washers [34].

SUBTASK 23-75-01-420-009

(4) For the secondary camera, M3383, with the shroud mounted on the outside of the panel, do these steps:

(a) Attach the secondary camera [31] to shroud [38] with screws [39] and washers [40].

(b) Attach the shroud [38] to the camera mounting bracket [41] with screws [39] and washers [40].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Put the Airplane Back to Its Usual Condition

SUBTASK 23-75-01-860-006

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	2	C04093	SURVEILLANCE CAMERA DC

SUBTASK 23-75-01-410-005

(2) For main camera, do this task: Forward Galley Door Doorway Header Panel Installation, TASK 25-21-34-400-801.

SUBTASK 23-75-01-410-006

(3) For secondary camera, do this task: Forward Entry Ceiling Panel Installation, TASK 25-21-71-400-806.

F. Installation Test

SUBTASK 23-75-01-730-001

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, TASK 23-75-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAMERA CONTROL PANEL ASSEMBLY- REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Camera Control Panel Removal

(2) Camera Control Panel Installation.

B. The camera control panel is installed on the aisle stand P8 panel, in the flight compartment.

TASK 23-75-02-000-801

2. Camera Control Panel Removal

(Figure 401)

A. General

(1) This task shows how to remove the camera control panel for the flight deck entry video surveillance system.

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Prepare for the Removal

SUBTASK 23-75-02-860-005

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	2	C04093	SURVEILLANCE CAMERA DC

D. Camera Control Panel Removal

SUBTASK 23-75-02-020-001

(1) Loosen the captive fasteners [2] on the camera control panel [1].

SUBTASK 23-75-02-020-002

(2) Lift the camera control panel [1] to get access to the electrical connector [3].

SUBTASK 23-75-02-020-003

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

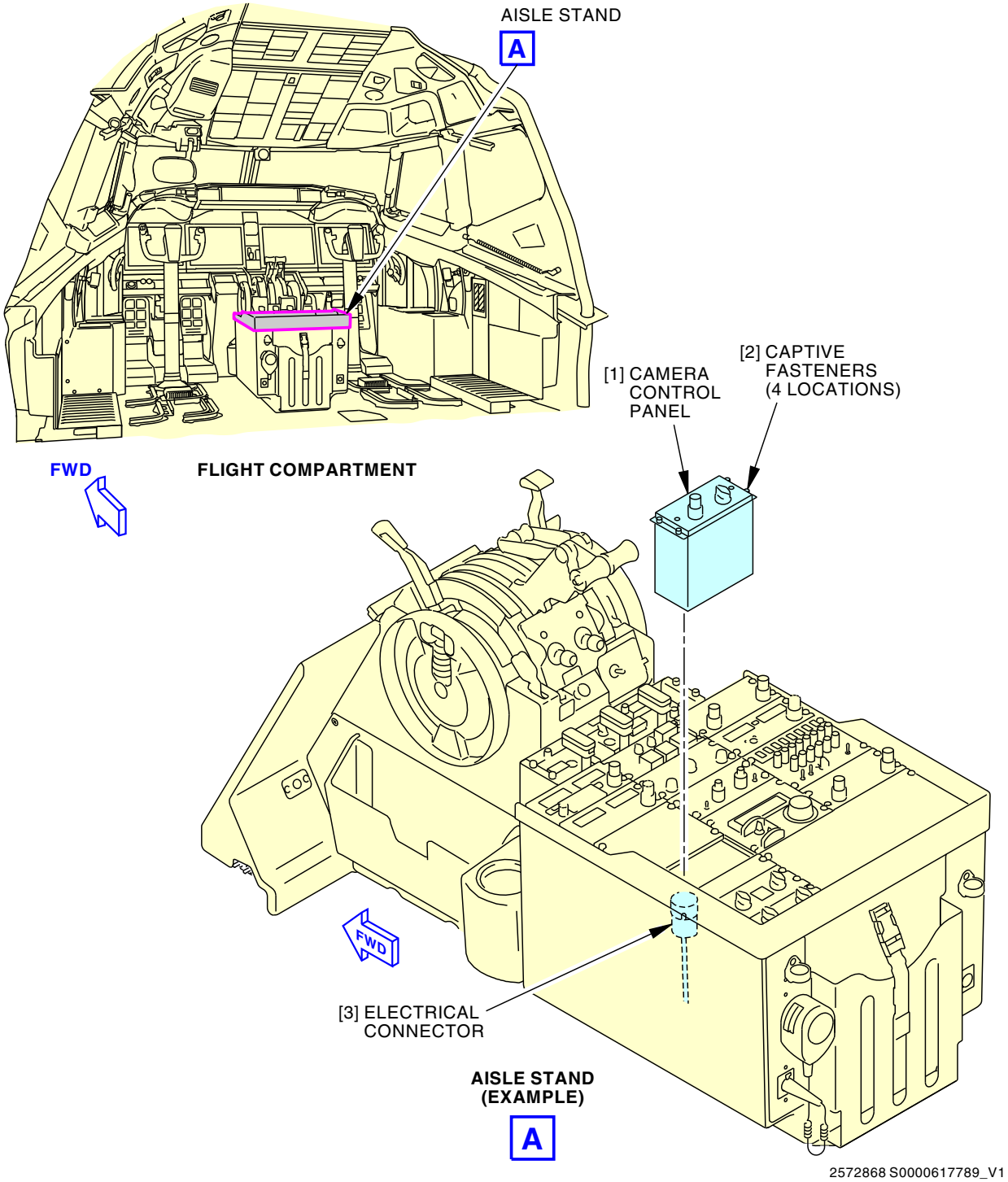
(3) Disconnect the electrical connector [3] from the camera control panel [1].

SUBTASK 23-75-02-860-002

(4) Put a protective cover on the electrical connector [3].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Camera Control Panel
Figure 401/23-75-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 23-75-02-400-801

3. Camera Control Panel Installation

(Figure 401)

A. General

(1) This task is for the installation of the camera control panel for the flight deck entry video surveillance system.

B. References

Reference	Title
23-75-00-730-801	Flight Deck Entry Video Surveillance System - System Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Camera control panel	23-75-02-02-010	PTF ALL

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

E. Camera Control Panel Installation

SUBTASK 23-75-02-420-001

(1) Install the camera control panel as follows:

(a) Remove the protective cover from the electrical connector [3].

(b) Examine the electrical connector [3] for dirt, broken pins, and damage.

(c) Connect the electrical connector [3] to the camera control panel [1].

(d) Lower the camera control panel [1] into the aisle stand.

(e) Tighten the captive fasteners [2] on the camera control panel [1].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 23-75-02-860-006

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
B	2	C04093	SURVEILLANCE CAMERA DC

G. Camera Control Panel Installation Test

SUBTASK 23-75-02-730-001

(1) Do this task: Flight Deck Entry Video Surveillance System - System Test, TASK 23-75-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details